A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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1 Topological Preliminaries

This section aims to set the playing ground of the thesis. Since Morse theory needs the topological spaces to be rather restricted and the goal is to relate Morse Homology to topological K-Theory, there is an urge to just not worry about topological nuances. However, for proving things we still need the vocabulary of topological notions. This chapter serves to provide the notions and theorems with references whilst omitting the technical proofs. Steenrod himself writes in his article "A convenient category of topological spaces" [Ste67]:

Steenrod1967

> "Our need is to be able to make a variety of constructions and to know that the results have good properties without the tedious spelling out at each step of lengthy hypotheses such as countably paracompact normal, completely regular, first axiom of countability, metrizable, and so forth. It may be good research technique and an enjoyable exercise to analyse the precise circumstances for which an argument works; but if a developing theory is to be handy for research workers and attractive to students, then simplicity of the fundamentals must be the goal."

Whilst Steenrod argues for the category of *compactly generated Hausdorff spaces* the standard nowadays is the category of *compactly generated weak Hausdorff spaces*. The latter has direct limits, whilst Steenrod's doesn't. One might wonder why we don't just use the category of CW-complexes, like Hatcher does. There are two reasons for this. The main goal is a Morse theoretic Spectral sequence, which relies on a Morse theoretic filtration. It would be a theoretical drawback to assume that all spaces already omit a CW filtration, as such a filtration naturally already leads to a Atiyah Hirzebruch Spectralsequence. The second reason is, that whilst a lot of pointset topological technicalities can be omitted using CW-complexes, not all of them vanish. This results in the necessity of a discussion of those technicalities, or ignoring the details. Hence we stay away from CW complexes. Let us start constructing this convenient category. All topological constructions that are taught in the compulsory section of a university are assumed to be known. Most proves are omitted but can be found in [Str].

strickland_cgwh

1.1 Compactly Generated Weak Hausdorff Spaces

Remark 1.1.1 (Notation and Convention)**.** The literature on topological spaces in geometric contexts is widely non-uniform. For this work, a topological space is *compact* if it is quasi-compact (every open cover has a finite subcover) and Hausdorff (every pair of points can be separated by disjoint open subsets).

def:topPre-compactlyGenerated

Definition 1.1.2 (Compactly Generated)**.** A topological space X is **compactly generated**, if a subset $A \subseteq X$ is closed, if and only if for all compact K and continuous $f : K \to X$ $f^{-1}(A)$ is closed.

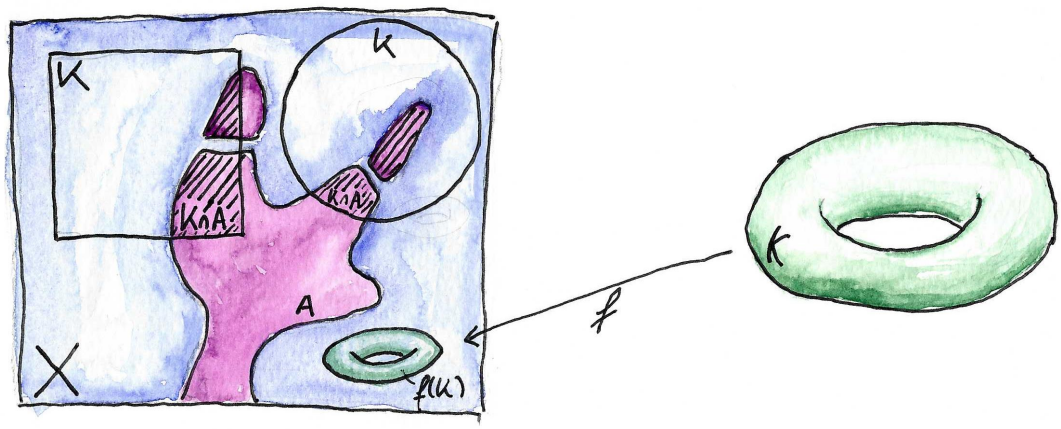


Figure 1: The definition of compactly generated weak Hausdorff spaces.

def:topPre-weakHausdorff

Definition 1.1.3 (Weak Hausdorff)**.** A topological space X is called weak Hausdorff, if for all compact spaces K and continuous map $f : K \to X$ the image $f(K)$ is closed in X.

Definition 1.1.4. We call the category of compactly generated weak Hausdorff spaces together with continuous functions as morphisms **CGWH**.

prop:topPre-spacesThatAreCGWH

Proposition 1.1.5.

1. Every closed subset of a compactly generated weak Hausdorff space is compactly generated weak Hausdorff in the subspace topology.
2. Every locally compact Hausdorff space, and hence every compact space, is compactly generated weak Hausdorff.
3. Every manifold is compactly generated weak Hausdorff.
4. Every metric space is compactly generated weak Hausdorff.
5. Every disjoint union of compactly generated weak Hausdorff spaces is compactly generated weak Hausdorff.
6. If X is compactly generated weak Hausdorff and Y locally compact Hausdorff, then $X \times Y$ is compactly generated weak Hausdorff in the product topology.
7. If X is compactly generated weak Hausdorff and $Y \subseteq X$ a closed subset of X. Then the quotient topology on X/Y is compactly generated and weak Hausdorff.

There is another caveat and reason for the usage of **CGWH**. If we view two CW-complexes as topological spaces, their product might not be a CW-complex. However as objects in **CGWH** their product is a CW-complex. The reason behind this is that the general product doesn't produce the final topology; when it fails to be compactly generated it is strictly coarser than the CW-topology. Passing to the product in **CGWH**

(hence making the topologgy finer to become compactly generated) repairs this, and the product of CW-complexes is again a CW-complex.

Remark 1.1.6. From now on all topological spaces will be compactly generated weak Hausdorff. For such spaces the product $X \times Y$ always means the product in **CGWH**.Proposition 1.1.5 (6) tells us: If one of the factors is also locally compact Hausdorff, the product in **CGWH** agrees with the classsical product.

For the construction of topological K-Theory we will heavily rely on the two theorems from pointset topology:

Theorem 1.1.7 (Tietze Extension Theorem)**.** *Let X be a normal space (and hence the theorem is true in* **CGWH***), $Y \subseteq X$ a closed subspace, V a finite dimensional real vector space, and $f : Y \to V$ a continuous map. Then there exists a continuous map $g : X \to V$ such that $g|_Y = f$.*

Theorem 1.1.8 (Existence of Partitions of Unity)**.** *Let X be a compact space, $\{U_i\}$ a finite open covering. Then there exist continuous maps $f_i : X \to \mathbb{R}$ such that:*

1. *$f_i(x) \geq 0$ for all $x \in X$.*
2. *$\operatorname{supp}(f_i) \subseteq U_i$.*
3. *$\sum_i f_i(x) = 1$ for all $x \in X$.*

Such a collection $\{f_i\}$ is called a ***partition of unity****.*

1.2 Topological Pairs

Definition 1.2.1 (Homotopy Extension Property)**.** Let (X, A) be a topological pair, i.e. $A \subseteq X$. The pair satisfies the homotopy extension property if for any map $f : X \to Y$ and any homotopy $f_t : A \to Y$ with $f_0 = f|_A$ there exists an extension $\tilde{f}_t : X \to Y$ with $\tilde{f}_t|_A = f_t$ and $\tilde{f}_0 = f$. The inclusion $i : A \to X$ of such a pair is called a **cofibration**. We denote the category of pairs of compactly generated weak Hausdorff spaces where the inclusion is a cofibration by $\mathbf{CGWH}^2_{\mathbf{cof}}$.

A more intuitive but (for A closed in X) equivalent definition is:

Definition 1.2.2. If (X, A) is a topological pair, A is closed in X and there is a continuous function $u : X \to I$ and a Homotopy $H : X \times I \to X$ such that

Falsche Definition !!

1. $H(x, 0) = x$ for all $x \in X$.
2. $H(a, t) = a$ for all $a \in A$ and $t \in I$.

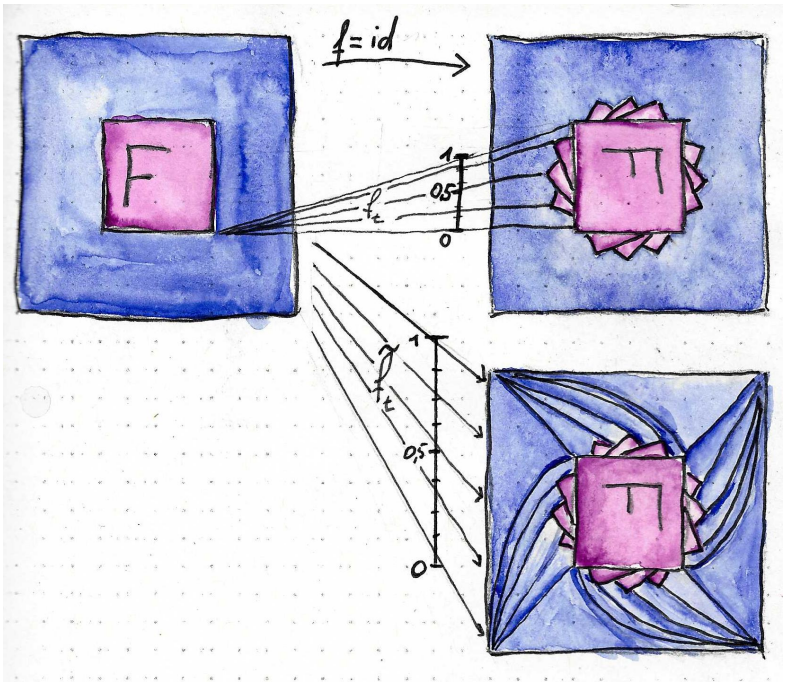


Figure 2: The homotopy extension property for the pair (X, F), where F gets rotated.

3. $H(x, 1) \in A$ for all $x \in X$ with $u(x) < 1$.
4. $u^{-1}(0) = A$.

Such a pair is called an **NDR**-pair which stands for neighbourhood deformation retract-pair. The name comes from the interpretation of $u^{-1}[0, 1)$ being a neighbourhood that deforms onto A under the global homotopy H.

Remark 1.2.3. If we encounter an NDR pair in this thesis that lives in a manifold, we will often omit proving the property. A proof would mostly consist of applying the tubular neighbourhood theorem and the collar neighbourhood theorem.

lem:topPre-hepImpliescontractingisHomotopicalInvariant

Lemma 1.2.4. *If the pair (X, A) satisfies the homotopy extension property and A is contractible, then the quotient map*

$$q : X \to X/A$$

is a homotopy equivalence.

Proof. Let $f_t : X \to X$ be a homotopy extending the contraction of A with $f_0 = \mathrm{id}$. Then the composition $q \circ f_t$ maps A to a point and hence factors over X/A for all t giving us the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f_t} & X \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle q} \\ X/A & \xrightarrow[\overline{f_t}]{} & X/A \end{array}$$

Here, $\overline{f_t}$ denotes the induced map on the quotient. Now $f_1(A) = pt$ where pt is a point in A. So f_1 factors over X/A and denote the induced map by $g : X/A \to X$. Now $g \circ q = f_1$ and hence

$$q \circ g(\overline{x}) = q \circ g \circ q(x) = q \circ f_1(x) = \overline{f_1}(\overline{x}) .$$

In other words we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f_1} & X \\ \downarrow{\scriptstyle q} & \overset{g}{\nearrow} & \downarrow{\scriptstyle q} \\ X/A & \xrightarrow[\overline{f_1}]{} & X/A \end{array}$$

This concludes the proof, since $g \circ q = f_1 \simeq f_0 = \mathrm{id}$ and $q \circ g = \overline{f_1} \simeq \overline{f_0} = \mathrm{id}$. □

Definition 1.2.5. We call a pointed topological space (X, x_0) well pointed, if the inclusion $x_0 \to X$ is a cofibration. We denote the category of well pointed compactly generated weak Hausdorff spaces by $\mathbf{CGWH}^*_{\mathbf{cof}}$.

1.3 Mapping Spaces

Definition 1.3.1 (Compact-open topology)**.** Let X, Y be topological spaces. Then the set of continuous maps $C(X, Y)$ can be equipped with the topology that has

$$\{V(K,U) | K \subseteq X \text{ compact }, U \subseteq Y \text{ open}\} \text{ where } V(K,U) := \{f \in C(X,Y) |\ f(K) \subseteq U\}$$

as a subbasis. For real vector spaces V, W in the standard topology, this coincides with the topology of $\mathrm{Hom}(V, W)$ as a subset of $\mathbb{R}^n$ after identification with the corresponding matrix.

Proposition 1.3.2. If X is compactly generated and Y compactly generated weak Hausdorff, then $C(X, Y)$ is compactly generated weak Hausdorff.

prop:kTheo-curryingIsContinuous

Proposition 1.3.3. Let X, Y, Z be objects in **CGWH**. The natural map

$$C(X, C(Y, Z)) \to C(X \times Y, Z)$$

is a homeomorphism.

2 Vector Bundles and Riemannian Geometry

sec:background-geometry

Due to differential geometry and the theory of vector bundles being areas of mathematics used by pure mathematicians interested in analytical aspects like curvature, by topologists interested in invariants like Euler characteristics, physicists using the spaces as playgrounds for general relativity theories, and many others, there are numerous competing notational conventions. This thesis lies at the tripoint of differential geometry, dynamical systems, and algebraic topology. Hence, there is no a priori preferred notational convention. In this chapter, we establish the necessary background in the theory of vector bundles and Riemannian manifolds whilst not proving all the basics; we establish the notation used throughout this thesis. We establish manifolds and vector bundles in parallel, as they are interdependent on one another. We begin by defining a topological manifold. Afterwards, we equip that space with increasingly more structure and hence reduce the generality until we arrive at the playground of the thesis:

def:vb-topologicalManifold

Definition 2.0.1 (Topological Manifold)**.** A second countable Hausdorff space M is called a **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, for all $p \in M$ there exists an open neighbourhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : U \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M, and φ^{-1} a **local coordinate system around** p **on** M.

def:vb-differentiableManifold

Definition 2.0.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension m.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$, the **transition function**:
   $$\varphi_{ij} : \varphi_j(U_i \cap U_j) \to \varphi_i(U_i \cap U_j)$$
   $$x \mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x)$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$, we call the pair (M, c) a smooth manifold.

Remark 2.0.3. The requirement of second countabilitys together with Hausdorff ensures that we have partitions of unitiy suborinate to an open cover of the manifolds. In the smooth case we can assume the continuous maps of the partition to be smooth.

Corollary 2.0.4. Every transition function φ_{ij}, $i,j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i\in I}$ is not merely a homeomorphism but also a diffeomorphism. This follows from the fact that $\varphi_{ij}^{-1} = \varphi_{ji}$ which provides smooth inverses.

def:vb-smooth Functions

Definition 2.0.5 (Smooth Functions)**.** Let (M,c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any chart (and hence for all) $(\varphi_i)_{i\in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f : U \to \mathbb{R} \text{ continuous} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 2.0.6. Let (M,c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}-$algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \operatorname{res}_V^U : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V \ . \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 2.0.7. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighbourhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p\in U} \mathcal{E}(U) \right) \Big/ \sim_p$$

for the set of germs, and call it the **stalk in** p. Where $\sum$ denotes the co-product (disjoint union) in **CGWH**.

Corollary 2.0.8. For a smooth manifold (M,c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}-$algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \operatorname{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\operatorname{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. We only need to verify the local ring structure of the stalks. An element $f_p \in \mathcal{E}_p(M)$ is invertible if and only if $f(p) \neq 0$, equivalently if $f_p \notin \ker(\mathrm{eval}_p)$. Thus $\mathcal{E}_p(M)/\ker(\mathrm{eval}_p)$ is a field, which proves that $\ker(\mathrm{eval}_p)$ is maximal. □

Definition 2.0.9. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibniz rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f_p, g_p \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it an $\mathbb{R}$-vector space structure by pointwise operations. We define the **tangent space of M at p** to be the vector space

$$T_pM := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})\,.$$

Corollary 2.0.10. Let (M, c) be a smooth manifold of dimension m and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ (where $\varphi \in \mathfrak{A} \in c$). Then we define the derivation

$$\begin{aligned} \partial_i(p) : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto \partial_i(p)(f_p) := \frac{\partial(f \circ \varphi^{-1})}{\partial x_i}\Big|_{x_0}\,. \end{aligned}$$

This is well-defined and hence gives a tangent vector. Moreover, the family

$$(\partial_1(p), \ldots, \partial_m(p))$$

defines a basis of T_pM. Hence, the dimension of T_pM is m.

This is a good point to take a break from smooth manifolds and introduce the structure of vector bundles.

Analogously to the definition of a smooth manifold, we start by defining families of vector spaces and equip them with increasingly more structure:

def:vb-vector Bundles

Definition 2.0.11 (Vector Bundles)**.** Let $\mathbb{K}$ be a field. A $\mathbb{K}$ **vector bundle** of rank m is a surjection

$$\pi : E \to X$$

such that E and X are topological spaces and π is continuous. Furthermore π satisfies:

1. For all $x \in X$, $\pi^{-1}(x)$ carries the structure of an m-dimensional $\mathbb{K}$-vector space.
2. For all $x \in X$, there exist a neighbourhood $U \subseteq X$ and a homeomorphism

   $$\psi : E|_U := \pi^{-1}(U) \to U \times \mathbb{K}^m\,,$$

   that is compatible in the following sense: Let π_1 denote the projection $U \times \mathbb{K}^m \to U$ onto the first factor. Then

   $$\pi_1 \circ \psi = \pi$$

   and

   $$\psi_x : E_x := \pi^{-1}(x) \to \{x\} \times \mathbb{K}^m$$

   is a vector space isomorphism.

We call E the **total space**, X the **base space**, E_x the **fibre over** x and ψ a **local trivialisation**.

Remark 2.0.12. In the context of differential geometry we will mostly consider $\mathbb{R}$ vector bundles. However, the K-theory that we will develop will depend on $\mathbb{C}$ vector bundles.

Definition 2.0.13. Let $\pi : E \to X$ be a vector bundle. A continuous function $s : X \to E$ such that $\pi \circ s = \mathrm{id}_X$ is called a (global) **section** (if $s : U \subseteq X \to E$ we call it a local section). We denote the space of global sections by $\Gamma(E)$ and by $\Gamma(U)$ the space of local sections over an open subset $U \subseteq X$.

def:vb-homomorphismOfVectorBundles

Definition 2.0.14 (Homomorphism of Vector Bundles)**.** A **homomorphism between two vector bundles** $\pi : E \to X$ and $\tilde{\pi} : F \to X$ over the same base space is a continuous function $\varphi : E \to F$ such that:

1. $$\begin{array}{ccc} E & \xrightarrow{\varphi} & F \\ & {\scriptstyle\pi}\searrow \quad \swarrow{\scriptstyle\tilde{\pi}} & \\ & X & \end{array}$$ commutes.

2. For all $x \in X$ the function $\varphi_x : E_x \to F_x$ is linear.

def:vb-pullbackOfVerctorBundles

Definition 2.0.15 (Pullback of Vector Bundles)**.** Let $\pi : E \to Y$ be a vector bundle, X be a topological space and $f : X \to Y$ be continuous. Then we define the **pullback** $f^*\pi : f^*(E) \to X$ as follows: The total space $f^*(E)$ is defined as the subspace of $(x, e) \in X \times E$ such that $f(x) = \pi(e)$, and the map $f^*\pi$ is the projection to the second factor restricted to said subspace.

Example 2.0.16. Let $\pi : E \to X$ be a vector bundle. For any $Y \subseteq X$ we have the vector bundle

$$\pi|_Y : \pi^{-1}(Y) \to Y\,,$$

which is defined as the pullback along the inclusion of Y into X.

rem:vb-restrictionOfBundles

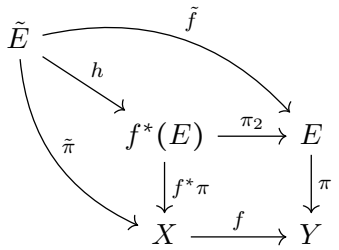
Remark 2.0.17. We will often omit the notation of the projection and denote a vector bundle by its total space. The restriction $\pi|_Y : \pi^{-1}(Y) \to Y$ is often denoted by $E|_Y$. The restriction may equivalently be defined by restricting the domain and codomain of the projection.

rem:vg-pullbackInducesBundleMap

Remark 2.0.18. Notice that this construction of the pullback $f^*\pi : f^*(E) \to X$ is the simplest space to make the square commute:

$$\begin{array}{ccc} f^*(E) & \xrightarrow{\pi_2} & E \\ \downarrow{\scriptstyle f^*\pi} & & \downarrow{\scriptstyle\pi} \\ X & \xrightarrow{f} & Y \end{array}$$

The notion of simplicity is captured by the universal property, that for any other vector bundle $\tilde{\pi} : \tilde{E} \to X$ and function $\tilde{f} : \tilde{E} \to E$ there exists a unique $h : \tilde{E} \to f^*(E)$ (which is given by $h(y) := (\tilde{\pi}(y), \tilde{f}(y))$ such that the following diagram commutes:



Definition 2.0.19 (Morphism of Vector Bundles over Different Base Spaces)**.** Let $\pi : E \to X$ and $\pi' : E' \to X'$ be two vector bundles and $f : X \to Y$ continuous. A **bundle morphism over** f is a map continouos map $\Phi : E \to E'$ such that

$$f \circ \pi = \pi' \circ \Phi$$

and for all x, the restriction $\varphi_x : E_x \to E'_{f(x)}$ is a linear map.

Remark 2.0.20. Notice how a bundle morphism φ always factors over the pullback $f^*(E')$. Furthermore, if φ_x is an isomorphism for all x, then $E \cong f^*(E')$.

Remark 2.0.21. In the context of K-Thery, bundle maps will always be homomorphisms between two vector bundles over the same base space. Hence, we omitt the *over the same base space*. If a bundle map appears that is not over the same base space, we will note that explicitly.

Remark 2.0.22. Let $\pi : E \to X$ be a vector bundle. Then the space of sections $\Gamma(E)$ is a (in general, infinite-dimensional) $\mathbb{R}$-vector space. Furthermore, and more relevant to the purposes of this thesis, we view $\Gamma(E)$ (and $\Gamma(U)$ for open subsets) as a $C(X)$ (or $C(U)$) module. The latter denotes the ring of continuous functions from E (or U) to $\mathbb{R}$.

Example 2.0.23. For any smooth manifold M of dimension m , we can equip the union

$$TM := \bigsqcup_{p \in M} T_pM$$

with a topology, such that $\pi : TM \to M$ defines a vector bundle which we call **tangent bundle**. For this we first define the trivialisations and then equip TM with a topology to make them homeomorphisms: Let $\varphi : U \to V$ be a chart on M. If we denote the i-th coordinate function of φ by φ_i we can define the bijection

$$\tilde{\varphi} : \pi^{-1}(U) \to V \times \mathbb{R}^m, \quad \xi_p \mapsto \left(\varphi(p), \begin{pmatrix} \xi_p(\varphi_1) \\ \vdots \\ \xi_p(\varphi_m) \end{pmatrix} \right).$$

Now we declare a subset $W \subseteq TM$ to be open, if and only if for all charts φ the set $\tilde{\varphi}(W \cap \pi^{-1}(U)) \subset V \times \mathbb{R}^m$ is open. We can now extend and give meaning to the notation of the local basis vectors $\partial_i(p) \in T_pM$: For any chart $\varphi : U \to V$ we have the local sections

$$\partial_i : U \to TU, \quad p \mapsto \partial_i(p).$$

Furthermore, TM is a $2m$-dimensional manifold. Using a smooth atlas on M yields a smooth atlas on TM making it a smooth manifold. Notice that the family $(\partial_i)_i$ is a basis of the $C(U)$-module $\Gamma(U)$. This follows by expressing any section in local coordinates via $\tilde{\varphi}$.

Notice that TM as a bundle satisfies more than just the bundle definition: Taking two different charts induces two different local trivialisations. Transitioning between them is not just fibrewise linear, but also smooth. This is captured in the definition:

Henceforth we suppress the differentiable structure from the notation.

Definition 2.0.24 (The Derivative)**.** Let M and M' be smooth manifolds. We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth map between the tangent bundles as manifolds as follows:

$$\mathrm{d}f : TM \to TM' \ , \mathrm{d}f_p(\xi_p)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi_p \in T_pM$, $g_p \in \mathcal{E}_p(M')$. The function $\mathrm{d}f$ is called the **derivative** or **pushforward** of f.

Remark 2.0.25. Notice how the pushforeward $\mathrm{d}f$ is a bundle morphism over f with the base spaces being M and M'.

Definition 2.0.26 (Smooth Vector Bundles)**.** A **smooth vector bundle of rank** r is a surjection $\pi : E \to M$ where the codomain (or base space) is a smooth manifold together with a family of continuous functions (or local trivialisations)

$$\Phi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{R}^r$$

such that $\bigcup_i U_i = M$ and for all overlapping $U_i \cap U_j \neq \varnothing$, the transition functions

$$\Phi_j \circ \Phi_i^{-1} : \pi^{-1}(U_i \cap U_j) \to (U_i \cap U_j) \times \mathbb{R}^r$$

are of the form

$$\Phi_j \circ \Phi_i^{-1}(p, v) = (p, g_{i,j}(v)), \quad \text{where} \quad g_{i,j} : (U_i \cap U_j) \to \mathrm{GL}(r, \mathbb{R})$$

is smooth.

Corollary 2.0.27. A smooth vector bundle is a smooth manifold and also a vector bundle. The latter is ensured by the codomain of $g_{i,j}$ which is also called the **structure group**. We can enforce structure on the bundle by reducing the structure group. As a preview: reducing it to $\mathrm{GL}^+(k, \mathbb{R})$ the linear maps with positive determinant means that we can induce the concept of orientations. Reducing it to $\mathrm{O}(k)$ lets us equip it with a Riemannian metric.

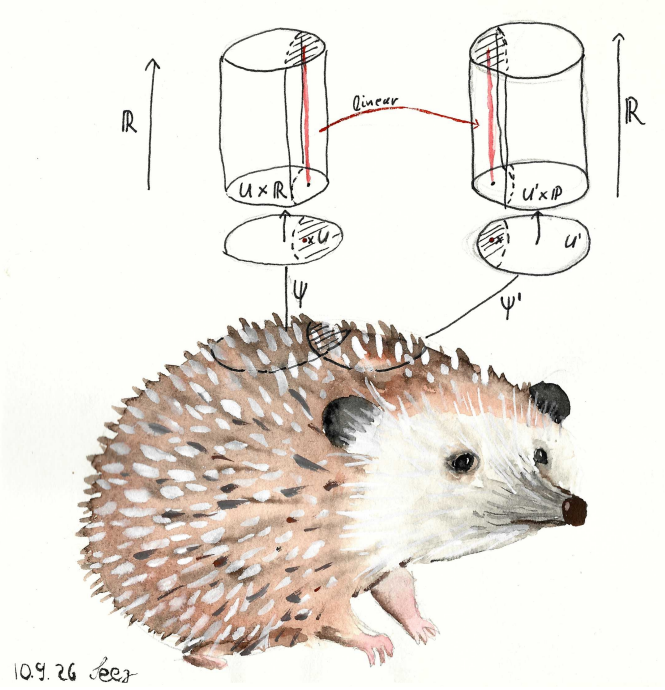


Figure 3: A smooth vector bundle (If we ignore the face and feet and assume the hedghogs spines to be infinitly tight and long)

def:vb-frames FromTrivialisations

Definition 2.0.28. Let $\pi : E \to M$ be a smooth vector bundle of rank r and let

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r$$

be a local trivialisation. Then a section is called smooth if it is a smooth map of manifolds. A family of smooth local sections $(s_1, \dots, s_r)$ with $s_i \in \Gamma(U)$ is called a **local frame** if they form a basis of the $C(U)$-module $\Gamma(U)$. Such a family is obtained from Φ by defining $s_i(p) := \Phi^{-1}(p, e_i)$. Similarly, from a local frame we can obtain a local trivialisation by defining:

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r\,, \quad v = \sum_{i=1}^{r} \lambda_i s_i(\pi(v)) \mapsto (\pi(v), \lambda_1, \dots, \lambda_r)\,.$$

Example 2.0.29. The local trivialisations ∂_i of the tangent bundle of a smooth manifold M form a local frame and make the tangent bundle into a smooth vector bundle.

def:vb-homomorphismBundle

Definition 2.0.30 (Homomorphism Bundle)**.** Let W, V be vector spaces and $E = V \times X$, $F = W \times X$ the product bundles over X. Now any bundle homomorphism $\varphi : E \to F$ determines a map $\Phi : X \to \hom(V, W)$ by sending $x \mapsto \varphi_x$. With respect to the compact-open topology, this Φ is continuous. Furthermore, a continuous map $\Phi : X \to \hom(V, W)$ determines a bundle homomorphism $\varphi : E \to F$.

Proof. Let $\{e_i\}$ and $\{f_i\}$ be bases of V and W. Then each $\Phi(x)$ is a matrix $\Phi_{i,j}(x)$, such that:

$$\Phi(x)e_i = \sum_j \Phi(x)_{i,j} f_j \,.$$

Now Φ is continuous if and only if $\Phi_{i,j}$ is continuous. Since $\varphi(x,v) = (x, \Phi(x)v)$ this is equivalent to φ being continuous. □

thm:vb-isomorphicVectorBundles

Lemma 2.0.31 (Isomorphic Vector Bundles)**.** *Let E and F be vector bundles and $\varphi : E \to F$ be a homomorphism. Then φ is an isomorphism if and only if all φ_x are isomorphisms.*

Proof. To show that it is an isomorphism we need a continuous inverse, equivalently, we need to show that φ is a homeomorphism. First of all we can deduce, that φ is bijective: Assume it wasn't injective, then there would be a fibre in which φ_x is not injective. Assume that it wasn't surjective. Then there would be a $w \in F$ that has no preimage which means that in the fibre containing w there is no isomorphism. Thus, there is a unique inverse and it remains to verify that it is continuous, which is a local condition. Therefore we can assume that E and F are product bundles. Then we have a continuous map $\Phi : X \to \mathrm{Hom}(V,W)$. Furthermore, since all φ_x are isomorphisms, we can restrict the image to $\mathrm{Iso}(V,W)$ which is an open subset of $\mathrm{Hom}(V,W)$. Now we can define the map $x \mapsto \Phi(x)^{-1}$ which is continuous since taking inverses is continuous. By the corollary above, we obtain a continuous map $\psi : F \to E$ by sending $(x,w) \mapsto (x, \Phi(x)^{-1}(w))$, which is the inverse. □

Example 2.0.32. Let M be a smooth manifold of dimension m, $\varphi : U \to V$ a chart and $\tilde{\varphi} : TU \to V \times \mathbb{R}^m$ be the corresponding bundle chart. Then it follows from the lemma, that TU is isomorphic to $V \times \mathbb{R}^m$. This is verified by observing that $\partial_i(p)$ is a basis of T_pU for all $p \in U$. In this case $\tilde{\varphi}(\partial_i) = e_i$ giving us isomorphisms in each fibre.

def:vb-subBundles

Definition 2.0.33 (Sub-bundles)**.** Let $\pi : E \to X$ be a vector bundle. A **sub-bundle** is a subset of E which is a vector bundle in the induced topology and fibrewise vector space structure.

def:vb-monoAndEpimorphisms

Definition 2.0.34 (Mono- and Epimorphisms)**.** A homomorphism $\varphi : E \to F$ is a **monomorphism** if all φ_x are monomorphisms (i.e. injective). Respectively, it is called an **epimorphism** if all φ_x are epimorphisms (i.e. surjective).

Corollary 2.0.35. If F is a sub-bundle of E, then the inclusion $\varphi : F \to E$ is a monomorphism.

lem:vb-monosInduceSubBundles

Lemma 2.0.36. *If $\varphi : E \to F$ is a monomorphism, then $\varphi(E)$ is a sub-bundle and $\varphi : E \to \varphi(E)$ is an isomorphism.*

Proof. Once we know that $\varphi(E)$ is a sub-bundle, the map $\varphi : E \to \varphi(E)$ is a bijective homomorphism and fibrewise an isomorphism, hence an isomorphism by Lemma 2.0.31. So we need to check that $\varphi(E)$ is a sub-bundle, which is a local property. Thereby we can assume that E and F are product bundles $E = X \times V$ and $F = X \times W$. Let $x \in X$ and let $W_x \subseteq W$ be a subspace complementary to $\varphi_x(V)$. Then $G := X \times W_x \subseteq F$ and we define

$$\begin{aligned} \theta : E \oplus G &\to F \\ (a, b) &\mapsto \varphi(a) + i(b) \quad \text{where } i \text{ denotes the inclusion } G \hookrightarrow F\,. \end{aligned}$$

In the fibre over x, the map θ_x is an isomorphism and thereby there exists an open neighbourhood U of x such that $\theta|_U$ is an isomorphism. That is because $\mathrm{Iso}(V \oplus W_x, W)$ is open in $\mathrm{Hom}(V \oplus W_x, W)$. Since E is a sub-bundle of $E \oplus G$ we have that $\theta(E)|_U = \varphi(E)|_U$ is a sub-bundle of $F|_U$. □

cor:vb-subBundlesAreLocalSummands

Corollary 2.0.37. In the proof, we have shown more: locally a sub-bundle is a direct summand. Furthermore, we have shown that the set of points x where φ_x is a monomorphism, is open.

def:vb-quotientBundle

Definition 2.0.38 (Quotient Bundle)**.** If F is a sub-bundle of E we define the **quotient bundle** E/F as the union of the spaces E_x/F_x equipped with the quotient topology.

Corollary 2.0.39. The quotient bundle is a vector bundle, because locally F is a direct summand and thereby E/F is locally trivial.

lem:vb-quotiendBundlesOfSmoothSubbundlesAreSmooth

Lemma 2.0.40. *Quotiend bundles of smooth subbundles are smooth.*

Proof. Let $F \subseteq E$ be a smooth subbundle of rank k, where E has rank n and M is the base manifold. We need to check that the transition functions consist of smooth maps into the structure group. First, we will show that there are smooth local frames of E that restricts to a smooth local frame of F:

Let $p \in M$. Since F is smooth there exists a local frame $(s_1, \ldots, s_k)$ of F arround p. In particular, the s_i are smooth local sections into E arround p. Let $(t_1, \ldots, t_k)$ be a smooth local frame of F near p. (without restrictions over the same domain as the sections s_i) Now we make use of steinitz exchange lemma to choos $n - k$ many t_i (Without restrictions $t_{k+1}, \ldots, t_n$) such that

$$\big(s_1(p), \ldots s_k(p), t_{k+1}(x), \ldots, t_n(p)\big)$$

is a basis of E_p. Being a basis over a point $q \in M$ near p is equivalent to the invertability of the matrix where the columns are the sections at q, and since invertability (or rather nonzero determinant) is an open condition, there exists a neighbourhood, such that the family

$$\big(s_1, \ldots s_k, t_{k+1}, \ldots, t_n\big)$$

is a local frame, whose first k sections span F. By Definition 2.0.28 it induces a smooth trivialisation Φ_U of E with $\Phi_U\big(F|_U\big) = U\times\mathbb{R}^k\times\{0\}$, and fibrewise passing to the quotient $\mathbb{R}^n/(\mathbb{R}^k\times\{0\}) \cong \mathbb{R}^{n-k}$ yields a trivialisation $\bar{\Phi}_U$ of $(E/F)|_U$.

Now we can cover M by such adapted trivializations. Their transition functions $g_{i,j} : U_i \cap U_j \to \mathrm{GL}(n,\mathbb{R})$ are smooth and pointwise preserve $\mathbb{R}^k\times\{0\}$: for $v \in \mathbb{R}^k\times\{0\}$ we have $\Phi_i^{-1}(p,v) \in F_p$ since Φ_i is adapted, and hence $\Phi_j(\Phi_i^{-1}(p,v)) \in \{p\}\times\mathbb{R}^k\times\{0\}$ since Φ_j is adapted. Hence they have block form

$$g_{i,j} = \begin{pmatrix} A_{i,j} & B_{i,j} \\ 0 & C_{i,j} \end{pmatrix},$$

where all coordniate entries and hence all blocks smoothly depend on the basepoint. Finally, the transition functions of the quotient with respect to the trivialization $\bar{\Phi}_U$ of $(E/F)|_U$ consist of the smooth maps $p \mapsto C(p) = (C_{i,j})_{i,j}(p)$ into the structure group $\mathrm{GL}(n-k,\mathbb{R})$. □

Now we want to construct new bundles from old ones. There are many operations to construct new vector spaces such as the tensor product, direct sum, Hom-Spaces and many more. Instead of individually proving that those operations applied fibrewise yield new bundles, we subsume these operations within the following definition:

Definition 2.0.41 (Continuous Functor)**.** Let T be a covariant endofunctor in the category of finite-dimensional vector spaces. We call T a **continuous functor**, if for all V,W the map $T : \hom(V,W) \to \hom(T(V),T(W))$ is continuous. All vector spaces are given the topology induced by their isomorphism to real Euclidean space. If E is a vector bundle we define the set:

$$T(E) := \bigcup_{x\in X} T(E_x)\,.$$

Now we want to equip this with a topology such that $T(E)$ is a bundle over the same base space as E.

def:vb-constructingNewBundles

Definition 2.0.42 (Constructing new bundles)**.** We start by assuming that $E = X\times V$. In that case we define the topology on $T(E)$ as the product topology $X\times T(V)$. Now suppose that $F = X\times W$ is another product bundle and $f : E \to F$ is a homomorphism. Let $\Phi : X \to \hom(V,W)$ be the corresponding map. Then $T\Phi : X \to \hom(T(V),T(W))$ is continuous by assumption. Thereby, $T(\varphi) : X\times T(V) \to X\times T(W)$ is continuous. If φ is an isomorphism, then $T(\varphi)$ is also an isomorphism. That follows because it is continuous and an isomorphism fibrewise.

Now assume that E is trivial. Then we can choose an isomorphism $\alpha : E \to X\times V$ and induce a topology on $T(E)$ by pulling it back via $T(\alpha) : T(E) \to X\times T(V)$. This topology does not depend on α by the above. For general vector bundles we do this construction locally.

If $f : X \to Y$ is continuous, there is a natural isomorphism

$$f^*T(E) \cong Tf^*(E)$$

This construction is carried out in detail by Atiyah in [Ati89, Chapter 1.2] and can be generalised for continuous functors with several variables both co- and contravariant. Hence this procedure gives us for vector bundles E and F over the same base space the new bundles:

(i) $E \oplus F$, their direct sum

(ii) $E \otimes F$, their tensor product

(iii) $\mathrm{Hom}(E, F)$

(iv) E^*, the dual bundle of E

(v) $\Lambda^i(E)$ the i-th exterior power

and by the natural isomorphism we have natural isomorphisms:

(i) $E \oplus F \cong F \oplus E$

(ii) $E \otimes F \cong F \otimes E$

(iii) $E \otimes (F \oplus F') \cong E \otimes F \oplus E \otimes F'$

(iv) $\mathrm{Hom}(E, F) \cong E^* \otimes F$

cor:vb-sectionsIntoHom

Corollary 2.0.43. Let $\pi_E : E \to X$ and $\pi_F : F \to X$ be two vector bundles over X. Then, the space $\Gamma(\mathrm{Hom}(E, F))$ can be identified with the space of bundle morphisms $\hom(E, F)$ since for any section $s : X \to \mathrm{Hom}(E, F)$ we can define the bundle homomorphism $\tilde{s} : E \to F$ by sending $v \mapsto s_{\pi_E(v)}(v)$. This is a homomorphism since $\pi_F(\tilde{s}(v)) = \pi_E(v)$.

Example 2.0.44 (Cotangent bundle)**.** Let M be a smooth manifold of dimension m. Then we can define the **cotangent bundle** T^*M to be the dual of the tangent bundle. A local frame of said bundle is given by the fibrewise dual of the frame (∂_i). Sections into the tangent bundle are called **vector fields** and sections into the cotangent bundle are called **one-forms**.

def:vb-linearIdentification

Definition 2.0.45. Let V be a linear manifold (i.e. a finite-dimensional vector space). Then for any $v \in V$ we have the canonical maps

$$\begin{aligned} R_v : V &\to T_v V \\ x &\mapsto \left(f_v \mapsto \left.\frac{\partial}{\partial t}\right|_{t=0} f(v + tx) \right) \end{aligned}$$

Let $L : V \to V$ be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$\mathrm{d}L_v \circ R_v = R_{L(v)} \circ L\,.$$

This is proved by a short calculation:

$$\Big(\mathrm{d}L_v\big(R_v(x)\big)\Big)(f_{L(v)}) = R_v(x)\,(f\circ L) = \frac{\partial}{\partial t}\Big|_{t=0} f\circ L(v+tx) = \underbrace{\frac{\partial}{\partial t}\Big|_{t=0} f\big(L(v)+tL(x)\big)}_{=R_{L(v)}\circ L(x)}$$

Now let $v, v' \in V$. Then we have the translation

$$tr_{v,v'} : V \to V; \quad x \mapsto (x + (v' - v))$$

This allows us to relate R_v and $R_{v'}$ as follows:

$$\big(\mathrm{d}tr_{v,v'}|_v\,(R_v(x))\big)\,(f_{v'}) = \frac{\partial}{\partial t}\Big|_{t=0}(f\circ tr_{v,v'}(v+tx)) = \frac{\partial}{\partial t}\Big|_{t=0}\big(f(v'+tx)\big) = R_{v'}(x)(f_{v'})\,.$$

Hence:

$$R_{v'} = \mathrm{d}tr_{v,v'}\circ R_v\,.$$

The **linear identification** is the map Q given on the whole tangent bundle:

$$Q : TV \to V; \quad \xi_p \mapsto R_p^{-1}(\xi_p)\,,$$

and satisfies for any linear $L : V \to V$

$$Q\circ \mathrm{d}L = L\circ Q\,.$$

cor:vb-vectorFieldsTakeFunctions

Corollary 2.0.46. Let $f : M \to \mathbb{R}$ be smooth. Then we can interpret $\mathrm{d}f$ as a one-form. To be precise, let $Q : T\mathbb{R} \to \mathbb{R}$ denote the linear identification with respect to the identity on $\mathbb{R}$. For $v \in \Gamma(TM)$ we have

$$(Q\circ \mathrm{d}f)(v) = v(f)$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We retain this notation so that vector fields can take functions as an input.

Proof. Let $\varphi : U \to V$ be a chart of M. We prove this by showing it for the local sections ∂_i and the general case follows by $C(U)$-linearity, since those sections form a local frame. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f_p(\partial_i(p))(g_p) = \partial_i((g\circ f)_p) = \frac{\partial}{\partial x_i}\Big|_{x_0}(g\circ f\circ\varphi^{-1}) = \frac{\partial}{\partial x}_{f_p(p)}(g_p)\cdot\partial_i(p)(f_p)$$

Hence, applying Q to the calculation yields $(Q\circ \mathrm{d}f)(v) = v(f)$. □

Definition 2.0.47 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(T^*M \otimes T^*M)$ into the tensor product of the dual tangent bundle with itself is a **Riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non-degenerate**, meaning if for a $v_p \in T_pM$ we have $g_p(v_p, w_p) = 0$ for all $w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ for all $v_p, w_p \in T_pM$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0$ for all v_p with equality only for $v_p = 0$.

We call a tuple (M, g) a **Riemannian manifold**.

Remark 2.0.48. Equipping a smooth manifold M of dimension m with a Riemannian metric is equivalent to reducing the structure group of the tangent bundle from $\mathrm{GL}(m)$ to $\mathrm{O}(m)$. To be precise: A Riemannian metric g lets us canonically choose an atlas with structure group $\mathrm{O}(m)$ and an atlas with such a structure group lets us canonically choose a Riemannian metric.

Proof. We start by assuming that we have an atlas with structure group $\mathrm{O}(m)$. In Definition 2.0.28 we saw that a covering of local trivialisations is equivalent to a covering of smooth frames. Now in a single trivialisation

$$\Phi : \pi^{-1}(U) =: TU \to U \times \mathbb{R}^m$$

we call the projection to the $\mathbb{R}^m$-part (the second factor) π_2 and define

$$\begin{aligned} g : U &\to T^*U \otimes T^*U \\ p &\mapsto \begin{aligned} g_p^\Phi : T_pU \times T_pU &\to \mathbb{R} \\ (\nu_p, \xi_p) &\mapsto g_p^\Phi(\nu_p, \xi_p) := \pi_2\left(\Phi(\nu_p)\right) \cdot \pi_2\left(\Phi(\xi_p)\right) , \end{aligned} \end{aligned}$$

where $\cdot$ denotes the standard scalar product. This gives us locally a Riemannian metric. If we do this for all trivialisations we have to make sure, that the result is well defined. So assume that two trivialisations Φ and Ψ overlap in V. The structure group is $\mathrm{O}(m)$, $\Phi = R\Psi$ for some function

$$R : V \to \mathrm{O}(m) .$$

Let p be in said overlapping and (ν_p, ξ_p) be from $T_pM \times T_pM$. Then since R is orthogonal we can calculate the independence from the trivialisation:

$$g_p^\Phi(\nu_p, \xi_p) = \pi_2\left(R\Psi(\nu_p)\right) \cdot \pi_2\left(R\Psi(\xi_p)\right) = \pi_2\left(\Psi(\nu_p)\right) \cdot \pi_2\left(\Psi(\xi_p)\right) = g_p^\Psi(\nu_p, \xi_p) .$$

To get an orthonormal atlas from a smooth one given a Riemannian metric, we start by building a covering of local frames from the trivialisations. We then apply Gram–Schmidt to the local frames to get orthonormal frames. Now a covering of orthonormal frames gives a covering of trivialisations, where the structure group consists of basis changes between orthonormal bases and hence is $\mathrm{O}(m)$. □

Definition 2.0.49 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a Riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function and $Q : T\mathbb{R} \to \mathbb{R}$ the

linear identification. We define its **gradient** to be a vector field $\operatorname{grad}(f) \in \Gamma(TM)$ such that for any vector field V:

$$Q \circ \mathrm{d}f(V) = g(\operatorname{grad}(f), V)\,.$$

Notice how the non-degeneracy of g ensures existence and uniqueness.

def:vb-localDescriptionOfGradientGij

Definition 2.0.50. Let (M, g) be a Riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be:

$$g_{ij}(p) = g_p(\partial_i(p), \partial_j(p))\,.$$

Then if two vector fields v, w over U are given with $v = \sum v_i \partial_i$ and $w = \sum_i w_i \partial_i$ we can calculate the metric locally:

$$g(v, w) = \sum_{i,j} g_{ij} v_i w_j \ : U \to \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure as can be seen by the construction via Cramer's rule) to get smooth functions

$$\begin{aligned} g^{ij} : U &\to \mathbb{R} \\ p &\mapsto g^{ij}(p) = \left(\left(g_{kl}(p)\right)^{-1}_{kl}\right)_{ij} \end{aligned}$$

Remark 2.0.51. Due to the rare appearance of big summations with vectors and covectors in this work, we omit the Einstein summation convention and assign all coordinate entries lower indices.

lem:vb-localDescriptionOfGradient

Lemma 2.0.52 (Local Description of the Gradient)**.** *Let (M, g) be a smooth Riemannian manifold of dimension m and $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\operatorname{grad}(f) = \sum_{i,j} g^{ij} \left(\partial_i(f)\right) \cdot \partial_j\,.$$

Proof. Assume that $v, w \in \Gamma(TU)$ such that $v = \sum_i v_i \partial_i$ and $w = \sum_i w_i \partial_i$ (Here v_i and w_i are smooth functions $U \to \mathbb{R}$). Then $g(v, w) = \sum_{i,j} g_{ij} v_i w_j$. Suppose that $\operatorname{grad}(f) = \sum_j G^j \partial_j$ and Q is the linear identification $T\mathbb{R} \to \mathbb{R}$ in each tangent space. Then by definition of the gradient and corollary 2.0.46 we have:

$$\partial_j(f) = Q \circ \mathrm{d}f(\partial_j) = g(\operatorname{grad}(f), \partial_j) = g(\sum_i G^i \partial_i, \partial_j) = \sum_i g_{ij} G^i$$

Reading this as a pointwise matrix product:

$$\begin{aligned} (G^1, \cdots G^m)(g_{ij}) &= (\partial_1(f), \cdots, \partial_m(f)) \\ \text{which gives } (G^1, \cdots G^m) &= (\partial_1(f), \cdots, \partial_m(f))\,(g^{ij})\,. \end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij} \partial_i(f)$.

□

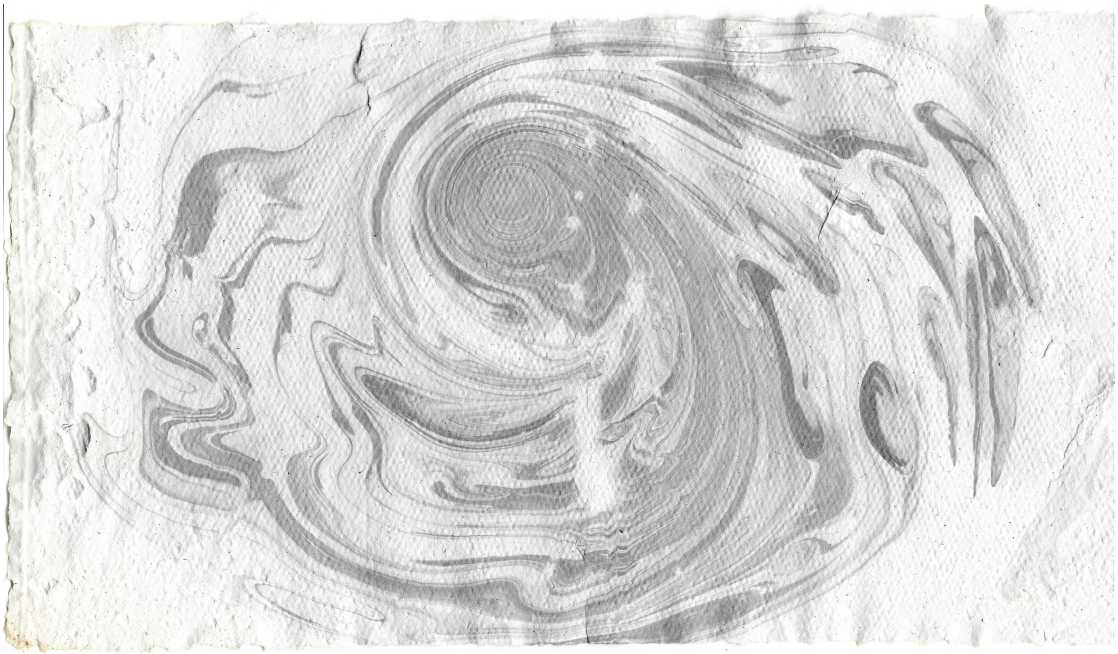


Figure 4: A dynamical system with a critical point created by printing a flow of inc.

Definition 2.0.53 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M (sometimes also called a **flow on** M) is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cap \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:
$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$
3. For all p we require $\varphi(0,p) = p$.

We will write $\varphi_t(p) := \varphi(t,p)$ to keep the notation manageable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 2.0.54. For a global dynamical system φ on a smooth manifold M the map
$$\rho : \mathbb{R} \to \mathrm{Diff}(M), \quad t \mapsto \varphi_t$$
is a homomorphism. To verify well-definedness notice how for all $t \in \mathbb{R}$, φ_{-t} is an inverse of φ_t. The map ρ is called a **one-parameter group of diffeomorphisms**.

Definition 2.0.55 (Vector Field of a Dynamical System)**.** Let $\varphi : \Omega \to M$ be a dynamical system on a smooth manifold M. For a chart $\psi : U \to V$ of M we have the chart $\mathrm{id} \times \psi$ of Ω. For this chart we denote the associated local frame by $(\partial_t, \partial_1, \ldots, \partial_m)$. Using this we define the mapping

$$\begin{aligned} M &\to TM \\ p &\mapsto X(p) := \mathrm{d}\varphi(\partial_t(0,p)) \end{aligned}$$

and call it the **associated vector field**. This is in fact a smooth vector field.

Definition 2.0.56 (Dynamical System of a Vector Field)**.** Let $X \in \Gamma(TM)$ be a global vector field on the smooth manifold M. Then we call a dynamical system φ on M the **associated dynamical system** to X, if for all $t \in \mathbb{R}$ and $x \in M$

$$\mathrm{d}\varphi\left(\partial_t(t,x)\right) = X(\varphi_t(x))\,.$$

Remark 2.0.57. Regarding the left-hand side of the definition $\mathrm{d}\varphi\left(\partial_t(t,x)\right)$ we could also differentiate the map

$$t \mapsto \varphi_t(x)\,,$$

for fixed x at t and evaluate this at the constant vector field $t \mapsto R_t(1) \in T_t\mathbb{R}$ from definition 2.0.45. The condition on the dynamical system reads then

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x)\big(R_t(1)\big) = X(\varphi_t(x))$$

Definition 2.0.58. Let $X \in \Gamma(TM)$ be a global vector field on the smooth manifold M and denote the associated flow by φ. Then we call $q \in M$ a **critical point of** X, if $X(q) = 0$.

prop:vb-onceCriticalAlwaysCritical

Proposition 2.0.59. Let $X \in \Gamma(TM)$ be a global vector field on the smooth manifold M and denote the associated flow by φ. Then the function

$$t \mapsto \varphi_t(x)$$

is constant if x is a critical point of X. Furthermore, if x is not critical, then for all t, $\varphi_t(x)$ is also not critical.

Proof. The first statement follows from Picard–Lindelöf, as the constant function $\mathrm{const}_x : t \mapsto x$ for all t satisfies the condition

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{const}_x = 0 = X(\mathrm{const}_x(t))\,.$$

Now assume that for some t_0, $y := \varphi_{t_0}(x)$ is critical. Then by the flow property $x = \varphi_{-t_0}(y)$ and hence x was already critical. □

prop:vb-ffDecreasesAlongGradientFlow

Proposition 2.0.60 (The Flow of the Gradient)**.** Let $f : M \to \mathbb{R}$ be smooth function on a Riemannian manifold (M, g) and φ be the dynamical system associated to $\operatorname{grad}(f)$. Then for any $x \in M$, the function

$$f \circ \varphi_{\square}(x) : \mathbb{R} \to \mathbb{R}, \quad t \mapsto f \circ \varphi_t(x)$$

is strictly increasing if x is not a critical point of $\operatorname{grad}(f)$ and constant if x is a critical point of $\operatorname{grad}(f)$. (If the function is only defined on an open subset of $\mathbb{R}$, the statement still holds.)

Proof. We just calculate the differential of $\varphi_\cdot(x)$ and make use of positive definiteness of g:

$$\frac{\mathrm{d}}{\mathrm{d}t} f \circ \varphi_t(x)(R_t(1)) = \mathrm{d}f_{\varphi_t(x)}\big(\operatorname{grad}_{\varphi_t(x)}(f)\big) = g(\operatorname{grad}_{\varphi_t(x)}(f), \operatorname{grad}_{\varphi_t(x)}(f)) \geq 0 \quad \forall t \in \mathbb{R}$$

where equality holds only if and only if $\operatorname{grad}_{\varphi_t(x)}(f) = 0$. Prop 2.0.59 tells us that the latter is equivalent to x being a critical point of our vector field. □

These two constructions are mutually inverse, giving the following theorem:

Theorem 2.0.61. *Let M be a smooth manifold. Then there is a one-to-one correspondence between global dynamical systems and vector fields given by the constructions above.*

3 Complex K-Theory

3.1 Mathematical Goals

Theorem 3.1.1 (The Theorems and Mathematical Structures)**.**

- *Definition of The K-Functor*
- *Cohomology structures*
  - *Additivity (pointed sum)*
  - *Long exact sequences*
  - *Excision*

3.2 Classification of Vector Bundles

The first goal is to classify all vector bundles over sufficiently nice spaces. The latter refers to spaces admitting partitions of unity for which Tietze's extension theorem holds. For simplicity we just consider compact spaces. Along the way we will discover that the set of isomorphism classes of vector bundles is a homotopy invariant.

Before we go back to the analytical reals, we will further develop the theory of vector bundles:

Definition 3.2.1. For a topological space X we define Vect_n to be the set of isomorphism classes of complex vector bundles of rank n and $\mathrm{Vect}(X)$ to be the set of isomorphism classes of complex vector bundles of any rank. By declaring $\mathrm{Vect}(f) = f^*$ to be the pullback we get a contravariant functor Vect from **CGWH** to **Set**. We can equip the set $\mathrm{Vect}(X)$ with the direct sum and tensor product operation. It is then easy to see that these operations descend to isomorphism classes and hence we get an abelian semiring $(\mathrm{Vect}(X), \oplus, \otimes)$ (abelian semigroup with associative and distributive multiplication). In fact we even get a contravariant functor from **CGWH** to the category of abelian semirings.

lem:kTheo-sectionExtension

Lemma 3.2.2. *Let X be a compact space, $Y \subseteq X$ a closed subspace and E a vector bundle over X. Then any section $s : Y \to E|_Y$ can be extended to a global section of X.*

Proof. For a detailed proof see Chapter 1.4 of [Ati89]. The general argument however is easy to follow: By Tietze's extension theorem we can extend the section to around every $x \in X$. By compactness we can pick finitely many of those extensions. Now using a partition of unity with respect to their domains we can build the sum of all the extensions to get a well defined global one. □

cor:kTheo-isomorphismExtension

Corollary 3.2.3. Let Y be a closed subspace of a compact space X, and let E, F be bundles over X. If $f : E|_Y \to F|_Y$ is an isomorphism, then there is an open neighbourhood $U \supset Y$ and an extension $f : E|_U \to F|_U$ which is an isomorphism.

Proof. By corollary 2.0.43 we can view f as a section into the bundle $\mathrm{Hom}(E|_Y, F|_Y)$, which can be extended to a global one in $\mathrm{Hom}(E,F)$. Let U be the open set for which the section maps to isomorphisms. Since U contains Y this yields the required extension. □

thm:kTheo-homotopyInvariance

Theorem 3.2.4 (Homotopy Invariance of the Pullback)**.** *Let X, Y be two compact spaces, $I = [0,1]$, and*

$$f : Y \times I \to X$$

be a homotopy. For a fixed $t \in I$, we define the function $f_t : Y \to X$, $f_t(y) = f(t,y)$. Let $\pi : E \to X$ be a vector bundle. Then

$$f_0^*(E) \cong f_1^*(E)\,.$$

Proof. Consider the following two vector bundles over $Y \times I$:

$$f^*E \quad \text{and} \quad \pi_1^* f_t^* E\,,$$

where $\pi_1 : Y \times I \to Y$ denotes the projection and $t \in I$. Since the restriction is by definition the pullback along the inclusion $i : Y \times \{t\} \to Y \times I$. The equality of the two maps:

$$f \circ i = f_t \circ \pi_1 \circ i : Y \times \{t\} \to X$$

together with the functoriality gives us an isomorphism

$$\Phi : f^*E|_{Y\times\{t\}} \to \pi_1^* f_t^* E|_{Y\times\{t\}}\,.$$

By Corollary 3.2.3 we can extend the isomorphism for some open neighbourhood of $Y \times \{t\}$ and by the product topology we can deduce that there exists a δ_t such that U contains the thickened line $Y \times (t-\delta_t, t+\delta_t)$. The isomorphism class of $f^*E|_{Y\times\{t\}}$ as a bundle over Y is locally constant in t: Let $0 < \varepsilon < \delta$ and $r \in I$. Then define the inclusions

$$\begin{aligned} i_t : Y &\to Y \times \{t\} \subseteq Y \times I \\ y &\mapsto (y,t) \end{aligned} \qquad \text{and} \qquad \begin{aligned} i_{t+\varepsilon} : Y &\to Y \times (t-\delta_t, t+\delta_t) \\ y &\mapsto (y, t+\varepsilon) \end{aligned}$$

Then we have the identities $f_t = f \circ i_t$ and $\pi_1 \circ i_{t+\varepsilon} = \mathrm{id}_Y$ and we can calculate:

$$f_{t+\varepsilon}^* E = i_{t+\varepsilon}^* f^* E = i_{t+\varepsilon}^* \pi_1^* f_t^* E = (f_t \circ \pi_1 \circ i_{t+\varepsilon})^* E = f_t^* E\,.$$

Finally, since I is connected, any locally constant map from I is globally constant and therefore $f_0^* E \cong f_1^* E$. □

lem:kTheo-contractibleBundlesAreTivial

Corollary 3.2.5. 1. If $f : X \to Y$ is a homotopy equivalence, $f^* : (Y) \to (X)$ is a semiring isomorphism.

2. If X is contractible, every bundle over X is trivial and $(X) \cong \mathbb{N}_0$

def:kTheo-trivialisation

Definition 3.2.6 (Trivialisation of a Vector Bundle)**.** Let Y be a closed subspace of X, E a vector bundle over X and $\alpha : E|_Y \to Y \times V$ be an isomorphism. Then we call α **trivialisation of E over Y**.

lem:kTheo-trivialisationDefinesABundle

Lemma 3.2.7. *A trivialisation α of a bundle E over $Y \subseteq X$ defines a bundle E/α over X/Y. The isomorphism class of E/α depends only on the homotopy class of α.*

Proof. We define the bundle E/α as follows: Let $\pi_2 : Y \times V \to V$ denote the projection and define an equivalence relation on $E|_Y$ by

$$e \sim e' \Leftrightarrow \pi_2\alpha(e) = \pi_2\alpha(e')\,.$$

We extend this relation by the identity on $E|_{X\setminus Y}$ and we define E/α to denote the quotient space of E defined by that relation. To see why this is a bundle we need to check the local triviality only around the base point Y/Y of X/Y. This follows from the fact that by Corollary 3.2.3 we can extend α to an isomorphism $\tilde{\alpha} : E|_U \to U \times V$. Then $\tilde{\alpha}$ induces an isomorphism

$$E/\alpha\big|_{U/Y} \cong (U/Y) \times V$$

proving the local triviality. Now for the homotopy invariance of said bundle suppose that α_0 and α_1 are homotopic trivialisations of E over Y. This means that we have a trivialisation β of $E \times I$ over $Y \times I \subset X \times I$ that induces the α_i at the endpoints. Let $f : (X/Y) \times I \to (X \times I)/(Y \times I)$ be the natural map. Then $f^*((E \times I)/\beta)$ is a bundle on $(X/Y) \times I$ whose restriction to $(X/Y) \times \{i\}$ is E/α_i $(i = 0, 1)$. Hence by the homotopy invariance of the pullback (Theorem 3.2.4) we have

$$E/\alpha_0 \cong E/\alpha_1\,.$$

□

cor:kTheo-trivializationIsRightInverse

Corollary 3.2.8. The trivialisation acts as a right inverse of the pullback along the collapse: If we call $f : X \to X/Y$ the collapse, and fE can be trivialised over Y via α, then

$$f^*(E/\alpha) \cong E\,.$$

Proof. Intuitively this is clear, as the pullback along the collapse builds a bundle, that is trivial over Y and agrees with E outside of Y. To prove the statement we will construct a bundle map, that is an isomorphism in each fibre: Let $q : E \to E/\alpha$ denote the quotient map from the construction and let $\pi_E : E \to X$, $\pi_{E/\alpha} : E/\alpha \to X/Y$ be the bundle projections. By construction of E/α we have $\pi_{E/\alpha} \circ q = f \circ \pi_E$, so

$$\Phi : E \to f^*\big(E/\alpha\big)\,, \qquad \Phi(e) = \big(\pi_E(e),\, q(e)\big)$$

is a well-defined bundle morphism. We show Φ is a fibrewise linear isomorphism, and hence a bundle isomorphism by Lemma 2.0.31.

Case $x \notin Y$: Here the equivalence relation defining E/α restricts to the identity on E_x, so

$$q_x : E_x \xrightarrow{\cong} (E/\alpha)_{f(x)} = (E/\alpha)_x$$

is a linear isomorphism.

Case $x \in Y$: By definition of the equivalence relation, two points $e, e' \in E_x$ are identified with $\pi_2\alpha(e) = \pi_2\alpha(e') \in V$, and all fibres E_x ($x \in Y$) are glued to the single fibre $(E/\alpha)_{y_0} \cong V$ over the basepoint $y_0 = Y/Y$. Hence

$$q_x = \pi_2 \circ \alpha_x : E_x \to V = (E/\alpha)_{y_0} ,$$

where $\alpha_x : E_x \to V$ is the (linear) restriction of α to the fibre over x. Since α_x is a linear isomorphism, so is q_x.

Thus Φ is a bundle morphism which is a linear isomorphism on every fibre. □

lem:kTheo-collapsingOfContractibleSubspace

Lemma 3.2.9. *Let $Y \subseteq X$ be a closed contractible subspace of a compact space X. Then*

$$f : X \to X/Y \quad \textit{induces a semiring isomorphism} \quad f^* : \mathrm{Vect}(X/Y) \to \mathrm{Vect}(X) .$$

Proof. Since Y is contractible, $E|_Y$ is trivial and thereby a trivialisation $\alpha : E|_Y \to Y \times V$ exists. Two such trivialisations only differ by the automorphism of $Y \times V$ which in turn corresponds to a map $Y \to \mathrm{GL}(V) \cong \mathrm{GL}_{\mathrm{rank}(E)}(\mathbb{C})$. Now since Y is contractible those two maps factor (up to homotopy) over any one point $y \in Y$ and are thereby homotopically equivalent to the constant map. Finally since $\mathrm{GL}_{\mathrm{rank}(E)}(\mathbb{C})$ is connected all constant maps are homotopically equivalent. Thereby α is unique up to homotopy. By Lemma 3.2.7 the isomorphism class of E/α only depends on the homotopy class of α we have a well defined map

$$\mathrm{Vect}(X) \to \mathrm{Vect}(X/Y) ,$$

which in turn is an inverse of f^* concluding the proof. □

def:kTheo-gluingAndClutching

Definition 3.2.10 (Gluing and Clutching of Vector Bundles)**.** Let

$$X = X_1 \cup X_2 , \quad A = X_1 \cap X_2 ,$$

and all be compact spaces. Assume that E_i is a vector bundle over X_i and $\varphi : E_1|_A \to E_2|_A$ is an isomorphism. Then we define a vector bundle $E_1 \cap_\varphi E_2$ on X as follows: We topologise the space $E_1 \cup_\varphi E_2$ as the quotient of $E_1 + E_2/\sim$, where $\sim$ is the identification of points with their image under φ. Identifying X with $X_1 + X_2$ in the quotient space we get the natural projection $p : E_1 \cap_\varphi E_2 \to X$ and the fibres have a natural vector space structure. It remains to show that this is locally trivial. This is obvious outside of A so let $a \in A$ and V_1 be a closed neighbourhood of a in X_1 such that $E_1|_{V_1}$ is trivial giving us the isomorphisms

$$\theta_1 : E_1|_{V_1} \to V_1 \times \mathbb{C}^n$$
$$\theta_1^A : E_1|_{V_1 \cap A} \to (V_1 \cap A) \times \mathbb{C}^n$$

Finally, define

$$\theta_2^A : E_2|_{V_1 \cap A} \to (V_1 \cap A) \times \mathbb{C}^n$$

by composition $\theta_1^A \circ \varphi^{-1}$. Let V_2 be a neighbourhood of a in X_2 such that

$$\theta_2 : E_2|_{V_2} \to V_2 \times \mathbb{C}^n$$

is an extension of θ_2^A. This extension exists again by extending the section into the homomorphism bundle corresponding to the isomorphism θ_2^A. Finally θ_1 and θ_2 define a well defined isomorphism

$$\theta_1 \cap_\varphi \theta_2 : E_1 \cap_\varphi E_2 \to (V_1 \cap V_2) \times \mathbb{C}^n .$$

This proves the local triviality.

We call such a triple (E_1, E_2, φ) a **clutching datum** and φ the **transition function**.

cor:kTheo-elementaryPropertiesGluingAndClutching

Corollary 3.2.11 (Elementary Properties of the Gluing and Clutching Construction)**.** Elementary properties of the gluing and clutching construction are:

1. If E is a bundle over X, then the identity defines an isomorphism and
$$E_1 \cup_{\mathrm{id}_A} E_2 \cong E.$$
2. If $\beta_i : E_i \to E_i'$ are isomorphisms on X_i and $\varphi' \beta_1 = \beta_2 \varphi$, then
$$E_1 \cup_\varphi E_2 \cong E_1' \cup_{\varphi'} E_2' .$$
3. If (E_i, φ) and (E_i', φ') are two "clutching data" on X_i, then
$$\begin{aligned} (E_1 \cup_\varphi E_2) \oplus (E_1' \cup_{\varphi'} E_2') &\cong E_1 \oplus E_1' \bigcup_{\varphi \oplus \varphi'} E_2 \oplus E_2' , \\ (E_1 \cup_\varphi E_2) \otimes (E_1' \cup_{\varphi'} E_2') &\cong E_1 \otimes E_1' \bigcup_{\varphi \otimes \varphi'} E_2 \otimes E_2' , \\ (E_1 \cup_\varphi E_2)^* &\cong E_1^* \bigcup_{(\varphi^*)^{-1}} E_2^* . \end{aligned}$$

   $(-)^*$ denotes the dual here.

lem:kTheo-homotopyInvarianceClutchingFunction

Lemma 3.2.12. *The isomorphism class of $E_1 \cup_\varphi E_2$ depends only on the homotopy class of the isomorphism $\varphi : E_1|_A \to E_2|_A$.*

Proof. A homotopy of isomorphisms is a map

$$\Phi : E_1|_A \times I \to E_2|_A ,$$

such that for all t the map

$$\Phi_t : E_1|_A \to E_2|_A \quad \Phi_t(\xi) := \Phi(\xi, t)$$

is a bundle isomorphism. We can encode this Φ into a bundle isomorphism as follows: Let $\pi : X \times I \to X$ be the projection. Then elements $\xi \in \pi^* E_1$ are by definition 2.0.15 of the form $\xi_{x,t} = ((x,t), \xi_x) \in (\pi^* E_1)_{x,t}$. With this notation define the bundle isomorphism:

$$\Psi : \pi^*(E_1)|_{A\times I} \to \pi^*(E_2)|_{A\times I}\,, \quad \Psi(\xi_{x,t}) := ((x,t), \Phi_t(\xi_x))$$

This is a bundle isomorphism, since Φ_t is one. Now denote the inclusion

$$i_t : X \to X \times I\,, \quad i_t(x) := (x,t)\,,$$

and notice that the pullback along i_t is the same as restricting to $X \times \{t\}$.

With those definitions it is clear by the construction of the gluing, that

$$E_1 \cup_{\Phi_t} E_2 \cong i_t^* \left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right)\,. \tag{1}$$

{eq:kTheo-gluingHomotopyInvariant}

The isomorphism can be seen by part (2) of Corollary 3.2.11 as follows: First notice the isomorphism $\pi^*(E_i)|_{A\times\{t\}} \cong E_i$ for $i = 1, 2$ and any t. By construction of the gluing this leads to the isomorphism

$$E_1 \cup_{\Phi_t} E_2 \cong \pi^*(E_1)|_{A\times\{t\}} \cup_{(\Phi_t,\mathrm{id})} \pi^*(E_2)|_{A\times\{t\}}\,.$$

Now again by construction we notice, that gluing restricted spaces is the same as gluing the whole space and then restricting. Furthermore the map (Φ_t, id) is the restriction of Ψ to $X \times \{t\}$. Therefore:

$$\pi^*(E_1)|_{A\times\{t\}} \cup_{(\Phi_t,\mathrm{id})} \pi^*(E_2)|_{A\times\{t\}} \cong (\pi^*(E_1) \cup_\Psi \pi^*(E_2))|_{X\times\{t\}}$$

Finally, pulling back via i_t is the same as restricting $X \times i$ to $X \times \{t\}$ and identifying X with $X \times \{t\}$. This concludes the chain of arguments:

$$(\pi^*(E_1) \cup_\Psi \pi^*(E_2))|_{X\times\{t\}} \cong i_t^* (\pi^*(E_1) \cup_\Psi \pi^*(E_2))\,.$$

Since the pullback is homotopy invariant we can conclude by the isomorphism (1):

$$E_1 \cup_{\Phi_0} E_2 \cong E_1 \cup_{\Phi_1} E_2\,.$$

□

def:kTheo-homotopyClassesOfMaps

Definition 3.2.13. Let X, Y be topological spaces. By $[X, Y]$ we denote the set of homotopy classes of maps from X to Y.

lem:kTheo-gluingAndHomotopyClassesToGLn

Lemma 3.2.14. *Let $X = X_1 \cup X_2$ with X_1 and X_2 closed contractible spaces. Then there is a canonical bijection*

$$\mathrm{Vect}_n(X) \to [A, \mathrm{GL}_n(\mathbb{C})]\,.$$

Proof. Let $[E]$ be any vector bundle over X of rank n. Then E can be represented by a vector bundle obtained by gluing that is trivial on both X_i: Since X_i is contractible we can choose two trivialisations

$$\alpha_1 : E|_{X_1} \to X_1 \times \mathbb{C}^n, \quad \text{and} \quad \alpha_2 : E|_{X_2} \to X_2 \times \mathbb{C}^n .$$

With part (2) of Corollary 3.2.11 we can conclude:

$$E \cong E|_{X_1} \cup_{\mathrm{id}_A} E|_{X_2} \cong X \times \mathbb{C}^n \cup_{\alpha_2 \circ \alpha_1^{-1}|_A} X \times \mathbb{C}^n$$

Since a map between the two trivial bundles $A \times \mathbb{C}^n \to A \times \mathbb{C}^n$ can be identified with a map $A \to \mathrm{GL}_n(\mathbb{C})$ we get an assignment from bundles over X together with the trivialisations to maps $A \to \mathrm{GL}_n(\mathbb{C})$. The next step is to show that two isomorphic vector bundles give homotopic maps: Assume that we are given two vector bundles E, F over X that are isomorphic via $T : E \to F$ and without loss of generality we assume that they are given as the gluing of two trivial bundles:

$$E : X_1 \times \mathbb{C}^n \cup_\varphi X_2 \times \mathbb{C}^n , \quad \text{and} \quad F : X_1 \times \mathbb{C}^n \cup_\psi X_2 \times \mathbb{C}^n .$$

We now want to define a homotopy between ψ and φ. For this we start by splitting the isomorphism T into its two parts over X_1 and X_2:

$$T|_{X_i} : X_i \times \mathbb{C}^n \to X_i \times \mathbb{C}^n .$$

These maps are given by maps

$$T_i : X_i \to \mathrm{GL}_n(\mathbb{C}) ,$$

where the gluing compatibility of the $T|_{X_i}$ yield for any $a \in A$:

$$\psi(a) T_1(a) = T_2(a) \varphi(a) , \quad \text{equivalently} \quad \psi(a) = T_2(a) \varphi(a) T_1^{-1}(a) .$$

By the contractibility of X_i we have for any fixed $x_i \in X_i$ a homotopy $h_t^i : X_i \to X_i$ such that $h_0^i = c_{x_i}$ is the collapsing map whilst $h_1^i = \mathrm{id}_{X_i}$. Furthermore, since $\mathrm{GL}_n(\mathbb{C})$ is connected we have a path $\gamma^i : I \to \mathrm{GL}_n(\mathbb{C})$ that connects $T_i(x_i) = \gamma^i(1)$ to $\mathbb{1}_n = \gamma^i(0)$. With this we can define the homotopies

$$\begin{aligned} T_{i,t} : X_i &\to \mathrm{GL}_n(\mathbb{C}) \\ x &\mapsto \begin{cases} T_i \circ h_{2t}^i(x) & \text{if } t \le \frac{1}{2} , \\ \gamma^i(2(t - \frac{1}{2})) & \text{if } t > \frac{1}{2} , \end{cases} \end{aligned}$$

that satisfy $T_{i,1} = T_i$ and $T_{i,0}(x) = \mathbb{1}_n$ for all x. Finally we define the homotopy $H_t : A \to \mathrm{GL}_n(\mathbb{C})$ by

$$H_t(a) := T_{2,t}(a) \varphi(a) T_{1,t}^{-1}(a)$$

to get a homotopy from $\varphi = H_0$ to $\psi = H_1$. Hence, we get a well defined map from isomorphism classes of bundles over X of rank n to the homotopy classes of maps from A to $\mathrm{GL}_n(\mathbb{C})$ that is the inverse of the gluing construction and hence is bijective. □

ex:kTheo-tautopogicalLineBundleOverCP1

Example 3.2.15. We now want to analyse the tautological line bundle l over $\mathbb{CP}^1$. For this we cut $\mathbb{CP}^1$ into the two pieces which are contractible as they come with homeomorphisms into the unit disc in $\mathbb{C}$:

$$H^+ := \{[x:y] \in \mathbb{CP}^1 \,|\, |x| \geq |y|\} \to D \subset \mathbb{C}\,, \quad [x:y] \mapsto \frac{y}{x}$$
$$H^- := \{[x:y] \in \mathbb{CP}^1 \,|\, |x| \leq |y|\} \to D \subset \mathbb{C}\,, \quad [x:y] \mapsto \frac{x}{y}$$

Those maps give us unique representations of elements in H^+ and H^-. We now want to evaluate the bijection $\mathrm{Vect}_1(\mathbb{CP}^1) \to [H^+ \cap H^-, \mathrm{GL}_1(\mathbb{C})]$ at $[l]$. The image of the intersection $H^+ \cap H^-$ under the two maps is the boundary of the disc and hence a circle. Furthermore, $\mathrm{GL}_1(\mathbb{C}) \cong C^*$. Hence, we know that the image is isomorphic to $[S^1, \mathbb{C}^*] \cong \pi_1(S^1) \cong \mathbb{Z}$. This tells us, that there must be nontrivial vector bundles on $\mathbb{CP}^1$. By the proof of 3.2.14 we know that $[l]$ gets mapped to the function gluing together trivialisations over H^+ and H^- along their intersection. To define the trivialisations we need to define local frames over H^+ and H^-. Those frames are given by a single section defined as:

$$s_+ : H^+ \to l\,, \quad [x:y] \mapsto (1, \frac{y}{x})\,,$$
$$s_- : H^- \to l\,, \quad [x:y] \mapsto (\frac{x}{y}, 1)\,.$$

With those we define the trivialisations:

$$t_+ : \left. l \right|_{H^+} \to H^+ \times \mathbb{C}\,, \quad ([x:y], \lambda \cdot s_+[x:y]) \mapsto ([x:y], \lambda)$$
$$t_- : \left. l \right|_{H^-} \to H^- \times \mathbb{C}\,, \quad ([x:y], \mu \cdot s_-[x:y]) \mapsto ([x:y], \mu)$$

in each fibre over the intersection we have

$$\lambda \cdot s_+([x:y]) = \lambda(1, \frac{y}{x}) = \lambda \cdot \frac{y}{x}(\frac{x}{y}, 1) = \lambda \cdot \frac{y}{x} s_-([x:y])$$

Hence in each fibre over $[x:y] \in H^+ \cap H^-$ the gluing function can be calculated:

$$\lambda \mapsto \left. t_- \right|_{[x:y]} \circ \left. t_+ \right|^{-1}_{[x:y]} (\lambda) = \left. t_- \right|_{[x:y]} (\lambda \cdot s_+([x:y])) = \left. t_- \right|_{[x:y]} (\lambda \cdot \frac{y}{x} \cdot s_-([x:y])) = \lambda \cdot \frac{y}{x}$$

Hence, as a map from $H^+ \cap H^- \to GL_1(\mathbb{C}) \equiv \mathbb{C}^*$ this map reads:

$$H^+ \cap H^- \to \mathbb{C}^* \quad [x:y] \mapsto \frac{y}{x}$$

viewing this as an element of $\pi(S^1)$ by identifying the intersection with a circle via the homeomorphism from $H^+ \to D \subseteq \mathbb{C}$ we notice that the gluing map corresponds to the identity in $\pi_1(S^1)$. If the bundle l would be trivial, the gluing function would be homotopically equivalent to the constant one-function $H^+ \cap H^- \to \mathbb{C}^*$ that maps all $[x:y] \mapsto 1$. This cannot be, because interpreted as elements in $\pi(S)$ they are different.

def:kTheo-metricsOnBundles

Definition 3.2.16 (Metrics on Bundles)**.** A **metric** on a complex bundle E is a section $h : X \to \mathrm{Herm}(E)$ such that $h(x)$ is positive definite for all x. $\mathrm{Herm}(E)$ denotes the bundle of hermitian forms (conjugate-symmetric sesquilinear) in each fibre. A bundle together with a metric is a **Hermitian bundle**.

def:kTheo-projectionOperator

Definition 3.2.17 (Projection Operator)**.** A **projection operator** $P : E \to E$ is a bundle homomorphism in E such that $P^2 = P$.

lem:kTheo-projectionsGiveSumdecomposition

Lemma 3.2.18. *Any projection operator $P : E \to E$ determines a direct sum decomposition $E = PE \oplus (1-P)E$.*

Proof. Observe that fibrewise $1 - P$ maps each ξ to the kernel of P, and hence fibrewise $P(E)$ and $(1-P)(E)$ are disjoint. Furthermore $\mathrm{im}(P)+\mathrm{im}(1-P) = E$, because $v = P(v)+(v - P(v))$. Therefore in each fibre we have the direct sum $E_x = P(E_x) \oplus (1-P)(E_x)$. Now the question is, if that gives a global splitting, meaning that $P(E)$ and $(1-P)(E)$ are continuous as subbundles of E. For this to be true, we need local triviality, which is clear as long as the rank of P (and $(1-P)$) is locally constant. In each fibre we know that $\mathrm{rank}(P_x) + \mathrm{rank}(1-P) = \dim(E_x)$. Furthermore, $\mathrm{rank}(P_x)$ and $\mathrm{rank}(1-P)$ are locally constant. To see this first notice that they are lower semi-continuous, meaning that for y from a small enough neighbourhood of x $\mathrm{rank}(P_x) \leq \mathrm{rank}(P_y)$. To see this, pick a $\mathrm{rank}(P_x)_x \times \mathrm{rank}(P_x)_x$ minor with nonzero determinant. The latter is an open condition and hence stable under small perturbation. But now we have the sum of two lower semi-continuous functions that give something locally constant and hence, each summand is locally constant. Finally the above shows that $\ker(P) = (1-P)(E)$ we have the decomposition. □

lem:kTheo-metricInducedComplement

Lemma 3.2.19 (Metric Induced Complement)**.** *Suppose that F is a sub-bundle of E. Then a metric provides a canonical complementary sub-bundle.*

Proof. Let h denote the metric. For each x we have the orthonormal projection $P_x : E_x \to F_x$ inducing a map $P : E \to F$. Let $(f_1, \dots f_n)$ be an orthonormal frame of F. Then fibrewise with $\xi \in E_x$ the orthogonal projection is given by

$$P_x(\xi) = \sum_i h_x(\xi, f_i(x)) f_i(x)\,.$$

And since h is continuous the whole map P is. Furthermore $P^2 = P$ and hence we get a canonical sum decomposition. □

Definition 3.2.20. We call a sequence of bundle homomorphisms

$$\cdots \longrightarrow E \longrightarrow F \longrightarrow \cdots$$

exact, if it is fibrewise exact.

lem:kTheo-splittingPrinzip le

Lemma 3.2.21 (Splitting Principle)**.** *Every short exact sequence of bundles over the same base space splits.*

Proof. Let X be the base space and assume that the sequence is given as

$$0 \longrightarrow E' \xrightarrow{\varphi'} E \xrightarrow{\varphi''} E'' \longrightarrow 0$$

Now it is well known that any short exact sequence of vector spaces splits. We just need to ensure that this splitting is continuous. For this we give E a metric. Now since φ' is injective, it is a monomorphism and $\varphi'(E')$ is a subbundle by Lemma 2.0.36. By Lemma 3.2.19 we can split $E \cong \varphi'(E') \oplus L$. However, $\varphi''|_L$ is fibrewise (and hence as a bundle map) an isomorphism. This can be seen as φ'' being surjective and trivial on $\varphi'(E')$ hence the restriction is still surjective. The injectivity is also clear as the kernel by the homomorphism theorem of linear maps. But then we clearly have $E = E' \oplus E''$ and the sequence is isomorphic as a short exact sequence to the splitting one:

$$0 \longrightarrow E' \xrightarrow{i_1} E' \oplus E'' \xrightarrow{\pi_2} E'' \longrightarrow 0$$

□

def:kTheo-ampleSubspace

Definition 3.2.22. A subspace $V \subseteq \Gamma(E)$, where $\Gamma(E)$ is viewed as a $\mathbb{C}$-vector space is called **ample**, if

$$\varphi : X \times V \to E, \quad \varphi(x, s) := s(x)$$

is a surjection.

Corollary 3.2.23. If the subspace V is ample and $\dim(V) = \operatorname{rank}(E)$, then E is trivialisable.

lem:kTheo-ampleSubspaceExists

Lemma 3.2.24. *For any vector bundle E over a compact space X, there exists a finite dimensional ample subspace of $\Gamma(E)$.*

Proof. Let $\{U_\alpha\}$ be a finite open cover of X such that $E|_{U_\alpha}$ is trivial. Let $\{s_{\alpha,i}\}$ be the corresponding local frames, and define V_α to be the finite dimensional ample subspace $V_\alpha \subseteq \Gamma(E|_{U_\alpha})$. Let ρ_α be a partition of unity subordinate to the cover $\{U_\alpha\}$. Define

$$\theta_\alpha : V_\alpha \to \Gamma(E), \quad \theta_\alpha(s_\alpha)(x) = \begin{cases} s_\alpha(x) \cdot \rho_\alpha(x) & \text{if } x \in U_\alpha, \\ 0 & \text{else.} \end{cases}$$

Then the θ_α give rise to a homomorphism

$$\theta : \Pi_\alpha V_\alpha \to \Gamma(E),$$

and the image of θ is finite dimensional and ample. The surjectivity is clear, as for any x there exists a $\rho_\alpha(x) > 0$ making the map

$$\theta_\alpha(V_\alpha) \to E_x$$

a surjection. □

cor:kTheo-epiFromTrivial

Corollary 3.2.25. If E is any bundle over X, there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$ for some integer m.

Proof. Just take a basis $(s_1, \dots, s_m)$ of an ample subspace V and define $\varphi(x, e_i) = s_i(x)$, where e_i is the i-th standard basis vector of $\mathbb{C}^m$. □

cor:kTheo-splittingTrivial

Corollary 3.2.26. If E is a bundle over X, there exists a bundle F such that $E \oplus F$ is trivial.

Proof. By Corollary 3.2.25 there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$. The kernel of φ is a subbundle of $X \times \mathbb{C}^m$ and hence by Lemma 3.2.21 we have a splitting of the short exact sequence

$$0 \longrightarrow \ker(\varphi) \longrightarrow X \times \mathbb{C}^m \xrightarrow{\varphi} E \longrightarrow 0$$

Hence, the middle term becomes isomorphic to the direct sum of the outer terms and we get $X \times \mathbb{C}^m \cong E \oplus \ker(\varphi)$. □

Begründung ergänzen, z.B.: Metriken auf $X \times \mathbb{C}^m$ und E wählen; dann ist $P := \varphi^*(\varphi\varphi^*)^{-1}\varphi$ ein stetiger Projektionsoperator mit $(1 - P)(X \times \mathbb{C}^m) = \ker(\varphi)$, und Lemma 3.2.18 liefert die Zerlegung. ($\varphi\varphi^*$ ist invertierbar, da φ Epimorphismus.)

def:kTheo-grassmanManifold

Definition 3.2.27 (Grassmann Manifold)**.** Let V be a vector space and n be any integer. The **Grassmann manifold** $\mathrm{Gr}_n(V)$ is the set of all subspaces of V with **codimension** n. If V is given a Hermitian metric, we can topologise $\mathrm{Gr}_n(V)$ as a subspace of the set of all projection operators on V and this defines a map

$$P : \mathrm{Gr}_n(V) \to \mathrm{End}(V)\,,$$

where $\mathrm{End}(V)$ is the space of all endomorphisms of V with the operator norm topology. We give $\mathrm{Gr}_n(V)$ the subspace topology of $\mathrm{End}(V)$.

def:kTheo-classifyingMap

Definition 3.2.28. Suppose that E is a bundle over X, V is a vector space and $\varphi : X \times V \to E$ is an epimorphism. Then we can define a map $f : X \to \mathrm{Gr}_{\dim(V)}(V)$ by sending x to the kernel of $\varphi_x : V \to E_x$. We call f **induced by** φ. This map is continuous for any metric on V. To see this notice that the topology on V is metric invariant, as V is finite dimensional. Now the map f is continuous if for the map

$$P : \mathrm{Gr}_n(V) \to \mathrm{End}(V)$$

the composition $P \circ f$ is continuous. But $P \circ f$ is given by sending x to the projection onto $\ker(\varphi_x)$, which is continuous as φ is.

def:kTheo-classifyingBundle

Definition 3.2.29 (Classifying Bundle)**.** Let V be a vector space and $F \subseteq \mathrm{Gr}_n(V) \times V$ be the subbundle consisting of all points (g, v) such that $v \in g$. We call F the **tautological bundle** over $\mathrm{Gr}_n(V)$. Furthermore, if $E = (\mathrm{Gr}_n(V) \times V)/F$ is the quotient bundle, we call E the **classifying bundle** over $\mathrm{Gr}_n(V)$.

Corollary 3.2.30. If E' is a bundle over X, and $\varphi : X \times V \to E'$ is an epimorphism, then φ induces a map $f : X \to \mathrm{Gr}_m(V)$ such that $E' \cong f^*E$, where E is the classifying bundle over $\mathrm{Gr}_m(V)$. The isomorphism follows from a fibrewise construction because $E_x \cong V \big/ \ker(\varphi_x)$. The latter however is exactly the fibre over $\ker(x)$ in the classifying bundle and hence the pullback via the φ induced map gives an isomorphism. That motivates the name "classifying bundle" for E and "classifying map" for f.

lem:kTheo-GrassmanTopologyMetricInvariant

Lemma 3.2.31. *The topology on* $\mathrm{Gr}_n(V)$ *is independent of the choice of the metric on* V.

Proof. Let h and h' be two metrics on V. Then we denote the topological space $\mathrm{Gr}_n(V)$ with the topology induced by h by $\mathrm{Gr}(V_h)$ and the one induced by h' by $\mathrm{Gr}(V_{h'})$. We need to show that the identity map $\mathrm{id} : \mathrm{Gr}(V_h) \to \mathrm{Gr}(V_{h'})$ is a homeomorphism. For this we need to show that id is continuous. The projection $\mathrm{Gr}_n(V_h) \times V_{h'} \to E$ (where E is the classifying bundle) is an epimorphism and hence induces a classifying map $f : \mathrm{Gr}_n(V_h) \to \mathrm{Gr}_n(V_{h'})$ such that $f^*E \cong E$. However that map is given by the identity on $\mathrm{Gr}_n(V)$ and hence the identity is continuous by Definition 3.2.28. □

Remark 3.2.32. To continue further we need direct limits of directed systems to construct general Grassmannians. However we will also need said construction for topological spaces and groups. Hence, we will give the definitions for arbitrary categories.

Definition 3.2.33 (Directed System)**.** Let $\mathcal{C}$ be a category and $(I, \leq)$ be a partially ordered set where every finite subset is bounded from above. Let $(A_i)_{i\in I}$ be a family of objects from $\mathcal{C}$ and $f_{ij} : A_i \to A_j$ be a morphism for all $i \leq j$ that satisfies:

1. $f_{ii} = \mathrm{id}_{A_i}$ for all $i \in I$.
2. $f_{kj} \circ f_{ik} = f_{ij}$ for all $i \leq k \leq j$.

The pair $\big((A_i)_{i\in I}, (f_{ij})_{i\leq j}\big)$ is called a **direct system**.

Definition 3.2.34 (Direct Limit)**.** Let $\big((A_i)_{i\in I}, (f_{ij})_{i\leq j}\big)$ be a direct system in a category $\mathcal{C}$. A pair $\big(X, (\varphi_i)_{i\in I}\big)$ where X is an object of $\mathcal{C}$ and $\varphi_i : A_i \to X$ are morphisms such that $\varphi_j \circ f_{ij} = \varphi_i$ for all $i \leq j$ is called a direct limit, if the following universal property holds: For any family of morphisms $g_i : A_i \to Y$ that satisfies $g_i \circ f_{ji} = g_j$ for all $i \leq j$ there exists a unique morphism $g : X \to Y$ such that $g_i = g \circ \varphi_i$. We denote the space X by $\mathrm{colim}_i A_i$ (or just $\mathrm{colim} A_i$), where the maps f_{ij} are omitted from the notation.

Corollary 3.2.35. As always, the direct limit is (if it exists) unique up to unique isomorphism.

Remark 3.2.36 (Construction of Direct Limits)**.** We will now see three constructions of direct limits. Let $\big((A_i)_{i\in I}, (f_{ij})_{i\leq j}\big)$ denote a directed system in the noted category.

In the category of sets we define the direct limit by

$$\mathrm{colim}_i A_i := \bigsqcup A_i \Big/ \sim,$$

where $x \sim y$ if there exists $i, j, k \in I$ such that $f_{ik}(x) = f_{jk}(y)$. This gives an equivalence relation. The maps $\varphi_i : A_i \to \mathrm{colim}_i A_i$ are the composition of the inclusion into the disjoint union with the projection onto the quotient space. The universal property is easy to verify. For a family of suitable maps $g_i : A_i \to B$ we define the map $g : \mathrm{colim}_i A_i \to B$

as $g(x) = g_m(x)$ for $x \in A_m$. This is well defined since the g_i are compatible with the f_{ij} and unique by definition.

In the category of topological spaces we define the direct limit to be the set-theoretic direct limit equipped with the final topology of the inclusions, i.e. a subset $U \subseteq \operatorname{colim}_i A_i$ is open if and only if $\varphi_i^{-1}(U)$ is open for all m. Again the universal property is easy to prove by using the construction of the universal map from the set-theoretic limit. To show the continuity we show that $g^{-1}(U)$ is open for any open U which is checked by verifying openness of $\varphi_i^{-1}\big(g^{-1}(U)\big)$ for all i, which in turn equals $g_i^{-1}(U)$ and is open as the g_i are continuous. Uniqueness follows from the uniqueness of g established already in set.

In an algebraic category like semigroups, groups, rings we define the direct limit again as the set-theoretic direct limit, where the algebraic operations are defined such that the φ_i are homomorphisms meaning for example (if + is an operation in the A_i) $a \in A_i$ and $b \in A_j$ we define $a + b$ in $\operatorname{colim} iA_i$ by including the sum $f_{ik}(a) + f_{jk}(b)$ which is defined in A_k. Again the universal morphism g is the one from the set-theoretic limit which now is a homomorphism by construction.

rem:kTheo-staircaseMapsClosedEmbedding

Remark 3.2.37. If we build the direct limit of sets and all the maps of the direct system are injective, then the inclusions into the direct limit are injective. If all the maps of the direct system are closed embeddings, then the inclusions into the direct limit are also closed embeddings.

Proof. Recall that the inclusions $\varphi_i : A_i \to \operatorname{colim} A_i$ are defined to be the composition of the inclusion into the disjoint union with the projection onto the quotient. Now assume that all maps f_{ij} in the direct system are injective and $\varphi_i(x) = \varphi_i(y)$. Then $x \sim y$ in the quotient and hence, there exists $i, j, k \in I$ such that $f_{ik}(x) = f_{jk}(y)$ which implies that $x = y$.

Now assume that all f_{ij} are closed embeddings. Then, the inclusions are injective by the set-theoretic argument. Furthermore, the images of $\varphi_i : A_i \to \operatorname{colim} A_i$ are closed. That follows from subsets in the final topology being closed, if and only if all their preimages are closed. Hence we check if $\varphi_j^{-1}(\varphi(A_i))$ is closed for all $j \in I$:

$$\begin{aligned} &\text{If } j \le i \quad \text{then:} \quad \varphi_j^{-1}(\varphi_i(A_i)) = (\varphi_i \circ f_{ji})^{-1}(\varphi_i(A_i)) = f_{ji}^{-1}(A_i) \quad \text{which is closed.} \\ &\text{If } i < j \quad \text{then:} \quad \varphi_j^{-1}(\varphi_i(A_i)) = \underbrace{\varphi_j^{-1}(\varphi_j}_{\varphi \text{ is injective}} f_{ij}(A_i)) = f_{ij}(A_i) \quad \text{which is closed.} \end{aligned} \tag{2}$$

{eq:kTheo-staircaseMapsClosedEmbedding}

To show that $\varphi_i^{-1} : \operatorname{im}(\varphi_i) \to A_i$ is continuous we assume that $B \subset A_i$ is open. Then $\varphi_i(B)$ is open if and only if for any j $\varphi_j^{-1}(\varphi_i(B))$ is open. This follows from the same calculations as in equation (2) using that the f_{ij} are homeomorphisms onto their image. □

cor:kTheo-directGrassmanSystem

Corollary 3.2.38. The so-called inverse system given by the projections

$$\pi : \mathbb{C}^m \to \mathbb{C}^{m-1}, \quad (x_1, \ldots, x_m) \mapsto (x_1, \ldots, x_{m-1}),$$

induces a direct system of Grassmann manifolds with the induced continuous maps

$$i_{m-1} : \mathrm{Gr}_n(\mathbb{C}^{m-1}) \to \mathrm{Gr}_n(\mathbb{C}^m)\,, \quad U \mapsto \pi^{-1}(U)\,.$$

If we denote the classifying bundle over $\mathrm{Gr}_n(\mathbb{C}^m)$ by E_m, then the pullback of E_m along i_{m-1} is isomorphic to E_{m-1}.

thm:kTheory-classifyingSpace

Theorem 3.2.39. *The map*

$$\mathrm{colim}_m[X, \mathrm{Gr}_n(\mathbb{C}^m)] \to \mathrm{Vect}_n(X)\,,$$

induced by $f \mapsto f^*E_m$ is a bijection for all compact spaces X.

Proof. We shall construct an inverse map. If E is a bundle over X, then by Corollary 3.2.25 there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$ for some integer m. Let $f : X \to \mathrm{Gr}_n(\mathbb{C}^m)$ be the map induced by φ. If we manage to show that the homotopy class of f as a map to $\mathrm{Gr}_n(\mathbb{C}^{m'})$ for some large enough m' is independent of the choice of φ and m, then we get a well defined map from $\mathrm{Vect}_n(X)$ to $\mathrm{colim}_m[X, \mathrm{Gr}_n(\mathbb{C}^m)]$ that is inverse to the classifying map.

Suppose that $\varphi_i : X \times \mathbb{C}^{m_i} \to E$ for $i = 1, 2$ are two epimorphisms. Let $g_i : X \to \mathrm{Gr}_n(C^{m_i})$ be the maps induced by φ_i. We want to show that g_1 and g_2 are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$. For this we define the map

$$\psi_t : X \times \mathbb{C}^{m_1} \times C^{m_2} \to E\,, \quad \psi_t(x, v_1, v_2) = t\varphi_1(x, v_1) + (1-t)\varphi_2(x, v_2)\,.$$

Since the φ_i are epimorphisms, this map is an epimorphism for any t. Let

$$f_t : X \to \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$$

be the maps induced by ψ_t. (where we identified $\mathbb{C}^{m_1} \times \mathbb{C}^{m_2}$ with $\mathbb{C}^{m_1+m_2}$ by mapping (v_1, v_2) to (v_1, v_2)). Then

$$f_0 = j_0 \circ g_1 \quad \text{and} \quad f_1 = T \circ j_1 \circ g_2\,,$$

where $j_i : \mathrm{Gr}_n(\mathbb{C}^{m_i}) \to \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$ is the map induced by the inclusion $\mathbb{C}^{m_i} \to \mathbb{C}^{m_1+m_2}$ and T is the map induced by the isomorphism $\mathbb{C}^{m_1+m_2} \to \mathbb{C}^{m_1+m_2}$ that sends (v_1, v_2) to (v_2, v_1). Since T is homotopic to the identity, we can conclude that $j_0 \circ g_1$ and $j_1 \circ g_2$ are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$. Hence, the homotopy class of the map induced by an epimorphism is independent of the choice of the epimorphism and the integer m. This concludes the proof. □

lem:kTheo-colimUndHomVertauschen

Lemma 3.2.40. *For any compact X and direct system $\big((Y_m)_{m\in\mathbb{N}}, (i_m)_{m\in\mathbb{N}}\big)$ where all Y_m are Hausdorff and all i_m are closed embeddings we have a canonical bijection between the set theoretic direct limit of homotopy classes of maps and the*

homotopy classes of maps to the topological direct limit:

$$\operatorname{colim}_m [X, X_m] \cong [X, \operatorname{colim}_m X_m] .$$

Proof. We denote $\operatorname{colim}_m Y_m = Y$ and the corresponding inclusion $\varphi_m : Y_m \to Y$. Since X is compact, any map $f : X \to Y$ has image that lies in some $\operatorname{im}(\varphi_m)$ for m big enough. Assume that this was not the case, then there would exist an increasing sequence $(j_i)_{i\in\mathbb{N}}$ such that for all i there is an x_i with $f(x_i) \in \operatorname{im}\varphi_{j_i} \setminus \operatorname{im}\varphi_{j_i-1}$. With this we define the set

$$A := \{ f(x_i) \,|\, i \in \mathbb{N} \} \subseteq \operatorname{im} f \subseteq Y .$$

Then A is closed, since for all m the intersection $\operatorname{im}\varphi_m \cap A$ is finite and since $\operatorname{im}(\varphi_m)$ is closed, said intersection is closed. By the final topology we can conclude that the whole A is closed and since it is contained in the image of f it is compact (Here we use that the Y_i are Hausdorff!). Furthermore, the same argument shows that every subset $B \subset A$ is closed, because again $B \cap \operatorname{im}\varphi_m$ is finite. This tells us that A as a subset of Y carries the discrete topology: Let $C \subseteq A$ be an arbitrary subset. Then $A \setminus C \subseteq A$ is closed in Y and hence its complement in Y is open and intersected with A yields C. Hence, C is open in the subspace topology. This however is a contradiction, as A cannot be infinite, compact and carry the discrete topology. Hence any map $f : X \to Y$ has image that lies in some $\operatorname{im}(\varphi_m)$ for m big enough. But then there exists a f_m such that $f = \varphi_m \circ f_m$. This assignment is well defined on homotopy classes because if $f, s : X \to Y$ are homotopic, then there exists a homotopy $H : X \times I \to Y$ between them: By the same argument as before, the image of H is compact and hence contained in some $\operatorname{im}\varphi_m$. Hence, f_m and s_m are homotopic via H_m as maps to $\operatorname{im}\varphi_m$. Hence, we get a well defined canonical map

$$\Phi : [X, \operatorname{colim}_m Y_m] \to \operatorname{colim}_m [X, Y_m] .$$

This map is injective, as if f_m and s_m are homotopic as maps to $\operatorname{im}\varphi_m$, then they are also homotopic as maps to the direct limit. Finally Φ is surjective, as any element of the direct limit is represented by some $f_m : X \to Y_m$ and hence $\Phi([\varphi_m \circ f_m]) = [f_m]$. □

Definition 3.2.41. We denote $\operatorname{colim}_m \operatorname{Gr}_n(\mathbb{C}^m)$ by $\mathrm{BU}(n)$ and call it the **classifying space** for complex vector bundles of rank n. The name originates from the functor B that sends a topological group G to the classifying space of vector bundles with structure group G.

cor:kTheo-classifyingSpaceVectorBundles

Corollary 3.2.42. For any compact space X, there is a canonical bijection between the homotopy classes of maps from X to $\mathrm{BU}(n)$ and the isomorphism classes of vector bundles of rank n over X:

$$[X, \mathrm{BU}(n)] \cong \mathrm{Vect}_n(X) .$$

Proof. This is a direct application of Lemma 3.2.40. □

3.3 K-Theory as a Cohomology Theory

Definition 3.3.1. We need the following categories of topological spaces:

- **CWH** is the category of compact spaces.
- $\mathbf{CWH}^*_{\mathbf{cof}}$ is the category of well-pointed compact spaces.
- $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ is the category of pairs of compact spaces, where the inclusions are cofibrations.

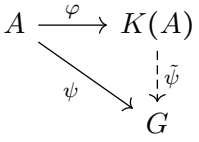
Here, we have the natural functors:

- $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}} \to \mathbf{CWH}^*_{\mathbf{cof}}$ by sending a pair $(X, Y) \mapsto (X/Y, [Y])$ with obvious morphisms. If $Y = \varnothing$ this is the disjoint union of X with a point.
- $\mathbf{CWH} \to \mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ by sending $X \mapsto (X, \varnothing)$ again with obvious morphisms. The inclusion of $\varnothing$ into X is a cofibration, as the empty set itself is an open neighbourhood and hence the pair is a NDR pair.
- $\mathbf{CWH} \to \mathbf{CWH}^*_{\mathbf{cof}}$ as the composition. Again, the inclusion of $[\varnothing]$ into the disjoint union $X \sqcup [\varnothing]$ is a cofibration, as $\{[\varnothing]\}$ is an open neighbourhood of itself.

If we descend to the category where the morphisms are homotopy classes of maps, we get the categories **hCWH**, $\mathbf{hCWH}^*_{\mathbf{cof}}$ and $\mathbf{hCWH}^{\mathbf{2}}_{\mathbf{cof}}$. Furthermore, the construction gives a functor from the category of abelian semigroups to the category of abelian groups.

def:kTheo-grothendiekGroupAndRing

Definition 3.3.2 (Grothendieck Group and Ring)**.** Let A be an abelian semigroup. Then the **Grothendieck group** of A is the abelian group $K(A)$ together with a homomorphism $\varphi : A \to K(A)$ such that for any other group G together with a homomorphism $\psi : A \to G$, there exists a unique homomorphism $\tilde{\psi} : K(A) \to G$ such that the following diagram commutes:



If G is a semiring, then $K(G)$ inherits a ring structure and we call $K(G)$ the **Grothendieck ring** of G.

cor:kTheo-ConstructionOfGrothendiekGroup

Corollary 3.3.3. The Grothendieck group exists for any abelian semigroup and is unique up to canonical isomorphism. If G is a semiring, then the Grothendieck ring exists and is unique up to canonical isomorphism.

Proof. The construction follows the following idea: We define tuples (a, b) that are to be interpreted as differences "$a - b$". For this construction to yield something useful we want to enforce the rule that "$a - b = a + c - (b + c)$" or said differently $(a + c, b + c) = (a, b)$. Finally, to account for the universal property, we want to use the smallest relation

that obeys said rule. This gives us the following construction: Let A be an abelian semigroup and $\Delta : A \to A \times A$ be the diagonal homomorphism that sends a to (a, a). Let $K(A) \coloneqq A \times A / \Delta(A)$. By this notation we mean: $\Delta(A)$ acts on $A \times A$ via

$$\Delta(A) \times (A \times A) \to A \times A \quad ((a, a), (b, c)) \mapsto (a + b, a + c)\,.$$

With this we have the set of orbits which gives rise to a relation:

$$\mathcal{R} \coloneqq \{((a, b), (a + c, b + c)) | c \in A\}$$

Now we take the smallest equivalence relation on $A \times A$ that contains $\mathcal{R}$ (This relation is made explicit in Remark 3.3.4). We define $K(A)$ to be $A \times A$ modulo said relation. Then $K(A)$ is a priori an abelian semigroup, but for $a, b \in A$ we have

$$(a, b) + (b, a) = (a + b, b + a) \in \Delta(A)\,,$$

and hence $K(A)$ is an abelian group. Also we can see that

$$\overline{(a, b)} = \overline{(a, 0)} - \overline{(b, 0)} = \alpha_A(a) - \alpha_A(b)$$

Together with the homomorphism $\alpha_A : A \to K(A)$ that sends a to the class of $(a, 0)$, we get the universal property of the Grothendieck group:

Let G be another abelian group and $\psi : A \to G$ a semigroup homomorphism. From this define

$$\tilde{\psi} : K(A) \to G\,, \quad \overline{(a, b)} \to \psi(a) - \psi(b)$$

The well definition of this map can be verified by a calculation. By calculating it also follows that $\tilde{\psi}$ is a group homomorphism and that $\tilde{\psi} \circ \alpha_A = \psi$. Hence, for the universal property we only need to verify uniqueness. So assume that $\theta : K(A) \to G$ is another group homomorphism. Then:

$$\theta\left(\overline{(a, b)}\right) = \underbrace{\theta(\alpha_A(a))}_{\psi(a)} - \underbrace{\theta(\alpha_A(b))}_{\psi(b)} = \tilde{\psi}\left(\overline{(a, b)}\right)$$

This concludes the construction of the Grothendieck group. To see the functoriality we first need to define what $K(\beta)$ means, when $\beta : A \to B$ defines a semigroup homomorphism between two abelian semigroups. For this we use the universal property and define $K(\beta)$ to be the unique group homomorphism coming from $\alpha_B \circ \beta : A \to K(B)$. Now the uniqueness of said definition shows the equalities $K(\mathrm{id}) = \mathrm{id}$ and $K(f \circ g) = K(f) \circ K(g)$. □

rem:kTheo-grothendieckEquivalence

Remark 3.3.4. In $K(A)$ we get the following equivalence for any two pairs $\overline{(a, b)}$ and $\overline{(a', b')}$:

$$\overline{(a, b)} = \overline{(a', b')} \Leftrightarrow \text{ There exists a } c \in A \text{ such that } (a + b' + c) = (a' + b + c)\,.$$

Proof. We start by proving "⇐". This direction follows from the properties of an equivalence relation, as then, every element of an equivalence class is also a representative:

$$\overline{(a,b)} = \overline{(a+(a'+b+c), b+(a+b'+c))} = \overline{(a'+(a+b+c), b'+(a+b+c))} = \overline{(a',b')}\,. \quad (3)$$

{eq:kTheo-theGrothendiekRelation}

The other direction follows from the minimality of the equivalence relation: To see this notice how the argument above works in every equivalence relation containing the orbits of $\Delta(A)$ and hence works in the smallest one. Furthermore,

$$(a,b) \sim (a',b') \Leftrightarrow \text{ There exists a } c \in A \text{ such that } (a+b'+c) = (a'+b+c)$$

defines an equivalence relation that contains the orbits and hence this is the smallest one. □

Remark 3.3.5 (A simpler Construction of the Grothendieck Group?)**.** One might be wondering, why we defined the relation on the pair $A \times A$ as the smallest equivalence relation starting by identifying the orbits of the operation of the diagonal, when Remark 3.3.4 might look and feel different. The reason for this detour is that I wanted to understand what Atiyah meant when he defined in Chapter II.1 [Ati89] atiyah1989k word for word:

A slightly different construction of $K(A)$ which is sometimes convenient is the following: Let $\Delta : A \to A$ be the diagonal homomorphism of semi-groups, and let $K(A)$ denote the set of cosets of $\Delta(A)$ in $A \times A$. It is a quotient semi-group, but the interchange of factors in $A \times A$ introduces an inverse in $K(A)$ so that $K(A)$ is a group.

Whilst this definition feels easy to the reader, it carries a lot of baggage. First of all, if one is not familiar with commutative monoid or abelian semigroup theory, one might want to generalise the notion of coset from group theory that states that for a subgroup $A \subseteq G$ of an abelian group G the cosets are of the form $g + A$. Whilst this notion can be generalised to semigroups, it would yield something useless. To see this just take $\mathbb{N}$. In the semigroup $\mathbb{N} \times \mathbb{N} / \Delta(\mathbb{N})$ the cosets $(2,0) + \mathbb{N}$ and $(3,1) + \mathbb{N}$ are different, as $(2,0)$ is not contained in the latter. To give Atiyah the benefit of the doubt we could generalise the definition of cosets as orbits of the right translation, which is what we did in our construction. However, the literature on semigroups from that time defines the notion of quotients in the setting of semigroups with the help of *congruence* instead of cosets. The latter refers to an equivalence relation that behaves well with the operation of said semigroup. Compare Chapter 1.5 [CP61]. CliffordPreston1961 It was even shown that the notion of congruence and coset are equivalent for groups (compare for example: Chapter 6.1 [Gre65]). Green1965 Regardless of what Atiyah meant by his notion of coset, the construction became known and cited. ([Bro96], [Lan05], [Cou04]) Brodzki1996 Landweber2005 Courtney2004 None of those references define coset in semigroups and hence this little detour into the history should function as a reminder, that we need to be careful when generalising things from the known categories. We are so used to cosets giving equivalence relations, that it's easy to overlook the needed care.

Remark 3.3.6. We will omit the inclusion maps α from now on and interpret (if not said different) the equivalence class $[E]$ of a vector bundle as an element in $K(X)$.

Corollary 3.3.7. In the proof above we have shown that the assignment K is a functor from the category of abelian semigroups to the category of abelian groups. Alternatively, we can view K as a functor from the category of semirings to the category of rings.

Definition 3.3.8. Let X be a compact Hausdorff space. We define the **K-group** of X to be the Grothendieck ring of the semiring $\mathrm{Vect}(X)$ of isomorphism classes of vector bundles over X with the direct sum as addition and the tensor product as multiplication. We denote the K-group of X by $K(X)$. As a whole we get a contravariant functor from the category of compact Hausdorff spaces to the category of rings

$$\mathbf{CWH} \to \mathbf{Ring}\,. \tag{4}$$

Remark 3.3.9. For a continuous map $f : X \to Y$ we denote the induced map $K(f^*) : K(Y) \to K(X)$ by just f^*.

Theorem 3.3.10. *The functor $K : \mathbf{CWH} \to \mathbf{Ring}$ factors through the homotopy category* **hCWH** *of compact Hausdorff spaces together with homotopy classes of maps.*

Proof. This directly follows from Theorem 3.2.4. □

def:kTheo-reducedKTheory

Definition 3.3.11 (Reduced K-Theory)**.** Let X be a compact Hausdorff space and $x_0 \in X$ be a base point. We define the **reduced K-group** $\tilde{K}(X)$ of X to be the kernel of the map $K(X) \to K(\{x_0\})$ induced by the inclusion $\{x_0\} \to X$.

cor:kTheo-splittingReducedGroupNatural

Corollary 3.3.12. If $c : X \to x_0$ is the collapsing map sending each point to the base-point, then we get a natural splitting

$$K(X) \cong \tilde{K}(X) \oplus K(x_0)\,.$$

Naturality here means, that for any homomorphism of pointed spaces $f : (X, x_0) \to (Y, y_0)$ we have a commuting diagram:

$$\begin{array}{ccc} K(X) & \longrightarrow & \tilde{K}(X) \oplus K(x_0) \\ \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle \tilde{K}(f)\oplus(\{x_0\}\to\{y_0\})^*} \\ K(Y) & \longrightarrow & \tilde{K}(Y) \oplus K(y_0) \end{array}$$

Proof. We have the exact sequence of rings:

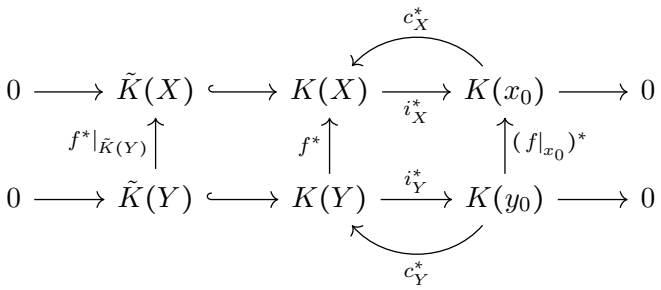
$$0 \longrightarrow \tilde{K}(X) \xrightarrow{inclusion} K(X) \xrightarrow{i_X^*} K(x_0) \longrightarrow 0$$

where $c_X^* : K(x_0) \to K(X)$ is a right inverse of i_X^*, and hence the sequence splits and $K(X) \cong \tilde{K}(X) \oplus K(x_0)$. The isomorphism is explicitly given by

$$K(X) \to \tilde{K}(X) \oplus K(x_0)\,, \quad \xi \mapsto (\xi - c_X^* \circ i_X^*(\xi), i_X^*(\xi))$$

and its inverse maps $(a, b) \mapsto a + c_X^*(b)$. The splitting is natural with respect to maps in $\mathbf{CWH}^*_{\mathbf{cof}}$, because for a map of pointed spaces $f : X \to Y$ we get commutativity:

diag:kTheo-NaturalSplitting



The well definition of $f^*|_{\tilde{K}(Y)}$ follows immediately from the commutativity. The commutativity itself follows from K being a contravariant functor and all topological maps commute. Finally, define $\tilde{K}(f) \coloneqq f^*|_{\tilde{K}(Y)}$ and we are done, since the naturality of the splittings, i.e. commutativity of the diagram in the corollary, then follows from commutativity of each square separately: the left square controls the $\tilde{K}$-summand and the right square (together with the sections c_X^*, c_Y^*) controls the $K(x_0)$-summand. □

Remark 3.3.13. For a map $f : (X, x_0) \to (Y, y_0)$ we denote $\tilde{K}(f)$ and $K(f)$ by f^*. The context will make clear if we work in the reduced theory or not.

Corollary 3.3.14. $\tilde{K}$ is a functor from $\mathbf{CWH}^*_{\mathbf{cof}}$ to **Ring**.

Proof. The proof follows immediately from the naturality of the splitting in Corollary 3.3.12. □

cor:kTheo-unpointedReducedIsNonReduced

Corollary 3.3.15. Let X be a compact Hausdorff space, and let

$$F : \mathbf{CWH} \to \mathbf{CWH}^*_{\mathbf{cof}}$$

be the functor that sends X to $(X, [\varnothing])$ with obvious homomorphisms. Then there is a natural isomorphism between the functors $\tilde{K} \circ F$ and K.

Proof. This follows from the fact that disjoint unions give rise to direct sums of semirings: $\mathrm{Vect}(X \sqcup \{\varnothing\}) \cong \mathrm{Vect}(X) \oplus \mathrm{Vect}(\{\varnothing\})$ and the isomorphism is given by gluing the bundles and hence natural. Furthermore, the Grothendieck ring of a direct sum gives the direct sum of rings (again in a natural manner). (This can be verified using the universal property). Hence we get the commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \tilde{K}(F(X)) & \xrightarrow{inclusion} & K(F(X)) & \xrightarrow{i^*} & K([\varnothing]) & \longrightarrow & 0 \\
\| & & & & \downarrow \Phi & & \| & & \| \\
0 & \longrightarrow & K(X) & \xrightarrow{i_1} & K(X) \oplus K([\varnothing]) & \xrightarrow{\pi_2} & K([\varnothing]) & \longrightarrow & 0
\end{array}$$

But then we can define the map $\Phi|_{\tilde{K}(F(X))}$ and diagram yoga shows that this is an isomorphism. Furthermore, since Φ is natural, the new map is also natural and defines a natural transformation between the two functors. □

Definition 3.3.16. Let $F : \mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}} \to \mathbf{CWH}^*_{\mathbf{cof}}$ be the functor that collapses the second space. Then the composition $\tilde{K} \circ F$ defines a contravariant functor.

Definition 3.3.17. Let (X, x_0) and (Y, y_0) be two pointed spaces. Then we define the **pointed sum** as:

$$X \vee Y := (X \sqcup Y)\big/(x_0 \sim y_0)\,.$$

With this, we can define the **smash product** as:

$$X \wedge Y := (X \times Y)\big/(X \vee Y)$$

Definition 3.3.18. For the sake of intuition, I want to give a bit of insight into tensor products without being too technical and opening Pandora's box of monoidal categories. We define a tensor product in a category simply as the space over which bilinear maps factor uniquely. To capture bilinearity in such a category we define the following. Let $\pi_i : X_0 \times X_1 \to X_i$ be the canonical projection for $i = 0, 1$. Any reasonable product should come with such surjective maps. Since those are surjective, the π_i admit a family of right inverses $r_i : X_i \to X_0 \times X_1$. We call such a right inverse "product preserving" if the compositions $\pi_2 \circ r_1$ and $\pi_1 \circ r_2$ are constant. Hence the right inverse maps into just one fibre. Now bilinearity in this general setting states, that for any product preserving right inverse of the projections r_i, the composition $f \circ r_i$ is structure preserving.

Remark 3.3.19 (What is the Smash Product)**.** We can think of the smash product as the tensor product in the pointed topological spaces as follows. Let $f : (X, x_0) \times (Y, y_0) \to (Z, z_0)$ be bilinear. For any $y \in Y$ we have the product preserving right inverse of π_1 given by

$$r_1^y : (X, x_0) \to (X, x_0) \times (Y, y_0), \quad x \mapsto (x, y)\,.$$

Analogously we define r_2^x for any $x \in X$ as

$$r_2^x : (Y, y_0) \to (X, x_0) \times (Y, y_0), \quad y \mapsto (x, y)\,.$$

Now for any $x \in X$ we have $f(x, y_0) = f \circ i_Y^x(y_0) = z_0$ as the composition is said to be basepoint preserving. Similarly we see that $f(x_0, y) = z_0$ for any $y \in Y$. Hence f factors through the quotient space

$$X \times Y\big/(X \times \{y_0\} \cup (Y \times \{x_0\}))\,,$$

where the numerator is the canonically embedded $X \vee Y$ into the product. Furthermore by the universal property of quotient spaces, this factorisation is unique.

Example 3.3.20. Let S^n be the n-dimensional sphere given by all points of norm one in $\mathbb{R}^{n+1}$. Then a general fact is that $S^n \cong I^n\big/\partial I^n$ and hence for all $n, m > 0$ we have $S^n \wedge S^m \cong S^{n+m}$. Later we will make this homeomorphism canonical. ref

def:orient-fromRelaSpaceToTheSphere

Definition 3.3.21 (From Real Space to the Sphere)**.** We now want to construct a canonical homeomorphism between S^m and $\overline{\mathbb{R}^m}$. First we use the stereographic projection from the South Pole: If $m = 0$ we define this to be

$$\pi_s : S^0 = \{-1, 1\} \setminus \{-1\} \to \mathbb{R}^0\,, \quad 1 \mapsto 0\,.$$

For $m > 0$ we define:

$$\begin{aligned}\pi_S : S^m \setminus \{S\} \subseteq \mathbb{R}^{m+1} &\to \mathbb{R}^m \\ (x_1, \dots, x_m, x_{m+1}) &\mapsto \left(\frac{x_1}{1+x_{m+1}}, \dots, \frac{x_m}{1+x_{m+1}}\right)\end{aligned}$$

with its inverse:

$$\begin{aligned}\pi_S^{-1} : \mathbb{R}^m &\to S^m \setminus \{S\} \subseteq \mathbb{R}^{m+1} \\ (y_1, \dots y_m) &\mapsto \left(\frac{2y_1}{1+\|y\|^2}, \dots, \frac{2y_m}{1+\|y\|^2}, \frac{1-\|y\|}{1+\|y\|}\right)\end{aligned}$$

Those are inverse maps and they give rise to maps between their one-point-compactifications. Hence $\overline{\mathbb{R}^m} \cong S^m$. Lets calculate $\pi_S^{-1} \circ \pi_s$, since we will later need a little bit of insight in those calculations:

$$\begin{aligned}\pi_S^{-1} \circ \pi_s(x_1, \dots, x_{m+1}) &= \pi_S^{-1}\left(\frac{x_1}{1+x_{m+1}}, \dots, \frac{x_m}{1+x_{m+1}}\right) \\ &= \left(\frac{2\frac{x_1}{1+x_{m+1}}}{1+\sum_{i=1}^m(\frac{x_i^2}{(1+x_{m+1})^2})}, \dots, \frac{2\frac{x_m}{1+x_{m+1}}}{1+\sum_{i=1}^m(\frac{x_i^2}{(1+x_{m+1})^2})}, \frac{1-\sum_{i=1}^m(\frac{x_i^2}{(1+x_{m+1})^2})}{1+\sum_{i=1}^m(\frac{x_i^2}{(1+x_{m+1})^2})}\right)\end{aligned}$$

Before we keep calculating, we notice the denominator can be rewritten as:

$$1 + \sum_{i=1}^m(\frac{x_i^2}{(1+x_{m+1})^2}) = \frac{(1+x_{m+1})^2 + \sum_{i=1}^m x_i^2}{(1+x_{m+1})^2}$$

And hence:

$$\begin{aligned}\frac{2\frac{x_j}{1+x_{m+1}}}{1+\sum_{i=1}^m(\frac{x_i^2}{(1+x_{m+1})^2})} &= \frac{2\frac{x_j}{1+x_{m+1}}}{\frac{(1+x_{m+1})^2+\sum_{i=1}^m x_i^2}{(1+x_{m+1})^2}} && (5)\\ &= \frac{2x_j(1+x_{m+1})}{(1+x_{m+1})^2+\sum_{i=1}^m x_i^2} && (6)\\ &= \frac{2x_j(1+x_{m+1})}{(1+x_{m+1})^2+\sum_{i=1}^m x_i^2} && (7)\\ &= \frac{2x_j(1+x_{m+1})}{1+2x_{m+1}+\underbrace{\sum_{i=1}^{m+1} x_i^2}_{=1}} && (8)\\ &= x_j && (9)\end{aligned}$$

And the calculation of the last term is:

$$\begin{aligned}
\frac{1-\sum_{i=1}^{m}\left(\frac{x_i^2}{(1+x_{m+1})^2}\right)}{1+\sum_{i=1}^{m}\left(\frac{x_i^2}{(1+x_{m+1})^2}\right)} &= \frac{\frac{(1+x_{m+1})^2-\sum_{i=1}^{m}x_i^2}{(1+x_{m+1})^2}}{\frac{(1+x_{m+1})^2+\sum_{i=1}^{m}x_i^2}{(1+x_{m+1})^2}} \\
&= \frac{(1+x_{m+1})^2-\sum_{i=1}^{m}x_i^2}{(1+x_{m+1})^2+\sum_{i=1}^{m}x_i^2} \\
&= \frac{1+2x_{m+1}+x_{m+1}^2-\sum_{i=1}^{m}x_i^2}{1+2x_{m+1}+\underbrace{x_{m+1}^2+\sum_{i=1}^{m}x_i^2}_{=1}} \\
&= \frac{1+2x_{m+1}+2x_{m+1}^2-\sum_{i=1}^{m+1}x_i^2}{2+2x_{m+1}} \\
&= \frac{x_{m+1}(2+2x_{m+1})}{2+2x_{m+1}} = x_{m+1}
\end{aligned}$$

This concludes the calculation of $\pi_S^{-1}\circ\pi_s = \mathrm{id}$ and $\pi_s\circ\pi_S^{-1} = \mathrm{id}$ can be calculated in a similar manner.

Definition 3.3.22 (One-Point Compactification)**.** For a topological space X we define $\overline{X} \coloneqq X\sqcup\{\infty\}$, where open sets are either open subsets of X, or sets of the form $\{\infty\}\sqcup X\setminus C$ where C is compact and closed in X. For non compact Hausdorff spaces this is the one-point compactification that is characterised by being the unique (up to homeomorphism) compactification that contains X and one more point. If X is already compact, then $\{\infty\}$ is an open subset of $\overline{X}$ (just choose $C = K$) and thereby isolated. Hence $\overline{X}$ is the pointing of X given by $(X,[\varnothing])$ from Corollary 3.3.15.

lem:orient-quotientIsCompactification

Lemma 3.3.23 (Open subspaces of compact spaces)**.** *Let Z be a compact Hausdorff space and $U\subseteq Z$ open. Then the map*

$$Z\big/(Z\setminus U)\to\overline{U}\,,\quad [z]\mapsto z \text{ for } z\in U\,,\quad [Z\setminus U]\mapsto\infty$$

is a homeomorphism.

Proof. The map is the quotient of the function $\varphi: Z\to\overline{U}$ that is the identity on U and sends $Z\setminus U$ constant to ∞. Hence, we only need to prove continuity for φ. Nor $O\subseteq\overline{U}$ is open, if it is open in U, or the complement of a compact set $C\subset U$ in $\overline{U}$. In both cases, $\varphi^{-1}(U)$ is open and hence φ is continuous. Since the quotiont map is now a continuouos bijection from a compact space to a Hausdorff space, it is a homeomorphism. □

lem:orient-compactificationOfProduct

Lemma 3.3.24 (Compactification of a product)**.** *Let X and Y be locally compact Hausdorff spaces, and point $\overline{X}$ and $\overline{Y}$ at their respective points at infinity. Then*

there is a homeomorphism

$$\Xi_{X,Y} : \overline{X \times Y} \xrightarrow{\cong} \overline{X} \wedge \overline{Y}$$

which is the identity on $X \times Y$ and sends ∞ to the base point. It is natural in the sense that for proper maps $f : X \to X'$ and $g : Y \to Y'$ the square

$$\begin{array}{ccc} \overline{X \times Y} & \xrightarrow{\Xi_{X,Y}} & \overline{X} \wedge \overline{Y} \\ {\scriptstyle \overline{f \times g}}\downarrow & & \downarrow{\scriptstyle \overline{f} \wedge \overline{g}} \\ \overline{X' \times Y'} & \xrightarrow[\Xi_{X',Y'}]{} & \overline{X'} \wedge \overline{Y'} \end{array}$$

commutes.

Proof. Let $Z := \overline{X} \times \overline{Y}$. Then Z is a compact Hausdorfspace that contains $U := X \times Y$ as an open subset. The complement of U consists of those points where one of the coordinates is infinit and hence

$$Z \setminus U = (\{\infty\} \times \overline{Y}) \cup (\overline{X} \times \{\infty\}) = \overline{X} \vee \overline{Y}\,.$$

Therefore, by definition of the smash product, $Z \big/ (S \setminus U) = \overline{X} \wedge \overline{Y}$ and by Lemma 3.3.23 we can conculde that

$$\overline{X \times Y} \cong (Z \big/ (Z \setminus U) = \overline{X} \wedge \overline{Y}\,.$$

For the naturality notice that the product of proper maps is propper and hence all maps are continouos. The commmutation then comes done to checking where infinity gets mapped which is always the basepoint of the wedge product. □

lem:kTheo-SmashIsAssociative

Lemma 3.3.25 (Associativity and Kommutativity)**.** *For pointed compact spaces $(A, a_0), (B, b_0), (C, c_0)$ there is a natural homeomorphism*

$$(A \wedge B) \wedge C \to A \wedge (B \wedge V)$$

That restricts to the identity over $(A \setminus \{a_0\}) \times (B \setminus \{b_0\}) \times (C \setminus \{c_0\})$. Furthermore, there is a natural homeomorphism

$$A \wedge B \to B \wedge A$$

that restricts to the change of coordinates over

$$(A \setminus \{a_0\} \times B \setminus \{b_0\}) \to (B \setminus \{b_0\} \times A \setminus \{a_0\})$$

Proof. Notice how for all compact pointed spaces (X, x_0) the identyty (and $x_0 \mapsto \infty$) is a natural homeomorphism $\overline{X \setminus \{x_0\}} \cong X$ and we denote $X^\circ := X \setminus \{x_0\}$. Furthermore,

the smash product of compact spaces is again compact (as it is a quotient of a closes space). Hence we can just calculate

$$\begin{aligned}(A \wedge B) \wedge C &\cong \overline{(A \wedge B)^\circ} \wedge \overline{C^\circ} \\ &\cong \overline{(A^\circ \times B^\circ)} \wedge \overline{C^\circ} \\ &\cong \overline{(A^\circ \times B^\circ) \times C^\circ} \\ &\cong \overline{A^\circ \times (B^\circ \times C^\circ)} \\ &\cong A \wedge (B \wedge V)\,.\end{aligned}$$

For the kommutativity we can proceed in the same way:

$$A \wedge B \cong \overline{(A \wedge B)^\circ} \cong \overline{(B \wedge A)^\circ} \cong B \wedge A \tag{10}$$

□

Remark 3.3.26. The statement about associativity can be extended to the category of $\mathbf{CGWH}^*_{\mathbf{cof}}$, but not all pointed topological spaces.

ref einfügen

def:ktheo-wedgeOfShoperesIsSphere

Definition 3.3.27 (Convention)**.** Let π_s^n denote the compactified stereographical projection $\overline{S^n} \to \overline{\mathbb{R}^n}$ With those methods we can now define the homeomorphism

$$\Upsilon_{n,m} := (\pi_s^{n+m})^{-1} \circ \Xi^{-1} \circ (\pi_s^n) \wedge (\pi_s^m) : S^n \wedge S^m \to S^{n+m}\,.$$

Often we will identify smash product of spheres and for those identification we will always use this homeomophism. For the composition to make sence we identify $\mathbb{R}^n \times \mathbb{R}^m$ with $\mathbb{R}^{n+m}$ (also for n or m equal zero.)

rem:kTheo-AssoziativityOfUspilon

Remark 3.3.28 (Assoziativity of Υ)**.** Let Υ be the map from Definition 3.3.27, and

$$AS : (S^k \wedge S^l) \wedge S^m \to S^k \wedge (S^l \wedge S^m)$$

denote the associativity map of the wedge product from Lemma 3.3.25. Then for all $k, l, m \in \mathbb{N}$ we have

$$\Upsilon_{k+l,m} \circ \left(\Upsilon_{k,l} \wedge \mathrm{id}_{S^m}\right) = \Upsilon_{k,l+m} \circ \left(\mathrm{id}_{S^k} \wedge \Upsilon_{l,m}\right) \circ AS\,.$$

Proof. We will pointwise compare the maps. Both sides map the basepoint to the basepoint. Now assume that $x \in (S^k \wedge S^l) \wedge S^m$ is not the basepoint. Then $x = [[u, v], w]$ with $u, v, w \neq S$ where S is the South pole. Let $a = \pi_S(u)$, $b = \pi_S(v)$, $c = \pi_s(w)$. Since Ξ is the identity outside of the base points we know that the left side reads

$$\pi_S^{-1}((a, b), c) = \pi_S^{-1}(a, b, c)\,,$$

whilst the right side reads

$$\pi_S^{-1}(a, (b, c)) = \pi_S^{-1}(a, b, c)\,.$$

(Remember, that we omitted the identification of $\mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^{n+m}$). □

Definition 3.3.29. Let (X, x_0) be a pointed topological space. We define the **reduced suspension** of X to be

$$\Sigma X := S^1 \wedge X\,.$$

Furthermore, we define inductively $\Sigma^n X := \Sigma(\Sigma^{n-1}X) \cong S^n \wedge X$ and using the principle of permanence we extend this definition to $\Sigma^0 X := S^0 \wedge X \cong X$. The last homeomorphism is canonical.

Definition 3.3.30. Let (X, x_0) be a pointed topological space. We define the **suspension** of X to be the space

$$S(X) := X \times I \big/ \sim \tag{11}$$

where $(x,t) \sim (x',t')$ if $t = 1$ or $t = 0$. Here we identify the top to a point and the bottom to a different point. $((x,1) \nsim (0,0))$. We view $(S(X),(x,1))$ as a pointed space.

In general the reduced and unreduced suspension differ. However, under the following condition there will be a natural homotopy equivalence:

lem:kTheo-wellpointedImpliesCylinderCofibration

Lemma 3.3.31. *If (X, x_0) is well-pointed, then the inclusion $\{x_0\} \times I \hookrightarrow S(X)$ is a cofibration.*

Proof. This proof is omitted but is Proposition 5.1.16 in [tom08]. tomdieck2008algebraic □

cor:kTheo-reducedSuspensionIsSuspension

Corollary 3.3.32. If (X, x_0) is well pointed, then $S(X)$ is homotopically equivalent to $\Sigma(X)$.

Proof. By lemma 3.3.31 we conclude that $(S(X), x_0 \times I)$ satisfies the homotopy extension property. Hence, by Lemma 1.2.4 we know that $S(X) \simeq S(X)\big/x_0 \times I$ and the homeomorphism

$$S(X)\big/x_0 \times I \cong \Sigma(X)$$

concludes the proof. □

def:kTheo-lowerKGroub

Definition 3.3.33 (Lower K-groups)**.** For $n \geq 0$ we define:

$$\begin{array}{lcll} \tilde{K}^{-n}(X, x_0) & := & \tilde{K}(\Sigma^n X) & \text{for} \quad (X, x_0) \in Ob(\mathbf{CWH}^*_{\mathbf{cof}}), \\ K^{-n}(X, Y) & := & \tilde{K}^{-n}(X/Y, [Y]) & \text{for} \quad (X, Y) \in Ob(\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}) \\ K^{-n}(X) & := & K^{-n}(X, \varnothing) & \text{for} \quad X \in Ob(\mathbf{CWH})\ . \end{array}$$

All those definitions give contravariant and homotopically invariant functors into the category of rings.

cor:kTheo-K(X)istK(X,nix)

Corollary 3.3.34. Let X be a topological space. Then we have a natural isomorphism between the functors $K^0(X)$ and $K(X)$. Furthermore, if (X, x_0) is a pointed topological space, we have a natural isomorphism between $\tilde{K}$ and K^0.

Proof. When disentangling the definitions, we see that

$$K^0(X) = \tilde{K}(\Sigma^0(X, [\varnothing])) \cong \tilde{K}(X, [\varnothing]) \cong K(X)$$

where the first isomorphism comes from Σ^0 being canonically isomorphic to the identity and the second isomorphism is given in Corollary 3.3.15. The natural isomorphism between K^0 and $\tilde{K}$ is given by definition and the canonical isomorphism $X \cong S^0 \wedge (X)$. □

Remark 3.3.35. If we have pointed topological spaces (X, x_0) or pairs (X, A) and want to denote $K^0(X, pt)$ or $K^0(X, A)$ we will often omit the exponent and just write $K(X, x_0)$ and $K(X, A)$.

Theorem 3.3.36 (Excision)**.** *For any pair (X, A) and $U \subseteq A$ open such that $\bar{U} \subseteq \mathring{A}$ the inclusion*

$$i : (X \setminus U, A \setminus U) \to (X, A)$$

induces an isomorphism

$$i^* : K^{-n}(X, A) \to K^{-n}(X \setminus U, A \setminus U)$$

Proof. Since U is open and A is closed, $(X \setminus U, A \setminus U)$ is a compact pair. The inclusion i induces a continuous bijection

$$\varphi : (X \setminus U) \Big/ (A \setminus U) \to X/A$$

which is a homeomorphism as a continuous bijection from a compact space to a Hausdorff space. Hence i^* is an isomorphism. □

3.4 Homotopy-Theoretic definition of K

def:kTheo-trivialBundleAndLimitInclusinos

Definition 3.4.1. For any natural number $n \in \mathbb{N}$ we define $\underline{n}$ to be the n-dimensional trivial bundle over a base space X. With this we define the inclusions

$$i_n : \mathrm{Vect}_n(X) \to \mathrm{Vect}_{n+1}(X), \quad \xi \mapsto \xi \oplus [\underline{1}],$$

and this gives rise to the direct limit $\mathrm{colim}_n \mathrm{Vect}_n(X)$.

Definition 3.4.2 (Stable Equivalence)**.** Two vector bundles E, F over the same base space are called **stably equivalent** if there exists an n such that E and F become isomorphic after adding the trivial bundle:

$$E \oplus \underline{n} \cong F \oplus \underline{n}.$$

Lemma 3.4.3. *In $K(X)$ two equivalence classes $[E]$ and $[F]$ are the same, if and only if E and F are stably equivalent.*

Proof. Assume that two vector bundles satisfy

$$E \oplus \underline{n} \cong F \oplus \underline{n}\,,$$

for some $n \in \mathbb{N}$. By abuse of notation we let E (and $\underline{n}$) denote a vector bundle and its isomorphism class simultaneously. Then by Remark 3.3.4

$$[E] = \overline{(E,0)} = \overline{(E+\underline{n},\underline{n})} = \overline{(F+\underline{n},\underline{n})} = \overline{(F,0)} = [F]\,.$$

On the other hand, if $[E] = [F]$, then by the same Remark we can conclude:

$$(E,0) \sim (F,0) \Leftrightarrow \text{ there exists } G \text{ such that} E \oplus G = F \oplus G\,.$$

Now by Corollary 3.2.26 we can find a vector bundle G' such that $G \oplus G' = \underline{n}$ for some n. Then

$$E \oplus G \oplus G' = F \oplus G \oplus G' \Rightarrow E \oplus \underline{n} = F \oplus \underline{n}$$

□

Definition 3.4.4. We call two vector bundles E, F over the same base space **eventually equivalent**, if there exist $n, m \in \mathbb{N}$ such that $E \oplus \underline{n} \cong F \oplus \underline{m}$.

Lemma 3.4.5. *The set of equivalence classes using the notion of eventual equivalence is the limit*

$$\operatorname{colim}_n \operatorname{Vect}_n(X)$$

Proof. Let $\operatorname{EVect}(X)$ be the set of equivalence classes using the notion of eventual equivalence. This satisfies the universal property of the direct limit. To see this let $f_m : \operatorname{Vect}_m(X) \to Y$ be a family of maps, that commute with the inclusions i_n from Definition 3.4.1. From those we define the map

$$f : \operatorname{EVect}(X) \to Y\,, \quad [\xi] \mapsto f_{\operatorname{rank}(\xi)}(\xi)\,.$$

This map is well defined, because f commuting with the inclusions i_n is equivalent to f being invariant under eventual equivalence. Uniqueness follows from every element of $\operatorname{EVect}(X)$ being represented by a vector bundle. □

Lemma 3.4.6. *The set of equivalence classes using the notion of eventual equivalence together with the direct sum is an abelian group*

Proof. Everything except inverses is clear. The neutral element is the class of $\underline{0}$. For inverses we take a bundle E and by Corollary 3.2.26 there exists a G such that $E \oplus G \cong \underline{n}$. Now $[E] + [G] = [\underline{n}] = [\underline{0}]$. □

lem:kTheo-reducedKisEventually

Lemma 3.4.7. *For a compact connected space X there is a natural group isomorphism.*

$$K(X) \cong \mathbb{Z} \times \mathrm{EVect}(X) \cong \mathbb{Z} \times \mathrm{colim}_n \mathrm{Vect}_n(X)\,.$$

Furthermore if (X,p) is pointed, there is a natural group isomorphism

$$\tilde{K}(X) \cong \mathrm{EVect}(X) \cong \mathrm{colim}_n \mathrm{Vect}_n(X)\,.$$

Proof. Let K be an element of $K(X)$. Then by Corollary 3.3.3 we can write K as the difference of two classes of vector bundles $K = [E] - [F]$. We abuse the notation of equivalence classes in this proof meaning we won't differentiate between isomorphism classes of vector bundles in Vect_n, eventually or stable isomorphism classes or equivalence classes of naturally included vector bundles in $K(X)$. The context should always be clear. With this we define the function

$$\mathrm{rank}(K) : K(X) \to \mathbb{Z}\,, \quad K = [E] - [F] \mapsto \mathrm{rank}(E) - \mathrm{rank}(F)\,.$$

This is well defined as if $K = [E] - [F] = [E'] - [F']$, then by the relation in $K(X)$ we know that there is an n such that

$$E \oplus F' \oplus \underline{n} \cong E' \oplus F \oplus \underline{n}\,.$$

and hence:

$$\mathrm{rank}(E) - \mathrm{rank}(F) = \mathrm{rank}(E') - \mathrm{rank}(F')$$

Furthermore, rank has a natural right inverse given by

$$r : \mathbb{Z} \to K(X)\,, \quad m \mapsto \begin{cases} [m] - [0] & \text{if } m \geq 0 \\ [0] - [-m] & \text{if } m < 0\,. \end{cases}$$

Now we claim that the kernel $\ker(\mathrm{rank}) \cong \mathrm{EVect}(X)$. For this we define the map

$$\varphi : \mathrm{EVect}(X) \to K(X)\,, \quad [E] \mapsto [E] - [\underline{\mathrm{rank}(E)}]$$

To see the well-definedness we assume that E and E' are eventually isomorphic and hence there are $n, n' \in \mathbb{N}$ such that $E \oplus \underline{n} \cong E' \oplus \underline{n'}$. But then in $K(X)$ we have:

$$[E] - [\underline{\mathrm{rank}(E)}] = [E \oplus \underline{n}] - [\underbrace{\underline{\mathrm{rank}(E)} \oplus \underline{n}}_{=\underline{\mathrm{rank}(F)} \oplus \underline{n'}}] = [E' \oplus \underline{n'}] - [\underline{\mathrm{rank}(F)} \oplus \underline{n'}] = [F] - [\underline{\mathrm{rank}(F)}]\,.$$

Obviously $\mathrm{rank} \circ \varphi \equiv 0$. To get the other inclusion we first notice the following: Let $K \in K(X) = [E] - [F]$ be arbitrary. By Corollary 3.2.26 there exists a G such that $F \oplus G \cong \underline{n}$ for some $n \in \mathbb{N}$. Hence if K is in the kernel of rank we have:

$$K = [E] - [F] = [E \oplus G] - [\underline{n}] = [E \oplus G] - [\underline{\mathrm{rank}(E \oplus G)}] = \varphi([E \oplus G])$$

This leads to the naturally split short exact sequence

$$0 \longrightarrow \mathrm{EVect}(X) \xrightarrow{\varphi} K(X) \underset{\text{rank}}{\overset{r}{\rightleftarrows}} \mathbb{Z} \longrightarrow 0$$

This already proves the isomorphisms

$$K(X) \cong \mathbb{Z} \times \mathrm{EVect}(X) \cong \mathbb{Z} \times \mathrm{colim}_n \mathrm{Vect}_n(X)\,.$$

Now we define the group isomorphism

$$l : \mathbb{Z} \to K(p)\,, \quad m \mapsto \begin{cases} [m] - [0] & \text{if } m \geq 0 \\ [0] - [-m] & \text{if } m < 0\,. \end{cases}$$

This gives the commutative diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{SVect}(X) & \xrightarrow{\varphi} & K(X) & \underset{\text{rank}}{\overset{r}{\rightleftarrows}} & \mathbb{Z} & \longrightarrow & 0 \\ & & & & \downarrow \mathrm{id} & & \downarrow l & & \\ 0 & \longrightarrow & \tilde{K}(X) & \xrightarrow{\text{inclusion}} & K(X) & \underset{c_p^*}{\overset{i^*}{\rightleftarrows}} & K(p) & \longrightarrow & 0 \end{array}$$

Where i is the inclusion $\{p\} \to X$ and c_p is the collapse $X \to \{p\}$. Then by some diagram chasing the map

$$\tilde{\varphi} : \mathrm{SVect}(X) \to \tilde{K}(X)$$

is well defined and by the five lemma (add zeros to the short sequences on the left) it is an isomorphism. □

rem:kTheo-elementsInReduced KtheoryAreGivenByEMionusN

Remark 3.4.8. In the proof above we have seen that every element $k \in K(X)$ is given as the difference of a vector bundle and a trivial bundle $k = [\xi] - [\underline{n}]$ for some vector bundle ξ and $n \in \mathbb{N}$. We also saw that $\tilde{K}(X)$ is in the kernel of rank and hence any $k \in \tilde{K}(X)$ is given by $k = [\xi] - [\underline{\mathrm{rank}(\xi)}]$ where ξ is a vector bundle.

Definition 3.4.9. We define the **unitary group** U to be the direct limit of the unitary group of rank n under the standard inclusions

$$U(n) \to U(n+1)\,, \quad A \mapsto A \oplus \mathbb{1}_1 = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition 3.4.10. We now want to make $BU(n)$ into a direct system. Notationally, this proof will have many maps with two indices. The convention is that lower indices show for which colimit the maps form a directed system. As an example, the maps i_m^n are used for the colimit over m. We start with the inclusion

$$j_m : \mathbb{C}^m \to \mathbb{C}^{m+1}\,, \quad (x) \mapsto (x, 0)\,,$$

which gives rise to the maps

$$f_n^m : \mathrm{Gr}_n(\mathbb{C}^m) \to \mathrm{Gr}_{n+1}(\mathbb{C}^{m+1}), \quad V \mapsto j_m(V).$$

Those inclusions commute with the maps

$$i_m^n : \mathrm{Gr}_n(\mathbb{C}^m) \to \mathrm{Gr}_n(\mathbb{C}^{m+1}), \quad V \mapsto V \oplus \langle e_{m+1} \rangle$$

from the direct system from Corollary 3.2.38. If we denote the inclusion by $\varphi_m^{n+1} : \mathrm{Gr}_{n+1}(\mathbb{C}^m) \to \mathrm{colim}_m(\mathrm{Gr}_{n+1}(\mathbb{C}^m))$ we get the following commuting diagram, for all $n \leq m$:

$$\begin{array}{ccccc} \mathrm{Gr}_n(\mathbb{C}^m) & \xrightarrow{f_n^m} & \mathrm{Gr}_{n+1}(\mathbb{C}^{m+1}) & \xrightarrow{\varphi_{m+1}^{n+1}} & \mathrm{colim}_m(\mathrm{Gr}_{n+1}(\mathbb{C}^m)) = BU(n+1) \\ \downarrow{\scriptstyle i_m} & & \downarrow{\scriptstyle i_m^n} & \nearrow_{\varphi_{m+2}^{n+1}} & \\ \mathrm{Gr}_n(\mathbb{C}^{m+1}) & \xrightarrow{f_n^{m+1}} & \mathrm{Gr}_{n+1}(\mathbb{C}^{m+2}) & & \end{array}$$

Now for all n we use the universal property of the direct limit $\mathrm{colim}_m(\mathrm{Gr}_n(\mathbb{C}^m))$ on the family

$$\left(\varphi_{m+1}^{n+1} \circ f_n^m : \mathrm{Gr}_n(\mathbb{C}^m) \to BU(n+1)\right)_m$$

to induce maps $f_n : BU(n) \to BU(n+1)$, which in turn give us a directed system and let us define:

$$BU := \mathrm{colim}_n BU(n) = \mathrm{colim}_n \left(\mathrm{colim}_m \mathrm{Gr}_n(\mathbb{C}^m) \right).$$

Lemma 3.4.11. *The maps $f_n : BU(n) \to BU(n+1)$ are closed embeddings.*

Proof. Since $\mathrm{Gr}_n(\mathbb{C}^m)$ is compact and Hausdorff for any $n \leq m$ the maps f_n^m are closed embeddings. Furthermore, the natural maps $\varphi_m^n : \mathrm{Gr}_n(\mathbb{C}^m) \to BU(n)$ are also closed embeddings. Now assume that $f_n(x) = f_n(x')$. Then for r big enough (omitting the inclusions) $f_n^r(x) = f_n^r(x')$ which contradicts the injectivity of the f_n^r. Now we want to show that f_n has a closed image. For this we let $C := \mathrm{im} f_n \subset BU(n+1)$ and call the natural inclusions $\varphi_m^{n+1} : \mathrm{Gr}_{n+1}(\mathbb{C}^m) \to BU(n+1)$. Then C is closed, if and only if $C \cap \mathrm{im}\varphi_m^{n+1}$ is closed for all $m > n$. To show this we first show that $C \cap \mathrm{im}\varphi_m^{n+1} = \mathrm{im}\, \varphi_m^{n+1} \circ f_n^{m-1}$. The first inclusion $\subseteq$ is clear. The other inclusion follows directly from the construction of f_n that satisfies $f_n \circ \varphi_{m-1}^n = \varphi_m^{n+1} \circ f_n^{m-1}$. By Remark 3.2.37 we know that $\varphi_m^{n+1} \circ f_n^{m-1}$ is a closed embedding hence $C \cap \mathrm{im}\varphi_m^{n+1}$ is closed for all $m > n$ and hence C is closed. Now finally we need to show that f_n are homeomorphisms onto their image. For this we show that they are closed maps, meaning they map closed sets to closed sets. This property together with injectivity implies the closed embedding property. Let $U \subset BU(n)$ be closed. By construction of the direct limit we can write $U = \bigcup_m (\mathrm{im}(\varphi_m^n) \cap U) = \bigcup_m (\varphi_m^n(U_m))$, where $U_m := (\varphi_m^n)^{-1}(U)$. Now by the same argument as before we have the equality $f_n(U) \cap \mathrm{im}\varphi_{m+1}^{n+1} = \varphi_{m+1}^{n+1} \circ f_n^m(U_m)$. This is closed for all m because the maps φ_{m+1}^{n+1} and f_n^m are closed embeddings and hence closed. But then $f_n(U)$ is closed in the final topology and hence f_n is a closed embedding. □

thm:kTheo-reducedKgroupAsHomotopy

Theorem 3.4.12. *For a connected pointed space X we have a natural homomorphism*

$$\tilde{K}(X) \cong [X, BU]$$

Proof. We can now interchange the homotopy classes of maps and the limit by Lemma 3.2.40:

$$\begin{aligned} [X, BU] &= [X, \operatorname{colim}_n \big(BU(n)\big)] \\ &\cong \operatorname{colim}_n \operatorname{Vect}_n(X) && \text{by Corollary 3.2.42,} \\ &\cong \tilde{K}(X) && \text{by Lemma 3.4.7.} \end{aligned}$$

□

lemma:kTheo-generalGroupContractsToUnitary

Lemma 3.4.13. *The general linear group $\mathrm{GL}_n(\mathbb{C})$ retracts onto the unitary group $U(n)$ and the retraction commutes with the inclusions $A \mapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$.*

Proof. Let $A \in \mathrm{GL}_n(\mathbb{C})$. Then A^*A is a positive-definite Hermitian matrix, since for $v \in \mathbb{C}^n$:

$$v^*A^*Av = (Av)^*(Av) = \|Av\|^2 \geq 0\,.$$

Hence, it is diagonalisable with real non-negative eigenvalues $\lambda_1, \dots \lambda_n > 0$. We define $P_A := \sqrt{A^*A}$ to be the matrix, that scales all eigenspaces by $\sqrt{\lambda_i}$. Then $P_A^2 = A^*A$, and $\sqrt{A^*A}$ is also positive definite Hermitian, as all eigenvalues are positive and hence, it is invertible. Notice, that

$$\sqrt{\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}^* \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}} = \begin{pmatrix} \sqrt{A^*A} & 0 \\ 0 & 1 \end{pmatrix}.$$

We then define $U_A := A(P_A)^{-1}$. By definition $A = U_A P_A$, and since P_A is Hermitian, P_A^{-1} is also Hermitian letting us conclude that U_A is unitary.

$$U_A^* U_A = ((P_A)^{-1})^* \underbrace{A^*A}_{P_A^2} (P_A)^{-1} = \mathbb{1}\,.$$

Notice again, that for $\tilde{A} = A \oplus \mathbb{1}_1 \in GL_{n+1}(\mathbb{C})$, we have $U_{\tilde{A}} = U_A \oplus \mathbb{1}_1 \in U(n+1)$. Now the space of positive-definite Hermitian matrices is convex. This is just a calculation of all the axioms for the matrices on the connecting line of two positive-definite Hermitian matrices. But as a convex space, it is contractible. □

lem:kTheo-reducedSuspendedKGroupHomotopically

Lemma 3.4.14. *For a connected pointed space X with well sitting basepoint, we have a natural homomorphism*

$$\tilde{K}(\Sigma X) \cong [X, U]$$

The group structure in $[X,U]$ comes from viewing $[X,U]$ as the direct limit of abelian groups $\operatorname{colim}[X, \mathrm{GL}_n(\mathbb{C})]$, where the group structure is derived from the one in $\mathrm{GL}_n(\mathbb{C})$.

Proof. If the base point is well sitting, we can apply Corollary 3.3.32, which gives us the homotopy equivalence

$$\Sigma X \simeq S(X)\,.$$

Now $S(X)$ itself can be cut into two contractible cones

$$\begin{aligned} C_+(X) &:= \{(x,t) \in S(X) \,|\, t \geq \tfrac{1}{2}\} \\ C_-(X) &:= \{(x,t) \in S(X) \,|\, t \leq \tfrac{1}{2}\}\,. \end{aligned}$$

Those cones intersect in $C_+ \cap C_- = X \times \{\frac{1}{2}\} \cong X$. Now denote the trivial n bundle over $C_+(X)$ by $\underline{n}_+$ and over $C_-(X)$ by $\underline{n}_-$. With this notation we know by the proof of Lemma 3.2.14 that the map

{eq:kTheo-SXtoHomotoy}

$$\begin{aligned} \varphi_n : [X, \mathrm{GL}_n(\mathbb{C})] &\to \mathrm{Vect}_n(\Sigma X) \\ f &\mapsto \underline{n}_+ \bigcup_f \underline{n}_- \end{aligned} \tag{12}$$

is a bijection for all n. Furthermore, the inclusion of $\mathrm{GL}_n(\mathbb{C})$ into $\mathrm{GL}_{n+1}(\mathbb{C})$ gives an inclusion $[X, \mathrm{GL}_n(\mathbb{C})]$ into $[X, \mathrm{GL}_{n+1}(\mathbb{C})]$ which commutes with the inclusion $\mathrm{Vect}_n(\Sigma X)$ into $\mathrm{Vect}_{n+1}(\Sigma X)$ under the bijection (12) meaning that for any $n \leq m$ the diagram

$$\begin{array}{ccc} [X, \mathrm{GL}_n(\mathbb{C})] & \xrightarrow{\varphi_n} & \mathrm{Vect}_n(\Sigma X) \\ \downarrow{\scriptstyle i_{nm}} & & \downarrow{\scriptstyle i_{nm}} \\ [X, \mathrm{GL}_m(\mathbb{C})] & \xrightarrow{\varphi_m} & \mathrm{Vect}_m(\Sigma X) \end{array}$$

commutes. Lemma 3.4.7 gives us

$$\tilde{K}(X) \cong \operatorname{colim}_n [X, \mathrm{GL}_n(\mathbb{C})] \cong [X, \operatorname{colim}_n \mathrm{GL}_n(\mathbb{C})]\,.$$

The latter bijection follows from Lemma 3.2.40. Finally, Lemma 3.4.13 gives us a retraction $\mathrm{GL}_n(\mathbb{C}) \to U(n)$ that commutes with the inclusions and hence we get the same direct limit.

$$[X, \operatorname{colim}_n \mathrm{GL}_n(\mathbb{C})] \cong [X, \operatorname{colim}_n U(n)] = [X, U]$$

concluding that we have a natural bijection. To get the group structure we need to look into said mapping. For this we inspect the mapping (12) closer. Let

$$\Phi_n : [X, \mathrm{GL}_n(\mathbb{C})] \to \operatorname{colim} \mathrm{Vect}_n(\Sigma X)$$

be the composition of φ_n with the inclusion $\mathrm{Vect}_n(\Sigma X) \to \operatorname{colim} \mathrm{Vect}_n(\Sigma X)$. By the commutative diagram above, the maps φ_n induce a map from $\operatorname{colim}[X, \mathrm{GL}_n(\mathbb{C})]$ which is a group homomorphism, if all Φ_n are one. Hence take two functions $f, g : X \to \mathrm{GL}_n(\mathbb{C})$. Then by definition

$$f \oplus g : X \to GL_n(\mathbb{C}), \quad x \mapsto f(x) \times g(x)$$

Now $f \oplus g$ can be included into $[X, \mathrm{GL}_{2n}(\mathbb{C})]$:

$$f \oplus g : X \to GL_{2n}(\mathbb{C}), \quad x \mapsto \begin{pmatrix} f(x) \times g(x) & 0 \\ 0 & \mathbb{1}_n \end{pmatrix}.$$

Now we have that $f \oplus g$ is homotopic to

$$\widetilde{f \oplus g} : X \to GL_{2n}(\mathbb{C}), \quad x \mapsto \begin{pmatrix} f(x) & 0 \\ 0 & g(x) \end{pmatrix}$$

via the homotopy $\rho_t(x)$ for $t \in (0, \frac{\pi}{2})$:

$$\rho_t(x) = \begin{pmatrix} f(x) & 0 \\ 0 & \mathbb{1}_n \end{pmatrix} \begin{pmatrix} \cos(t) & \sin(t) \\ \sin(t) & \cos(t) \end{pmatrix} \begin{pmatrix} \mathbb{1}_n & 0 \\ 0 & g(x) \end{pmatrix} \begin{pmatrix} \cos(t) & \sin(t) \\ \sin(t) & \cos(t) \end{pmatrix} \tag{13}$$

{eq:kTHeo-homotopyBetweendirectSumAndProduct}

Finally we can look at the image of $[f \oplus g] = [\widetilde{f \oplus g}]$ under Φ_{2n} which is given by inclusion of

$$\varphi_n([\widetilde{f \oplus g}]) = \underline{n}_+ \oplus \underline{n}_+ \bigcup_{\widetilde{f \oplus g}} \underline{n}_- \oplus \underline{n}_- = [\varphi_n(f) \oplus \varphi_n(g)]$$

into the colimit, where the equation

$$[\varphi_n(f) \oplus \varphi_n(g)] = [\varphi_n(f)] \oplus [\varphi_n(g)]$$

holds. The second last equality follows from the properties of the gluing construction given in Corollary 3.2.11. □

Remark 3.4.15. The proof of Lemma 3.4.14 gives a slightly more general statement: For a connected well pointed topological space (X, x_0) with a splitting into two contractible parts X^+ and X^- with $x_0 \in A := X^+ \cap X^-$ and $X^+ \cup X^- = X$ and the inclusion of A into the two contractible parts are cofibrations, there is a natural group isomorphism

$$[A, U] \cong \tilde{K}(X)$$

lem:kTheo-inversesInKGivenByHomotopicallyMethods

Corollary 3.4.16. Let Y be a pointed space. Define

$$T : S^1 \to S^1$$
$$t \mapsto 1 - t$$

and let $T \wedge 1 : SY \to SY$ be the map induced by T on S^1 and the identity on Y. Then:

$$(T \wedge 1)^* : \tilde{K}(SY) \to \tilde{K}(SY)$$
$$y \mapsto -y$$

Proof. We have the following commutative diagram:

$$\begin{array}{ccc} \tilde{K}(SX) & \longleftarrow & \operatorname{colim}[Y, \mathrm{GL}(n,\mathbb{C})] \\ {\scriptstyle (\Sigma T)^*}\downarrow & & \downarrow{\scriptstyle f\mapsto f^{-1}(pointwise)} \\ \tilde{K}(SX) & \longleftarrow & \operatorname{colim}[Y, \mathrm{GL}(n,\mathbb{C})] \end{array}$$

To see why this commutes, notice how we defined the map by gluing via f, where ΣT changes the place of the upper and lower vector bundle. Hence the gluing function needed is now the inverse. The horizontal maps are natural group homomorphisms and on the right side the inverse is taken. Thereby the commutativity tells us that the left side is also the same as taking the inverse. □

ex:kTheo-LineBundlesSatlquadequalslPlus1

Example 3.4.17. Let l and l' be line bundles over S^2. Then the clutching function for the vector bundle $l \otimes l'$ is given by the tensor product of clutching functions. So let f and f' be two clutching functions

$$f, f' : S^1 \to \mathrm{GL}_1(\mathbb{C})$$

Then the product bundle $l \otimes l'$ is given by the clutching function

$$f \otimes f' : S^1 \to \mathrm{GL}(\mathbb{C} \otimes \mathbb{C}) \cong \mathrm{GL}_1(\mathbb{C}), \quad x \mapsto f(x)f'(x)$$

Hence, the product in $\tilde{K}(S^2)$ is given by $f \otimes f' = f \cdot f'$. Now, the homotopy (13) of the proof above shows that $f \cdot f' \oplus \mathrm{id}_1 \simeq f \oplus f'$ and hence $l \otimes l' \oplus \underline{1} \cong l \oplus l'$.

Proposition 3.4.18. For compactly generated pointed Hausdorff spaces $(X, x_0), (Y, y_0)$ there is a natural bijection

$$[SX, Y] \to [X, \Omega Y],$$

where Ω denotes the loop space

$$\Omega Y := \{f : S^1 \to Y \mid f \text{ continuous and } f(1) = y_0\}.$$

together with the compact open topology.

Proof. For any three spaces $(X, x_0), (Y, y_0), (Z, z_0)$ we have that a basepoint preserving map $f : X \wedge Y \to Z$ can be lifted to a map $\tilde{f} : X \times Y \to Z; f(z) = \tilde{f}(\overline{z})$ that maps the whole $X \vee Y$ to z_0. This map is continuous by the universal property of the quotient topology. Hence we get a bijection

$$C(X \wedge Y, Z) \to \{f \in C(X \times Y, Z) \mid f|_{X \vee Y} \equiv z_0\}$$

its inverse is given by the quotient induced maps. We can compose this map with the natural currying map $C(X \times Y, Z) \to C(X, C(Y, Z))$. For $Y = S^1$ we gain a map

$$C(SX, Z) \to C(X, \Omega Z)$$

that is again a bijection, since any $f : X \to \Omega Z$ gives a map $\tilde{f} : X \times S \to Z$ $\tilde{f}(x,s) \mapsto f(x)(s)$. If $(x,s) \in X \vee S \subseteq X \times S$, then either $x = x_0$ or $s = 1$. In the first case, $f(x_0)$ is the constant loop and hence $f(x_0)(s) = z_0$ and in the second case we evaluate a based loop at its base and hence $f(x)(1) = z_0$. Hence, the $\tilde{f}$ factors through $X \vee S$ giving a well defined element in $C(SX, Z)$. This is its inverse. The continuity part of the argument follows from Proposition 1.3.3. □

Remark 3.4.19. By Theorem 3.4.12, and the Proposition above we know that for a connected pointed compact space we have

$$\tilde{K}(SX) \cong [SX, BU] \cong [X, \Omega BU]$$

On the other hand we have Lemma 3.4.14 which tells us that

$$\tilde{K}(SX) \cong [X, U]\,.$$

This hints at a connection that states that ΩBU is homotopically equivalent to U:

3.5 K-Theory as a Cohomology Theory

This chapter aims to give the definitions and theorems that make K theory an axiomatic cohomology theory as defined by Eilenberg and Steenrod. For this we start trying to find a long exact sequence and on the way we will prove the additivity:

lem:kTheo-exactShortPartForThePair

Lemma 3.5.1. *Let $(X,Y) \in ob(\mathbf{CWH^2_{cof}})$ and define $i : (Y,\varnothing) \to (X,\varnothing)$ and $j : (X,\varnothing) \to (X,Y)$ to be the inclusions. Then we have an exact sequence:*

$$K(X,Y) \xrightarrow{j^*} K(X) \xrightarrow{i^*} K(Y)$$

Here, we identify $K(X) = K(X,\varnothing)$ and the same with $K(Y) = K(Y,\varnothing)$ such that we have well defined induced functions from the same functor in the sequence. This is just a formal note, as $K(X,\varnothing) \cong K(X)$ by Corollary 3.3.34.

Proof. First, we show that $\mathrm{im}(i^*) \subseteq \ker(j^*)$. To see this, notice that $i^* \circ j^* = (j \circ i)^*$, and $j \circ i$ factors through (Y,Y):

$$\begin{array}{ccc} (Y,\varnothing) & \xrightarrow{i} & (X,\varnothing) \\ \downarrow & & \downarrow j \\ (Y,Y) & \longrightarrow & (X,Y) \end{array}$$

Hence, $j^* \circ i^*$ factors through the trivial ring $K(Y,Y)$ and thereby is trivial. Now suppose $k \in \ker(i^*)$. The intuition now is that if we have a vector bundle that gets mapped to 0 this means that it is trivial over Y which lets us define a vector bundle over the quotient X/Y by a trivialisation: We write $k = [\xi] - [\underline{n}]$ for ξ being a vector bundle over X.

(Compare Remark 3.4.8) Since $i^*\xi = 0$ it follows that $i^*([\xi] - [\underline{n}]) = [\xi|_Y] - [\underline{n}|_Y] = 0$ and hence ξ and $\underline{n}$ are stably equivalent over Y giving us an m such that

$$(\xi \oplus \underline{m})|_Y \cong \xi|_Y \oplus \underline{m} \cong \underline{n} \oplus \underline{m}$$

Hence, there is a trivialisation $\alpha : (\xi \oplus \underline{m})|_Y \to Y \times \mathbb{C}^{\operatorname{rank}(\xi)+m}$ that defines a bundle $(\xi \oplus \underline{m})|_Y \big/ \alpha$ as shown in the proof of Lemma 3.2.7 over $X\big/Y$. Now we define

$$\eta = [\,(\xi \oplus \underline{m})|_Y \big/ \alpha] - [\underline{n+m}] \in \tilde{K}(X\big/Y) = K(X,Y)$$

and since the trivialisation acts as a right inverse of the pullback of the restriction (compare Corollary 3.2.8)

$$j^*(\eta) = [\xi \oplus \underline{m}] - [\underline{n+m}] = k\,.$$

□

lem:kTheo-pointedUnionExactSequence

Lemma 3.5.2. *Let X, Y be pointed topological spaces and define the maps*

$$i : X \to X \vee Y \;\text{ the inclusion, and } j : X \vee Y \to Y \;\text{ the collapse of } X.$$

Then the sequence

$$0 \longrightarrow \tilde{K}(X) \xrightarrow{j^*} \tilde{K}(X \vee Y) \xrightarrow{i^*} \tilde{K}(Y) \longrightarrow 0$$

is exact and naturally splits via the map q^*, that is a right inverse of j^*, where $q : X \vee Y \to Y$ is the collapse.

Proof. Notice how there is a natural homeomorphism $X \cong (X \vee Y)\big/Y$. Hence, Lemma 3.5.1 provides the exactness at $K(X \vee Y)$. Now the map q is a left inverse of j as continuous maps, and hence q^* is a right inverse of j^* in the K group, proving the injectivity of j^*. Now only the surjectivity of i^* is missing. However the inclusion $l : Y \to X \vee Y$ is a topological right inverse of i and thereby l^* is a left inverse of i^* giving us the injectivity and hence proving the exactness and splitting of the short exact sequence. □

thm:kTheo-pointedAdditivity

Theorem 3.5.3 (Pointed Additivity)**.** *Let X, Y be pointed topological spaces and denote the inclusions of X (resp. Y) into $X \vee Y$ by i_X (resp. i_Y). Then the map*

$$i_X^* \oplus i_Y^* : \tilde{K}(X \vee Y) \to \tilde{K}(X) \oplus \tilde{K}(Y)$$

is a natural isomorphism with inverse

$$\psi : \tilde{K}(X) \oplus \tilde{K}(Y) \to \tilde{K}(X \vee Y)\,, \quad (x, y) \mapsto c_X^*(x) + c_Y^*(y)\,,$$

where c_X (resp. c_Y) denotes the collapse of X (hence onto Y) respectively the collapse of X and hence onto Y.

Proof. This is a direct consequence of Lemma 3.5.2 and the algebraic machinery of splitting short exact sequences. □

thm:kTheo-unpointedAdditivity

Corollary 3.5.4 (Unpointed Additivity)**.** Let X, Y be topological spaces, then there is a natural isomorphism

$$K(X \sqcup Y) \to K(X) \oplus K(Y)$$

Proof. This follows from $K(A) \cong \tilde{K}(A, [\varnothing])$ for all topological spaces. Furthermore, $(X \sqcup Y, [\varnothing]) \cong ((X, [\varnothing]) \vee (Y, [\varnothing]))$. Hence, using those isomorphisms, the statement is a direct consequence of the pointed additivity. □

cor:kTheo-exactnesOfTheReducedInclusionSequence

Corollary 3.5.5. If $(X, Y) \in ob(\mathbf{CWH^2_{cof}})$ and $Y \in ob(\mathbf{CWH^*_{cof}})$ such that X is pointed by using the base point y_0 from Y. Then the sequence

$$K(X, Y) \xrightarrow{j^*} \tilde{K}(X) \xrightarrow{i^*} \tilde{K}(Y)$$

is exact.

Proof. We have the commutative diagram with exact top row:

$$\begin{array}{ccccc} K(X,Y) & \longrightarrow & K(X) & \longrightarrow & K(Y) \\ & \searrow & \downarrow \cong & & \downarrow \cong \\ & & \tilde{K}(X) \oplus K(y_0) & \longrightarrow & \tilde{K}(Y) \oplus K(y_0) \\ & & \downarrow & & \downarrow \\ & & \tilde{K}(X) & \longrightarrow & \tilde{K}(Y) \end{array}$$

The middle horizontal map is a map that respects the grading, since we have the same base point. With this we can do a bit of diagram yoga concluding the statement. □

def:kTheo-coneOverSubspace

Definition 3.5.6. Let X be a topological space and $I = [0, 1]$ be the unit interval. We define the cone over X to be the space

$$CX := (X \times I) \big/ X \times \{0\}\,.$$

Let (X, Y) be a pair of compact topological spaces. We define the cone over the subspace Y to be

$$\Big(X \sqcup CY\Big) \Big/ Y \sim Y \times \{1\}$$

and denote that with $(X \cup CY)$.

prop:kTheo-coneIsFunctorAndCYoverYIsRedSuspension

Proposition 3.5.7. If we define for $f : X \to Y$ the induced maps on the cones Cf by

$$Cf : CX \to CY \quad \overline{(x,t)} \mapsto \overline{(f(x),t)},$$

the cone defines a functor $\mathbf{CGWH} \to \mathbf{CGWH}^*$. The cone over $Y \subseteq X$ defines in similar fashion a functor from $\mathbf{CGWH}^2_{\mathbf{cof}}$ to $\mathbf{CGWH}^*$. There is a natural isomorphism:

$$T : (X \cup CY)/X \to CY/Y$$

Here, the left and right hand side both define functors from $\mathbf{CGWH}^2_{\mathbf{cof}}$ to $\mathbf{CGWH}^*$ with the obvious morphisms. Furthermore, we have that CY/Y is homotopically equivalent to ΣY (via the map induced from $\overline{(x,t)} \mapsto \overline{(t,x)}$) which is homotopically equivalent to SY as long as (Y, y_0) is well pointed. All the equivalences are natural up to homotopy.

Corollary 3.5.8. From the considerations above we get a natural isomorphism

$$\theta : K(X \cup CY, Y) \to \tilde{K}(SY) = \tilde{K}^{-1}(Y)\,.$$

def:kTheo-connectingHomomorphism

Definition 3.5.9 (Connecting Homomorphism)**.** Let (X, Y) be a topological pair of compact spaces and Y be pointed. Define the two maps

$$m : (X \cup CY, \varnothing) \to (X \cup CY, X) \tag{14}$$

$$\pi : (X \cup CY, \varnothing) \to X/Y \tag{15}$$

where m is the inclusion in the second space, and π is the collapse of CY. Since CY is contractible, the induced map $\pi^* : \tilde{K}(X/Y) \to K(X \cup CY, \varnothing)$ is an isomorphism by Lemma 3.2.9. Together with the isomorphism

$$\theta : K(X \cup CY, X) \to \tilde{K}(SY) = \tilde{K}^{-1}(Y)\,.$$

We define the connecting homomorphism

$$\delta := (\pi^*)^{-1} \circ m^* \circ \theta^{-1} : \tilde{K}^{-1}(Y) \to K(X, Y)$$

This map defines also the higher degrees connecting homomorphisms by replacing the spaces with their suspensions.

Remark 3.5.10. The map $\delta : K^{-n}(A) \to K^{-n+1}(X, A)$ is a natural transformation between the functors

- $K^{-n} \circ R$ from $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ to $\mathbf{Ab}$, where R is an endofunctor in $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ sending $(X, A) \mapsto (A, \varnothing)$.
- K^{-n+1} from $\mathbf{CWH}^{\mathbf{2}}_{\mathbf{cof}}$ to $\mathbf{Ab}$.

Proof. Let $f : (X, A) \to (Y, B)$ be a map of pairs. Then we need to check if the diagram

$$\begin{array}{ccc} (X,A) & K^{-n}(A,\varnothing) & \xrightarrow{\delta} K^{-n+1}(X,A) \\ \downarrow f & (f|_A)^* \uparrow & f^* \uparrow \\ (Y,B) & K^{-n}(B,\varnothing) & \xrightarrow{\delta} K^{-n+1}(Y,B) \end{array}$$

commutes. This however is the case since δ is defined via the pullback of the inclusion (m^*) together with the natural isomorphisms θ and π^*. □

thm:kTheo-longExactSequence

Theorem 3.5.11 (Long Exact Sequence)**.** *For $(X,Y) \in ob(\mathbf{CWH^2_{cof}})$ we have a natural exact sequence that is infinite to the left:*

$$\cdots \longrightarrow K^{-2}(Y) \xrightarrow{\delta} K^{-1}(X,Y) \xrightarrow{j^*} K^{-1}(X) \xrightarrow{i^*} K^{-1}(Y)$$

$$\xrightarrow{\delta} K^0(X,Y) \xrightarrow{j^*} K^0(X) \xrightarrow{i^*} K^0(Y)$$

The maps i^* are induced by the inclusion $i : Y \to X$, j^* are induced by the inclusion $j : (X,\varnothing) \to (X,Y)$ and δ is the connecting homomorphism. For the lower degrees, δ is the connecting homomorphism where X and Y are replaced by the reduced suspensions of X and Y.

Proof. We will prove the pointed version of the theorem, that states that for a compact pair (X,Y) such that Y is itself compact pointed (and X is pointed with the same basepoint) we have the exact sequence (infinite to the left):

$$\cdots \longrightarrow \tilde{K}^{-2}(Y) \xrightarrow{\delta} K^{-1}(X,Y) \xrightarrow{j^*} \tilde{K}^{-1}(X) \xrightarrow{i^*} \tilde{K}^{-1}(Y)$$

$$\xrightarrow{\delta} K^0(X,Y) \xrightarrow{j^*} \tilde{K}^0(X) \xrightarrow{i^*} \tilde{K}^0(Y)$$

The pointed sequence induced the statement for unpointed spaces because for any topological space A we have a natural isomorphism $\tilde{K}(A,[\varnothing]) \cong K(A)$.

We will furthermore just prove the exactness of the segment:

$$\tilde{K}^{-1}(X) \xrightarrow{i^*} \tilde{K}^{-1}(Y) \xrightarrow{\delta} K^0(X,Y) \xrightarrow{j^*} \tilde{K}^0(X) \xrightarrow{i^*} \tilde{K}^0(Y) \tag{16}$$

{eq:kTheo-proofSesShortAndreducedSequence}

From this we regain the long sequence by replacing (X) and Y with $S^n(X)$ and S^nY.

In $\tilde{K}^0(X)$, the exactness was shown in Corollary 3.5.5. To get the exactness at the other spots we apply the same corollary to the pairs:

- $(X \cup CY, X)$
- $((X \cup CY) \cup CX, X \cup CY)$

For the middle part we have the commutative diagram where the naming of the maps is the same as in the definition of the connecting homomorphisms (Definition 3.5.9).

$$\begin{array}{ccccc} K(X \cup CY, X) & \xrightarrow{m^*} & \tilde{K}(X \cup CY) & \xrightarrow{k^*} & \tilde{K}(X) \\ \downarrow{\scriptstyle \theta} & & \uparrow{\scriptstyle \pi^*} & & \downarrow{\scriptstyle \mathrm{id}} \\ K^{-1}(Y) & \xrightarrow{\delta} & K(X,Y) & \xrightarrow{j^*} & \tilde{K}(X) \end{array}$$

The commutativity in the left square is the definition of δ and in the right square follows from $\pi \circ k = j : X \to X/Y$. Since all vertical maps are isomorphisms, Corollary 3.5.5 implies exactness in the bottom row $K(X,Y)$. Now only the exactness in the first three terms is missing. For this we look at the pair $((X \cup C_1Y) \cup C_2X, X \cup C_1Y)$, where the C_i are cones and denoted with the index to distinguish the cones. First we get again an exact sequence:

$$K\big((X \cup C_1Y) \cup C_2X, X \cup C_1Y\big) \xrightarrow{l^*} \tilde{K}\big((X \cup C_1Y) \cup C_2X\big) \xrightarrow{s^*} \tilde{K}(X \cup C_1Y)$$

Our goal is now to show that we commute with the start of the exact sequence (16), hence to find vertical maps between the top rows (denoted with ψ in the diagram below) such that the diagram commutes:

diag:kTheo-proofConnectingHomo

$$\begin{array}{ccccc} K\big((X \cup C_1Y) \cup C_2X, X \cup C_1Y\big) & \xrightarrow{l^*} & \tilde{K}\big((X \cup C_1Y) \cup C_2X\big) & \xrightarrow{s^*} & \tilde{K}(X \cup C_1Y) \\ \downarrow{\scriptstyle \psi_1} & & \downarrow{\scriptstyle \psi_2} & & \downarrow{\scriptstyle \psi_3} \\ \tilde{K}^{-1}(X) & \xrightarrow{i^*} & \tilde{K}^{-1}(Y) & \xrightarrow{\delta} & K(X,Y) \end{array}$$

Diagram 17

We start with the right square and notice that by definition we get a commutative diagram:

$$\begin{array}{ccc} K(X \cup C_1Y \cup C_2X) & \xrightarrow{s^*} & \tilde{K}(X \cup C_1Y) \\ \downarrow{\scriptstyle (\tilde{\pi}^*)^{-1}} & & \downarrow{\scriptstyle \mathrm{id}} \\ K(X \cup C_1Y, X) & \xrightarrow{m^*} & \tilde{K}(X \cup C_1Y) \\ \downarrow{\scriptstyle \theta} & & \downarrow{\scriptstyle (\pi^*)^{-1}} \\ K^{-1}(Y) & \xrightarrow{\delta} & K(X,Y) \end{array}$$

The bottom square commutes by definition. In the top square, the map $\tilde{\pi}$ denotes the collapsing map $X \cup C_1Y \cup C_2X \to (X \cup C_1Y)/X$ which is invertible, as the quotiented out space C_2X is contractible. To see the commutativity in the top square notice how $\tilde{\pi}^* \circ s^* = (s \circ \tilde{\pi})^*$ and the maps m and $s \circ \tilde{\pi}$ coincide and hence the top square commutes. With $\psi_2 := \theta \circ (\tilde{\pi}^*)^{-1}$ and $\psi_3 := (\pi^*)^{-1} \circ \mathrm{id}$. For the left square, a natural choice is the

The bottom row is added for clarification on what inclusion induces i^*.. besser erklähren

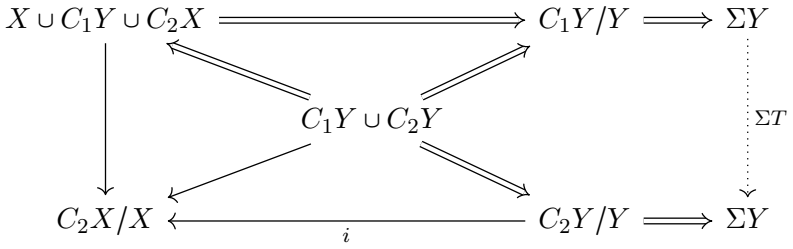
homotopical equivalence given by quotienting the second space of the pair $\big((X \cup C_1Y) \cup C_2X, X \cup C_1Y\big)$ to get:

$$\Big((X \cup C_1Y) \cup C_2X \Big/ X \cup C_1Y\Big) \cong (C_2X/X) \simeq S(X)\,.$$

We now have all the unknown maps, but we run into a problem, since the map l^* induces a map λ by the "wrong" inclusions (compared to i^* which is induced by the inclusion $C_2Y \to C_2X$, whereas we have C_1Y). This is depicted in the non-commutative diagram:

$$\begin{array}{ccc}
K\big((X \cup C_1Y) \cup C_2X, X \cup C_1Y\big) & \xrightarrow{l^*} & \tilde{K}\big((X \cup C_1Y) \cup C_2X\big) \\
\downarrow{\scriptstyle \psi_1} & & \downarrow{\scriptstyle \psi_2} \\
\tilde{K}(C_2X/X) \cong \tilde{K}^{-1}(X) & \xrightarrow{\lambda} & \tilde{K}(C_1Y/Y) \cong \tilde{K}^{-1}(Y) \\
\uparrow & & \uparrow \\
K(C_2X, X) & \xrightarrow{i^*} & K(C_2Y, Y)
\end{array}$$

To resolve that problem we introduce the double cone on Y namely $C_1Y \cup C_2Y$, that fits into the commutative diagram where all unnamed maps are the natural ones:



The inclusion induced map i is the one, that induces the bottom left map in Diagram 17. The double arrows always induce isomorphisms in the K-rings. To see that all those maps induce isomorphism we note that we always, with one exception, collapse something contractible. The exception is the map $C_1Y \cup C_2Y \to X \cup C_1Y \cup C_2X$. Here, we get that the induced map is invertible, by noticing the commutative triangle on the top, where two maps induce isomorphisms and thereby also the third. The map ΣT is one we want to construct below. For said construction we formulate the following Lemma:

lem:kTheo-doubleConeFlip

Lemma 3.5.12. *Let Y be a well-pointed space and denote the points of CY and ΣY by (x,t) such that CY has the apex $[\{0\} \times Y]$ and base $\{1\} \times Y \cong Y$. Let*

$$D := C_1Y \cup C_2Y = \big(C_1Y \sqcup C_2Y\big) \Big/ \big((x,1)_1 \sim (x,1)_2\big)$$

denote the double cone glued along a common base. Now define the maps

- *$c_i : D \to C_iY/Y$ the collapse of the complementary cone C_jY $j \neq i$.*

- $\overline{\pi_1} : C_iY/Y \to \Sigma Y \quad [(x,t)] \mapsto (x,t)$.
- $p_i := \overline{\pi_1} \circ c_i : D \to \Sigma Y$.
- $R : D \to D \quad (x,t)_i \mapsto (x,t)_j$ *where* $j \neq i$, *the cone swap.*
- $\Sigma T : \Sigma Y \to \Sigma Y\,, \quad (x,t) \mapsto (x, 1-t)$ *the suspension flip.*
- $\Phi : D \to \Sigma Y\,, \quad (x,t)_1 \mapsto (x, 1-\frac{t}{2}), \quad (x,t)_2 \mapsto (x, \frac{t}{2})$.

Then we have that

$$p_2 = p_1 \circ R \quad \textit{and} \quad \Phi \circ R = \Sigma T \circ \Phi\,.$$

Proof of Lemma 3.5.12. The first statement $p_2 = p_1 \circ R$ is clear, as:

$$\begin{aligned} p_2(x,t)_2 &= [(x,t)] = p_1(x,t)_1 = p_1 \circ R(x,t)_2\,, \quad \text{and} \\ p_2(x,t)_1 &= [\{1\} \times Y] = p_1(x,t)_2 = p_1 \circ R(x,t)_1\,. \end{aligned}$$

Similarly, the second statement is also just a calculation:

$$\begin{aligned} \Sigma T \circ \Phi(x,t)_1 &= \Sigma T(x, 1-\frac{t}{2}) = (x, \frac{t}{2}) = \Phi(x,t)_2 = \Phi \circ R(x,t)_1 \\ \Sigma T \circ \Phi(x,t)_2 &= \Sigma T(x, \frac{t}{2}) = (x, 1-\frac{t}{2}) = \Phi(x,t)_1 = \Phi \circ R(x,t)_2\,. \end{aligned}$$

□

By the lemma above, we know that ΣT makes the diagram commute and by Lemma 3.4.16 we know that

$$(\Sigma T)^* : \tilde{K}(SY) \to \tilde{K}(SY)$$

is multiplication by minus one. Hence, the diagram where all maps are either topologically induced or topologically induced and inverted:

$$\begin{array}{ccccc} \tilde{K}(X \cup C_1Y \cup C_2X) & & \xrightarrow{\quad\psi_2\quad} & & \tilde{K}(C_1Y/Y) \longrightarrow \tilde{K}^{-1}(Y) \\ & \searrow & & & \\ \uparrow \rho & & \tilde{K}(C_1Y \cup C_2Y) & & \Vert\, (-1)^* \\ & & & \searrow & \\ \tilde{K}^{-1}(X) & & \xrightarrow{\quad i^*\quad} & & \tilde{K}(C_2Y/Y) \longrightarrow \tilde{K}^{-1}(Y) \end{array}$$

is commutative. By construction (just composition of collapsing maps) we have that

$$l^* = \rho \circ \psi_1\,, \quad \text{and by the diagram} \quad \psi_2 \circ \rho = -i^*\,.$$

Which gives us in total:

$$i^* \circ \psi_1 = -\psi_2 \circ \rho \circ \psi_1 = -\psi_2 \circ l^*$$

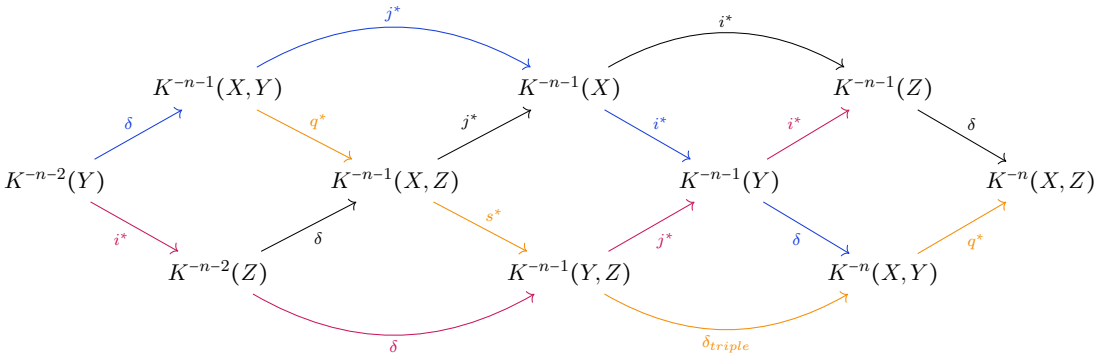
and thereby the diagram commutes up to sign:

$$\begin{array}{ccccc} K\big((X\cup C_1Y)\cup C_2X, X\cup C_1Y\big) & \xrightarrow{l^*} & \tilde{K}\big((X\cup C_1Y)\cup C_2X\big) & \xrightarrow{s^*} & \tilde{K}(X\cup C_1Y) \\ \downarrow{\scriptstyle \psi_1} & & \downarrow{\scriptstyle \psi_2} & & \downarrow{\scriptstyle \psi_3} \\ \tilde{K}^{-1}(X) & \xrightarrow{\quad i^*\quad} & \tilde{K}^{-1}(Y) & \xrightarrow{\quad \delta\quad} & K(X,Y) \end{array}$$

The exactness in the top row now proves exactness in the bottom. □

ktheo:cor-triple Sequence

Corollary 3.5.13 (Triple Sequence)**.** Let (X,Y,Z) be a topological triple, meaning that all inclusions are cofibrations. Then we can inspect the three long exact sequences from the pairs (X,Y), (Y,Z) and (X,Z) that are in the diagram below. The map δ_{triple} is defined to be the composition $\delta \circ j^*$, $q*$ is defined to be induced by the inclusion $(X,Z) \to (X,Y)$ and s^* is also given by the inclusion $(Y,Z) \to (X,Z)$. Then, the whole diagram becomes commutative by the naturality of the maps appearing in the long exact sequences.



Furthermore, the yellow sequence is a chain complex which we need to verify at three locations:

1. At $K^{-n-1}(X,Z)$ we can see that the composition $s^* \circ q^*$ is the pullback of the inclusion $(Y,Z) \to (X,Y)$ which factors through (Y,Y) and thereby the pullback factors through the trivial ring $K^{-n-1}(Y,Y)$.
2. At $K^{-n-1}(Y,Z)$ we get that the composition is zero as it factors through two consecutive maps in the sequence of (X,Y): $\delta_{triple} \circ s^* = \delta \circ i^* \circ j^* = 0$.
3. At $K^{-n}(X,Y)$ we get $q^* \circ \delta = 0$ as it factors through $i^* \circ j^*$.

Then, by the braid lemma (which is typical diagram yoga), we know that the orange sequence is exact. We call this sequence the **triple exact sequence** and it is natural with respect to maps of triples, meaning that for two such triples and a map of triples, we get two sequences and an induced mao of sequences.

referenz einfügen: Munkres, p. 148 of Elements of algebraic topology?

cor:kTheo-retractionsInduceSplittingSequences

Corollary 3.5.14. If (X,Y) is a compact pair such that $Y \subseteq X$ is a retract, i.e. the inclusion has a continuous up to homotopy left inverse. Then for all $n \geq 0$, the sequence

$$K^{-n}(X,Y) \xrightarrow{j^*} K^{-n}(X) \xrightarrow{i^*} K^{-n}(Y)$$

is a splitting short exact sequence giving us an isomorphism

$$K^{-n}(X) \cong K^{-n}(X,Y) \oplus K^{-n}(Y)\,.$$

This splitting is naturally dependent on the retraction map.

3.6 New Perspective on the K-Connecting Homomorphism

Hier hätte ich gerne die Puppe Sequence, und i.A. puppe seuqence + halb exacter Functor der über hTop factorisiert -¿ long exact sequence

3.7 Product Structures

Corollary 3.7.1. Let X, Y be objects in $\mathbf{CWH}^*_{\mathbf{cof}}$. Then we have the projection maps

$$\pi_X : X \times Y \to X\,, \quad \pi_Y : X \times Y \to Y\,.$$

Those maps induce isomorphisms for all $n \geq 0$:

$$\tilde{K}^{-n}(X \times Y) \cong \tilde{K}^{-n}(X \wedge Y) \oplus \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(Y)$$

Proof. X is a retract of $X \times Y$ as the projection yields a continuous left inverse of the natural inclusion, and Y is a retract of $(X \times Y)/X$ (for the quotient to be natural, we include X at the base point of Y). With this we apply corollary 3.5.14 twice:

$$\begin{aligned} \tilde{K}^{-n}(X \times Y) &\cong \tilde{K}^{-n}(X \times Y/X) && \oplus \tilde{K}^{-n}(X) \\ &\cong \tilde{K}^{-n}(Y) \oplus \underbrace{\tilde{K}^{-n}((X \times Y/X)/Y)}_{\tilde{K}^{-n}(X \wedge Y)} && \oplus \tilde{K}^{-n}(X) \end{aligned}$$

□

cor:kTheo-splittingAndConnectionWedgeAndProduct

Corollary 3.7.2. Let's inspect the isomorphism induced by the splitting: Since we apply the splitting twice, we know that the final isomorphism

$$\Phi : \tilde{K}^{-n}(X \times Y) \to \underbrace{\tilde{K}^{-n}((X \times Y/X)/Y)}_{\tilde{K}^{-n}(X \wedge Y)} \oplus \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(Y)$$

is of the form

$$\Phi = (\, g \circ l\,,\, \tilde{i_Y}^{\,*} \circ l\,,\, {i_X}^{*}\,)\,,$$

where:

- $i_Y : Y \to X \times Y$
- $\tilde{i}_Y : Y \to (X \times Y/X)$.

- $j : (X \times Y) \to (X \times Y/X)$.
- $l : \tilde{K}^{-n}(X \times Y) \to \tilde{K}^{-n}(X \times Y/X)$ is the unique left inverse of j^* such that $j^* \circ l = \mathrm{id} - \pi_X^* \circ i_X^*$.
- $g : \tilde{K}^{-n}(X \times Y/X) \to K(X \times y/X, Y)$ is a left inverse of the map $K(X \times Y/X, Y) \to \tilde{K}^{-n}(X \times Y/X)$.

We now want to show that $\tilde{i_Y}^* \circ l$ is induced by the inclusion of Y into $X \times Y$.
We have that $\tilde{i}_Y = j \circ i_Y$ and with this we can deduce:

$$\tilde{i_Y}^* \circ l = {i_Y}^* \circ j^* \circ l = {i_Y}^* \big(\mathrm{id} - \pi_X^* \circ i_X^* \big) = i_Y^* - {i_Y}^* \circ \pi_X^* \circ i_X^* = i_Y^* - \underbrace{(\pi_X \circ i_Y)}_{\text{constant}}^* \circ {i_X}^* = i_Y^*$$

Hence, in the proof above the isomorphism has the form:

$$\Phi : \tilde{K}^{-n}(X \times Y) \to \tilde{K}^{-n}(X \wedge Y) \oplus \tilde{K}^{-n}(Y) \oplus \tilde{K}^{-n}(X)$$
$$\Phi = g \oplus i_Y^* \oplus i_X^*$$

or said differently, the short exact sequence that splits looks like:

$$0 \longrightarrow \tilde{K}^{-n}(X \wedge Y) \xrightarrow{q^*} \tilde{K}^{-n}(X \times Y) \xrightarrow{(i_X)^* \oplus (i_Y)^*} \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(Y) \longrightarrow 0$$

Before we start constructing a cohomology theory, we extend the K groups to be defined not only over the negatives, but over the whole $\mathbb{Z}$ which is done via an extension. The extension relies on periodic behaviour. So-called Bott periodicity.

Definition 3.7.3 (External Product)**.** The **external product** $\mu : K(X) \otimes K(Y) \to K(X \times Y)$ is defined via the two projections $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ by:

$$\mu : K(X) \otimes K(Y) \to K(X \times Y)$$
$$(a \otimes b) \mapsto (\pi_X)^*(a) \cdot (\pi_Y)^*(b)\,,$$

where $\cdot$ denotes the multiplication in the K-group, which was induced by the tensor product. This μ is a ring homomorphism, since:

$$\begin{aligned}
\mu((a \otimes b)(c \otimes d)) &= \mu(ac \otimes bd) \\
&= \pi_X^*(ac)\pi_Y^*(bd) \\
&= \pi_X^*(a)\pi_X^*(c)\pi_Y^*(b)\pi_Y^*(d) \\
&= \pi_X^*(a)\pi_Y^*(b)\pi_X^*(c)\pi_Y^*(d) \\
&= \mu(a \otimes b)\mu(c \otimes d)
\end{aligned}$$

prop:kTheo-reducedExternalProduct

Proposition 3.7.4. Let (X, x_0) and (Y, y_0) be well-pointed compact spaces and let $q : X \times Y \to X \wedge Y$ denote the quotient map that appears in the exact sequence of Corollary 3.7.2. Then:

1. For $a \in \tilde{K}(X)$ and $b \in \tilde{K}(Y)$ we have $\mu(a \otimes b) \in q^*\tilde{K}(X \wedge Y)$. Since q^* is injective, this defines the **reduced external product**

$$\tilde{\mu} := (q^*)^{-1} \circ \mu|_{\tilde{K}(X)\otimes\tilde{K}(Y)} : \tilde{K}(X) \otimes \tilde{K}(Y) \to \tilde{K}(X \wedge Y)\,.$$

2. With respect to the decompositions

$$\begin{aligned} K(X) \otimes K(Y) &\cong \big(\tilde{K}(X) \otimes \tilde{K}(Y)\big) \oplus \tilde{K}(X) \oplus \tilde{K}(Y) \oplus \mathbb{Z}\,, \\ K(X \times Y) &\cong q^*\tilde{K}(X \wedge Y) \oplus \pi_X^*\tilde{K}(X) \oplus \pi_Y^*\tilde{K}(Y) \oplus \mathbb{Z}\cdot 1_{X\times Y} \end{aligned}$$

the external product is block diagonal:

$$\mu = (q^* \circ \tilde{\mu}) \oplus \pi_X^*|_{\tilde{K}(X)} \oplus \pi_Y^*|_{\tilde{K}(Y)} \oplus \mathrm{id}_{\mathbb{Z}}\,.$$

In particular μ is an isomorphism if and only if $\tilde{\mu}$ is.

Proof. (i): Let $i_X : X \to X \times Y$, $x \mapsto (x, y_0)$, and i_Y analogously. Then

$$i_X^*\,\mu(a \otimes b) = i_X^*\big(\pi_X^*(a) \cdot \pi_Y^*(b)\big) = (\underbrace{\pi_X \circ i_X}_{=\mathrm{id}_X})^*(a) \cdot (\underbrace{\pi_Y \circ i_X}_{=\text{constant}})^*(b) = 0\,,$$

since $b \in \ker\big(i_{y_0}^* : K(Y) \to K(\{y_0\})\big)$ and the constant map $\pi_Y \circ i_X$ factors through $i_{y_0} : \{y_0\} \hookrightarrow Y$. Analogously $i_Y^*\,\mu(a \otimes b) = 0$, so by exactness of the sequence from Corollary 3.7.2

$$\mu(a \otimes b) \in \ker\big((i_X^*, i_Y^*)\big) = \mathrm{im}(q^*) = q^*\tilde{K}(X \wedge Y)\,.$$

(ii): The first decomposition arises by tensoring the splittings $K(X) \cong \tilde{K}(X) \oplus \mathbb{Z}\cdot 1_X$, the second by combining the analogous splitting of $K(X \times Y)$ with the splitting $\pi_X^* \oplus \pi_Y^*$ of the exact sequence from Corollary 3.7.2. On the first summand μ equals $q^* \circ \tilde{\mu}$ by (i) and the definition of $\tilde{\mu}$. Further $\mu(a \otimes 1_Y) = \pi_X^*(a) \cdot 1_{X\times Y} = \pi_X^*(a)$, and $\pi_X^*|_{\tilde{K}(X)}$ is an isomorphism onto $\pi_X^*\tilde{K}(X)$. The third summand is symmetric, and $\mu(1_X \otimes 1_Y) = 1_{X\times Y}$. Finally, a direct sum of homomorphisms is an isomorphism if and only if each block is one, and $q^* \circ \tilde{\mu}$ is an isomorphism onto $q^*\tilde{K}(X \wedge Y)$ if and only if $\tilde{\mu}$ is one. □

By example 3.4.17 we know that for any line bundle l, $[l \otimes l \oplus \underline{1}] = [l]^2 + [\underline{1}] = [l] + [l]$ which is equivalent to $([l] - 1)^2 = [l^2] - [l] + [l] + [\underline{1}] = 0$. Now $\{[\underline{1}], [l]\}$ is an additive basis of $\mathbb{Z}\big/([l]^2 - [\underline{1}])$. The mapping that is defined on a basis of the tensor product as follows

$$\begin{aligned} \Big(\mathbb{Z}([l])\Big/([l]^2 - [\underline{1}])\Big) \otimes K(X) &\to K(S^2 \times X) \\ [\underline{1}] \otimes [\xi] &\mapsto \mu([\underline{1}], [\xi]) \\ [l] \otimes [\xi] &\mapsto \mu([l], [\xi]) \end{aligned}$$

is well defined.

Definition 3.7.5. We denote the external product of two elements by

$$\zeta \boxtimes \xi := \tilde{\mu}(\zeta \otimes \xi)\,.$$

Proposition 3.7.6 (Naturality)**.** Let $f : X \to X'$ and $g : Y \to Y'$ be pointed maps, and $a \in \tilde{K}(X')$, $b \in \tilde{K}(Y')$. Then

$$f^*(a) \boxtimes g^*(b) = (f \wedge g)^*(a \boxtimes b)\,.$$

Proof. We start unreduced. Let $f : X \to X'$ and $g : Y \to Y'$ be continuous maps. then $\pi_{X'} \circ (f \times g) = f \circ \pi_X$ and similar $\pi_{Y'} \circ (f \times g) = g \circ \pi_Y$. Using that $(f \times g)^*$ is a ring homomorphism we calculate

$$\begin{aligned}(f \times g)^*\mu(a \otimes b) &= (f \times g)^*\Big((\pi_{X'})^*(a) \cdot (\pi_{Y'})^*(b)\Big)\\ &= \big((f \times g)^*(\pi_{X'})^*(a)\big) \cdot \big((f \times g)^*(\pi_{Y'})^*(b)\big)\\ &= \big(\pi_X^* \circ f^*(a)\big) \cdot \big(\pi_Y^* \circ g^*(b)\big)\\ &= \mu(f^*(a) \otimes g^*(b))\,.\end{aligned}$$

Now assume that f and g are pointed and denote the projections $q' : X' \times Y' \to X' \wedge Y'$ and $q : X \times Y \to X \wedge Y$. Then $q' \circ (f \times g) = (f \wedge g) \circ q$ by definition of the wedge product of maps. But then we can calculate

$$q^*(f \wedge g)^*(a \boxtimes b) = (f \times g)^* q'^*(a \boxtimes b) = (f \times g)^*\mu(a \otimes b) = \mu(f^*a \otimes g^*b) = q^*(f^*a \boxtimes g^*b)\,.$$

finally since q^* is injective we are done. □

prop:kTheo-associativityExternalProduct

Proposition 3.7.7 (Associativity of the Reduced External Product)**.** Let A, B, C be pointed spaces and $a \in \tilde{K}(A)$, $b \in \tilde{K}(B)$ and $c \in \tilde{K}(C)$. Let $\psi : A \wedge (B \wedge C) \to (A \wedge B) \wedge C$ be the homeomorphism from Lemma 3.3.25. Then

$$\psi^*((a \boxtimes b) \boxtimes c) = a \boxtimes (b \boxtimes c)\,.$$

Proof. Write $q_{X,Y} : X \times Y \to X \wedge Y$ for the quotient maps and π_A, π_B, π_C for the projections of $A \times B \times C$. Let

$$Q := q_{A \wedge B, C} \circ (q_{A,B} \times \mathrm{id}_C) : A \times B \times C \to (A \wedge B) \wedge C$$

and analogously $Q' := q_{A, B \wedge C} \circ (\mathrm{id}_A \times q_{B,C})$. Both are quotient maps, and by the construction of ψ in Lemma 3.3.25 we have

$$\psi \circ Q' = Q\,, \tag{18}$$

{eq:kTheo-assocQuotients}

since both sides are induced by the identity on $A \times B \times C$. Furthermore Q^* is injective, being the composite of the injective maps $q^*_{A \wedge B, C}$ and $(q_{A,B} \times \mathrm{id}_C)^*$.

We first compute Q^* of the left hand side. By definition of $\boxtimes$ we have the equialities

$$q^*_{A\wedge B,C}\big((a\boxtimes b)\boxtimes c\big) = \mu\big((a\boxtimes b)\otimes c\big) \in \tilde{K}((A\wedge B)\times C) \tag{19}$$

{eq:kTheo-associativeExPro1}

$$\pi_C \circ (q_{A,B}\times \mathrm{id}_C) = \pi_C : (A\times B)\times C \to C \tag{20}$$

{eq:kTheo-associativeExPro2}

$$\pi_{A\wedge B}\circ (q_{A,B}\times \mathrm{id}_C) = q_{A,B}\circ \pi_{A\times B} : (A\times B)\times C \to C \tag{21}$$

{eq:kTheo-associativeExPro3}

hence

$$\begin{aligned}
Q^*\big((a\boxtimes b)\boxtimes c\big) &= (q_{A,B}\times \mathrm{id}_C)^*\Big(\pi^*_{A\wedge B}(a\boxtimes b)\cdot \pi^*_C(c)\Big) && \text{using (19)}\\
&= (q_{A,B}\times \mathrm{id}_C)^*\pi^*_{A\wedge B}(a\boxtimes b)\cdot \pi^*_C(c) && \text{using (20) and (21)}\\
&= (q_{A,B}\times \mathrm{id}_C)^*\big(\pi^*_A(a)\cdot \pi^*_B(b)\big)\cdot \pi^*_C(c) && \text{similar to (19)}\\
&= (\pi^*_A(a)\cdot \pi^*_B(b))\cdot \pi^*_C(c) && \pi_A\circ\pi_{A\times B}=\pi_A \text{ and for } \pi_B
\end{aligned}$$

The same computation on the other side gives

$$Q'^*\big(a\boxtimes (b\boxtimes c)\big) = \pi^*_A(a)\cdot \pi^*_B(b)\cdot \pi^*_C(c)\,,$$

using associativity of the multiplication in $K(A\times B\times C)$. Combining this with (18),

$$Q'^*\psi^*\big((a\boxtimes b)\boxtimes c\big) = Q^*\big((a\boxtimes b)\boxtimes c\big) = Q'^*\big(a\boxtimes (b\boxtimes c)\big)\,,$$

and by corollary 3.7.2 we know that $q_{A,B}$ is injective. Hence, Q'^* is injective, which proves the claim. □

def:kTheo-lowerExternalProduct

Definition 3.7.8 (Lower External Product)**.** Let $a\in \tilde{K}^{-n}(X) = \tilde{K}(S^n\wedge X)$ and $b\in \tilde{K}^{-m}(Y) = \tilde{K}(S^m\wedge Y)$. Then we define $a\boxtimes_{n,m} b\in \tilde{K}^{-n-m}(X\wedge Y)$ in three steps:

1. First we apply the classical reduced external product:
$$a\boxtimes b \in \tilde{K}(S^n\wedge X\wedge S^m\wedge Y)$$
2. Let $\tau : S^n\wedge X\wedge S^m\wedge Y\to S^n\wedge S^m\wedge X\wedge Y$ be the map that swaps S^m and X. Then:
$$\tau^*(a\boxtimes b)\in \tilde{K}(S^n\wedge S^m\wedge X\wedge Y)$$
3. Lastly we apply $\Upsilon^{-1}_{n,m}\wedge \mathrm{id}_{X\wedge Y}$ where $\Upsilon : S^n\wedge S^m\to S^{n+m}$ is the map from Definition 3.3.27, to get:
$$a\boxtimes_{n,m} b := \big(\Upsilon^{-1}_{n,m}\wedge \mathrm{id}_{X\wedge Y}\big)^*\circ \tau^*(a\boxtimes b)\in \tilde{K}^{-(n+m)}(X\wedge Y)$$

Remark 3.7.9. The lower external product is associative , meaning that for $a\in \tilde{K}^{-k}(X)$, $b\in \tilde{K}^{-l}(Y)$, $c\in \tilde{K}^{-m}(Z)$ we have

$$(a\boxtimes_{k,l} b)\boxtimes_{k+l,m} c = a\boxtimes_{l,k+m}(b\boxtimes_{k,m} c)\,,$$

and natural, meaning that for $f : X\to X'$ and $g : Y\to Y'$

$$f^*(a)\boxtimes_{n,m} g^*(b) = (f\wedge g)^*(a\boxtimes_{n,m} b)\,.$$

Proof. We start with the associativity: Let $a \in \tilde{K}^{-k}(X)$, $b \in \tilde{K}^{-l}(Y)$, $c \in \tilde{K}^{-m}(Z)$. Let

$$\Phi_{k,l} := \tau \circ \left(\Upsilon_{k,l}^{-1} \wedge \mathrm{id}_{X\wedge Y}\right) : (S^{k+l} \wedge X \wedge Y) \to (S^k \wedge X \wedge S^l \wedge Y)\,.$$

Then we get the shorter form of the lower external product $a \boxtimes_{k,l} b = \Phi_{k,l}^*(a \boxtimes b)$, helping us to calculate:

$$\begin{aligned}(a \boxtimes_{k,l} b) \boxtimes_{k+l,m} c &= \Phi_{k+l,m}^*\big(\Phi_{k,l}^*(a \boxtimes b) \boxtimes (\mathrm{id}_{S^m\wedge Z})^* c\big) \\ &= \Big(\underbrace{(\Phi_{k,l} \wedge \mathrm{id}_{S^m\wedge Z}) \circ \Phi_{k+l,m}}_{=:L}\Big)^*((a \boxtimes b) \boxtimes c)\,.\end{aligned}$$

The last equiality is the naturality of the reduced external product. Similarly, we rewrite the other order:

$$a \boxtimes_{k,l+m} (b \boxtimes_{l,m} c) = \Big(\underbrace{(\mathrm{id}_{S^k\wedge X} \wedge \Phi_{l,m}) \circ \Phi_{k,l+m}}_{=:R}\Big)^*(a \boxtimes (b \boxtimes c))\,.$$

By the associativity of the reduced external product we know that $(a \boxtimes b) \boxtimes c = AS^*(a \boxtimes (b \boxtimes c))$. So what we need now is to show that $AS \circ L = R$ as maps

$$S^{k+l+m} \wedge X \wedge Y \wedge Z \to S^k \wedge X \wedge S^l \wedge Y \wedge S^m \wedge Z$$

modulo the identification $(X \wedge X) \wedge Z = X \wedge (Y \wedge Z)$. This can be done pointwise by calculating the image and using the associativity of Υ from Remark 3.3.28. Both sides of the equation map a point $[p, x, y, z]$ that is not the basepoint with $\pi_S(p) = (\alpha, \beta, \gamma) \in \mathbb{R}^k \times \mathbb{R}^l \times \mathbb{R}^m$ to

$$[\pi_S^{-1}(\alpha), x, \pi_S^{-1}(\beta), y, \pi_S^{-1}(\gamma), z]\,.$$

Finally, the naturality is shorter and follows from the naturality of $\boxtimes$ and $\Phi_{k,l}$, which in turn follows from the naturality of Ξ, together with the naturalitys of the swapping τ. □

rem:kTheo-neutralELementExternalProduct

Remark 3.7.10. Let ζ be the vector bundle over $S^0 = \{-1, +1\}$ that is $\underline{1}$ over -1 and $\underline{0}$ over $+1$. Then $[\xi]$ is a generator of $\tilde{K}(S^0)$, as any bundle that is in the cernal of the inclusion of $+1$ is $\underline{0}$ over $+1$, and trivial over $\{-1\}$. Let $\xi \in \tilde{K}(Y)$ be represented by $\xi = [\xi_1] - [\xi_2]$ such that ξ_i are vector bundles ($i = 1, 2$) over a pointed topological space Y. Then for $i = 1, 2$:

$$[\zeta] \boxtimes [\xi_i] = (q^*)^{-1}(\mu([\zeta] \otimes [\xi_i])) = (q^*)^{-1}\pi_{S^0}^*(\zeta) \cdot \pi_Y^*([\xi_i]) = (q^*)^{-1}([\pi_{S^0}^*(\zeta) \otimes \pi_Y^*(\xi_i]))$$

The bundle $\pi_{S^0}^*(\zeta)$ is $\underline{1}$ over $\{-1\} \times Y$ and $\underline{0}$ over $\{1\} \times Y$. $\pi_Y^*(\xi_i]$ on the otherhand is the pullback of ξ_i via the homeomorphism $h_- : \{-1\} \times Y \to Y$ over $\{-1\} \times Y$ and similar over $\{+1\} \times Y$. Hence, the tensor product $\pi_{S^0}^*(\zeta) \otimes \pi_Y^*(\xi_i]$ is given by $h_-^*(Y)$ over

$$\{-1\} \times Y$$

and $\underline{0}$ over $\{+1\} \times Y$. This bundles equivalence class in $\tilde{K}$ is given by the bundle h_-^* viewed over the pointed $S^0 \wedge Y = \{-1\} \times Y$ with basepoint $[(-1, 1)]$.

thm:kTheo-fundamentalProductTheorem

Theorem 3.7.11 (The Fundamental Product Theorem)**.** *For any compact X, and the tautological line bundle H on $\mathbb{CP}^1 \cong S^2$, the external product map*

$$\Big(\mathbb{Z}([H])\Big/([H]-[\underline{1}])^2\Big) \otimes K(X) \to K(S^2 \times X)$$

is an isomorphism of rings.

Proof. The proof is omitted here. There are elementary proofs given in Chapter 2.1 [Hat02] or Chapter 2.3 [Ati89]. □

hatcher2002algebraic atiyah1989k

Corollary 3.7.12 (The Easy Spaces)**. ??** We already know that $\mathrm{Vect}(pt) \cong \mathbb{N}$ as a semiring, where $n \mapsto [\underline{n}]$. Hence, as a Ring $K(\{pt\}) \cong \mathbb{Z}$. Since pointing the unpouinted space $\{pt\}$ results in S^0 we know by Corollary 3.3.34 that $\mathrm{K}^0(S^0) \cong \mathbb{Z}$ and generated by the trivial bundle of degree one over $\{-1\}$ and degree 0 over $\{+1\}$. We call this generator β_0. With the fundamental product Theorem, this tells us, that $K(S^2) \cong \Big(\mathbb{Z}([H])\Big/([H]-[\underline{1}])^2\Big)$ as a ring. Hence, we are isomorphic to the ring $\mathbb{Z} \times \mathbb{Z}$ with componentwise multiplication. $K(S^2)$ is generated by $\{[\underline{1}], [H]\}$. Furthermore, together with the map

$$\mathrm{rank} : \Big(\mathbb{Z}([H])\Big/([H]-[\underline{1}])^2\Big) \to \mathbb{Z}$$

that maps the generators to their rank, and

$$\nu : \mathbb{Z} \to K(pt);\ , \quad n \mapsto [\underline{n}]$$

we get the commutative diagram:

$$\begin{array}{ccc} \mathbb{Z}([H])\Big/([H]^2-[\underline{1}]) & \xrightarrow{\mathrm{rank}} & \mathbb{Z} \\ \downarrow & & \downarrow \nu \\ K(S^2) & \xrightarrow{i_{pt}^*} & K(pt) \end{array}$$

By the commutativity, we get an isomorphism induced by the map from the product theorem:

$$\tilde{K}(S^2) = \ker(i_{pt}^*) \cong \ker(\mathrm{rank}) = \mathrm{span}_{\mathbb{Z}}([H]-[\underline{1}])$$

Hence the Abelian group $\tilde{K}(S^2) \cong \mathbb{Z}$.

def:kTheo-BottGenerator

Definition 3.7.13. We call the generator $([H]-[\underline{1}])$ that generates $\tilde{K}(S^2)$ the **Bott generator** denoted by β.

thm:Ktheo-BottPeriodicity

Theorem 3.7.14 (Bott Periodicity)**.** *Let X be a well-pointed compact space. The isomorphism from the fundamental product theorem applied in its reduced form*

gives a natural isomorphism:

$$\beta \boxtimes (-) : K^{-n}(X) \to K^{-n}(S^2 \wedge X) \cong K^{-n-2}(X)$$

Proof. We start with $n = 0$. By Proposition 3.7.4 we know that the reduced external product applied to the spaces $\tilde{K}(S^2) \cong \mathrm{span}_{\mathbb{Z}}([H] - [\underline{1}])$ and $\tilde{K}(X)$ is an isomorphism, as applying the non-reduced external product gives an isomorphism by the fundamental product theorem. Hence, we have that

$$\mathbb{Z}([H] - [\underline{1}]) \otimes \tilde{K}(X) \to K^{-2}(X)$$

is an isomorphism. And since $\tilde{K}(X) \cong \mathbb{Z}(\beta) \otimes \tilde{K}(X)$ via $[\xi] \mapsto (\beta) \otimes [\xi]$ we have the isomorphism

$$\beta \boxtimes (-) : K^{-2}(X) \otimes \tilde{K}(X), \quad [\xi] \mapsto \tilde{\mu}(\beta \otimes [\xi]).$$

Naturality follows from μ being natural, q^* being natural (and thereby its inverse) and $\tilde{\mu}$ being the composition of both of them. Furthermore, for Replacing X with $S^n X$ proves the lower degree. □

natürlich im zweiten faktor mit festem ersten

cor:kTheo-kGroupsOfTheSpheres

Corollary 3.7.15 (Spheres)**.** Let S^n be pointed spheres. We already saw, that $K^0(S^2) = \tilde{K}(S^2) \cong \mathbb{Z}$. Hence, the Bott periodicity together with the homotopy equivalence $S^2 \wedge S^{2k-1} \simeq S^{2k}$ lets us deduce that $K^0(S^2) \cong K^0(S^4) \cong \cdots \cong K^0(S^{2k}) \cong \mathbb{Z}$, and a generator of $K^0(S^{2k})$ is given by $\underbrace{\beta \boxtimes \cdots \boxtimes \beta}_{k\text{-times}}$.

On the other hand we can inspect $\tilde{K}(S^1)$. By Lemma 3.4.14 we know that

$$\tilde{K}(S^1) \cong \tilde{K}(\Sigma\{pt\}) \cong [\{pt\}, U] = \pi_0(U).$$

However U is pathconnected and thereby the reduced K group of the one sphere is trivial. The path connectedness of U follows from all $U(n)$ being path connected which again can be deduced from $\mathrm{GL}_n(\mathbb{C})$ being path connected and contracting onto $U(n)$. Then, by the Bott periodicity we can conclude that $\tilde{K}(S^{2k+1}) = 0$ for all $k \in \mathbb{N}$.

def:ktheo-higherBottGenerators

Definition 3.7.16 (Convention)**.** Let $\beta_0 \in \tilde{K}^0(S^0)$ be the canonical generator from Corollary **??**, $\beta_1 \in \tilde{K}^{-1}(S^1)$ be the Bott generator of $\tilde{K}(S^2)$, viewed in $\tilde{K}^{-1}(S^1) = \tilde{K}^0(S^1 \wedge S^1)$ via Definition 3.3.27. Writing

$$\Upsilon_{n,m} := \left(\overline{\pi_S^{n+m}}\right)^{-1} \circ \Xi^{-1}_{\mathbb{R}^n,\mathbb{R}^m} \circ \left(\overline{\pi_S^n} \wedge \overline{\pi_S^m}\right) : S^n \wedge S^m \to S^{n+m}$$

for the homeomorphism of Definition 3.3.27, we define inductively

$$\beta_n := (\mathrm{id}_{S^n} \wedge \Upsilon^{-1}_{1,n-1})\beta_1 \boxtimes_{1,n-1} \beta_{n-1} \in \tilde{K}^{-n}(S^n)$$

Or alternatively, using the associativity of Υ from Remark 3.3.28 we can rewrite this to be

$$\beta_n = \Upsilon^n(\underbrace{\beta_1 \boxtimes \cdots \boxtimes \beta_1}_{n\text{-times}}),$$

where $\Upsilon^n : \underbrace{S^2 \wedge \cdots \wedge S^2}_{n\text{-times}} \to S^{2n}$ denotes iterative application of $S^1 \wedge S^l \to S^{1+l}$ that keeps the order of the coordinates.

By Theorem 3.7.14, the external product with β_1 is an isomorphism, so each β_n is a generator.

lem:ktheo-bottGeneratorsAreMultiplicative

Lemma 3.7.17. *For all $n, m \geq 0$ we have $\Upsilon^*_{n,m}(\beta_{n+m}) = \beta_n \boxtimes \beta_m$.*

Proof. Set $b_i := \left(\overline{\pi^i_S}\right)^* (\beta_i) \in \tilde{K}^{-i}\left(\overline{\mathbb{R}^i}\right)$.

Reduction. Since $\overline{\pi^n_S} \wedge \overline{\pi^m_S}$ is a homeomorphism, naturality of $\boxtimes$ gives $\left(\overline{\pi^n_S} \wedge \overline{\pi^m_S}\right)^* (\beta_n \boxtimes \beta_m) = b_n \boxtimes b_m$. Applying $\left(\overline{\pi^n_S} \wedge \overline{\pi^m_S}\right)^*$ to the claim and using $\left(\overline{\pi^n_S} \wedge \overline{\pi^m_S}\right)^* \Upsilon^*_{n,m} = \left(\Upsilon_{n,m} \circ \left(\overline{\pi^n_S} \wedge \overline{\pi^m_S}\right)\right)^* = \left(\left(\overline{\pi^{n+m}_S}\right)^{-1} \circ \Xi^{-1}_{n,m}\right)^*$, the claim becomes

$$\Xi^*_{n,m}(b_n \boxtimes b_m) = b_{n+m}\,, \tag{22}$$

eq:kTheo-multInR

and by definition $b_n = \Xi^*_{1,n-1}(b_1 \boxtimes b_{n-1})$.

Base case $n = 0$. Here $\Xi_{0,m}$ is the canonical homeomorphism $\overline{\mathbb{R}^m} \to \overline{\mathbb{R}^0} \wedge \overline{\mathbb{R}^m}$ and (22) is Remark 3.7.10.

Induction step. By definition of ψ in Lemma 3.3.25 the diagram above commutes, i.e.

$$(\Xi_{1,n-1} \wedge \mathrm{id}) \circ \Xi_{n,m} = \psi \circ (\mathrm{id} \wedge \Xi_{n-1,m}) \circ \Xi_{1,(n-1)+m}\,. \tag{23}$$

eq:kTheo-coherence

Now

$$\begin{aligned}
\Xi^*_{n,m}(b_n \boxtimes b_m) &= \Xi^*_{n,m}\left(\Xi^*_{1,n-1}(b_1 \boxtimes b_{n-1}) \boxtimes \mathrm{id}^*(b_m)\right) && \text{definition of } b_n \\
&= \Xi^*_{n,m}(\Xi_{1,n-1} \wedge \mathrm{id})^*((b_1 \boxtimes b_{n-1}) \boxtimes b_m) && \text{naturality of } \boxtimes \\
&= ((\Xi_{1,n-1} \wedge \mathrm{id}) \circ \Xi_{n,m})^*((b_1 \boxtimes b_{n-1}) \boxtimes b_m) && \text{functoriality} \\
&= \Xi^*_{1,(n-1)+m}(\mathrm{id} \wedge \Xi_{n-1,m})^* \psi^*((b_1 \boxtimes b_{n-1}) \boxtimes b_m) && \text{using (23)} \\
&= \Xi^*_{1,(n-1)+m}(\mathrm{id} \wedge \Xi_{n-1,m})^*(b_1 \boxtimes (b_{n-1} \boxtimes b_m)) && \text{Proposition 3.7.7} \\
&= \Xi^*_{1,(n-1)+m}\left(\mathrm{id}^*(b_1) \boxtimes \Xi^*_{n-1,m}(b_{n-1} \boxtimes b_m)\right) && \text{naturality of } \boxtimes \\
&= \Xi^*_{1,(n-1)+m}\left(b_1 \boxtimes b_{(n-1)+m}\right) && \text{induction hypothesis} \\
&= b_{n+m} && \text{definition of } b_{n+m}
\end{aligned}$$

□

4 Orientations

The idea of Morse homology is now to find an invariant under manipulations of the Morse–Smale function. Obviously neither the number of critical points nor the number of flow lines stays invariant. However certain linear combinations stay invariant, when one establishes a sophisticated way to count flow lines. For this count we want to introduce signs to the flow lines between critical points with index one apart where the signs rely on orientations of over the flow lines.

4.1 Oriented Vector Spaces and Bundles

Definition 4.1.1 (Orientation)**.** Let V be a finite dimensional $\mathbb{R}$ vector space and $\det(V) := \Lambda^{\dim(V)}$ the maximal exterior algebra (or Grassmann algebra). By the principle of permanence we define $\Lambda^0(V) = \mathbb{R}$. An orientation of V is a choice of one of the two connected components of $\det(V) \setminus \{0\}$. we call the touple $(V, [\mathfrak{o}])$ where $\mathfrak{o}$ is a non-zero element of $\det(V)$ and hence $[\mathfrak{o}]$ is an orientation, an **oriented vector space**. Often we will just write V oriented, and omitt the orientation.

Definition 4.1.2 (Convention)**.** If we use $\mathbb{R}^n$ as an oriented vector space we fix $e := e_1 \wedge \cdots \wedge e_n$ to induce the orientation.

Remark 4.1.3. Classically, an orientation of a vector space V of dimension m is given by an ordered basis $\mathfrak{b} = (b_1, \dots b_m)$ of V and hence the product $b_1 \wedge \cdots \wedge b_m$ is an element of the maximal exterior product $\det(V) = \Lambda^m V$. Two ordered bases define different elements in $\det(V)$ and their difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence a choice of equivalence class of bases modulo base change matrix of positive determinant is the same as a choice of one connected component of $\det(V) \setminus \{0\}$.

Definition 4.1.4 (Determinant of an endomorphism)**.** Let $f : V \to V$ be an endomorphism. Then we define

$$\Lambda^{\dim(V)}(f) : \det(V) \to \det(V) \tag{24}$$

$$v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots \wedge f(v_{\dim(V)}) . \tag{25}$$

Since $\dim(\det(V)) = 1$, the function $\Lambda^{\dim(V)}(f)$ is given by a scaler multiplication with a nonzero element $\lambda \in \mathbb{R}$. We define the determinant $\det(f) := \lambda$ to be said scalar. Obviously, if $\det(V)$ is positive, then f is orientation preserving in the sence, that f applied to the ordered basis gives again an ordered basis that is compatible.

rem:orient-calculateDeterminatViaMatrixDeterminant

Remark 4.1.5. From linear algebra we know, that we can calculate $\det(f) = \det(M_{\mathfrak{b}}^{\mathfrak{b}}(f))$, where $\mathfrak{b}$ is an ordered basis of V. Since the determinant of a matrix commutes with taking products, we have that the determinant is invariant under conjugation of linear endomorphisms.

def:oeient-SignOfAAbstractLinearMap

Definition 4.1.6. Let $(V, [\mathfrak{o}])$ and $(W, [\mathfrak{p}])$ be oriented vector spaces. We call a linear isomorphism $f : V \to W$ between two orineted vector spaces **orientation preserving**, if $[\Lambda^{\dim(V)}(f)(\mathfrak{o})] = [\mathfrak{v}]$. Let o and p denote the basis induced from $\mathfrak{o}$ resp. $\mathfrak{p}$. Then by the Remark 4.1.5 the map f is orientation preserving, if and only if $\det(M^{\mathfrak{o}}_{\mathfrak{p}}(f))$ is positive. We define the sing of f (and sometimes even the seign of $\det(f)$) to be the sign of $\det(M^{\mathfrak{o}}_{\mathfrak{p}}(f))$.

Definition 4.1.7 (Orientation and Exact Sequences)**.** A short exact sequence of vector FIX spaces

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

induces a canonical isomorphism

$$\begin{aligned} \det(A) \otimes \det(C) &\to \det(B) \\ (a_1 \wedge \cdots \wedge a_l) \otimes (c_1 \wedge \cdots \wedge c_k) &\mapsto (f(a_1) \wedge \ldots f(a_l) \wedge \tilde{c_1} \wedge \cdots \wedge \tilde{c_k}) , \end{aligned}$$

where $\tilde{c_j} \in g^{-1}(c_j)$. The well-definedness of this map (invariance under chosen lift of c) can be calculated, since two different choices differ in an element in the kernel of g which equals the image of f and hence the different choices vanish in the wedge product. By this isomorphism we get an orientation induced in the middle from the outside. Notice the convention, that the basis from A is placed in front of the basis from C. Notice how the definition is consistent with the edge cases of trivial vector spaces A or C. If $A = 0$ we get an isomorphism $B \cong C$ and a basis from C induced to B, that is either the induced one from g, or exactly the opposite one, if the trivial vector space A is negatively oriented.

Remark 4.1.8. Let

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be a short exact sequence of vector spaces and $\mathfrak{a}$ be a basis of A, $\mathfrak{c}$ be a basis of C and $\tilde{\mathfrak{c}}$ be a lift of $\mathfrak{c}$ via g. Replacing the basis $\mathfrak{a}$ with $P\mathfrak{a}$ and $\mathfrak{c}$ with $Q\mathfrak{c}$ where P, Q are invertible matrices, gives us a new induced basis on B, which differs from the original one by a matrix $\begin{pmatrix} P & * \\ 0 & Q \end{pmatrix}$, where $*$ depends on the chosen lift. Hence the corresponding elements in $\det(B)$ differ by the product $\det(P)\det(Q)$.

def:orient-orientedVectorBundle

Definition 4.1.9. For a rank r vector bundle $E \to X$ we set $\det(E) := \Lambda^r(E)$. An orientation is an equivalence class of a non-vanishing section into the determinant bundle, where two sections are equivalent if they differ by multiplication of a positive scalar valued function. E is orientable if such a section exists, i.e. if $\det(E)$ is trivial. An orientation of a bundle restricts to an orientation in each fibre.

lem:morse-OrientationsInAPointInduceOrientationsOfBundle

Lemma 4.1.10. *Let $E \to X$ be an orientable vector bundle and X be connected. For each $x \in X$ we get the bijection induced from the restriction to the fibre:*

$$\{\text{Orientation on } E\} \to \{\text{Orientation on } E_x\}\,.$$

Hence, an orientation in one fibre can introduce an orientation in every other fibre.

Proof. Two orientations over E are given as nowhere-vanishing sections s, s'. The quotient $x \mapsto \frac{s}{s'}$ is continuous and its image is contained in $\mathbb{R} \setminus \{0\}$. Hence the sign of the quotient is a continuous function into $\{-1, 1\}$. Since X is connected we conclude that the sign of said quotient is a constant function. Thereby, if the sections induce the same orientation, they differ by multiplication of the positive number $\frac{s(x)}{s'(x)}$ and then the sign of $\frac{s}{s'}$ is globally one. The surjectivity is clear by mapping the sections s and $-s$. □

Definition 4.1.11. Let $\pi : E \to X$ and $\pi' : E' \to X'$ be two oriented vector bundles, $f : X \to X'$ continouos, and $\Phi : E \to E'$ a morphism of vector bundles over f such that for all x, Φ_x is an isomorphism of vector bundles. Then we call Φ orientation preserving, if and only if $\Phi_x : E_x \to E'_{f(x)}$ is an orientation preserving map between oriented vector spaces, that inherit their orientation from the bundles via Lemma 4.1.9 for all $x \in X$.

def:orient-orientedMorphism0fBundlesEquivalentDef

Proposition 4.1.12 (Equivalent Definition)**.** Let s and t be non-vanishing sections into the determinant bundle that represent orientations of the oriented vector bundles $\pi : E \to X$ and $\pi' : E' \to X'$, $f : X \to Y$ be continuous and $\Phi : E \to E'$ be a bundle morphism over f. Then for all $x \in X$ there exists a $\lambda(x) \in \mathbb{R} \setminus \{0\}$ such that $\Lambda^{\operatorname{rank}(E)}(\Phi)(s(x)) = \lambda(x)t(f(x))$. The map $x \mapsto \lambda(x)$ is continouos, as $\lambda = \frac{\Lambda^{\operatorname{rank}(E)}(\Phi)\circ s}{t \circ f}$ is continouos (one can check this in a local frame of $\det(E')$) and non-vanishing. Thereby its sign is locally constand. Hence, if X is connected, then Φ is oriented, if for **any** $x \in X$ Φ_x is an orientation preserving map between oriented vector spaces, that inherit their orientation from the bundles via Lemma 4.1.9

Proposition 4.1.13. Let

$$0 \longrightarrow E' \xrightarrow{f} E \xrightarrow{g} E'' \longrightarrow 0$$

be a short exact sequence of orientable vector bundles. Then the fibrewise isomorphisms $\det(E'_p) \otimes \det(E''_p) \to \det(E_p)$ from the vector space case assemble into an isomorphism of line bundles

$$\det(E') \otimes \det(E'') \xrightarrow{\cong} \det(E)\,.$$

In particular, orientations of E' and E'' induce a canonical orientation of E.

Proof. Fibrewise the map is the canonical isomorphism from the vector space case: a lift $\tilde{c} \in g_p^{-1}(c)$ exists since g_p is surjective, and the resulting element $f(a_1) \wedge \cdots \wedge f(a_l) \wedge \tilde{c}_1 \wedge \cdots \wedge \tilde{c}_k$ is independent of the choice of lift.

It remains to check that this bundle map is continuous (resp. smooth) to ensure that it maps sections smoothly into sections, which is a local statement: locally choose a frame $(a_1, \dots, a_l)$ of E' and a frame $(c_1, \dots, c_k)$ of E''; since g is an epimorphism it admits local sections, so we may choose local lifts $\tilde{c}_j$ of the c_j which are (smooth) local sections of E. Then the map is given in the induced local frames of the determinant bundles by wedging these sections, hence continuous (resp. smooth). By this, two non-vanishing sections get mapped to a non-vanishing section and hence, orientations on E' and E'' which are represented by sections give a orientation over E. □

Definition 4.1.14. We call a manifold orientable, if its tangent bundle is **orientable**. We call a touple $(M, [\omega])$ where $[omega]$ denotes an orientation of TM an **oriented manifold**. We will often omitt the orientation and just call M oriented.

Definition 4.1.15 (Orientation Preserving Charts and Maps)**.** Let $(M, [\omega])$ be an oriented n-dimensional manifold and let $\varphi : U \to \mathbb{R}^n$ be a chart. We call φ **orientation preserving** if for every $x \in U$ $\partial_1(x) \wedge \cdots \wedge \partial_n(x)$ and $\omega(x)$ lie in the same connected component of $\det(T_x M) \setminus \{0\}$. More generalyly, let $(N, [\eta])$ be another oriented n-dimensional manifold and let $h : M \to N$ be differentiable at x with $\mathrm{d}h_x$ invertible. We define

$$\operatorname{sign}\det(\mathrm{d}h_x) := \begin{cases} +1 & \text{if } \Lambda^n(\mathrm{d}h_x)(\omega(x)) \text{ and } \eta(x) \text{ induce the same orientation in } T_h(x)N \\ -1 & \text{otherwise.} \end{cases}$$

Corollary 4.1.16. A chart $\varphi : U \to \mathbb{R}^n$ is orientation preserving, if for all $x \in U$, and $Q_{f(x)} : T_{\varphi(x)}\mathbb{R}^m \to \mathbb{R}^m$ (from Definition 2.0.45) the map $Q_{f(x)} \circ \mathrm{d}\varphi|_x : T_x U \to \mathbb{R}^m$ is an oriented map of vector spaces.

Remark 4.1.17. One easyly checks that $\operatorname{sign}\det(\mathrm{d}h_x)$ is well-defined and reversing both orientations doesn't affect the sign. In particular, for $N = M$ the number $\operatorname{sign}\det(\mathrm{d}h_x)$ does not depend on the choosen orientation.

lem:orient-computingSignInCharts

Lemma 4.1.18 (Computing the Sign in Charts)**.** *Let $(M, [\omega])$ and $(N, [\eta])$ be orinted n-dimensional manifolds and φ be an orinetation preserving chart arround x in M and ψ one arround $h(x)$. Then*

$$\operatorname{sign}\det(\mathrm{d}h_x) = \operatorname{sign}\det\left(\mathrm{d}\left(\psi \circ h \circ \varphi^{-1}\right)\big|_{\varphi(x)}\right),$$

wherre the right hand side calculates the sign of the determinant of a matrix.

Proof. Define $A := \mathrm{d}\left(\psi \circ h \circ \varphi^{-1}\right)\big|_{\varphi(x)} = \mathrm{d}\psi|_{h(x)} \circ \mathrm{d}h|_x \circ (\mathrm{d}\varphi|_x)^{-1}$ and denote the standart orientation of $\mathbb{R}^n$ by e. Since φ is orientation preserving and $\Lambda^n(\mathrm{d}\varphi|_x)(\partial_1(x) \wedge \cdots \wedge \partial_n(x)) = e$, we have by linearity $\Lambda^n(\mathrm{d}\varphi|_x)(\omega(x)) = a \cdot e$ where $a > 0$. Similar $\Lambda^n(\mathrm{d}\psi|_{h(x)})(\eta(h(x))) = b \cdot e$ with $b > 0$. Hence

$$\Lambda^n\left(\mathrm{d}\psi|_{h(x)}\right) \circ \Lambda^n\left(\mathrm{d}h|_x\right)(\omega(x)) = \Lambda^n(A) \circ \underbrace{\Lambda^n\left(\mathrm{d}\varphi|_x\right)(\omega(x))}_{=a\cdot e} = a \det(A) \cdot e$$

Now by definition we need to check if $\Lambda^n(\mathrm{d}h_x)(\omega(x)) = c \cdot \eta(x)$ for $c > 0$, which is equivalent to

$$\underbrace{\Lambda^n\left(\mathrm{d}\psi|_{h(x)}\right) \circ \Lambda^n\left(\mathrm{d}h|_x\right)(\omega(x))}_{=a\det(A)\cdot e} = c \cdot \underbrace{\Lambda^n\left(\mathrm{d}\psi|_{h(x)}\right)(\eta(x))}_{=b\cdot e}.$$

which is given if and only if $\det(A) > 0$. □

cor:orient-existsOrientationPreservingAtlas

Corollary 4.1.19. A smooth manifold M of dimension n is orientable, if and only it has an atlas such that the determinants of the Jacobians of the transition functions are positive.

Proof. Let $[\omega]$ be an orientation on M and let U_α be an open cover of M with all U_α connected and the charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$. Let $\partial_1^\alpha \wedge \cdots \wedge \partial_n^\alpha$ be the induced local section into the determinant bundle. Then

$$\omega|_{U_\alpha} = g_\alpha \partial_1^\alpha \wedge \cdots \wedge \partial_n^\alpha$$

whith g_α non vanishing and U_α connected. If g_α is negative, replace φ_α with $\tilde{\varphi_\alpha}(x) = (-(\varphi_\alpha(x))_1, \ldots, \varphi_\alpha(x)_n)$, which changes the sign og g_α. Doing this in all charts gives a orientaion preserving atlas. Hence, by defininition $\operatorname{sign}\det(\mathrm{d}\,\mathrm{id}_M|_x) = 1$ and by Lemma 4.1.18 we can calculate $\operatorname{sign}\det(\mathrm{d}\,\mathrm{id}_M|_x)$ in this atlass by the determinants of the Jacobians of the transition functions. Hence those determinants are positive.

Fot the other direction we want to glue together the local sections $\partial_1^\alpha \wedge \cdots \wedge \partial_n^\alpha$ using a smooth partition of unity subordinate to the open cover $\{U_i\}$. This yealds something non-vanishing, as all local section are pointwise positive multiples of each other where they intersect. □

cor:orient-transitionFunctionPositiveDeterminantAlsoOrientationPreserving

Corollary 4.1.20. The proof of Corollary 4.1.19 also shows that if one chart is orientation preserving, then also another chart that intersects with the first and has positive determinant of the Javobian of the transition function.

Lemma 4.1.21. *Let M be a smooth manifold that can be covered with two charts such that their intersection is connected. Then M is orientable.*

Proof. Just ensure that the two charts are compatible such that they build an atlas such that the determinants of the Jacobians of the transition functions are positive. To achive this one just manipulates one of the charts. □

Corollary 4.1.22. S^n for $n \geq 2$ is orientable, as it is coverd by the two stereographic projections. S^1 and S^0 are also orientable which can be seen by picking charts such that they fit into corollary 4.1.19.

4.2 K-theoretic Orientation

K theoretic orientation relies on bases induced generators of the K group of the sphere. Hence, we want to fix a canonical map from R^n to S^n:

lem:reflection-is-minus-one

Lemma 4.2.1 (Reflection in one variable)**.** *Let $R_1 : \mathbb{R} \to \mathbb{R}$, $x \mapsto -x$, and let Z be a pointed compact Hausdorff space. Then*

$$\left(\overline{R_1} \wedge \mathrm{id}_Z\right)^* = -\mathrm{id} \qquad \text{on } \tilde{K}\left(\overline{\mathbb{R}} \wedge Z\right).$$

Proof. Realise S^1 as $\mathbb{R}/\mathbb{Z}$ with base point $[0]$, so that the map T of Corollary 3.4.16 is $T([t]) = [-t]$. Define

$$h : \overline{\mathbb{R}} \to \mathbb{R}/\mathbb{Z}, \quad h(x) = \left[\tfrac{1}{2} + \tfrac{1}{\pi}\arctan(x)\right], \quad h(\infty) = [0].$$

Since $x \mapsto \frac{1}{2} + \frac{1}{\pi}\arctan(x)$ is a strictly increasing bijection onto $(0,1)$, the map h is bijective; it is continuous on $\mathbb{R}$ and also at ∞, because $\arctan(x) \to \pm\frac{\pi}{2}$ gives $h(x) \to [1] = [0]$ resp. $h(x) \to [0]$. Being a continuous bijection from a compact space to a Hausdorff space, h is a pointed homeomorphism. It intertwines the two reflections:

$$h(-x) = \left[\tfrac{1}{2} - \tfrac{1}{\pi}\arctan(x)\right] = \left[-\tfrac{1}{2} - \tfrac{1}{\pi}\arctan(x)\right] = T(h(x)),$$

using $[-\frac{1}{2}] = [\frac{1}{2}]$, and $h(\overline{R_1}(\infty)) = [0] = T(h(\infty))$. Hence $h \wedge \mathrm{id}_Z$ is a pointed homeomorphism with

$$(h \wedge \mathrm{id}_Z) \circ \left(\overline{R_1} \wedge \mathrm{id}_Z\right) = (T \wedge 1) \circ (h \wedge \mathrm{id}_Z).$$

Applying $\tilde{K}$ and Corollary 3.4.16 gives

$$\left(\overline{R_1} \wedge \mathrm{id}_Z\right)^* = (h \wedge \mathrm{id}_Z)^* \circ (T \wedge 1)^* \circ ((h \wedge \mathrm{id}_Z)^*)^{-1} = -\mathrm{id},$$

as all maps are ring homomorphisms and thereby commute with multiplicatino by minus one. □

Definition 4.2.2 (K-theoretic orientations)**.** We define a **K-theoretic orientation** of a finite dimensional real vector space V to be a generator of the group $\tilde{K}^{-m}(\overline{V})$. As a sainity check we can confirm: since $\overline{V} \cong S^m$ we are looking at a generator of a group isomorphic to $\tilde{K}(S^{2m}) \cong \mathbb{Z}$, which has two possible generators.

lem:orient-linearMapsOnSmashProducts

Lemma 4.2.3 (Linear Maps on Smash Products)**.** *Let $A \in \mathrm{GL}(V)$ and let Z be a pointed compact space. Then*

$$(\overline{A} \wedge \mathrm{id}_Z)^* = \operatorname{sign}\det(A) \cdot \mathrm{id} \qquad \text{on } \tilde{K}(\overline{V} \wedge Z).$$

In particular, if $\overline{\mathbb{R}^{m_1}} \wedge \cdots \wedge \overline{\mathbb{R}^{m_r}}$ is identified with $\overline{R^m}$ by Lemma 3.3.24, a permutation od the smash factors acts on $\tilde{K}(\overline{\mathbb{R}^{m_1}} \wedge \cdots \wedge \overline{\mathbb{R}^{m_r}} \wedge Z)$ by the sign of the

induced permutation of the $m = m_1 + \cdots m_r$ coordinates.

Proof. Since the determinant of an endomorphism is invariant under conjugation we can choose a basis and assume that $V = \mathbb{R}^m$. The group $\mathrm{GL}_n(\mathbb{R})$ as exactly two path components, seperated by the sign of the determinant. Hence, there is a contiuous path $A_\bullet : [0,1] \to \mathrm{GL}_m(\mathbb{R})$ with $A_0 = A$ and $A_1 = \mathrm{id}$ if $\det(A) > 0$ and R if $\det(A) < 0$ with R being the reflection $(x_1, \ldots, x_m) \mapsto (-x_1, x_2 \ldots, x_m)$. The map

$$H : \mathbb{R}^m \times [0,1] \to \mathbb{R}^m, \quad H(v,t) := A_t v$$

is a proper homomorphism into a locally compact space: If $K \subseteq \mathbb{R}^m$ is compact and $A_t v \in K$, then by definition of the operator norm we have that $\|v\| \leq (\sup_t \|A_t^{-1}\|) \cdot \max_{k \in K} \|k\|$. Hence $H^{-1}(K)$ is bounded and closed by continuity. Thereby the the extension $\tilde{H} : \overline{V \times I} \to \overline{V}$ is continuous. Let $q : \overline{V} \times I \to \overline{V \times I}$ be the collaps of $\{\infty\} \times I$ to ∞. Then we have that $\overline{H} = \tilde{H} \circ q$ and hence $\overline{H}$ is continuous. If $\det(A) > 0$, we are done. Let $\overline{R}$ denote the extension of R to the compactifications. What is left to show is that $(\overline{R} \wedge \mathrm{id}_Z)^*$ is given by multiplication with minus one.

For this we write $R = R_1 \times \mathrm{id}_{\mathbb{R}^{n-1}}$ where R_1 denotes the reflection $x \mapsto -x$ on $\mathbb{R}$. Let $\Xi : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}} \wedge \overline{\mathbb{R}^{n-1}}$ be the natural homeomorphism from Lemma 3.3.24. By the naturality for proper maps we have that $\Xi \circ (\overline{R}) = (\overline{R_1} \wedge \overline{\mathrm{id}_{\mathbb{R}^{n-1}}}) \circ \Xi$, giving us:

$$(\Xi \wedge \mathrm{id}_Z)(\overline{R} \wedge \mathrm{id}_Z) = (\Xi \circ \overline{R}) \wedge \mathrm{id}_Z = \left((\overline{R_1} \wedge \overline{\mathrm{id}_{\mathbb{R}^{n-1}}}) \circ \Xi\right) \wedge \mathrm{id}_Z$$

If we now define $L := \overline{R^{m-1}} \wedge Z$ and let $AS : (\overline{\mathbb{R}^1} \wedge \overline{\mathbb{R}^{m-1}}) \wedge Z \to \overline{\mathbb{R}^1} \wedge (L)$ be the assozoiativity of the wedge product. Then we have

$$AS \circ (\Xi \wedge \mathrm{id}_Z)(\overline{R} \wedge \mathrm{id}_Z) = (\overline{R_1} \wedge \mathrm{id}_L) \circ AS \circ \Xi \wedge \mathrm{id}_Z$$

Hence,if we define $\Theta := AS \circ \Xi \wedge \mathrm{id}_Z$ we get

$$\overline{R} \wedge \mathrm{id}_Z = \Theta^{-1} \circ (\overline{R_1} \wedge \mathrm{id}_L) \circ \Theta$$

or as maps of K-groups:

$$(\overline{R} \wedge \mathrm{id}_Z)^* = \Theta^* \circ \underbrace{(\overline{R_1} \wedge \mathrm{id}_L)^*}_{= -\mathrm{id} \text{ by Lemma 4.2.1}} \circ (\Theta^{-1})^* = -\mathrm{id}\,.$$

□

prop:linearyTheDeterminandDertermines

Proposition 4.2.4. Let V be an n-dimensinoal real vector space and $A \in \mathrm{GL}(V)$. Then

$$\overline{A}^* = \mathrm{sign}\det(A) \cdot \mathrm{id} \quad \text{on } \tilde{K}^{-n}(\overline{V})\,.$$

Proof. The statement is almost a corollary of Lemma 4.2.3: We care for the map $\overline{A}^*$ which is defined as the map

$$(\mathrm{id}_{S^n} \wedge A)^* : \tilde{K}^{-n}(V) = \tilde{K}(S^n \wedge V) \to \tilde{K}(S^n \wedge V) = \tilde{K}^{-n}(V)$$

Let $\Theta : S^n \wedge V \to V \wedge S^n$ be the reordering from Lemma 3.3.25. Then we have that

$$\mathrm{id}_{S^n} \wedge A = \Theta^{-1} \circ (A \wedge \mathrm{id}\, S^n) \circ \Theta\,.$$

Hence, lemma 4.2.3 gives us:

$$\overline{A}^* = (\Theta^{-1} \circ (A \wedge \mathrm{id}\, S^n) \circ \Theta)^* = \Theta^* \circ \underbrace{(A \wedge \mathrm{id}\, S^n)^*}_{=\mathrm{sign}\det(A)\cdot\mathrm{id}} \circ (\Theta^{-1})^* = \mathrm{sign}\det(A) \cdot \mathrm{id}\ .$$

□

cor:orient-vectorspaceOrientationInducesKtheoreticOrientation

Corollary 4.2.5. Any finite dimensional real vector space V is homeomoprhic to $\mathbb{R}^n$ via the coordinate map κ_v that maps each vector to its coordinate representation. With this we can introduce a K-theoretic orientation from an ordered basis via the composition:

$$\tilde{K}^{-n}(S^n) \xrightarrow{\overline{(\pi_S^{-1})}^*} \tilde{K}^{-n}(\overline{\mathbb{R}^n}) \xrightarrow{\overline{\kappa_v}^*} \tilde{K}^{-n}(\overline{V})$$

With this we can map the bott generator β from Definition 3.7.13 to a generator of $\tilde{K}^{-n}(\overline{V})$ and hence induce a K-Theoretic orientation. This orientation only depends on the orientation of V and two different orientations induce different K-theoretic orientations

Proof. The statement follows imidiatly from Proposition 4.2.4. □

cor:orient-mapsbetweenabstractVectorSpacesOnKOrientations

Corollary 4.2.6. Let $f : V \to W$ be a linear isomorphism between oriented vector spaces $(V, \mathfrak{a})$ and $(W, \mathfrak{b})$. Let β_V be the K-Theoretic orientation of V and β_W of W. Then

$$\overline{f}^*(\beta_W) = \mathrm{sign}\det(f) \cdot (\beta_V)\,,$$

where $\mathrm{sign}\det(f)$ denotes the sign of f as in Definition 4.1.6.

Proof. Let $\kappa_V : V \to \mathbb{R}^n$ and $\kappa_W : W \to \mathbb{R}^n$ be positive coordinate maps. By Proposition 4.2.4 we know that

$$(\overline{\kappa_W} \circ \overline{f} \circ \overline{\kappa_V}^{-1})^* = \mathrm{sign}\det(f) \cdot \mathrm{id}$$

Now $\beta_V = \overline{\kappa_V}^*(\beta_{\mathbb{R}^n})$ and $\beta_W = \overline{\kappa_W}^*(\beta_{\mathbb{R}^n})$. This concludes the proof, as

$$\begin{aligned}\overline{f}^*(\beta_W) &= (\overline{\kappa_V}^* \circ (\overline{\kappa_V}^*)^{-1}) \circ \overline{f}^* \circ \left(\overline{\kappa_W}^*(\beta_{\mathbb{R}^n})\right) \\ &= \overline{\kappa_V}^* \circ (\overline{\kappa_W} \circ \overline{f} \circ \overline{\kappa_V}^{-1})^* \beta_{\mathbb{R}^n} \\ &= \mathrm{sign}\det(f)\, \overline{\kappa_V}^*\, \beta_{\mathbb{R}^n} = \mathrm{sign}\det(f)\, \beta_V\,.\end{aligned}$$

□

lem:orient-multiplication

Lemma 4.2.7 (Multiplication)**.** *Let V, W be oriented real vector spaces of dimension n, m and give $V \oplus W$ the orientation by placing the orientation of V infront of the one of W. Let $\beta_{\mathfrak{o}}$ be the K Theoretic orientation ov V induced by the oreintation of V and similar $\beta_{\mathfrak{p}}$ be the one on W, and finaly $\beta_{\mathfrak{o}\oplus\mathfrak{p}}$ be the K-thoretic*

orientation of $V \oplus W$. Then under the homeomorphism

$$\Xi : \overline{V \oplus W} \to \overline{V} \wedge \overline{W}$$

from Lemma 3.3.24 we have

$$\Xi^* (\beta_{\mathfrak{o}} \boxtimes \beta_{\mathfrak{p}}) = \beta_{\mathfrak{o}\oplus\mathfrak{p}} .$$

Proof. Let $\kappa_v : V \to \mathbb{R}^n$ and $\kappa_w : \mathbb{R}^n \to W$ be two basis maps from ordered basis compatible with the orientations ov V and W. By definition and naturality of the reduced external product:

$$\begin{aligned}\Xi^* (\beta_{\mathfrak{o}} \boxtimes \beta_{\mathfrak{p}}) &= \Xi^* \circ \left(\left(\overline{(\pi_S^n)}^{-1} \circ \overline{\kappa}_v \right)^* (\beta_n) \boxtimes \left(\overline{(\pi_S^m)}^{-1} \circ \overline{\kappa}_w \right)^* (\beta_m) \right) \\ &= \Xi^* \circ \left(\left(\overline{(\pi_S^n)}^{-1} \circ \overline{\kappa}_v \right) \wedge \left(\overline{(\pi_S^m)}^{-1} \circ \overline{\kappa}_w \right) \right)^* (\beta_n \boxtimes \beta_m)\end{aligned}$$

Now, the function $(\left(\overline{(\pi_S^n)}^{-1} \circ \overline{\kappa}_v \right) \wedge \left(\overline{(\pi_S^m)}^{-1} \circ \overline{\kappa}_w \right) : \overline{V} \to S^n$ is the compactification of $(\pi_S^n)^{-1} \circ \kappa_v$ Hence, by naturality of Ξ we conclude:

$$\begin{aligned}\Xi^* \circ \left(\left(\overline{(\pi_S^n)}^{-1} \circ \overline{\kappa}_v \right) \wedge \left(\overline{(\pi_S^m)}^{-1} \circ \overline{\kappa}_w \right) \right)^* &= \overline{\left((\pi_S^n)^{-1} \circ \kappa_v \right) \times \left(\pi_S^m)^{-1} \circ \kappa_w \right)}^* \circ \Xi^* \\ &= \overline{\kappa_v \times \kappa_w}^* \circ \overline{\left((\pi_S^n)^{-1} \times (\pi_S^m)^{-1} \right)}^* \circ \Xi^*\end{aligned}$$

Next we will apply Lemma to replace 3.7.17 $(\beta_n \boxtimes \beta_m)$ with $\Upsilon_{n-m}^*(\beta_{n+m})$ where

$$\Upsilon_{n,m} := \overline{(\pi_S^{n+m})}^{-1} \circ I \circ \Xi^{-1} \circ \left(\overline{\pi_S^n} \wedge \overline{\pi_S^m} \right) .$$

In the definition of Υ in Definition 3.7.16 we supressed the map I that identifies $\overline{\mathbb{R}^n \times \mathbb{R}^m}$ with $\overline{\mathbb{R}^{n+m}}$. For this proof we will write it out. This leads to

$$\begin{aligned}\Xi^* (\beta_n \boxtimes \beta_m) &= \left(\overline{(\pi_S^{n+m})}^{-1} \circ I \circ \Xi^{-1} \circ \left(\overline{\pi_S^n} \wedge \overline{\pi_S^m} \right) \circ \Xi \right)^* (\beta_{n+m}) \\ &= \left(\overline{(\pi_S^{n+m})}^{-1} \circ I \circ \left(\overline{\pi_S^n \times \pi_S^m} \right) \right)^* (\beta_{n+m})\end{aligned}$$

Now pulling everything together we get:

$$\begin{aligned}\Xi^* (\beta_{\mathfrak{o}} \boxtimes \beta_{\mathfrak{p}}) &= \overline{\kappa_v \times \kappa_w}^* \circ \overline{\left((\pi_S^n)^{-1} \times (\pi_S^m)^{-1} \right)}^* \circ \Xi^* (\beta_n \boxtimes \beta_m) \\ &= \overline{\kappa_v \times \kappa_w}^* \circ \overline{\left((\pi_S^n)^{-1} \times (\pi_S^m)^{-1} \right)}^* \circ \left(\overline{(\pi_S^{n+m})}^{-1} \circ I \circ \left(\overline{\pi_S^n \times \pi_S^m} \right) \right)^* (\beta_{n+m}) \\ &= \left((\overline{(\pi_S^{n+m})}^{-1}) \circ \underbrace{I \circ \overline{\kappa_v \times \kappa_w}}_{\kappa_{v\oplus w}} \right)^* (\beta_{n+m}) \\ &= \beta_{\mathfrak{o}\oplus\mathfrak{p}}\end{aligned}$$

□

4.3 Local K-Theoric Orientation

lem:orient-Minkowsky Gauge

Lemma 4.3.1 (Minkowsky Gauge)**.** *Let V be a finite dimensional real vector space with a norm $\|\cdot\|$ and let $U \subseteq V$ be open, bounded, convex and contain the origin of V. Choose $\varepsilon, r > 0$ such that $\{v \in V \mid \|v\| < \varepsilon\} \subseteq U \subseteq \{v \in V \mid \|v\| < r\}$ and define the Minkowsky gauge*

$$\mu : V \to [0, \infty], \quad \mu(v) := \inf\{s > 0 \mid s^{-1}v \in U\}.$$

Then:

(a) $\lambda U \subseteq U$ for $\lambda \in [0,1]$, and $s^{-1}v \in U$ for every $s > \mu(v)$.

(b) $\mu(\lambda v) = \lambda(\mu(v))$ for all $\lambda > 0$.

(c) $\frac{\|v\|}{r} \leq \mu(v) \leq \frac{\|(\|v)}{\varepsilon}$. In particular $\mu(v) > 0$ for $v \neq 0$.

(d) $\mu(v + w) \leq \mu(v) + \mu(w)$.

(e) $|\mu(v) - \mu(w)| \leq \frac{\|v-w\|}{\varepsilon}$, so μ us Lipschitz and hence continuous;

(f) $U = \{v \in V \mid \mu(v) < 1\}$ and $\overline{U} = \{v \in V \mid \mu(v) \leq 1\}$.

Proof. (a) For $x \in U$ and $\lambda \in [0,1]$ we have $\lambda x = \lambda x + (1-\lambda)0 \in U$ by convexity and since $0 \in U$. Now let $s > \mu(v)$. By definition of the infimum there exists a $s_0 \in (\mu(v), s]$ with $s_0^{-1}v \in U$ and $s^{-1}v = \frac{s_0}{s} s_0^{-1} v \in U$ by the first oriperty (since $\frac{s_0}{s} \in [0,1]$).

(b) Follows directly from the definition if one substitutes $s \mapsto \lambda s$.

(c) If $s^{-1}v \in U \subseteq \{v \in V \mid \|v\| \leq r\}$ then $\|v\| < sr$,i.e. $s > \|v\| r^{-1}$. After the infimum the reads $\mu(v) \geq \|v\| r^{-1}$. Conversly, if $s > \|v\| \varepsilon^{-1}$ then $\left\|s^{-1}v\right\| < \varepsilon$, and hence $s^{-1}v \in U$. Thereby, $\mu(v) \leq \|v\| \varepsilon^{-1}$.

(d) Let $s > \mu(v)$ and $t > \mu(w)$. Then by (a) both $s^{-1}v$ and $t^{-1}w$ lie in U, hence by convexity

$$\frac{v+w}{s+t} = \frac{s}{s+t} \cdot \underbrace{\frac{v}{s}}_{\in U} + \underbrace{\frac{w}{t}}_{\in U} \in U.$$

So $\mu(v+w) \leq s+t$ and letting s approach $\mu(v)$ and t approach $\mu(w)$ gives the claim.

(e) By (b) and (c), for all $v, w \in V$ we have: $\mu(v) \leq \mu(w) + \mu(v-w) \leq \mu(w) + \|v-w\| \varepsilon^{-1}$. Hence, $\mu(v) - \mu(w) \leq \|v-w\| \varepsilon^{-1}$. Interchangeing v and w gives the symetric statement for $-(\mu(v) - \mu(w)) = \mu(w) - \mu(v) \leq \|v-w\| \varepsilon^{-1}$ proofing the statement.

(f) If $v \in U$, then $s^{-1}v \in U$ for some $s < 1$, because U is open and $s \mapsto s^{-1}v$ is continuous at $s = 1$. Hence $\mu(v) < 1$. The other inclusion follows directly from (a): if $\mu(v) < 1$ then $1^{-1}(v) \in U$. Thus $U = \{v \in V \mid \mu(v) < 1\}$. Now, by continuity of μ the set $\{v \in V | \mu(v) \leq 1\}$ is closed and containes U and thereby also contains $\overline{U}$. conversely if $\mu(v) \leq 1$ then $\mu(tv) = t\mu(v)$ for $t \in (0,1)$ by (b), so $tv \in U$ and if t approaches one, tv approaches v. Hence v is in the closure of U. □

lem:orient-admissableNeighbourhoodsGiveCompactPairs

Lemma 4.3.2. *Let V be a finite dimensional real vector space and let $U \subseteq V$ be open, bounded , convex and non empty. Then $\overline{V} \setminus U$ is compact and homeomorphic to the closure $\overline{U} \subseteq V$. In pasrticular $\overline{V} \setminus U$ is contractible and contains ∞.*

Proof. The compactnes dolloes from U being open and $\overline{V}$ being comapct. After a translation we may assume that U is open , bounded convex neighbourhood of the origin and hence we can deffine the Minkowsky gauge with resoect to U and after fixing a norm, we may use all properties from Lemma 4.3.1 (and denote them with just (a)-(f)). Now we define

$$\iota : \overline{V} \setminus U \to \overline{U}\,, \quad \iota(v) := \frac{v}{\mu(v)^2}\,(v \neq \infty), \quad \iota(\infty) := 0.$$

The well-definition as a map follows from $\mu(v) \geq 1$ if $v \in V \setminus U$ and thereby $\mu(\iota(v)) = \mu(v)\mu(v)^2 = \mu(v)^{-1} \leq 1$ which concludes the well definition by (f). The map $w \mapsto w\mu(w)^{-2}$ $(0 \mapsto \infty)$is a two sided inverse of ι which can be seen by computation and realizing that ι maps ∞ to zero lem:orient-MinkowskyGauge (c). Finally, continuity of ι in $V \setminus U$ follows from μ beeing continuous and in ∞ the continuity follows from (c):

$$\|\iota(v)\| = \frac{\|v\|}{\mu(v)^2} \xrightarrow[\|v\|\to\infty]{} 0\,.$$

The same argument gives us continuouty of our inverse function. Alternitively we can argue that ι is a continouos bijection from a compact space to a Hausdorr space and thereby a homeomorphism. Finally, $\overline{U}$ is convex and compact and thereby contractible. □

Definition 4.3.3 (Admissible Neighbourhoods)**.** Let V be a real finite dimensional vector space. For $x \in V$ let

$$\mathcal{U}(x) := \{U \subset V \,|\, U \text{ open, bounded, convex neighbourhood of } x\},$$

and for $U \in \mathcal{U}(x)$ put $L(U) := \overline{V} \setminus U$. By Lemma 4.3.2 $(\overline{V}, L(U))$ is a compact pair with $L(U)$ contractible. Furthermore, $(\overline{V}, L(U))$ is an NDR pair. To see this, note that in therms of the gauge μ of U centered at x, the space subspace $L(U)$ consists of the points of "radius" greater than or equal to one, together with ∞. A radial push towards $L(U)$, damped across a clolor so as to match the identity on $\{v \in V \,|\, \mu(v) \leq 0.5\}$ and on $\{\infty\}$, gives the required homotopy. The function u from the definition of an NDR pair is adapted to said damping. Notice, how $\mathcal{U}$ is a directed system with respeect to inclusion.

def:orient-localKtheoreticOrientation

Definition 4.3.4 (Local K-theoretic orientation)**.** For $U \in \mathcal{U}(x)$ we call $K^{-n}(\overline{V}, L(U))$ the local group of V at x of size U. A **local K-theoretic orientation of V at** x is a generator of this group.

Next we want to make sure that the naming of said generator is deserved,i.e. the generator is induced and detects orientations, and that the size U is irrelevant.

lem:orient-comparisonOfSizes

Lemma 4.3.5 (Comparison of Sizes)**.** *Let $U \subseteq U'$ in $\mathcal{U}(x)$. Then $\mathrm{id}_{\overline{V}}$ gives a map of pairs $(\overline{V}, L(U')) \to (\overline{V}, L(U))$ and the induced map $\lambda_{U'U} : K^{-n}(\overline{V}, L(U)) \to K^{-n}(\overline{V}, L(U'))$ is an isomorphis. Furthermore let j and j' denote the inclusion of the pair $(\overline{V}, \varnothing)$ into $(\overline{V}, L(U))$ and $(\overline{V}, L(U'))$. Then, the triangle*

$$\begin{array}{ccc} K^{-n}(\overline{V}, L(U)) & \xrightarrow{\quad\lambda_{U'U}\quad} & K^{-n}(\overline{V}, L(U')) \\ & {\scriptstyle j^*}\searrow \quad \swarrow{\scriptstyle j'^*} & \\ & \tilde{K}^{-n}(\overline{V}) & \end{array}$$

commutes, and all maps are isomorphisms and all groups are isomorphic to $\mathbb{Z}$.

Proof. The commutation follows from the topological commutation of the maps that induce the diagram. The maps j and j^* induce isomorphisms by Lemma 3.2.9, as they are induced by collapsing contractible spaces where contractability follows from Lemma 4.3.2. Hence $\lambda_{U'U}$ is also an isomorphism. Finally $K^{-\dim(V)}(\overline{V}) \cong \mathbb{Z}$ gets a generator as a K-theoretic orientation. □

Definition 4.3.6. For $x \in V$ and $U \in \mathcal{U}(x)$ define the isomorphism

$$\Theta_{x,U} : \tilde{K}^{-n}(\overline{V}) \to K^{-n}(\overline{V}, L(U))$$

to be the inverse of the map induced by the pair inclusion $(\overline{V}, \varnothing) \to (\overline{V}, L(U))$.

Proposition 4.3.7 (A K-theoretic Orientation Induces a Local Orientation)**.** Let β be a K-theoretic orientation ov V, i.e. a generator of $\tilde{K}^{-n}(\overline{V})$. Then for every $x \in V$ and every $U \in \mathcal{U}(x)$ the element $\Theta_{x,U}(\beta)$ is a local K-theoretic orientation of V at X and these are compatible: $\lambda_{U'U}\Theta_{x,U}(\beta) = \Theta_{x,U'}(\beta)$ for $U' \subseteq U$.

Proof. All statements follow from Lemme 4.3.5, as $\Theta_{x,U}$ is an isomorphism and thereby maps generators to generators and the compatibility statement follows from the commutatuve triangle. □

Corollary 4.3.8. An orientation of V (i.e. an equivalence class of ordered bases) induce via Corollary 4.2.5 a K-theoretic orientation which in turn induces a local K-Theoretic orientation at every $x \in V$.

prop:orient-localKOrientationIsNatural

Proposition 4.3.9 (Naturality)**.** Let $A : V \to W$ be an isomorphism of n-dimenisonal real vector spaces, $x \in V$ and $U \in \mathcal{U}(x)$. Then $A(U) \in \mathcal{U}(Ax)$, the extension $\overline{A} : \overline{V} \to \overline{W}$ is a homeomorphism with $\overline{A}(LU) = L(AU)$, and the square

$$\begin{array}{ccc} \tilde{K}^{-n}(\overline{W}) & \xrightarrow{\overline{A}^*} & \tilde{K}^{-n}(\overline{V}) \\ \downarrow{\scriptstyle \Theta_{Ax,AU}} & & \downarrow{\scriptstyle \Theta_{x,U}} \\ K^{-n}(\overline{W}, L(AU)) & \xrightarrow{\overline{A}^*} & K^{-n}(\overline{V}, L(U)) \end{array}$$

commutes. The same holds for affine isomorphisms.

Proof. A linear isomorphism carries open bounded convex sets to open bounded convex sets, so $A(U) \in \mathcal{U}(AX)$. A is proper, so $\overline{A}$ is a homeomorphism that maps $L(U)$ onto $L(AU)$. If we replace the maps Θ with the inverses, the corresponding wquare commutes, as its inducing topological square commutes, and thereby our square also commutes. □

Corollary 4.3.10. For $A \in \mathrm{GL}(V)$, $x \in V$ and $U \in \mathcal{U}(x)$ the induced map $\overline{A}^* : K^{-n}(\overline{V}, L(AU)) \to K^{-n}(\overline{V}, L(Q))$ equals $\operatorname{sign}\det(A)$ after the canonical identification $\Theta_{x,U} \circ \Theta^{-1}_{Ax,AU}$.

Proof. This follows directly from Propositions 4.2.4 and 4.3.9. □

4.4 Orientations Calculated by a Jacobean Sign

We now write $B_r := \{v \in \mathbb{R}^n \mid \|v\| < r\}$ and $D_r := \overline{B_r}$ for the euclidean balls around 0, and $L_r := \overline{R^n} \setminus B_r = L(B_r)$.

lem:orient-Linearisation

Lemma 4.4.1 (Linearisation)**.** *Let $\Omega \subseteq \mathbb{R}^n$ be open with $0 \in \Omega$ and let $0 \in \Omega$ and let $g : \Omega \to \mathbb{R}^n$ be C^1 with*

$$g(0) = 0\,, \quad g(w) \neq 0 \text{ for } w \in \omega \setminus \{0\}\,, \quad A := \mathrm{d}g_0 \in \mathrm{GL}_n(\mathbb{R})\,.$$

Choose $\delta > 0$ with $D_\delta \in \Omega$ and let $\rho \in (0, \delta)$. Then there are $c > 0$ and $R > c$ such that both $g|_{D_\delta}$ and $A|_{D_\delta}$ are maps of pairs

$$(D_\delta, D_\delta \setminus B_\rho) \to (D_R, D_r \setminus B_c)$$

are homotopic as such. In particular $(g|_{D_\delta})^* = (A|_{D_\delta})^*$ on $K^{-n}(D_R, D_R \setminus B_c)$. Moreover c may be taken arbitrarily small and R arbitrarily large.

Proof. First we define the map

$$M : D_\delta \to \mathrm{Mat}_n(\mathbb{R})\,, \quad M(w) := \int_0^1 \mathrm{d}g|_{s\cdot w}\,\mathrm{d}s\,.$$

Then M is continuous, $M(0) = A$ and

$$g(w) = g(w) - g(0) = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}s}(g(s \cdot w))\,\mathrm{d}s = M(w)w$$

With this we define the mapping

$$H : D_\delta \times [0,1] \to \mathbb{R}^n\,, \quad H(v,t) := M(tv)v\,,$$

which is continuous, whith $H(\cdot, 1) = g|_{D_\delta}$ and $H(\cdot, 0) = A|_{D_\delta}$. For $v \neq 0$ and $t \in (0,1]$ we can calculate

$$H(v,t) = \frac{M(tv)(tv)}{t} = \frac{g(tv)}{t} \neq 0$$

because g only vanishes at the origin. At $t = 0$ we have $H(v,0) = Av \neq 0$. Hence, the set $\{H(v,t) \,|\, \rho \leq \|v\| \leq \delta,\, t \in [0,1]\}$ is compact and does not contain 0. Thereby the number

$$c_0 := \min\{\|H(v,t)\| \,|\, \rho \leq \|v\| \leq \delta,\, t \in [0,1]\}$$

is greater then zero. Now let $c \in (0, c_0]$. Since $H(D_\delta \times [0,1])$ is bounded we may choose R big enough such that $H(D_\delta \times [0,1]) \subseteq B_R$. With those constants we can conclude that $H(v,t) \in D_R$ for all (v,t) and if $\|v\| \geq \rho$ we have $\|H(v,t)\| \geq c_0 \geq c$ meaning that $H(v,t) \in D_r \setminus B_c$. Thereby we have our homotopy of pairs. □

prop:orient-1ocalIndex

Proposition 4.4.2 (Local Index)**.** Let $n \geq 1$ and let $\Psi : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ be continuous with

1. $\Psi^{-1}(0) = \{0\}$.
2. There is an open $\Omega \subseteq \mathbb{R}^n$ with $0 \in \Omega$, $\Psi(\Omega) \in \mathbb{R}^n$ and $\Psi|_\Omega$ is of class C^1.
3. $A := \mathrm{d}(\Psi|_\omega)_0$ is invertible.

Then

$$\Psi^* = \operatorname{sign}\det(A) \cdot \mathrm{id}\,, \quad \text{on } \tilde{K}^{-n}(\overline{\mathbb{R}^n})\,.$$

Proof. Choose $\delta > 0$ such that $D_\delta \subseteq \Omega$ and fix $\rho \in (0,\delta)$. Choose c, R as in Lemma 4.4.1 applied to $g|_\omega$ and assume c to be small enough such that

$$\Psi(\overline{R^n} \setminus B_\rho) \cap B_c = \varnothing\,, \tag{26}$$

which is possible, as the left side of the intersection doesn't contain the origin and is closed. Hence, there exist a small ball arround the origin that isn't contained. Furthermore, define $\sigma := \min\{\|Au\| \mid \|u\| = 1\} > 0$. Now assume $c \leq \sigma \cdot \rho$. Then we get two maps of pairs

$$\Psi : (\overline{R^n}, L_\rho) \to (\overline{R^n}, L_c)$$
$$\overline{A} : (\overline{R^n}, L_\rho) \to (\overline{R^n}, L_c)$$

For the well definition of the second map assume $\|v\| \geq \rho$. Then:

$$\|Av\| \geq \sigma \|v\| \geq \sigma\rho \geq c\,.$$

Now we apply Lemma 4.3.2 to get the two isomorphisms induced by inclusions:

$$j_\rho^* : K^{-n}(\overline{\mathbb{R}^n}, L_\rho) \to K^{-n}(D_\delta, D_\delta \setminus B_\rho)\,, \quad j_c^* : K^{-n}(\overline{\mathbb{R}^n}, L_c) \to K^{-n}(D_\delta, D_\delta \setminus B_c)\,.$$

Because $\Psi(D_\delta) \subseteq B_R$ and $\overline{A}(D_\delta) \subseteq B_R$ (by choice of R) we get the commuting square

$$\begin{array}{ccc} K^{-n}(\overline{R^n}, L_c) & \xrightarrow{F^*} & K^{-n}(\overline{R^n}, L_\rho) \\ \downarrow{\scriptstyle j_c^*} & & \downarrow{\scriptstyle j_\rho^*} \\ K^{-n}(D_R, D_R \setminus B_c) & \xrightarrow{F|^*_{D_\delta}} & K^{-n}(D_\delta, D_\delta \setminus B_\rho) \end{array}$$

for both $F = \Psi$ and $F = \overline{A}$. (Commutativity follows from the commutativity of the topological square that induces). Now we can apply Lemma 4.4.1 to deduce that both maps in the bottom row agree. As the vertical maps are isomorphisms we can deduce, that also the top two maps agree, namely $\Psi^* = \overline{A}^*$. Finally by lemma 3.3.23 we now that L_ρ and L_c are contractible and contain ∞. Hence the collaps $c_i : \overline{R^n} \to \overline{\mathbb{R}^n}/L_i$ for $i = \rho, c$ induces isomorphism that become the vertical maps in the commutative square:

$$\begin{array}{ccc} K^{-n}(\overline{R^n}, L_c) & \xrightarrow{F^*} & K^{-n}(\overline{R^n}, L_\rho) \\ \downarrow{\scriptstyle c_c^*} & & \downarrow{\scriptstyle c_\rho^*} \\ \tilde{K}^{-n}(\overline{R^n}) & \xrightarrow{F^*} & \tilde{K}^{-n}(\overline{R^n}) \end{array}$$

Again for $F = \Psi$ and $F = \overline{A}$. Hence, again we can conclude that $\Psi^* = \overline{A}^*$ now as maps in $\tilde{K}^{-n}(\overline{\mathbb{R}^n})$, where Proposition 4.2.4 tells us that $\overline{A}^* = \operatorname{sign}\det(A) \cdot \mathrm{id}$. □

conv:orient-orientationOfTheSphere

Remark 4.4.3 (Orientation convention)**.** Let $n \geq 1$. We orient S^n such that the stereographic projection π_S from Definition 3.3.21 is orientation preserving.

lem:orient-rotationsPreserveOrientation

Lemma 4.4.4. *Every $C \in \mathrm{SO}(n+1)$ acts on S^n as an orientation preserving diffeomorphism, and $\pi_S \circ C$ is an orientation preserving chart on $C^{-1}(S^n \setminus \{S\})$.*

Proof. Choose a path $\gamma : [0,1] \to \mathrm{SO}(n+1)$ with $\gamma(0) = \mathrm{id}$ and $\gamma(1) = C$, which exists since $\mathrm{SO}(n+1)$ is path-connected. Fix $x \in S^n$. The map $t \mapsto \operatorname{sign}\det\left(\mathrm{d}\gamma(t)|_x\right)$ takes values in $\{-1, 1\}$ and is continuous, because in charts it is the sign of the determinant of a matrix depending continuously on t, which never vanishes. Hence it is constant, and at $t = 0$ it equals +1. The second claim follows from the chain rule together with Corollary 4.1.20. □

lem:orient-singleRegularPreimage

Lemma 4.4.5 (Single Regular Preimage)**.** *Let $n \geq 1$ and let $\xi : S^n \to S^n$ be continuous. Suppose that there are $y, z \in S^n$ with*

$$\xi^{-1}(z) = \{y\},$$

such that ξ is C^1 near y and $\mathrm{d}\xi|_y$ is invertible. Then

$$\xi^* = \operatorname{sign}\det(d\xi_y) \cdot \mathrm{id}, \qquad \text{on } \tilde{K}^{-n}(S^n).$$

Proof. Let S be the south pole, N be the north pole and

$$\overline{\pi_S} : S^n \to \overline{\mathbb{R}^n}$$

be the compactifications of π_S to the one-point-compactifications, meaning $\overline{\pi_S}(S) = \infty$. Remember that $\pi_S(N) = 0$ is. Now choose two rotations $B, C \in \mathrm{SO}(n+1)$ such that $B(z) = N$ and $C(y) = N$. Then define the two homeomorphisms

$$\overline{\varphi} := \overline{\pi_S} \circ C\,, \quad \overline{\psi} := \overline{\pi_S} \circ B$$

that restrict charts over $S^n \setminus C^{-1}(S)$ respectivly over $S^n \setminus B^{-1}(S)$. Those charts are orientation preserving by Lemma 4.4.4. Now define

$$\Psi : \overline{\psi} \circ \xi \circ \overline{\varphi}^{-1} : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}\,,$$

Then Ψ satisfies the hypotheses of Proposition 4.4.2:

1. $\Psi(w) = 0$ if and only if $\xi \circ \overline{\varphi}^{-1}(w) = z$ if and only if $\overline{\varphi}^{-1}(w) = y$ which is only the case for $w = 0$.

2. Near 0 all maps that compose to Ψ are diffeomorphisms and hence Ψ is a diffeomorphism near 0.

3. By constructiuon $\mathrm{d}\Psi|_0$ is the coordinate representative of $\mathrm{d}\xi|_y$ in the smooth charts φ and ψ. Invertibility of the latter in the hypothesus exactly states the invertibility of the coordinate representation.

Hence, by Proposition 4.4.2 we have that $\psi^* = \operatorname{sign}\det(\mathrm{d}\Psi|_0) \cdot \mathrm{id}$ on $\tilde{K}^{-n}(\overline{\mathbb{R}^n})$. Finally, since φ and ψ are orientation preserving, this sign equials $\operatorname{sign}\det(d\xi_y)$ by Lemma 4.1.18. Finally, since $SO(n+1)$ is path connected there is a path $\gamma : [0,1] \to \mathrm{SO}(n+1)$ with $\gamma(0) = C$ and $\gamma(1) = B$. Then the map

$$S^n \times [0,1] \to \overline{\mathbb{R}^n}\,, \quad (x,t) \mapsto \overline{\pi_S}(\gamma(t)x)$$

is a homotopy from $\overline{\varphi}$ to $\overline{\psi}$ givin us that $\overline{\varphi}^* = \overline{\psi}^*$ and both maps are isomorphisms. Hence:

$$\xi^* = (\overline{\varphi}^*) \circ \Psi^* \circ (\overline{\psi}^*)^{-1} = \operatorname{sign}\det(d\xi_y) \circ \underbrace{(\overline{\varphi}^*) \circ (\overline{\psi}^*)^{-1}}_{=\mathrm{id}}\,.$$

□

5 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse theory, we will always deal with a smooth function $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

banyaga2004lectures

Definition 5.0.1 (Critical Points)**.** A point $p \in M$ at which the smooth map $\mathrm{d}f_p = 0$ is called **critical**. Equivalently, a point is critical, if it is a critical point of the vectorfied $\mathrm{grad}(f)$. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

def:morse-the Hessian

Definition 5.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear form

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangent vectors. Now choose two vector fields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. We then define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = \Xi\left(Z(f)\right)(p)$$

See 2.0.46 for the notation of applying a function to a vector field. In corollary 5.0.5 we will give a local description telling us that this definition is well defined, meaning independent of the extensions Ξ and Z.

Definition 5.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point, the Hessian is a symmetric bilinear and non-degenerate form.

For such a form there exists the following theorem:

thm:morse-sylvestersLawOfInertia

Theorem 5.0.4 (Sylvester's Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that:*

$$T^\top \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^\top \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^\top \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^\top \circ A \circ T') = k' + l' \,. \tag{27}$$

Hence it suffices to show that $k = k'$. Which we will do by proving the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^\top A x > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing "≤": Denote the first k columns of T with $x_1, ..., x_k$. They form a basis of the subspace $\mathbb{R}^k \subseteq \mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda_i x_i$ we have that by bilinearity

$$x^\top Ax = \sum_{i=1}^k \lambda_i x_i^\top Ax = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^\top Ax_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0\,. \tag{28}$$

This concludes the first inequality.

Now let U be any k-dimensional subspace such that for all non zero $x \in U$, $x^\top Ax > 0$. By a calculation analogous to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots, x_n)$ the number $x^\top Ax$ is less than or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U+W) - \dim(W) + \dim(U \cap W) \leq (k + (n-k)) - (n-k) + 0 = k\,. \tag{29}$$

This completes the proof. □

cor:morse-local Hessian

Corollary 5.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corresponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)_{i,j}$$

Proof. This follows by direct calculation of $\operatorname{Hess}_p(f)\,(\partial_i(p), \partial_j(p))$. Denote $x := \varphi(p)$, then:

$$\begin{aligned}
\left(\partial_i(p)\,(\partial_j(f))_p\right) &= \left(\partial_i(p)\left(p \mapsto \left.\frac{\partial f \circ \varphi^{-1}}{\partial x_j}\right|_{\varphi(p)}\right)_p\right) \\
&= \frac{\partial}{\partial x_i}\left(y \mapsto \left.\frac{\partial f \circ \varphi^{-1}}{\partial x_j}\right|_y\right)\Bigg|_x \\
&= \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\,.
\end{aligned}$$

□

Remark 5.0.6. If we have two different charts, then the matrix $M_p(f)$ of the Hessian transforms from one basis to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = (\mathrm{d}\varphi_{12})^\top \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

def:morse-morseIndex

Definition 5.0.7. By Sylvester's Law of inertia, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is an invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each critical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted by λ_p. If omitting the function would cause ambiguity, we write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

thm:morse-morseLemma

Theorem 5.0.8 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse function $f : M \to \mathbb{R}$. There exists a centred chart $\varphi : U \to \mathbb{R}^m$ around p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2 \,.$$

Proof. We refer to Lemma 3.11 in [BH04] for a proof. banyaga2004lectures □

def:morse-gradienFlowFlowLine

Definition 5.0.9. Let $\varphi_t : M \to M$ be the one-parameter group of diffeomorphisms generated by $-\operatorname{grad}(f)$, i.e. if $R_t(1) : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$ (compare definition 2.0.45), we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x)\big(R_t(1)\big) = -\operatorname{grad}(f)_{(\varphi_t(x))} \,, \quad \text{and}$$
$$\varphi_0(x) = x \,.$$

We call that group the **negative gradient flow** of f with respect to the metric g. For compact M we get that the flow φ_t is global. (also we would hit the boundary (of which there is non), or leave every compact set, which is impossible). The integral curve $\gamma_x : (a, b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Proposition 5.0.10. Let (M, g) be a smooth compact Riemannian manifold without boundary, let $f : M \to \mathbb{R}$ be a Morse function and let $\varphi_t : M \to M$ be the flow of $-\operatorname{grad}(f)$. Then the limits

$$\lim_{t \to \pm\infty} \gamma_x(t)$$

exist and are critical points.

Proof. We will show that the flow line has a limit as t tends to $+\infty$ and that said limit is a critical point. (The statement for $-\infty$ is proven analogously.) We define the functions:

$$\begin{aligned}
h &: \mathbb{R} \to \mathbb{R}, & t &\mapsto f \circ \gamma_x(t) \\
\psi &: \mathbb{R} \to \mathbb{R}, & t &\mapsto \left\| \operatorname{grad}_{\gamma_x(t)}(f) \right\|^2 \\
G &: \mathbb{R} \times M \to \mathbb{R}, & (t, x) &\mapsto \left\| \operatorname{grad}_{\varphi_t(x)}(f) \right\|^2
\end{aligned}$$

Then we have that $\lim_{t\to\infty} h(t)$ exists and is finite, due to its monotonicity (compare Proposition 2.0.60) and boundedness in $\mathbb{R}$. Furthermore,

$$-\frac{\mathrm{d}}{\mathrm{d}t}h = \mathrm{d}f_{\gamma_x(t)}(\operatorname{grad}_{\gamma_x(t)}(f)) = g(\operatorname{grad}_{\gamma_x(t)}(f), \operatorname{grad}_{\gamma_x(t)}(f)) = \psi \,,$$

and for any $t, \tau \in \mathbb{R}$:

$$\psi(t + \tau) = G(\tau, \varphi_t(x)) \,.$$

Let $t_n \to \infty$. Then $\gamma_x(t_n)$ has an accumulation point (as M is compact) and after passing to a subsequence we assume that $\gamma_x(t_n) \to y$. Assume that y is not critical and hence there exists an $\varepsilon > 0$ such that $G(0,y) = \left\|\operatorname{grad}_y(f)\right\|^2 \geq \varepsilon$. Now continuity of G in $(0,y)$ gives us a $\delta > 0$ and a neighbourhood $W \subseteq M$ of y such that $G(s,z) \geq \frac{\varepsilon}{2}$ for all $(s,z) \in [0,\delta] \times W$. Now we can calculate for all n big enough, such that $\gamma_x(t_n) \in W$:

$$h(t_n) - h(t_n+\delta) = \int_0^\delta \psi(t_n+\tau)\,\mathrm{d}\tau = \int_0^\delta G\underbrace{(\tau, \gamma_x(t_n))}_{\in[0,\delta]\times W}\,\mathrm{d}\tau \geq \delta\frac{\varepsilon}{2}$$

This contradicts the convergence of h, if we apply the limit $n \to \infty$ to the calculation. By this argument we see, that any accumulation point of $(\gamma_x(t_n))$ is critical.

We now want to show that the limit $\lim_{t\to\infty}\gamma_x(t)$ exists, meaning that all sequences approach the same critical point. For the sake of a contradiction we assume that $\lim_{t\to\infty}\gamma_x(t)$ does not exist. However, by the compactness of M we can assume that there is a sequence $t_n \to \infty$ such that $\gamma_x(t_n) \to q$ and by the argument above, q has to be critical. The Morse lemma gives us a chart (and hence a neighbourhood U) around q, where q is the only critical point.

Since $\gamma_x(t)$ does not converge to q for $t \to \infty$ we may assume that there exists an open neighbourhood V around q with $\overline{V} \subseteq U$ and another sequence $t_n' \to \infty$ such that $\gamma_x(t_n') \notin V$. Assume on the other hand, that $\gamma_x(t_n) \in V$ for all n (again, pass to a subsequence, if this is only possible for large n). Now passing to subsequences we may assume that $t_n < t_n'$ for all n. Then, we define $\tau_n := \inf\{t \geq t_n \mid \gamma_x(t) \notin V\} \leq t_n'$. By continuity of $\gamma_x(t)$ in t we know that $\tau_n > t_n$. This tells us two things: Firstly, $\tau_n \to \infty$ and secondly since $\gamma_x(t) \in V$ for $t \in [t_n, \tau_n)$ and $\gamma_x(\tau_n) \notin V$ we know that $\gamma_x(\tau_n) \in \partial V$. Since ∂V is compact, $\gamma_x(\tau_n)$ has an accumulation point which again is a critical point in $U \setminus V$ (by the same argument as above) giving us a contradiction, as q was the only critical point in U. Thereby, $\gamma_x(t) \to q$ as $t \to \infty$ where q is a critical point.

□

Next we will deepen our **local** understanding of the negative gradient. Notice how the Morse lemma is a statement about smooth manifolds, no metric needed nor mentioned. In fact there are many Morse charts and we can find one where the Riemannian metric in p is diagonal with non-zero entries. To prove such a statement we first need a somewhat obvious lemma:

lem:morse-linearChartChangeAndTangendBasisChange

Lemma 5.0.11. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the tangent bundle in a point p that come from a basis in said point induced from a chart χ by q_χ. Then the diagram*

$$\begin{array}{ccc} T_pM & \xrightarrow{q_\psi|_p} & \mathbb{R}^m \\ & \searrow^{q_{L\circ\psi}|_p} & \downarrow L \\ & & \mathbb{R}^m \end{array}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi\circ L}|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x_0}$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_pM$:

$$\begin{aligned}
\left[(q_{\psi\circ L}|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}|_p(f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}|_{\tilde{x}_0}(f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}|_p(f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[q_\psi|_p^{-1}(\sum_{i,j} v_i \lambda_{ji} e^j)\right](f_p) \\
&= \left[q_\psi|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. □

Now we can prove that a Morse chart exists where the gradient is of a particularly easy form:

thm:morse-simultaneousDiagonalisationOfQuadraticForms

Theorem 5.0.12 (Simultaneous Diagonalisation of Quadratic Forms)**.** *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ around p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a diagonal matrix* $\operatorname{diag}(\mu_1, \dots \mu_m)$ *with $\mu_i > 0$ for all i.*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precise this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_\psi|_p \oplus q_\psi|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalised eigenvalue problem $Hv = \lambda Gv$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and corresponding eigenvectors $v_1, \dots, v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i)\,.$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram–Schmidt to make the bases of the eigenspaces orthogonal and, by rescaling, orthonormal.

Let $L = [v_1 | \dots | v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G (Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^{m} z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H (Lz) = z^T L^T H L z \end{aligned}$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m-l$ positive). This is true by Sylvester's law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\operatorname{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \operatorname{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0 \ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^{l} -y_i^2 + \sum_{i=l+1}^{m} y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p\right)$. By the lemma 5.0.11

$$g(q_{L^{-1}\circ\psi}\big|_p^{-1}, q_{L^{-1}\circ\psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 5.0.13. We call a Morse chart where the matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulas around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

def:morse-jacobeanOfTheGradient

Definition 5.0.14 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\operatorname{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\operatorname{grad}(f))_\psi(x) := J\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial (q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}.$$

lem:morse-jacobianOfTheGradient

Lemma 5.0.15. *Let ψ be a coordinate system around a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of T_xM. Then the Jacobian of the gradient reads*

$$J(\operatorname{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij}, \tag{30}$$

where g^{ij} is defined as in definition 2.0.50.

Proof. Let q_ψ denote the coordinate function $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$. Then the gradient in ψ reads:

$$q_\psi \circ \operatorname{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned}\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\ &= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j\, \partial x_k}\right]\right)_{ij}.\end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j\, \partial x_k}\right]\right)_{ij}.$$

□

Now that we know how the gradient locally looks, it is reasonable to assume that the flow locally looks like its exponential:

lem:morse-differentialOfTheFlowInCriticalPoint

Lemma 5.0.16. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ is a critical point. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \mathrm{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceding Lemma 5.0.15 now finishes the proof.

□

Hence in a refined Morse chart we have:

rem:morse-jacobianOfTheFlowInRefinedMorsechart

Remark 5.0.17. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}\frac{\partial}{\partial x}\psi\circ\varphi_t(\psi^{-1}(x)) &= \exp(-J(\operatorname{grad}(f))(p)\cdot t)\\ &= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f\circ\psi^{-1}}{\partial x_j\partial x_k}}_{=s_k 2\delta_{jk}}\cdot t\right)\\ &= \operatorname{diag}\left(e^{2\mu_1 t},\dots,e^{2\mu_{\lambda_p}t},-e^{2\mu_{\lambda_p+1}t},\dots,-e^{2\mu_m t}\right)\end{aligned}$$

In a refined Morse chart we have that the (un-)stable manifold embedding:

Definition 5.0.18 (Stable and Unstable Manifold)**.** Let $p\in M$ be a critical point of $f: M\to\mathbb{R}$. Then we define:

1. The **stable set** of p as
$$W(\to p) := \{x\in M | \lim_{t\to\infty}\varphi_t(x) = p\}.$$
That is the points that flow towards p.
2. The **unstable set** of p as
$$W(p\to) := \{x\in M | \lim_{t\to-\infty}\varphi_t(x) = p\}.$$
That is the points that came from p.

Those sets will turn out to be manifolds, which explains the names **(un-)stable manifolds**.

thm:morse-stableAndUnstableManifoldTheorem

Theorem 5.0.19 (Stable and unstable manifold theorem)**.** *Let $f: M\to\mathbb{R}$ be a smooth function on a smooth compact Riemannian manifold (M,g). Let p be a critical point of f. Then the tangent space splits as:*

$$T_pM = T_p^s(M)\bigoplus T_p^uM$$

such that the Hessian is positive definite on T_p^sM and negative definite on T_p^uM. Moreover, the stable and unstable manifolds are submanifolds and images of smooth embeddings:

$$\begin{aligned}E_p^s : T_p^sM &\to W(\to p)\subset M,\\ E_p^u : T_p^uM &\to W(p\to)\subset M.\end{aligned}$$

Proof. I will summarise the proof given in Chapter 4.3 [BH04]. banyaga2004lectures We work in the discrete setting meaning that we take the negative gradient flow φ_t for a fixed positive t and the limit-behaviour of the flow is captured by $\lim_{n\to\infty}\varphi_t^n$ instead of the t-limit. First, one shows the splitting of spaces locally in a small neighbourhood U around a critical point p and then extends the splitting globally. Around U we already know by remark 5.0.17 that the tangent space T_pM splits into $T_p^sM \oplus T_p^uM$ such that the differential of the flow is a contraction around T_p^sM and an expansion in T_p^uM. We can now map a small neighbourhood U around p into T^pM using the map $\Psi : U \to T_pM$ given by a refined Morse chart and the identification of $\mathbb{R}^n$ with T_pM via $e_i \mapsto \partial_i(p)$, where the latter uses the same refined Morse chart. Now inspect the image of the stable manifold $W(\to p)$ under said map. This looks to be a small perturbation of the subspace T_p^sM and the local stable manifold theorem supports that intuition:

lem:morse-local UnStableManifoldTheorem

Lemma 5.0.20 (Local (Un-)stable Manifold Theorem)**.** *We define:*

$$T := \mathrm{d}\varphi_t\big|_p, \quad T_s := T\big|_{T_p^sM}, \quad T_u := T\big|_{T_p^uM}.$$

Let $\lambda < 1$ such that $\|T_s\| < \lambda$ and $\left\|T_u^{-1}\right\| < \lambda$, where $\|T\| = \sup\{T(x) | \|x\| \le 1\}$ denotes the operator norm. Now for this setting there exists an ε_λ depending only on λ, and there exists a $\delta > 0$ for every $r > 0$ which satisfy: For all $\theta : T_p^sM \times T_p^uM \to T_pM$ that satisfies $\theta - T$ is Lipschitz with Lipschitz constant

$$\mathrm{Lip}(\theta - T) < \varepsilon, \text{ and } \|\theta(0)\| < \delta.$$

In this setting,

$$W_r^s(\theta) := \{x \in T_pM \mid \forall n \ge 0, \theta^n(X) \text{ is defined and } \|\theta^n(x)\| \le r\}$$

is the graph of a Lipschitz function $g : T_p^sM \to T_p^uM$ with $Lip(g) \le 1$ and:

1. *If θ is C^k (k-times continuously differentiable), then so is g.*
2. *If θ is smoothly differentiable, $\theta(0) = 0$ and $\mathrm{d}\theta\big|_0 = T$, then $g(0) = 0$ and $\mathrm{d}g|_0 = 0$. Hence, $T_pW_r^s(\theta) = T_p^sM$.*

Proof. For a proof we refer to Chapter 4.2 [BH04]. banyaga2004lectures □

We can now define $\theta := \Psi \circ \varphi \circ \Psi^{-1}$. The local (un-)stable manifold theorem now tells us that $W_r^s(\theta)$ is the graph of a smooth function $g : T_p^sM \to T_p^uM$. Furthermore, it is easy to see that in a small neighbourhood of the origin we have the the equiality of $W_r^s(\theta)$ and $Psi(W(\to p) \cap U)$. Hence the projection π of said graph to its domain gives us a chart of the stable manifold around p. A whole atlas of $W(\to p)$ is obtained by $U^n := \varphi^{-n}(U)$ which covers the whole $W(\to p)$ and with this the maps $\chi^n : \pi \circ \Psi \circ \varphi^n$ give us an atlas. The smooth embeddings are obtained as follows: We define $h := \chi \circ \varphi \circ \chi^{-1}$ and $B := W(\to p) \cap U$. Now $\mathrm{d}\left(\varphi_p|_{T_p^sM}\right)$ having an operator norm less than one translates

to $\mathrm{d}h_p$ as they are similar as matrices. Let $0 < \alpha < 1$ and $B_0 \subseteq B$ be a ball centred at p such that $\|\mathrm{d}h_x\| \leq \alpha$ for all $x \in B_0$. Then $h|_{B_0}$ is a contraction, which can be extended to a global contraction $\tilde{h} : T_p^s M \to T_p^s M$. Now we define the embedding:

$$E^s : T_p^s M \to W(\to p)$$
$$x \mapsto \varphi^{-n} \circ \chi^{-1} \circ \tilde{h}^n(x) \quad n\text{such that } \tilde{h}^n(x) \in B_0 \,.$$

All required properties and the well-definedness of E^s are now standard calculations. The stable manifold theorem follows from the unstable one by replacing f with $-f$. □

rem:morse-tangendspaceOfStableManifold

Remark 5.0.21. The proof also gives us the equality $T_p W(\to p) = T_p^s M$: near p, the stable manifold agrees with $\Psi^{-1}(W_r^s(\theta))$, and by part 2 of Lemma 5.0.20 (applicable since $\theta(0) = 0$ and $\mathrm{d}\theta|_0 = T$) the set $W_r^s(\theta)$ is the graph of a function g with $g(0) = 0$ and $\mathrm{d}g|_0 = 0$, hence has tangent space $T_p^s M$ at the origin.

We want to refine the unstable embedding $E_q^u : T_q^u M \to W(q \to)$. For this we will make use of the Hartman–Grobman conjugation theorem:

Theorem 5.0.22 (Hartman-Grobman Theorem)**.** *Let E be an open subset of R^n containing the origin, let $f \in \mathcal{C}$, and let φ_t be a flow given by f. (i.e. $\frac{\partial}{\partial t}(\varphi_t(x)) = f(x)$). Assume that $f(0) = 0$, i.e. that 0 is a fixed point of φ_t and furthermore assume that fixed point to be hyperbolic, i.e. Df has no eigenvalue with zero real part. Then there exists a homeomorphism H of an open set U containing the origin onto an open set V containing the origin such that for each $x_0 \in I$, there is an open interval $I_0 \subseteq \mathbb{R}$ containing zero such that for all $x_0 \in U$ and $t \in I_0$ the diagram*

$$\begin{array}{ccc} U & \xrightarrow{H} & V \\ \downarrow{\scriptstyle \varphi_t} & & \downarrow{\scriptstyle \mathrm{e}^{At}} \\ U & \xrightarrow{H} & V \end{array}$$

commutes, meaning that locally the flow is topologically conjugate to its linearisation.

Proof. A proof is given in chapter 2.8 in [Per01]. Perko2001 □

thm:morse-refinedUnstableEmbedding

Theorem 5.0.23 (Refined Stable Embedding)**.** *In the setting of the stable manifold theorem, we can refine the embedding*

$$E_q^u : T_q^u M \to W(q \to) \,,$$

such that it resembles the flow, meaning that for $x = E^s_p(v)$ the flow is given by

$$\varphi_t(x) = E^s_p(A_t v)\,,$$

where A_t is a linear map defined as: $A_t = B_{\mathfrak{b}} \circ \operatorname{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}\right) \circ B_{\mathfrak{b}}^{-1}$, where $\mathfrak{b}$ is the basis of $T^u_q M$ that comes from the basis of $T_q M$ which is induced by a refined Morse chart. The diagonal matrix is the stable part of the Jacobian of the flow in said refined Morse chart ψ.

Proof. In the proof of the unstable manifold theorem from [BH04] which was summarised banyaga2004lectures above, we started with a local map $\rho' : U \to V$, where $U \subseteq W(q \to)$ is a neighbourhood of q and $V \supseteq T^u_q$. (This was a refined Morse chart together with a projection once realised that the stable part gets mapped to the graph of a smooth function). Now we define

$$\begin{aligned} \theta_t : V &\to V \\ v &\mapsto \rho' \circ \varphi_t \circ \rho'^{-1}(v) \end{aligned}$$

for all t such that the map is well defined. Now with a basis $\mathfrak{b}^s_p$ of $T^s_p M$ we can push these considerations into $\mathbb{R}^n$ and by abuse of notation assume that ρ maps to $\mathbb{R}^n$ and hence θ is a flow around the origin of $\mathbb{R}^n$. Now by the Hartman–Grobman Theorem we find a homeomorphism H around the origin, such that $H \circ \theta_t = e^{At} \circ H$, where $A = \mathrm{d}\theta_t =$ To calculate said differential we look into the map ρ' again. For this we give names to the functions:

- $\psi : U \to V$ is the refined Morse chart around q.
- $g : T^s_p M \to T^u_p M$ is the map that defines the image of the stable part under ψ to be a graph. I.e. $\psi\left(W(\to p)\right) = \Gamma_g$. We know that $g(0) = 0$ and $\mathrm{d}g\big|_0 = 0$.
- $\pi : \Gamma_g \to T^s_p M$ is the projection. Its inverse is obviously $\mathrm{id} \times g$.

We know from the local stable manifold theorem 5.0.20, that $\mathrm{d}g\big|_0 = 0$. With this we can calculate:

$$\begin{aligned} \mathrm{d}\theta_t\big|_0 &= \mathrm{d}\left(\pi \circ \psi \circ \varphi_t \circ \psi^{-1} \circ \pi^{-1}\right)\big|_0 \\ &= \mathrm{d}\pi\big|_{(0,0)} \circ \mathrm{d}\psi \circ \varphi_t \circ \psi^{-1}\big|_{0,0} \circ \mathrm{d}\,\mathrm{id} \times g\big|_0 \\ &= \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right) \circ \operatorname{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}, e^{2\mu_{\lambda_q+1} t}, \dots, e^{2\mu_m t}\right) \circ \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right)^T \\ &= \operatorname{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}\right) \end{aligned}$$

Now define for a small neighbourhood $V \subseteq \mathbb{R}^{m-\lambda_p}$ the map

$$\rho := \rho' \circ H : V \to U\,.$$

U is chosen such that the map is onto. Then we have that $\varphi_t \circ \rho^{-1} = \rho^{-1} \circ e^{At}$, in that small neighbourhood of p. Now finally we can extend this property to be true for the whole stable set: For this we define the embedding

$$E_p^s : T_p^s M \to W(\to p)$$
$$x \mapsto \varphi_{-n} \circ \rho^{-1} \circ A_n(x) \quad \text{where } n \in \mathbb{N} \text{ is big enough such that } A_n(x) \in V\,.$$

This map is in fact well defined since

$$\begin{aligned}\varphi_n \circ \rho^{-1} \circ A_n(x) &= \varphi_{-n} \circ \varphi_{-1} \circ \varphi_1 \circ \rho^{-1} \circ A_n(x)\\ &= \varphi_{-n} \circ \varphi_{-1} \circ \rho^{-1} \circ A_{n+1}(x)\\ &= \varphi_{-(n+1)} \circ \rho^{-1} \circ A_{n+1}(x)\end{aligned}$$

And obviously the flow compatibility that is given locally by ρ is now true globally: Let $x \in W(\to p)$ and assume $x = E_p^s(y)$. Then:

$$\begin{aligned}\varphi_t(x) = \varphi_t(E_p^s(y)) &= \varphi_t \circ \varphi_{-n} \circ \rho^{-1} \circ A_n(y)\\ &= \varphi_{-n} \circ \varphi_t \circ \rho^{-1} \circ A_n(y)\\ &= \varphi_{-n} \circ \circ A_n(A_t(y))\end{aligned}$$

The details of the proof are exactly the same as in the standard (un-)stable manifold theorem. □

Definition 5.0.24. We say that a flow line γ with respect to the negative gradient flow of a Morse function f starts in q, if $\lim_{t\to-\infty}\gamma(t) = q$, and ends in p, if $\lim_{t\to\infty}\gamma(t) = p$. We denote the space of flow lines starting in q by $\mathcal{M}(q \to)$, the ones ending in p by $\mathcal{M}(\to p)$ and the flow lines from q to p $\mathcal{M}(q \to p)$.

5.1 Signed Flow Lines

Definition 5.1.1 (Morse-Smale)**.** A Morse function $f : M \to \mathbb{R}$ such that all stable and unstable manifolds intersect transversally is called a **Morse-Smale** function.

Definition 5.1.2. Let $f : M \to \mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M, g). Furthermore let $\mathfrak{o}$ be a choice of orientations for all unstable tangent spaces ($T_q^u M$ for all $q \in \mathrm{Crit}(f)$). The tuple $\mathfrak{A} := (g, f, \mathfrak{o})$ is a **Morse datum**.

prop:morse-OrientedUnstableSpaces

Proposition 5.1.3 (Oriented Unstable Manifolds)**.** All unstable manifolds of a manifold with Morse function are orientable which can be deduced from the (un-)stable embedding theorem. Hence, the orientations from a Morse datum $\mathfrak{A} := (g, f, \mathfrak{o})$ induce orientations in all unstable manifolds by enforcing Lemma 4.1.10.

lem:morse-tangendBundleSMoothEmbeddingIsSmooth

Lemma 5.1.4. *Let M, N be smooth manifolds and $f : N \to M$ be a smooth embedding. Then the tangent bundle $T(f(N))$ of the smooth submanifold $f(N)$ is a smooth subbundle of $TM|_{f(N)}$.*

Proof. Write $S \coloneqq f(N)$, and let $n = \dim N$ and $m = \dim M$. Since f is an embedding, S satisfies the local slice condition: around every $p \in S$ there is a chart $\varphi : U \to \mathbb{R}^m$ of M with $\varphi(S \cap U) = \varphi(U) \cap (\mathbb{R}^n \times \{0\})$, and the restriction $\varphi|_{S\cap U} : S \cap U \to \mathbb{R}^n \times \{0\} \cong \mathbb{R}^n$ is a chart for S. With respect to these two charts, the inclusion $S \cap U \hookrightarrow U$ has the coordinate representation $v \mapsto (v, 0)$, hence for every $q \in S \cap U$ we have

$$T_qS \;=\; \operatorname{span}\big(\partial_1(q), \dots, \partial_n(q)\big) \;\subset\; T_qM,$$

where $\partial_1, \dots, \partial_m$ denotes the coordinate frame of φ. Hence the trivialisation of $TM|_U$ induced by φ maps $TS|_{S\cap U}$ onto $(S \cap U) \times (\mathbb{R}^n \times \{0\})$, i.e. TS is locally the standard subbundle. Moreover, any transition map between two slice charts sends slices to slices, so its Jacobian along S is block upper triangular; the transition functions of TS are the smooth upper-left $n \times n$ blocks. Thus TS is a smooth subbundle of $TM|_S$. □

prop:morse-orientTheQuotient

Proposition 5.1.5. Let $\mathfrak{A} \coloneqq (g, f, \mathfrak{o})$ be a Morse Datum. Then for all critical points p the quotient bundles $TM|_{W(\to p)} \big/ TW(\to p)$ inherit canonical orientations from .

Proof. By lemma 5.1.4 we know that the bundel $TW(\to p)$ is a smooth subbundle of $TM|_{W(\to p)}$ and by Lemma 2.0.40 we know that the quotient bundle $TM|_{W(\to p)} \big/ TW(\to p)$ is smooth. Finally, we have a natural vectorspace isomorphism

$$\Big(TM|_{W(\to p)} \big/ TW(\to p)\Big)_p = T_pM \big/ T_pW(\to p) \cong T_p^u M\,,$$

beacuse $T_pM \cong T_p^uM \oplus T_p^sM$ and by Remark 5.0.21 $T_p^sM = T_pW(\to p)$. This lets us induce an orientation in the quotient bundle over p and by Lemma 4.1.10 this orientation orients the whole bundle. □

def:morse-moduliSpaceOfFlowLines

Definition 5.1.6. For a Morse datum $(g, f, \mathfrak{o})$ and two critical points p, q we define the spaces

$$W(p \to q) \coloneqq W(p \to) \cap W(\to q) \quad \text{and} \quad \mathcal{M}(p \to q) \coloneqq W(p \to q) \big/ \mathbb{R}\,,$$

where the quotient is defined to be the set of orbits of the $\mathbb{R}$-action $t \mapsto \varphi_t(x)$ for all $x \in W(p \to q)$. Hence, $\mathcal{M}(p \to q)$ is the set of flow lines and is called the **moduli space of flow lines**.

def:morse-signedFlowLines

Definition 5.1.7 (Signed Flow Lines)**.** Let $\mathfrak{A} \coloneqq (g, f, \mathfrak{o})$ be a Morse Datum. Let q, p be two critical points with $\operatorname{ind}(q) - 1 = \lambda(p)$. Then we have for all $\gamma \in \mathcal{M}(q \to p$ the exact sequence of bundles

$$0 \longrightarrow T\gamma \xrightarrow{\;i\;} TW(q \to)|_\gamma \xrightarrow{\;p\;} TM \big/ TW(\to p)\big|_\gamma \longrightarrow 0$$

where the surjectivity of p follows from the smale condition. Let $T\gamma$ be oriented by the section $x \mapsto -\mathrm{grad}_x(f)$, and the quotient be oriented as in Propostion 5.1.5. Then the short exact sequence orients $TW(q\to)$ from the orientation in T^u_pM and on the other hand we can orient $TW(q\to)$ from T^u_qM by Proposition 5.1.3. We define the Mapping

$$\varepsilon : \bigcup_{\substack{p,q\in \mathrm{Crit}(f)\\ \mathrm{ind}(q)-1=\mathrm{ind}(p)}} \mathcal{M}(q\to p) \to \{-1,1\}$$

$$\gamma \mapsto \begin{cases} +1 & \text{If the two induced orientations agree.}\\ -1 & \text{If the two orientations disagree.}\end{cases}$$

Furthermore, we define the counting of signed flow lines:

$$n : \{(q,p)\in \mathrm{Crit}(f)^2 \mid \mathrm{ind}(q)-1=\mathrm{ind}(p)\} \to \mathbb{Z}$$
$$(q,p) \mapsto \sum_{\gamma\in\mathcal{M}(q\to p)} \varepsilon(\gamma)$$

def:Morsehomology

Definition 5.1.8 (Morse Homology)**.** Let $\mathfrak{A} := (g,f,\mathfrak{o})$ be a Morse datum. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ):

$$C_k(M,\mathfrak{A},\Lambda) := \bigoplus_{\mathrm{Crit}(f)} \Lambda$$

The boundary operatpr is given by counting the flow lines:

$$\partial : C_k(M,\mathfrak{A},\Lambda) \to C_{k-1}(M,\mathfrak{A},\Lambda)$$
$$q \mapsto \sum_{p\in\mathrm{Crit}_{k-1}(f)} n(q,p)\cdot p$$

Theorem 5.1.9. *The Morse chain groups form a chain complex, meaning that* $\partial^2 = 0$.

Proof. This will be a corollary of theorem . □

reference zu commuting boundary homomorphisms einfügen

5.2 A Morse-Theoretik Künneth Formula

def:Definitionproduce Morse datum

Definition 5.2.1. Let $(M,\mathfrak{A},\Lambda)$ and $(M',\mathfrak{A}',\Lambda)$ be two Morse data with $\mathfrak{A} = (g,f,\mathfrak{o})$ and $\mathfrak{A}' = (g',f',\mathfrak{o}')$. On $M\times M'$ we define the Morse function $f+f'$ and the metric $g\oplus g'$. Then, $\mathrm{d}_{(x,x')}(f+f') = \mathrm{d}_x(f) + \mathrm{d}'_x(f')$ and hence the critical points of $f+f'$ are pairs (p,p') such that p is critical for f and p' is critical for f'. By corollary 5.0.5 we conclude that the Hessian splits into block diagonal form and hence the property of non degenerayc follows. Since we use the product metric $g\oplus g'$ we also get a splitting of the gradient of $f+f'$:

$$\begin{aligned}\mathrm{d}_{(x,x')}(f+f')(V,V') &= \mathrm{d}_x(f)(V) + \mathrm{d}_{x'}(f')(V')\\ &= g(\mathrm{grad}_x(f),V) + g'(\mathrm{grad}_{x'}(f'),V')\\ &= (g\oplus g')(\underbrace{(\mathrm{grad}_x(f),\mathrm{grad}_{x'}(f'))}_{\mathrm{grad}_{(x,x')}(f+f')},(V,V'))\,.\end{aligned}$$

Hence, the flow also splits in the following sense: Assume $\tilde{\varphi}$ is the flow of $f+f'$ on $M\times M'$, φ is the flow of f on M and φ' is the flow of f' on M'. Then $\varphi_t(x,x') = (\varphi_t(x),\varphi_t(x'))$, as the right sight satisfies the differential equation. Thereby, the (un-)stable manifolds split, letting us orient all unstable manifolds via that splitting which gives us $\mathfrak{o}\oplus\mathfrak{o}'$. Finally, transversal intersection is a property that carries over to products and hence the Morse datum $(M\times M', \mathfrak{A}\times\mathfrak{A}', \Lambda)$ with $\mathfrak{A}\times\mathfrak{A}' := (g\oplus g', f+f', \mathfrak{o}\oplus\mathfrak{o}')$ indeed constitutes a valid Morse datum. The splitting of the flow gives a splitting of the connecting flow lines and hence of the moduli spaces for two critical points (q,q') and (p,p'):

$$\mathcal{M}\left((q,q')\to(p,p')\right) = \mathcal{M}(q\to p)\times\mathcal{M}(q'\to p')$$

If $p\neq q$ and $p'\neq q'$, then for the moduli space to be non-empty we need

$$\operatorname{ind}(q) > \operatorname{ind}(p) \quad \text{and} \quad \operatorname{ind}(q') > \operatorname{ind}(p')\,.$$

This results in the index of (q,q') being at least two bigger than the index of (p,p'). Hence, if we want to count flow lines between two consecutive indices, either q equals p or q' equals p'. In this case we have the moduli space

$$\mathcal{M}\left((q,q')\to(p,p')\right) = \begin{cases} p\times\mathcal{M}(q'\to p') & \text{if } p=q \\ \mathcal{M}(q\to p)\times p' & \text{if } p'=q'\,. \end{cases}$$

Now, we want to analyse how the splitting affects the counting of flow lines. For this we denote the flow lines $\tilde{\gamma}\in\mathcal{M}\left((q,q')\to(p,p')\right)$, $\gamma\in\mathcal{M}q\to p$ and $\gamma'\in\mathcal{M}(q'\to p')$ such that $\tilde{\gamma} = \gamma\times\gamma'$. Then we get a short exact sequence of bundles:

$$0 \longrightarrow T\tilde{\gamma} \longrightarrow TW\left((q,q')\to\right)|_{\tilde{\gamma}} \longrightarrow \left(T(M\times M')\Big/TW(\to(p,p'))\right)\Big|_{\tilde{\gamma}} \longrightarrow 0$$

This sequence, however, is graded in the sense that all spaces (with the exception of the 0) are direct sums and all maps respect them, meaning it is the direct sum of the two short exact sequences:

$$0 \longrightarrow T\gamma \text{ or } 0 \longrightarrow TW\left(q\to\right)|_{\gamma} \longrightarrow \left(TM\Big/TW(\to p)\right)\Big|_{\gamma} \longrightarrow 0 \tag{31}$$

{eq:morse-firstExactSequence}

and

$$0 \longrightarrow T\gamma' \text{ or } 0 \longrightarrow TW\left(q'\to\right)|_{\gamma'} \longrightarrow \left(TM'\Big/TW(\to p')\right)\Big|_{\gamma'} \longrightarrow 0 \tag{32}$$

{eq:morse-secondExactSequence}

Now, we have the two orientations of $TW\left((q,q')\to\right)|_{\tilde{\gamma}}$: The first one comes from the orientation of $T^u_{(q,q')}M\times M'$ and is given as the wedge product of two non-vanisching sections into the determinant bundle from the orientations of T^u_qM and $T^u_{q'}M'$ and thereby represented by $s_q\otimes s_{q'}$(Remember that the determinant of a direct sum is the tensor product of the determinants; this is also true for bundles. Furthermore, we omit

the needed inclusions or lifts to pair them with a $\wedge$.). The other way of orienting comes from the exact sequences. Let $-\mathrm{grad} \wedge s_p$ denote a representative of the orientation induced by the short exact sequence (31) and similarly denote the orientation induced in the middle of the second short exact sequence (32) by $-\mathrm{grad}' \wedge s_{p'}$. By definition, the comparison of s_q with $-\mathrm{grad} \wedge s_p$ gives the sign $\varepsilon(\gamma)$ and similarly the comparison of $s_{q'}$ with $-\mathrm{grad}' \wedge s_{p'}$ gives the sign $\varepsilon(\gamma')$. For ease of notation we assume that the sections live in the right space.

Now we assume that $q' = p'$ resulting in the first element of the short exact sequence (32) being trivial and hence giving us the need to compare those two sections:

$$(-\mathrm{grad}) \wedge (s_p \wedge s_{p'}) .$$

Hence we need to compare the two orientations:

$$s_q \wedge s_{q'} \quad \text{with} \quad (-\mathrm{grad}) \wedge (s_p \wedge s_{q'}) = (-\mathrm{grad} \wedge s_p) \wedge s_{q'} .$$

It is now clear that the sign we get from comparing can be expressed as $\varepsilon(\gamma)$.

If $q = p$, the first short exact sequence (31) has a trivial first bundle and hence we need to compare the two orientations:

$$s_q \wedge s_{q'} \quad \text{with} \quad (-\mathrm{grad}') \wedge (s_q \wedge s_{q'}) = (-1)^{\mathrm{ind}_q(f)} (s_q \wedge -\mathrm{grad} \wedge s_{p'}) ,$$

where the sign comes from moving s_q to the front. Is is now clear that the sign we get from comparing can be expressed as $(-1)^{\mathrm{ind}_q(f)} \varepsilon(\gamma')$.

This results for critical points (q, q') and (p, p') with consecutive indices in the formula:

$$n\big((q,q'),(p,p')\big) = \begin{cases} (-1)^{\mathrm{ind}_q(f)} n(q',p') & \text{if } q = p \\ n(q,p) & \text{if } q' = p' \end{cases}$$

Remark 5.2.2. Let $(M, \mathfrak{A}, \Lambda)$ and $(M', \mathfrak{A}', \Lambda)$ be two Morse data with $\mathfrak{A} = (g, f, \mathfrak{o})$ and $\mathfrak{A}' = (g', f', \mathfrak{o}')$. Then the map (given on the generators of the pure tensors)

$$\Phi : \bigoplus_{i+j=k} C_i(M, \mathfrak{A}, \Lambda) \otimes_\Lambda C_j(M', \mathfrak{A}', \Lambda) \to C_k(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda) , \quad (p \otimes p') \mapsto (p, p')$$

is an isomorphism of groups.Furthermore, if we denote the boundary morphism of the chain complexes corresponding to M, M' and $M \times M'$ by ∂_f, $\partial_{f'}$ and $\partial_{f+f'}$ respectifely, we obtain the following isomorphism of chain complexes induced by Φ:

$$\left(\bigoplus_{i+j=*} C_i(M, \mathfrak{A}, \Lambda) \otimes_\Lambda C_j(M', \mathfrak{A}', \Lambda),\ \partial_f \otimes 1 + (-1)^i 1 \otimes \partial_{f'} \right) \\ \to \left(C_*(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda), \partial_{f+f'} \right)$$

Here, the left side is,by definition, the tensor product of chain complexes. Hence, we get a natural isomorphism of chain complexes:

$$C_*(M, \mathfrak{A}, \Lambda) \otimes_\Lambda C_*(M', \mathfrak{A}', \Lambda) \to C_*(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda) .$$

Proof. The isomorphism of groups is easy to see, since we have a bijection of the generators

$$(p \otimes q) \leftrightarrow (p, q)\,.$$

Since both sides are free Λ-modules we can apply the universal property of free modules, giving us a canonical isomorphism of Λ-modules and hence a group isomorphism. Now we want to show that we get an isomorphism of chain complexes and hence need to check if the boundary map commutes with Φ. Let q be a critical point of index i of f and let q' be a critical point of index j of f'. On the one hand, we can calculate

$$\begin{aligned}
&\Phi\left(\partial_f \otimes 1 + (-1)^i 1 \otimes \partial_{f'}\right)(q \otimes q') \\
&\qquad = \Phi\left(\partial_f(q) \otimes q' + (-1)^i q \otimes \partial_{f'}(q')\right) \\
&\qquad = \Phi\left(\sum_{p \in \mathrm{Crit}_{i-1}(f)} n(q,p)\, p \otimes q' + (-1)^i q \otimes \sum_{p' \in \mathrm{Crit}_{j-1}(f')} n(q',p')\, p'\right) \\
&\qquad = \sum_{p \in \mathrm{Crit}_{i-1}(f)} n(q,p)(p,q') + \sum_{p' \in \mathrm{Crit}_{j-1}(f')} (-1)^i n(q',p')(q,p')\,,
\end{aligned}$$

and on the other hand,

$$\begin{aligned}
\partial_{f+f'}\left(\Phi(q \otimes q')\right) &= \partial_{f+f'}(q,q') \\
&= \sum_{(p,p') \in \mathrm{Crit}_{i+j-1}(f+f')} n\left((q,q'),(p,p')\right)\,(p,p') \\
&= \underbrace{\sum_{p \in \mathrm{Crit}_{i-1}(f)} n(q,p)(p,q')}_{q'=p'} + \underbrace{\sum_{p' \in \mathrm{Crit}_{j-1}(f')} (-1)^i n(q',p')(q,p')}_{q=p}\,.
\end{aligned}$$

□

lem:morse-Alg KnForm

Lemma 5.2.3 (Algebraic Künneth Formula)**.** *If Λ is a principal ideal domain, and C and C' are chain complexes over Λ such that all C_n are free Λ-modules, then for each n there is a natural short exact sequence*

$$0 \longrightarrow \oplus_i \left(H_i(C) \otimes_\Lambda H_{n-i}(C')\right) \longrightarrow H_n(C \otimes_\Lambda C') \longrightarrow \oplus_i \left(\mathrm{Tor}_\Lambda\left(H_i(C), H_{n-i-1}(C')\right)\right) \longrightarrow 0$$

that splits.

Proof. This is a standard result of homological algebra, found in most algebraic topology textbooks. Compare Theorem 3B.5. in [Hat02] for example. hatcher2002algebraic □

Theorem 5.2.4 (Morse-theoretic Künneth Formula)**.** *Let Λ be a principal ideal domain, and let $(M,\mathfrak{A},\Lambda)$ and $(M',\mathfrak{A}',\Lambda)$ be two Morse data. Then for each n there is a splitting short exact sequence*

$$0 \to \bigoplus_i \big(H_i(M,\mathfrak{A},\Lambda) \otimes_\Lambda H_{n-i}(M',\mathfrak{A}',\Lambda)\big) \to H_n(M\times M', \mathfrak{A}\times\mathfrak{A}', \Lambda) \to \bigoplus_i \big(\mathrm{Tor}_\Lambda\big(H_i(M,\mathfrak{A},\Lambda), H_{n-i-1}(M',\mathfrak{A}',\Lambda)\big)\big) \to 0$$

Proof. The theorem follows directly from Lemma 5.2.3 above together with the isomorphism of chain complexes

$$C_*(M,\mathfrak{A},\Lambda) \otimes_\Lambda C_*(M',\mathfrak{A}',\Lambda) \to C_*(M\times M', \mathfrak{A}\times\mathfrak{A}', \Lambda)\,.$$

□

Remark 5.2.5. One might wonder, why there is no *natural* short exact sequence in the Künneth Formula for Morse homology. This is due to the missing functoriality of Morse homology at the moment.

Definition 5.2.6 (Morse Cohomology)**.** Let G be a commutative ring with one. To get the cohomology we have to dualice the chain complex. Let $\mathrm{Crit}_k(f) = \{p_1,\dots p_l\}$ be a basis of the k-th chain group $C_k(M,\mathfrak{A},G)$. Then, the functionals

$$\eta_i : C_k(M,\mathfrak{A},G) \to G; p_j \mapsto \begin{cases} 1 & \text{if j=i} \\ 0 & \text{else.} \end{cases}$$

are a basis of the dual group $C^k(M,\mathfrak{A},G)$. The coboundary map is easy to obtain too: Let q_j for $j = 1,\dots,r$ be the critical points with index $k{+}1$ and let ξ_i be the corresponding forms. Let $n(q_j,p_i)$ be the number of flow lines form q_j to p_i counted with the sign. Then the coboundary map is given as:

$$\partial^k : C^k(M,\mathfrak{A},G) \to C^{k+1}(M,\mathfrak{A},G)$$
$$\eta_i \mapsto \partial^k\eta_i = \eta_i \circ \partial_{k+1} = \sum_j n(q_j,p_i)\xi_j\,.$$

Corollary 5.2.7 (Algebraic Künneth Formula for Cohomology)**.** If Λ is a principal ideal domain, and C and C' are cochain complexes over Λ such that all C^n are free Λ-modules, then for each n there is a natural short exact sequence

$$0 \longrightarrow \oplus_i \big(H^i(C) \otimes_\Lambda H^{n-i}(C')\big) \longrightarrow H^n(C\otimes_\Lambda C') \longrightarrow \oplus_i \big(\mathrm{Tor}_\Lambda\big(H^i(C), H^{n-i+1}(C')\big)\big) \longrightarrow 0$$

that splits.

Proof. This is a corollary of the Algebraic Künneth Formula. For this we define the chain complexes

$$D_{-i} \coloneqq C^i \quad \text{and} \quad D'_{-i} \coloneqq C'^i$$

with the same differentials

$$d_i : D_i \to D_{i-1}\,, c \mapsto d^{-i}(c) \quad \text{and} \quad d'_i : D'_i \to D'_{i-1}\,, c' \mapsto d'^{-i}(c')$$

Then we have the equalities

$$H_i(D) = H^{-i}(C)\,, \quad \text{and} \quad H_i(D') = H^{-i}(C')\,.$$

Furthermore, the redefinition is well behaved with tensor products in the sense that

$$D \otimes_\Lambda D' = \bigoplus_{i+j=m} D_i \otimes D_j = \bigoplus_{i+j=m} C^{-i} \otimes C^{-j} = (C \otimes_\Lambda C')^{-m}$$

and the boundary map of the tensor product chain complex is the same as the coboundary map of the tensor product of the cochain complexes (with the inverted degree). This leads to the equality

$$H_m(D \otimes_\Lambda D') = H^{-m}(C \otimes_\Lambda C')\,.$$

Now we can apply the Algebraic Künneth Formula to D, D' to get the short splitting exact sequence

$$0 \longrightarrow \oplus_i \left(H_i(D) \otimes_\Lambda H_{m-i}(D')\right) \longrightarrow H_m(D \otimes_\Lambda D') \longrightarrow \oplus_i \left(\mathrm{Tor}_\Lambda\left(H_i(D), H_{m-i-1}(D')\right)\right) \longrightarrow 0$$

We can now define $n \coloneqq -m$ and reindex the summations from i to $-i$, giving us the desired sequence. □

Theorem 5.2.8 (Morse-theoretic Künneth Formula for Cohomology)**.** *Let Λ be a principal ideal domain, and let $(M, \mathfrak{A}, \Lambda)$ and $(M', \mathfrak{A}', \Lambda)$ be two Morse data. Then for each n there is a splitting short exact sequence*

$$\begin{aligned} 0 \;\to\; \bigoplus_i \left(H^i(M, \mathfrak{A}, \Lambda) \otimes_\Lambda H^{n-i}(M', \mathfrak{A}', \Lambda)\right) \;\to\; & H^n(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda) \\ \to\; & \bigoplus_i \left(\mathrm{Tor}_\Lambda\left(H^i(M, \mathfrak{A}, \Lambda), H^{n-i+1}(M', \mathfrak{A}', \Lambda)\right)\right) \;\to\; 0 \end{aligned}$$

Proof. We show this by giving an isomorphism of the cochain complexes

$$C^*(M, \mathfrak{A}, \Lambda) \otimes_\Lambda C^*(M', \mathfrak{A}', \Lambda) \to C^*(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda)\,.$$

First we need to give a group isomorphism

$$\Phi : \bigoplus_{i+j=k} C^i(M, \mathfrak{A}, \Lambda) \otimes_\Lambda C^j(M', \mathfrak{A}', \Lambda) \to C^k(M \times M', \mathfrak{A} \times \mathfrak{A}', \Lambda)$$

This isomorphism ,however, is clear and given on the generators: Let p and p' be critical points of index i and j with respect to f and f'. Let η and η' be the corresponding generators of $C^i(M, \mathfrak{A}, \Lambda)$ and $C^j(M', \mathfrak{A}', \Lambda)$. Then we have the bijection

$$\eta \otimes \eta' \leftrightarrow (\eta, \eta') .$$

To get a chain isomorphism we need to check the compatibility with the coboundary maps. For each $q_l \in \mathrm{Crit}_{i+1}(f)$ let ξ_l be the corresponding form and similarly for each ${q'}_k \in \mathrm{Crit}_{j+1}(f')$ let ${\xi'}_k$ be the corresponding form. Then we calculate

$$\begin{aligned}
\Phi\left(\partial^f \otimes 1 + (-1)^i 1 \otimes \partial^{f'}\right)(\eta \otimes \eta') &= \Phi\left(\partial^f(\eta) \otimes \eta' + (-1)^i \eta \otimes \partial^{f'}(\eta')\right) \\
&= \Phi\left(\sum_l n(q_l, p)\xi_l \otimes \eta' + \sum_k (-1)^i n({q'}_k, p')\eta \otimes {\xi'}_k\right) \\
&= \sum_l n(q_l, p)(\xi_l, \eta') + \sum_k (-1)^i n({q'}_k, p')(\eta, {\xi'}_k)
\end{aligned}$$

and the other way around, if we define for (q, q') with $\mathrm{ind}_{f+f'}(q, q') = \mathrm{ind}_{f+f'}(p, p') + 1$ the corresponding form to be ν we have:

$$\begin{aligned}
\partial^{f+f'}\Phi(\eta \otimes \eta') &= \partial^{f+f'}(\eta, \eta') \\
&= \sum_{(q,q') \in \mathrm{Crit}_{i+j+1}(f+f')} n\left((q, q'), (p, p')\right)(\nu) \\
&= \underbrace{\sum_l n(q_l, p)(\xi_l, \eta')}_{q'=p'} + \underbrace{\sum_k (-1)^i n({q'}_k, p')(\eta, {\xi'}_k)}_{q=p}
\end{aligned}$$

Now finally we can apply the Algebraic Künneth Formula for Cohomology and we are done. The latter formular is almost the same as the homological one but with a different index shift in the Tor functor. Again this is standard homological algebra. □

6 A Morse-Theoretic Filtration

6.1 Conley index

def:morseFiltr-compactInvariantIsolatedSubset

Definition 6.1.1 (Compact Invariant Isolated Subset)**.** For a flow $\varphi_t : M \to M$ on a locally compact metric space, we call a subset $S \subseteq M$ an **invariant subset** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t \in \mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated** if a compact neighbourhood N exists such that $I(N) = S$.

def:morseFiltr-indexPairs

Definition 6.1.2 (Index Pairs)**.** Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of** S, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq \operatorname{int}(N \setminus L)$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in** N. Here $\varphi_{[0,t]}(x) := \{\varphi_{\tilde{t}}(x) | \tilde{t} \in [0, t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S. Due to the third property we call L the **exit set**.

def:morseFiltr-regularIndexPairs

Definition 6.1.3 (Regular Index Pairs)**.** We call an index pair (N, L) **regular** if and only if the inclusion $i : L \hookrightarrow N$ is a cofibration.

thm:morseFiltr-homotopyEquivalenceOfIndexPairs

Theorem 6.1.4 (Homotopy Equivalence of Index Pairs)**.** *If S is an isolated compact invariant subset and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S, then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces. This homotopy equivalence is canonical, and transitive up to homotopy.*

Proof. For a proof of the homotopy equivalence we refer to [BH04], Theorem 7.15. The constructed homotopy equivalence is given as follows: Choose $T \geq 0$ such that the following implications hold for $t \geq T$:

$$\begin{aligned} \varphi_{[-t,t]}(x) \subseteq N \setminus L \quad &\Rightarrow \quad x \in \tilde{N} \setminus \tilde{L}\,, \\ \varphi_{[-t,t]}(x) \subseteq \tilde{N} \setminus \tilde{L} \quad &\Rightarrow \quad x \in N \setminus L\,, \end{aligned}$$

where $\varphi_{[a,b]}(x) \coloneqq \{\varphi_t(x) \mid t \in [a,b]\}$. Then the map:

$$h : N/L \times [T,\infty) \to \tilde{N}/\tilde{L}$$
$$([x],t) \mapsto \begin{cases} [\varphi_{3t}(x)] & \text{if } \varphi_{[0,2t]}(x) \subseteq N \setminus L \text{ and } \varphi_{[t,3t]}(x) \subseteq \tilde{N} \setminus \tilde{L} \\ [\tilde{L}] & \text{otherwise,} \end{cases}$$

is continuous and for any fixed t this gives our homotopy equivalence. Obviously, such a map is canonical and transitive up to homotopy. □

def:morseFiltr-connectedSimpleSystem

Definition 6.1.5 (Connected Simple System)**.** A **connected simple system** consists of a set I_0 of pointed spaces along with a set I_m of homotopy classes of maps between these spaces such that

(i) $\hom(X,Y) \coloneqq \{[f] \in [X,Y] \mid [f] \in I_m\}$ consists of exactly one element for each pair of spaces $(X,Y) \in I_0^2$, where $[X,Y]$ denotes the set of homotopy classes of maps between X and Y.

(ii) If $X,Y,Z \in I_0, [f] \in \hom(X,Y)$, and $[g] \in \hom(Y,Z)$, then $[g \circ f] \in \hom(X,Z)$.

(iii) $\hom(X,X) = \{[\mathrm{id}_X]\}$ for all $X \in I_0$.

rem:morseFiltr-connectedSimpleSystemSal

Remark 6.1.6. This definition comes from Salamon in [Sal85]. Salamon1985

rem:morseFiltr-indexPairConnectedSimpleSystem

Remark 6.1.7. Let S be an isolated compact invariant subset of M with respect to the Morse flow φ. Then the collections

$$I_0 \coloneqq \{N/L \mid (N,L) \text{ is an index pair of } S\}$$
$$I_m \coloneqq \{[f] \mid f \text{ is a homotopy equivalence from Theorem 6.1.4}\},$$

make up a connected simple system. We denote this by $\mathcal{IP}_S$. If $S = \mathrm{Crit}_k(f)$ for some $k \in \mathbb{N}$, we denote $\mathcal{IP}_{\mathrm{Crit}_k(f)} = \mathcal{IP}_k$.

def:morseFiltr-morphismsOfConnectedSimpleSystems

Definition 6.1.8 (Morphisms of Connected Simple Systems)**.** A **morphism** $\Phi : I \to J$ between two connected simple systems $I = (I_0, I_m)$ and $J = (J_0, J_m)$ is a set of homotopy classes of maps between spaces in I_0 and J_0 such that

(i) For every $X \in I_0$ and $Y \in J_0$ the set $\Phi(X,Y) \coloneqq \{[\varphi] \in [X,Y] \mid [\varphi] \in \Phi\}$ consists of exactly one element.

(ii) If $X, \bar{X} \in I_0$ and $Y, \bar{Y} \in J_0$ and if $[\varphi] \in \Phi(X,Y)$, $[f] \in \hom(\bar{X},X)$, $[g] \in \hom(Y,\bar{Y})$, then $[g \circ \varphi \circ f] \in \Phi(\bar{X},\bar{Y})$

rem:morseFiltr-morphismInducedBySingleMap

Remark 6.1.9. By the second property of a morphism between connected simple systems I and J, we can induce such a morphism by a single map $\varphi : X \to Y$ where $X \in I_0$ and $Y \in J_0$.

def:morseFiltr-canonicalIsomorphismClass

Definition 6.1.10 (Canonical Isomorphism Class)**.** A **canonical isomorphism class** of Abelian groups (or other algebraic structures) is a set of groups R_0 together with a set of group homomorphisms R_m such that

(i) For $R, L \in R_0$ the set $\hom(R, L) \coloneqq \{f : R \to L : f \in R_m\}$ consists of exactly one element.

(ii) If $R, L, S \in R_0$ and $f \in \hom(R, L)$ and $g \in \hom(L, S)$, then $g \circ f \in \hom(L, S)$.

(iii) $\hom(R, R) = \{\mathrm{id}_R\}$

A **morphism** between two canonical isomorphism classes of Abelian groups $\Psi : (R_0, R_m) \to (S_0, S_m)$ is a collection of homomorphisms such that

(i) For any $R \in R_0$ and $S \in S_0$ the set $\Psi(R, S) \coloneqq \{f : R \to S : f \in \Psi\}$ contains exactly one element.

(ii) If $R, \bar{R} \in R_0$ and $S, \bar{S} \in S_0$, $f \in \hom(\bar{R}, R)$, $g \in \hom(S, \bar{S})$ and $\psi \in \Psi(R, S)$, then $g \circ \psi \circ f \in \Psi(\bar{R}, \bar{S})$.

Remark 6.1.11. Again the second property lets us induce a morphism between two canonical isommorphism classes of Abelian Groups $\Psi : (R_0, R_m) \to (S_0, S_m)$ by a single isomorphism

$$\psi : R \to S$$

for $R \in R_0$ and $S \in S_0$. We denote the induced morphism by $\psi_\# \coloneqq \Psi$.

Example 6.1.12. Any pointed topological space X defines a simple connected system $(\{X\}, [\mathrm{id}_x])$. Similar, any Abelian group G defines a caninical isomorphism class of Abelian groups $(\{G\}, \{\mathrm{id}_R\})$. Let $\varphi : S \to L$ be a second homomorphism that induces a morphism of canonical isomorphism classes. It is then obvious that

$$(\varphi \circ \psi)_\# = \varphi_\# \circ \psi_\# .$$

def:morseFiltr-systemCategories

Definition 6.1.13. We denote the category of connected simple systems with their morphisms by **CSS** and similar we call the category of canonical isomorphism classes of Abelian groups by **CIAB**, respectively, the category for rings by **CIRing**.

Definition 6.1.14. Let $a \in A$ be an element of an abelien group A, which is part of a canonical isomorphism class $A \in A_\#$. Then we a collection of elements

$$a_\# \coloneqq \{f(a) \mid f \in \hom(A, A') \text{ for all } A' \in A_\#$$

an **induced element of** a **in** $A_\#$. Notice that for all $A' \in A_\#$ there is exactly one $a' \in a_\#$ that is contained in A'.

def:morseFiltr-postcomposedFunctor

Definition 6.1.15. Let G be an Abelian group and X be any set. We can **tag** G with X by defining the set $G \times \{X\}$ and endowing it with the algebraic information from G. (for example we define $(g, X) \cdot (g', X) = (g \cdot_G g', X)$). Let $f : G \to L$ be a homomorphism. Then f induces a homomorphism between the tagged $G \times \{X\}$ and $L \times \{Y\}$ by sending $(g, X) \mapsto (f(g), Y)$. Let $F : \mathbf{CGWH}^* \to \mathbf{Ab}$ be a (contravariant) functor from pointed topological spaces to Abelian groups that factors over the homotopy category, i.e. is

homotopy invariant. We first define the tagged functor F^{tag} to map spaces to groups that are tagged with the spaces they come from:

$$\begin{aligned} F^{tag} : \mathbf{CGWH}^* &\to \mathbf{Ab} \\ X &\mapsto F(X) \times \{X\} \\ \big(f : X \to Y\big) &\mapsto \big(F^{tag}(f) : F(Y) \times \{Y\} \to F(X) \times \{X\}\big) \end{aligned}$$

For covariant functors the procedure is analogous. Now we can define the **postcomposed functor** $F_\#$ from the category of connected simple systems to the canonical isomorphism classes as follows:

$$\begin{aligned} F_\# : \mathbf{CSS} &\to \mathbf{CIAB} \\ I_0 &\mapsto \{F^{tag}(X) \mid X \in I_0\} \\ I_m &\mapsto \{F^{tag}(g) \mid [g] \in I_m\} \end{aligned}$$

Let $\Phi : (I_0, I_m) \to (J_0, J_m)$ be a morphism in **CSS**. Then we define

$$F_\#(\Phi) : F_\#(J_0, J_m) \to F_\#(I_0, I_m)\,, \quad F_\#(\Phi) \coloneqq \{F^{tag}(g) \mid [g] \in \Phi\}\,.$$

This obviously becomes a new functor, where the tagging removed uniqueness problems for non-injective (in the literal sense not up to isomorphism) functors F. If $F(X) = F(Y)$ for different $X, Y \in I_0$ with $g : X \to Y \in I_m$, then without tagging there could be two different morphisms $F(X) \to F(X)$: the identity and $F(g)$. Since our functors are almost always injective, we will ignore the tagging from now on.

Remark 6.1.16. Obviously, the procedure of postcomposing a functor works for any homotopy invariant functor from pointed topological spaces into any algebraic category and not just Abelian groups.

Definition 6.1.17 (Direct Sum in the Category of Canonical Isomorphism Classes)**.** Let $A_\#$ and $B_\#$ be canonical isomorphism classes of Abelien groups. Then we define the **direct sum** $A_\# \oplus B_\#$ as follows: The groups are given as all direct sums:

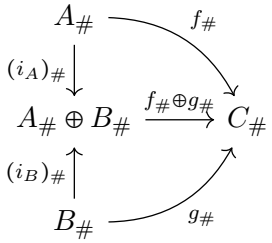
$$\{A \oplus B \mid A \in A_\#\,, B \in B_\#\}$$

and for two groups $(A_1 \oplus B_1)$ and $(A_2 \oplus B_2)$ together with the morphisms $f_{A,A'} \in \hom(A, A')$ and $g_{B,B'} \in \hom(B, B')$ we define the morphism

$$f = f_{A,A'} \oplus f_{B,B'} : (A_1 \oplus B_1) \to (A_2 \oplus B_2)$$

that makes up the homomorphism set $\hom((A_1 \oplus B_1), (A_2 \oplus B_2))$. The direct sum comes together with the inclusions $(i_A)_\#$ and $(i_B)_\#$ that are defined as follows: let $1_B \in B \in B_\#$. Then we define the morphism $(i_A)_\# = (a \mapsto (a, 1_B))_\#$ to be the morphism induced by the natural inclsion of A into $A \oplus B$. This is independent of the choosen $B \in B_\#$.

Let $1_{B'}$ be the neutral element of another $B' \in B_\#$. Then the induced morphisms of canonical isomorphism classes agree, as all individual morphisms agree. This follows from the internal homomorphisms (for example $g_{B,B'}$) mapping $1_B \mapsto 1_{B'}$. This constuction together with the inclusions makes up the coproduct in **CIAB**, as it satisfies the universal property: Let $C_\#$ be another canonical isomorphism class of Abelien groups, together with two morphisms $f_\# : A_\# \to C_\#$ and $g_\# : B_\# \to C_\#$. Then there exists a unique morphism $f_\# \oplus g_\#$ such that the diagram comutes:

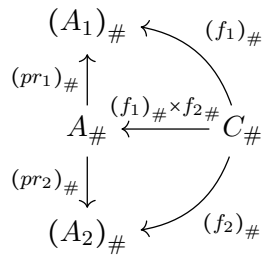


This morphism is induced by the direct sum of morphisms on all elements of the canonical isomorphism classes, meaning that $f_\# \oplus g_\# = (f \oplus g)_\#$.

Remark 6.1.18. Let $A_\# = (A_1)_\# \oplus (A_2)_\#$ be the direct sum of two canonical isomorphism classes. Then $A_\#$ comes with two canonical projection morphisms

$$(pr_i)_\# : A_\# \to (A_i)_\#, \quad (i = 1, 2)$$

that are induced by the projection $A_1 \oplus A_2 \to A_i$ for $i = 1, 2$ for any $A_1 \oplus A_2 \in A_\#$. Then, for any canonical isomorphism class $C_\#$ together with maps $(f_i)_\# : C_\# \to A_{i\#}$ there is a unique morphism $(f_1)_\# \times f_{2\#} : A_\# \to C_\#$ such that the diagram commutes



Again, the map $(f_1)_\# \times f_{2\#}$ is induced by the product of the maps that induce the $(f_i)_\#$ and hence $(f_1)_\# \times f_{2\#} = (f_1 \times f_2)_\#$.

Corollary 6.1.19. Let $A_\# = \bigoplus_{i=1}^n (A_i)_\#$ be the direct sum of finitly many canonical isomorphism classes and let $C_\#$ be another canonical isomorphism class. If we are given a morphism $f_\# : A_\# \to C_\#$, we call the morphisms

$$(f_i)_\# : (A_i)_\# \to C_\#, \quad i = 1, \dots n$$

given by $f_\# \circ (i_i)_\#$ the **summand functions of** f. By the uniqueness of the universal property of the coproduct we know that

$$f_\# = \bigoplus_{i=1}^{n} (f_i)_\# = \Big(\bigoplus_{i=1}^{n} f_i\Big)_\# .$$

Similar, if we are given a function $g_\# : C_\# \to A_\#$ we call the functions

$$(g_i)_\# := (pr_i)_\# \circ g_\# : C_\# \to (A_i)_\#$$

the **factor functions of** g. Again by uniqueness we have

$$g_\# = \bigtimes_{i=1}^{n} (g^i)_\# = \Big(\bigtimes_{i=1}^{n} g^i\Big)_\# .$$

Now assume that we are given a function $f : \bigoplus_{i=1}^{n} (A_i)_\# \to \bigoplus_{j=1}^{m} (B_j)_\#$. Then we can look at all the factor functions of all summand functions to get

$$((f_i)_\#)^j{}_\# = ((f_i)^j)_\# : (A_i)_\# \to (B_{j\#}) .$$

Similary, we could have looked at all the summands of all factor functions to get $((f^j)_i)_\#$. By definition we then have that

$$f = \bigoplus_{i=1}^{n} \Big(\bigtimes_{j=1}^{m} (f_i)^j\Big) = \bigtimes_{j=1}^{m} \Big(\bigoplus_{i=1}^{n} (f^j)_i\Big)$$

rem:morseFiltr-criticalPointsAsIsolatedCompactInvariantSubsets

Remark 6.1.20 (Critical points as isolated compact invariant subsets)**.** Let $f : M \to \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$. Around q we have refined Morse charts $\varphi : U \to \mathbb{R}^m \cong \mathbb{R}^{m-\lambda_q} \oplus \mathbb{R}^{\lambda_p}$ where the splitting of $\mathbb{R}^m$ is taken such that $\mathrm{d}\varphi_q$ is expanding in $\mathbb{R}^{m-\lambda_q}$ and contracting in $\mathbb{R}^{\lambda_p}$. With this we can define the balls:

$$\begin{aligned} D^s_\varepsilon &:= \{v \in \mathbb{R}^{m-\lambda_q} \mid \|v\| \le \varepsilon\}, \\ D^u_\varepsilon &:= \{v \in \mathbb{R}^{\lambda_q} \quad\ \mid \|v\| \le \varepsilon\}. \end{aligned}$$

For epsilon small enough, they give rise to the index pair $N_q := \varphi^{-1}(D^s_\varepsilon \times D^u_\varepsilon)$ and $L_q := \varphi^{-1}(D^s_\varepsilon \times \partial D^u_\varepsilon)$. To see this, notice that in such a refined Morse chart φ, the differential of the flow in the critical point splits the tangent space into $T^u_q M \times T^s_q M$. Finally, since the property of $\mathrm{d}\varphi_x$ being a contraction on $T^s_q M$ is an open condition and thereby there is an $\varepsilon > 0$ such that $\mathrm{d}\varphi_x$ is a contraction on $T^s_q M$ for all x with $\left\|\varphi^{-1}(x)\right\| < \varepsilon$. Thereby, the norm of $\varphi_t(x)$ projected to $T^s_q M$ is strictly decreasing in t. This proves that L_q is an exit set of N_q. Similarly, one sees that L_q is positively invariant in N_q.

This index pair is in fact regular, since $(N_q, L_q) \simeq (D^s_\varepsilon \times D^u_\varepsilon, D^s_\varepsilon \times \partial D^u_\varepsilon) \simeq (D^u_\varepsilon, \partial D^u_\varepsilon)$ and the latters regularity is easy to see by showing that it is an NDR pair. We will cal such an index pair a **standart index pair of** q.

Furthermore, the index pair (N_q, L_q) is homotopically equivalent to $\overline{T_q^u M}$, which is clear, as $(D_\varepsilon^u, \partial D_\varepsilon^u) \simeq \overline{T_q^u M}$ by mapping $\mathbb{R}^{\lambda_q} \to T_q^u M$ via $\mathrm{d}\varphi^{-1}\big|_0 \circ R_0$, where $R_0 : \mathbb{R}^m \to T_0\mathbb{R}^m$ is the linear identification from Derfinition 2.0.45. Finally, we just need to strech the ball to infinity.

We know the K-group of such a pair for any $l \in \mathbb{N}$ to be:

$$K^{-i}(N_q, L_q) \cong \tilde{K}^{-i}(\overline{T_q^u M}) \cong \begin{cases} \mathbb{Z} & \text{if } i-k \in 2\mathbb{Z}, \\ 0 & \text{else.} \end{cases} \tag{33}$$

{eq:homologyofindexpairofcriticalpoint}

Notice that for $k = 0$ we need to consider $\partial D^k = \varnothing$ to get an index pair. That's why we let ∂ denote the manifold boundary instead of the topological boundary. However, now we can identify for all k:

$$C^k(M, \mathfrak{A}, \tilde{K}^{-l}(pt)) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^{-k}(N_q, L_q). \tag{34}$$

{kriticalpointsasdirectsumofhomologygroups}

Or using the language of connected simple systems, we have the identification

$$C^k(M, \mathfrak{A}, \tilde{K}^{-l}(pt)) \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K_\#^{-k}(\mathcal{IP}_q)\,, \tag{35}$$

where the isomorphism can be read as an isomorphism of canonical isomorphism classes with the left-hand side viewed as an object of **CIAB** consists of only one group.

rem:morseFiltr-indexPairForDifferentChart

Remark 6.1.21. For any chart χ that is centered at a critical point q and $\varepsilon > 0$ we define the topological pair

$$(N_\chi, L_\chi) := (\chi^{-1}(D_\varepsilon^s \times D_\varepsilon^u), \chi^{-1}(D_\varepsilon^s \times \partial D_\varepsilon^u))\,.$$

The argument that shows that (N_ψ, L_ψ) is a reglar index pair of q only needed that $\mathrm{d}\psi|_q$ splitts. Hence if $\mathrm{d}\chi|_q = \mathrm{d}\psi|_q$, we can conclude, that there is an $\varepsilon > 0$ such that (N_χ, L_χ) is also a regular index pair.

Definition 6.1.22. Let $a_\#$ be a canonical isomorphism class if Abelien groups. Let $a \in a_\#$, and let $S \subseteq a$ be a generating set of a. Then we call the set

$$S_\# := \{G \mid G = f(S)\,, \text{where } f \in \hom(a, a') \text{ for all } a' \in a_\#\}$$

a generating set of $a_\#$. Notice, that for all $a' \in a_\#$ there is a $G \in S_\#$ such that G is a generating set of a'.

rem:MorseFiltr-ProductStructureOfKIPk

Remark 6.1.23. In Remark 6.1.20 we established an isomorphism for any $l \in \mathbb{N}$

$$C^k(M, \mathfrak{A}, \tilde{K}^{-l}(pt)) \to \bigoplus_{q \in \mathrm{Crit}_k(f)} K_\#^{-k}(\mathcal{IP}_q)\,.$$

Now notice, that the disjoint union of all index pairs $\sqcup_{q \in \mathrm{Crit}_k(f)}(N_q, L_q)$ is an index pair for $\mathrm{Crit}_k(f)$. Hence the group isomorphism given by the additivity of K-theory by Theorem 3.5.3 gives us a natural isomorphism

$$\bigoplus_{q \in \mathrm{Crit}_k(f)} \tilde{K}^{-k}(N_q/L_q) \to \tilde{K}^{-k}\Big(\bigvee_{q \in \mathrm{Crit}_k(f)} (N_q/L_q)\Big)$$

which in turn defines a natural isomorphism of isomorphism classes of groups

$$\bigoplus_{q\in \mathrm{Crit}_k(f)} K_{\#}^{-k}(\mathcal{IP}_q) \to K_{\#}^{-k}(\mathcal{IP}_k)$$

Hence, the class $K_{\#}^{-k}(\mathcal{IP}_k)$ satisfies the propertie of a coproduct with the maps

$$\begin{aligned} i_{q\#} &:= (c_q^*)_{\#} : \tilde{K}_{\#}^{-k}(\mathcal{IP}_q) \to K_{\#}^{-k}(\mathcal{IP}_k) \quad \text{where} \\ c &: \bigvee_{r\in \mathrm{Crit}(k)} (N_r/L_r) \to N_q/L_q \quad \text{is the collapse,} \end{aligned}$$

and the properties of the product with the mapsto

$$\begin{aligned} pr_{q\#} &:= (i_q)^*{}_{\#} : K_{\#}^{-k}(\mathcal{IP}_k) \to \tilde{K}_{\#}^{-k}(\mathcal{IP}_q) \quad \text{where} \\ & i_q : N_q/L_q \to \bigvee_{r\in \mathrm{Crit}(k)} (N_r/L_r) \text{ is the inclusion.} \end{aligned}$$

rem:MorseFiltr-canonicalIdentificationOfMorseChainGrou

Remark 6.1.24 (Generators for Canonical Isomorphism Classes)**.** Hence, there is an isomorphism

$$C^k(M, \mathfrak{A}, \tilde{K}^{-l}(pt)) \to K_{\#}^{-k}(\mathcal{IP}_k)\,.$$

This isomorphism can be made natural. For this we need to establish a natural generating set of $K_{\#}^{-k}(\mathcal{IP}_k)$, which is done by defining a generating set for any (and thereby for all) groups in $K_{\#}^{-k}(\mathcal{IP}_k)$. We pick the group $\tilde{K}^{-k}(\bigvee_{q\in \mathrm{Crit}_k(f)}(N_q/L_q))$. Additivity tells us that a generating set is given by including generators of the summands $\tilde{K}^{-k}(N_q/L_q)$. Now for each summand we have the isomorphism $\tilde{K}^{-\lambda_q}(N_q/L_q) \cong \tilde{K}^{-\lambda_q}(\overline{T_q^u M})$, which is natural and described in Remark 6.1.20. Hence, if $T_q^u M$ is oriented, Corollary 4.2.5 gives us just what we need, making the map canonical. The generator is then given by pulling back the generator of $K^{-\mathrm{ind}_p}(\overline{T_q^u M})$ along the map of pairs:

$$\kappa_q := \mathrm{d}\varphi^{-1}\big|_0 \circ R_0 \circ s \circ \mathrm{pr}_u \circ \varphi : (\varphi^{-1}(D_\varepsilon^s \times D_\varepsilon^u), \varphi^{-1}(D_\varepsilon^s \times \partial D_\varepsilon^u)) \to (\overline{T_q^u M}, \{\infty\})$$

where pr_u is the projection onto the unstable coordinates and s is the stretching $x \mapsto (\frac{x}{\varepsilon - \|x\|})$ with $\mathrm{d}s|_0 = \frac{1}{\varepsilon}\,\mathbb{1}$. So for every critical point $q \in \mathrm{Crit}_k(f)$ we get a generator of $\alpha_q \in \tilde{K}^{-\lambda_q}(N_q/L_q)$ which pulls back along the collapsoiing map $c_q^*(\alpha_q) \in \tilde{K}^{-k}(\bigvee_{r\in \mathrm{Crit}(k)}(N_q/L_r))$and then induces an element in $K_{\#}^{-k}(\mathcal{IP}_k)$. We denote said element as $(\alpha_q^{(N_q,L_q)})_{\#}$ and call it the **canonical generator corresponding** to q.

Notice the abuse of notation here, where $(\alpha_q^{(N_q,L_q)})_{\#}$ can either be an element of $K_{\#}^{-k}(\mathcal{IP}_k)$, or a generator of $K_{\#}^{-k}(\mathcal{IP}_{\{q\}})$. Those two meanings are related in the following sense:

$$\begin{aligned} (c_q^*)_{\#}((\alpha_q^{(N_q,L_q)})_{\#}) &= (\alpha_q^{(N_q,L_q)})_{\#} \in K_{\#}^{-k}(\mathcal{IP}_k) \\ (i_q^*)_{\#}((\alpha_q^{(N_q,L_q)})_{\#}) &= (\alpha_q^{(N_q,L_q)})_{\#} \in K_{\#}^{-k}(\mathcal{IP}_{\{q\}}) \end{aligned}$$

We could have picked another refined Morse chart, or another ε and hence another index pair (N'_q, L'_q) of q. This would give another generator $\alpha_q'^{(N'_q,L'_q)} \in \tilde{K}^{-\lambda_q}(N'_q/L'_q)$. However, as canonical generators, they would agree: For this we need to check the following: let $h : N_q/L_q \to N'_q, L'_q$ be the homotopy equivalence from Theorem 6.1.4. In their summands, $\alpha_q'^{(N'_q,L'_q)}$ and $\alpha_q^{(N_q,L_q)}$ are generators, and as h^* induces an isomorphism we have $h^*(\alpha'_q(N'_q, L'_q)) = \pm\alpha_q^{(N_q,L_q)}$. So we need to show, that the sign here is +. Whilst we could brute force a calculation here using the single preimage Lemma 4.4.5 we will later in Remark 6.2.6 proof said statement a bit more elegant.

Remark 6.1.25. The independence of the chosen Morse chart was the main motivation for introducing the language of canonical isomorphism classes. The book [BH04] proofes the homotopy equivalence of all index pairs and proceeds much as we do to show that singular homology and Morse homology are isomorphic, albeit in singular homology rather than complex K-theory. This is a written out version of Salamons proof of said statement in [Sal90]. Similar to us, they also introduce a generator for an index pair by orientation and a Morse chart, but do not show the invariance of said chart (compare [BH04, Example 7.17]). Whilst this is no gap in the theory (the invariance is to be expected and the authors presumably considered it evident), without the language of canonical isomorphism classes, such an invariance is cumbersome even to state precisely.

banyaga2004lectures
Salamon1990
banyaga2004lectures

Definition 6.1.26 (From Morse to K-Theory)**.** Let M be a smooth manifold wit the Morse datum $\mathfrak{A}$ and denote the one point pointed space by pt. Then for any $l \in \mathbb{N}$, on the generators $\mathrm{Crit}_k(f)$ we define the morphism of canonical Isomorphism calses, where the left side consists of only one group:

$$\mathcal{X} : C^k(M, \mathfrak{A}, \tilde{K}^{-l}(pt)) \to \tilde{K}_{\#}^{-(k+l)}(\mathcal{IP}_k), \quad q \mapsto (\alpha_q^{(N_q,L_q)})_{\#}.$$

Remark 6.1.24 ensures invaraiance of the choosen ε and refined Morse chart, that made up $(\alpha_q^{(N_q,L_q)})_{\#}$.

def:morseFilt-adaptedLyapunov

Definition 6.1.27 (Adapted Lyapunov Function)**.** An **adapted Lyapunov function** for the Morse-Smale pair (f, g) (where g denotes the metric and f the Morse function) is a smooth function $\ell : M \to \mathbb{R}$ such that

(i) $\mathrm{d}\ell(-\mathrm{grad}(f)) < 0$ on $M \setminus \mathrm{Crit}(f)$,

(ii) $\ell(r) = \mathrm{ind}_r(f)$ for all $r \in \mathrm{Crit}(f)$,

(iii) every $r \in \mathrm{Crit}(f)$ has a neighbourhood on wich $\ell - f$ is constant.

prop:morseFiltr-existenceAdaptedLyapunov

Proposition 6.1.28. An adapted Lyapunov function exists.

Proof. We apply [Sma61, Theorem B] to the vector field $X = \mathrm{grad}(f)$, which satisfies the needed conditions: For such a setup, Smales Theorem gives us a function F that satisfies

Smale1961OnGradient

(a) The critical points of F are the critical points of X (and thereby, in our case of f) and F coincides with $f + c$ for $c \in \mathbb{R}$ in small neigbhourhoods of those critical points,

(b) X is transversal to F at levelsets outside of critical points.

(c) $F(r) = \mathrm{ind}_f(r)$ for all $r \in \mathrm{Crit}(F) = \mathrm{Crit}(f)$.

By (a), $\mathrm{d}F(\mathrm{grad}(f)) = \mathrm{d}f(\mathrm{grad}(f)) = \|\mathrm{grad}(f)\|^2$ near critical points. Outside of critical points we have that $\mathrm{d}F(-\mathrm{grad}(f)) \neq 0$ and as for $\dim(M) \geq 2$ this set is connected, we conclude that $\mathrm{d}F(-\mathrm{grad}(f)) > 0$ there also. If $\dim(M) = 1$ we can argue similarly in each connected component that contains a point close to a critical point. Hence, all properties of an adapted Lyapunov function are satisfied by $\ell := F$. □

def:morseFilt-sublevelTriple

Definition 6.1.29 (Sublevel Index Triple)**.** Let $p \in \mathrm{Crit}_{k-1}(f)$, $q \in \mathrm{Crit}_k(f)$ and $S_{pq} := W(q \to p) \cup \{p, q\}$. An **index triple** for (p, q) is a triple $Z \subseteq Y \subseteq X$ of compact sets such that (X, Z) is an index pair of S_{qp}, (X, Y) is an index pair of $\{q\}$ and (Y, Z) is an index pair of $\{p\}$. It is called a **sublevel triple** if moreover

$$Y = Z \cup \left(X \cap \ell^{-1}(-\infty, c]\right) \qquad \text{for some } c \in (k-1, k)\,.$$

morseFiltr-sublevelTripoleGivePairs

Lemma 6.1.30. *Let (X, Z) be an index pair of S_{qp}, $c \in (k-1, k)$ and$Y = Z \cup \left(X \cap \ell^{-1}(-\infty, c]\right)$. Then (X, Y, Z) is a sublevel triple.*

Proof. We need to check the three axioms for the two pairs (X, Y) and (Y, Z). We start checking the axioms for (X, Y):

1. Assume that $x \in I(\overline{X \setminus Y})$. Then $x \in I(\overline{X \setminus Z})$ and thereby $x = q$, or $x \in W(q \to p)$. But the latter cannot be true, as for all $x \in W(q \to p)$, there is a $t > 0$ such that $f(\varphi_t(x)) < c$. Hence $x = q$.

2. Assume that Y would not be positively invariant in X. Then there would be an $x \in \ell^{-1}(-\infty, c]$ (since Z is positively invarient), and such that $f(\varphi_t(x)) > c \geq f(x)$ which cannot happen, as f decreases along flow lines.

3. Finally, as Y contains Z, the exit set property is trivially satisfied.

Now for (Y, Z):

1. This is shown by the same argument as for (X, Y).

2. Z is positively invariant in X and $Y \subseteq X$. Hence Z is positively invariant in Y.

3. Z is an exit set of X and thereby also of Y, as the latter is again contained in X.

□

lem:morseFiltr-flowMapOfTriples

Lemma 6.1.31 (Flow Maps and Triples)**.** *Let (X,Y,Z) be an index triple and $(\tilde{X},\tilde{Y},\tilde{Z})$ be a sublevel triple for (p,q), with level $\tilde{c}$. Dentote by h_t, h_t^p, h_t^q the maps of Theorem 6.1.4for the index pairs $(X,Z)\to(\tilde{X},\tilde{Z})$ of S_{qp}, $(Y,Z)\to(\tilde{Y},\tilde{Z})$ of $\{p\}$ and $(X,Y)\to(\tilde{X},\tilde{Y})$ of $\{q\}$. Then there exists a T_0 such that for all $t\geq T_0$:*

(i) $h_t(Y/Z)\subseteq\tilde{Y}/\tilde{Z}$,

(ii) $h_t|_{Y/Z}=h_t^p$,

(iii) the map $X/Y\to\tilde{X}/\tilde{Y}$ iunduced by h_t equals h_t^q.

def:morseFiltr-aMapOfKGroups

Definition 6.1.32 (A Map of K-Groups)**.** Let $f:M\to\mathbb{R}$ be a Morse-Smale function on a smooth compact Riemannian manifold (M,g). Let $q\in\mathrm{Crit}_k(f)$ and $p\in\mathrm{Crit}_{k-1}(f)$. Assume for a moment we already have a regular index pair (N_2,N_0) of the isolated compact subset $S=W(p\to q)\cup\{p,q\}$. For a $c\in(f(p),f(q))$, we set $N_1=N_0\cup(N_2\cap M^c)$, where M^c is the sublevel set. Clearly now (N_2,N_1) is a regular index pair of q and (N_1,N_0) is a regular index pair of p.

Furthermore, we have the triple connecting homomorphism $\delta^{p,q}_{triple}:K^{-\mathrm{ind}_p}(N_1,N_0)\to K^{-\mathrm{ind}_p+1}(N_2,N_1)$, which we can compose with the Bott periodicity map $b:=\beta\boxtimes(-)$. This lets us induce the map:

$$\left(b\circ\delta^{p,q}_{triple}\right)_{\#}:K_{\#}^{-\mathrm{ind}_p}(\mathcal{IP}_{\{p\}})\to K_{\#}^{-\mathrm{ind}_q}(\mathcal{IP}_{\{q\}})$$

Now the disjoint union of the index pairs (N_q,L_q) for all $q\in\mathrm{Crit}_k(f)$ is an index pair of $\mathrm{Crit}_k(f)$ itself. The collapsing maps $c_p:\bigvee_{p\in\mathrm{Crit}_{k-1}}(N_p/L_p)\to(N_p/L_p)$ together with the inclusions $i_q:(N_q/L_q)\to\bigvee_{q\in\mathrm{Crit}_k}(N_q/L_q)$ let us define the map

$$\Delta^{k-1}:=\Big(\sum_{q\in\mathrm{Crit}_k}\sum_{p\in\mathrm{Crit}_{k-1}}c_p^*\circ\left(b\circ\delta^{p,q}_{triple}\circ i_q^*\right)\Big)_{\#}:K_{\#}^{-(k-1)}(\mathcal{IP}_{k-1})\to K_{\#}^{-k}(\mathcal{IP}_k)\,.$$

Due to the naturality of the triple connecting homomorphism, the inclusion and the collapsing, any list of topological triples (X,Y,Z) such that (X,Y) is a regular index pair of $\{q\}$, (Y,Z) of $\{p\}$ and (X,Z) of $W(q\to p)\cup\{q\}\cup\{p\}$ for all $q\in\mathrm{Crit}_k(f),p\in\mathrm{Crit}_{k-1}(f)$ gives rise to the same morphism of canonical isomorphism classes.

alternative Definition using the language of sum and products of functions

Remark 6.1.33 (Invariance of the Choosen Triple)**.**

6.2 Commuting Boundary Maps

thm:morseFiltr-commutingMorsecoboundary

Theorem 6.2.1. *Let M be a smooth manifold with the Morse datum $\mathfrak{A}$, denote the one pointed topological space by pt. Then, for any $l\in\mathbb{Z}$ the following diagram*

comutes:

$$\begin{array}{ccc} C^{k-1}(M,\mathfrak{A},\tilde{K}^{-l}(pt)) & \xrightarrow{\partial^{k-1}} & C^{k}(M,\mathfrak{A},\tilde{K}^{-l}(pt)) \\ \Big\downarrow \mathcal{X} & & \Big\downarrow \mathcal{X} \\ K_{\#}^{-(k-1)-l}(\mathcal{IP}_{k-1}) & \xrightarrow{\Delta^{k-1+l}} & K_{\#}^{-(k)-l}(\mathcal{IP}_{k}) \end{array} \tag{36}$$

{eq:morseFilt-commutingDiagramTheorem}

Before we can proof this theorem we need the following Lemma:

def:morseFilt-setupCommutingDiff

Definition 6.2.2 (Definitions and Setup)**.** We work in a compact, Riemanian, smooth manifold (M,g) of dimension m together with a Morse-Smale function

$$f : M \to \mathbb{R} \tag{37}$$

and orientations $\mathfrak{o}$ of all unstable tangent spaces. We denote the flow corresponding to the negative gradient by $\varphi_t(x)$. We assume that q is a critical point of index k and p is a critical point of index $k-1$. We define $f(q) = b$ and $f(p) = a$ and assume that there is no critical point in $f^{-1}(a,b) \subseteq M$. We define the sublevle and superlevel sets

$$M^t := \{x \in M | f(x) \leq t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \geq t\}\,, \tag{38}$$

and the constants

$$c \in (a,b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big}\,. \tag{39}$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}\,, \tag{40}$$
$$L_q := \{x \in N_q | f(x) = c\}\,, \tag{41}$$
$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}\,, \tag{42}$$
$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}\,, \tag{43}$$

and finally:

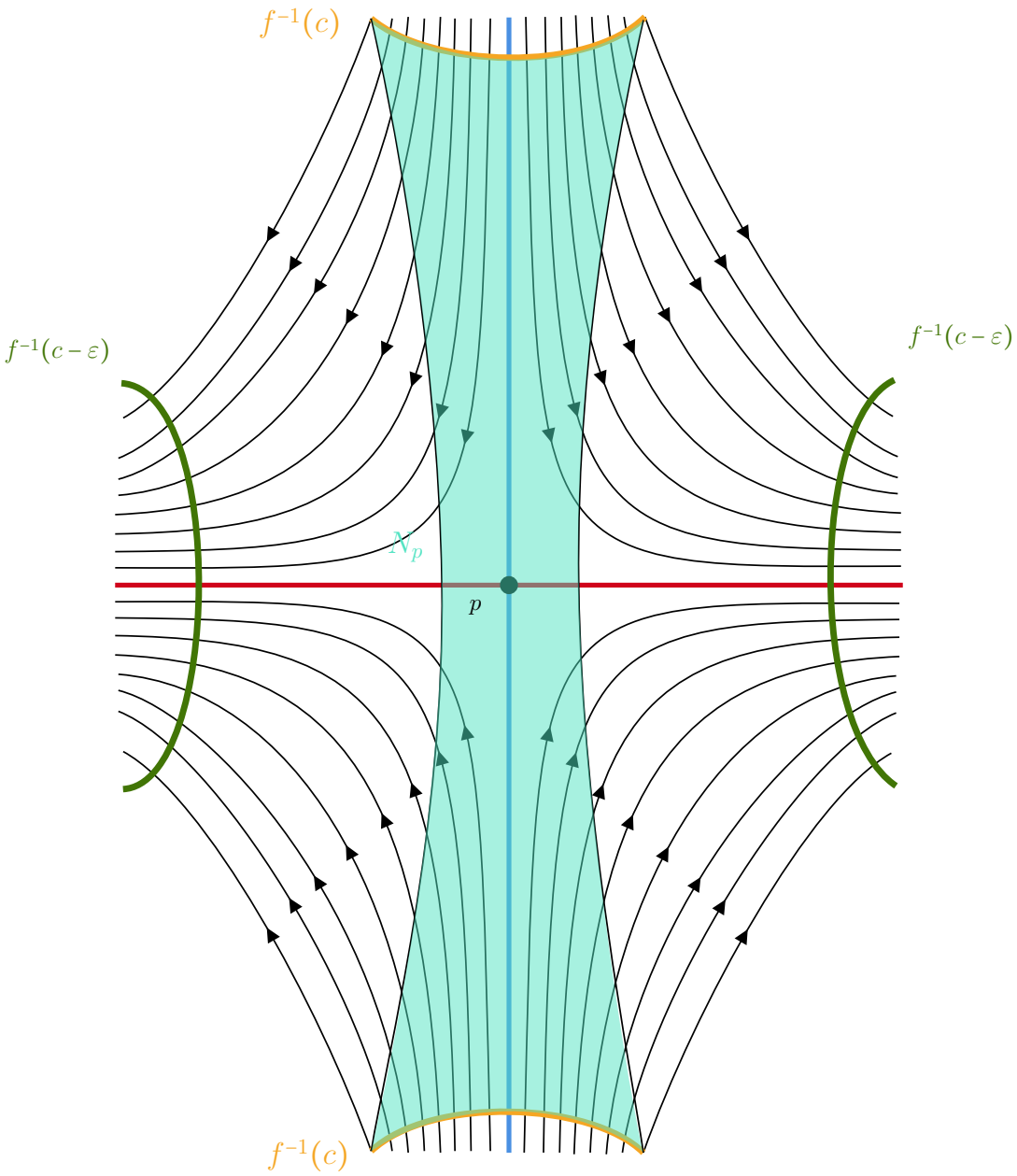
$$C := N_p \cup N_q\,, \tag{44}$$
$$B := N_p \cup L_q\,, \tag{45}$$
$$A := L_p \cup \operatorname{int}(L_q \setminus N_p) \tag{46}$$
$$S(q\to) := W(q\to) \cap f^{-1}(c) \tag{47}$$
$$D_q := W(q\to) \cap M_c\,. \tag{48}$$

Furthermore, we assume $W(q\to)$ to be oriented with respect to the orientation of $T_q^u M$ as in Proposition 5.1.3. Let $Q^p := TM|_{W(\to p)} \Big/ TW(\to p)$ be oriented with respect to $T_p^u M$ as in Proposition 5.1.5. For an oriented real vector space V of dimension n, let $\beta_V \in K^{-n}(\overline{V})$ denote its K-theoretic orientation from Corollary 4.2.5. For an index pair

(N, L) of $\{p\}$ let $\alpha_p^{(N,L)} \in K^{-\mathrm{ind}_p}(N, L)$ denote the canonical generator from Remark **??**, i.e. the pullback of $\kappa_p^*(\beta_{T_p^u M})$ along the homotopy equivalence of Theorem 6.1.4 from (N, L) to the standart index pair and $\kappa_v = \kappa_q := R_0 \circ \mathrm{d}\varphi^{-1}\big|_0 \circ s \circ \mathrm{pr}_u \circ \varphi$.

später

Let $(\alpha_p)_\#$ be a generator of $K_\#^{\lambda_p}(\mathcal{IP}_p)$ meaning that $(\alpha_p)_\#$ is a family of generators for all groups in $K_\#^{\lambda_p}(\mathcal{IP}_p)$ that transform into each other under the canonical group isomorphisms. Notice that $(\alpha_p)_\#$ is induced by any of its elements.



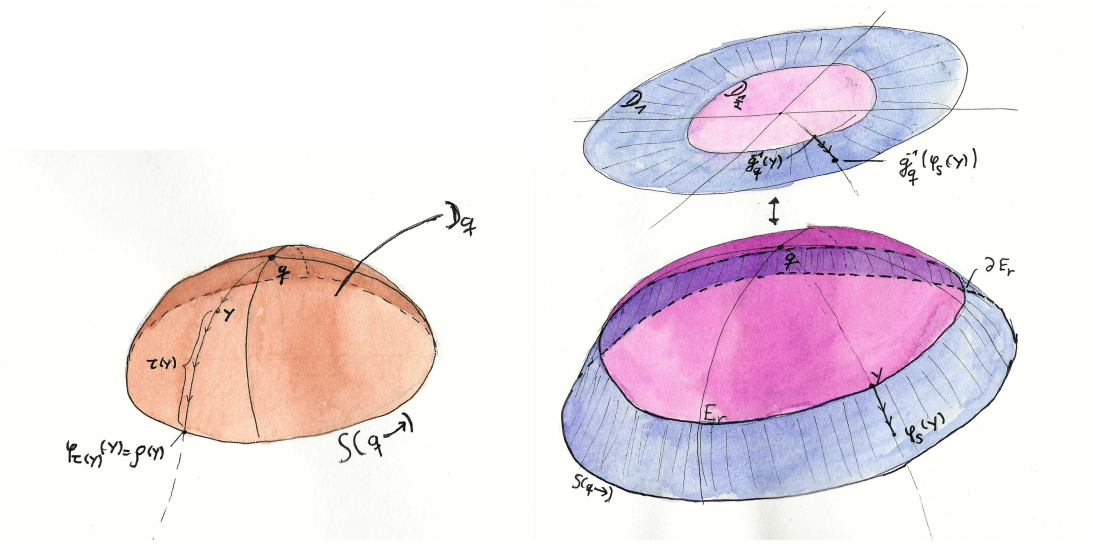


Figure 5: The map τ and ρ from Lemma 6.2.4 (a) is shown on the left and the map G from (c) is shown on the right.

lem:morseFilt-npIsTubular

lem:enum:1

lem:enum:2

lem:enum:3

lem:enum:4

lem:enum:5

Lemma 6.2.3. *In the setting of Definition 6.2.2, we claim that*

1. (N_q, L_q) *is a regular index pair for* q .
2. (C, B) *is an index pair for* q .
3. (N_p, L_p) *is a regular index pair for* p .
4. (B, A) *is an index pair for* p .
5. N_p *is a tubular neighbourhood of* $W(\to p) \cap M^c$.

lem:morseFilt-flowTimeAndProjectionToLevelSets

Lemma 6.2.4. *Let* $\operatorname{ind}_q(f) \geq 1$.

(a) For $x \in D_q \setminus \{q\}$ *there is a unique* $\tau(x) \geq 0$ *with* $f(\varphi_{\tau(x)}(x)) = c$. *The resulting map* $\tau : D_q \setminus \{q\} \to [0, \infty)$ *is smooth,* $\tau^{-1}(0) = S(q \to)$, $\tau(\varphi_s(x)) = \tau(x) - s$, $\mathrm{d}\tau|_x(-\operatorname{grad}_x(f)) = -1$ *and* $\tau(x) \to \infty$ *for* $x \to q$. *We set* $\tau(q) = \infty$.

(b) The map $\rho : D_q \setminus \{q\} \to S(q \to)$, $\rho(x) := \varphi_{\tau(x)}(x)$ *is a smooth retraction with* $\rho \circ \varphi_s = \rho$. *In particular,* $\mathrm{d}\rho|_x(\operatorname{grad}_x(f)) = 0$ *and for* $y \in S(q \to)$, $w \in T_y(S(q \to))$ *and* $s \geq 0$:

$$\mathrm{d}\rho\big(\mathrm{d}\varphi_{-s}|_y\, w\big) = w, \quad \mathrm{d}\tau\big(\mathrm{d}\varphi_{-s}|_y\, w\big) = 0\,.$$

(c) *$S(q\to)$ is a closed submanifold of $W(q\to)$ of dimension $\operatorname{ind}_q - 1$. There are homeomorphisms $g_q : S^{\operatorname{ind}_q} \to D_q/S(q\to)$ with $g_q(S) = [S(q\to)]$ and a constant $T_q \geq 0$ such that, with the open neighbourhood*

$$W_q := \{x \in D_q \mid \tau(x) > T_q\} \ni q$$

of q, the map g_q restricts to an orientation preserving diffeomorphism from $U_q := g_q^{-1}(W_q)$ onto W_q.

Bild

Proof. (a) The function f strictly decreases along flow lines, which approach critical points. As there is no critical point in $f^{-1}[c,b)$ our flow line needs to pass through $f^{-1}(c)$ giving us a $\tau(x)$, where uniqueness follows again from strict monotonicity. This uniqueness lets us deduce that $\tau(\varphi_s(x)) = \tau(x) - s$. The smoothness follows from the implicit function theorem applied to $G(x,t) := f(\varphi_t(x)) - c$, since $\partial_t G(x,t) = -\left\|\operatorname{grad}_{\varphi_t(x)}(f)\right\|^2 \neq 0$. (compare Proposition 2.0.60 for the calculation) Differentiating $\tau(\varphi_s(x)) = \tau(x) - s$ in $s = 0$ gives $\mathrm{d}\tau(-\operatorname{grad}(f)) = -1$. Finally assume that $x_n \to q$ whilst $\tau(x_n) \leq T$ for some $T > 0$. Then we can pass to a subsequence such that $\tau(x_n) \to t$ for $t \leq T$. In this subsequence we then have that $\lim_{n\to\infty} f(\varphi_{\tau(x_n)}(x_n)) = f(\varphi_t(q)) = b$ contradicting $f\big(\varphi_{\tau(x_n)}(x_n)\big) = c < b$ for all $n \in \mathbb{N}$.

(b) By (a) we have that $\rho \circ \varphi_s = \rho$, and differentiating it in s gives us $\mathrm{d}\rho(\operatorname{grad}(f)) = 0$. The last two identities are the differentials of $\rho \circ \varphi_{-s} = \mathrm{id}$ and $\tau \circ \varphi_{-s} = \mathrm{const}_s$ on $S(q\to)$.

(c) $\operatorname{grad}(f)$ is tangent to $W(q\to)$ (meaning that it is contained in the tangent bundle of $W(q\to)$ as a subbundle of TM) and non-zero away from q. Hence, c is a regular value of $f|_{W(q\to)}$, ($\mathrm{d}\, f|_{W(q\to)}(\operatorname{grad}(f))$ is nonzero outside of critical points) hence $S(q\to)$ is a submanifold of $W(q\to)$.

Let $\psi = (\psi_s, \psi_u)$ be a refined Morse chart around q such that the unstable coordinates correspond to a positively oriented basis of $T_q^u M = T_q W(q\to)$. Here, ψ_u denotes $\pi_u \circ \psi$ where π_u projects to the unstable coordinates (and π_s is defined analogously). By Lemma 5.0.20 the unstable manifold is, in these coordinates the graph of a smooth function g with $g(0) = 0$, that is Lipschitz with Lipschitz constant less than one. (This g does not denote the Riemannian metric, which doesnt appear by name in this proof.) Hence, ψ_u restricts to a diffeomorphism from a neighbourhood O of q in $W(q\to)$ onto an open ball around 0 and $\|\psi(x)\| \leq \sqrt{2}\,\|\psi_u(x)\|$ on O (The latter is deduced from the lipschitz constant of g being less than or equal to one). Write $u(t) := \psi_u(\varphi_t(x))$ for $x \in O$ (and t small enough such that $\varphi_t(x) \in O$) and denote pullback of the gradient via the chart ψ by $X := -q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}$, where q_ψ denotes the coordinate map on the tangent bundle induced by the frame ∂_i (which comes from the morse chart. Compare Definition 5.0.14). By Lemma 5.0.15 together with Remark 5.0.17 we know that $\mathrm{d}X|_0 = J(\operatorname{grad}(f))_\psi(0)$ is diagonal. Now we define

$$Y : \mathbb{R}^{\operatorname{ind}_q} \supseteq B \to \mathbb{R}^{\operatorname{ind}_q}, \quad v \mapsto Y(v) := \pi_u \circ (X_{g(v),v})$$

and Taylor-expand said function in the origin. By the chain rule and splitting of the tanget space at the origin we get that $Y(v) = \Lambda v + R(v)$ with $\Lambda = \operatorname{diag}(2\mu_1, \ldots, 2\mu_{\operatorname{ind}_q})$

and $\|R(v)\| = o(\|v\|)$. Now

$$\frac{\mathrm{d}}{\mathrm{d}t}(u(t)) = \pi_u \circ q_\psi \circ \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) = \pi_u \circ q_\psi \circ -\mathrm{grad}_{\varphi_t(x)}(f) = Y(u(t))\,.$$

Thus, with Cauchy-Schwarz we can calculate that for v close enough to zero:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|u(t)\|^2 = \big\langle u(t), \Lambda u(t)\big\rangle \underbrace{+\big\langle u(t), R(u(t))\big\rangle}_{\geq -\|u(t)\|\cdot \underbrace{\|R(u(t))\|}_{\leq \mu_{min}\|u(t)\|}} \geq 2\mu_{\min}\|u(t)\|^2 - \mu_{\min}\|u(t)\|^2 > 0\,.$$

Now choose r small enough such that the estimate holds and the ball $B_r(0) \subseteq \psi_u(O)$, and $f > c$ on $E_r := (\psi_u|_O)^{-1}(B_r(0))$. Then:

(i) For $y \in \partial E_r$ and $t > 0$ we have $\varphi_{-t}(y) \in \mathrm{int}(E_r)$, since $\psi_u(\varphi_{-t}(y))$ strictly decreases in t, by the estimate above.

(ii) Every flow line in $D_q \setminus \{q\}$ meets ∂E_r exactly once: It meets the boundary at least once, as it starts in $q \in \mathrm{int}(E_r)$ and ends in $S(q \to)$ which is disjoint from E_r. If a flow line met the boundary at multiple times, there would be a time t_0 where the flow line reenters E_r, which forces the t differential of said flow line in the chart ψ_u to be less than or equal to 0 (the norm needs to be non-increasing), which contradicts the estimate above.

Now we come to the homeomorphism: By (ii), the map

$$\Theta : \{(y,s) \mid y \in \partial E_r, 0 \leq s \leq \tau(y)\} \to D_q \setminus \mathrm{int}(E_r)\,, \quad \Theta(y,s) := \varphi_s(y)$$

is a continuous bijection from a compact set into a Hausdorff space and hence a homeomorphism. Now we define

$$G : D_q \to D_1 \subseteq \mathbb{R}^{\mathrm{ind}_q}\,, \quad \begin{cases} G(x) := \frac{1}{2r}\psi_u(x) & \text{for } x \in E_r, \\ G(\varphi_s(y)) := \left(\frac{1}{2} + \frac{s}{2\tau(y)}\right)\frac{\psi_u(y)}{r} & \text{for } y \in \partial E_r,\ 0 \leq s \leq \tau(y)\,. \end{cases}$$

Both formulas define something continuous (even smooth on their domain) and agree on ∂E_r. Hence G is continuous. Furthermore, G is bijective and the domain is compact, whilst the image is Hausdorff, giving us a homeomorphism that maps ∂D_q onto ∂D_1. Now let $\overline{\pi_S} : S^{\mathrm{ind}_q} \to \overline{\mathbb{R}^{\mathrm{ind}_q}}$ denote the function from Definition 3.3.21 and with this define

$$g_q^{-1} := \overline{\pi_S^{-1}}\left(\frac{G}{1 - \|G\|}\right) : D_q / S(q \to) \to S^{\mathrm{ind}_q} \quad \text{where } \frac{c}{0} := \infty.\ \text{for } c \in \mathbb{R}$$

Now only the statement about the orientation preservation of g_q at a neighbourhood of q is missing: Define $T_q := \max_{\partial E_r} \tau$. If $\tau(x) > T_q$ and $x \neq q$, let y be the point, where the flow line through x passes through the boundary ∂E_r. Then $x = \varphi_{-t}(y)$

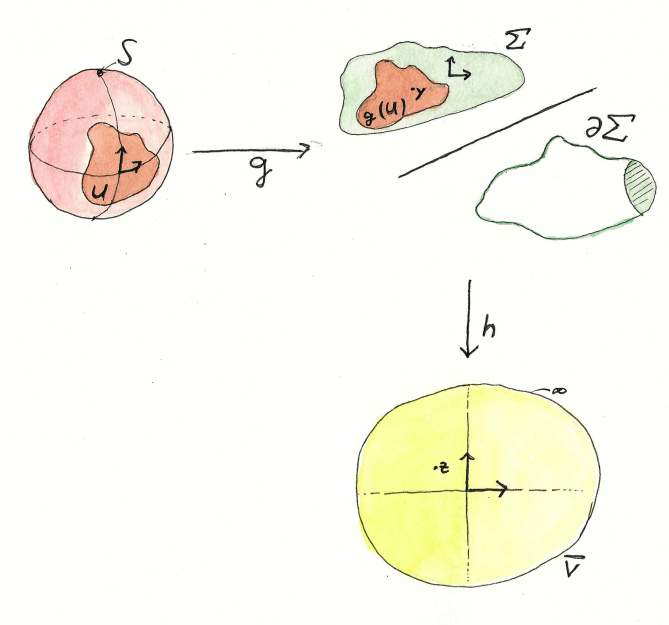


Figure 6: The setup of Lemma 6.2.5.

with $t = \tau(x) - \tau(y) > 0$ and thereby $x \in \operatorname{int}(E_r)$ by (i). Hence, $W_q \subseteq \operatorname{int}(E_r)$ and in said interior of E_r G is a diffeomorphism that is orientation preserving at q by the definition of ψ_u and by Lemma 4.1.18 we know that the sign of the determinant of a diffeomorphism of oriented manifolds is computed by a smooth procedure, and thereby it is locally constant, making G orientation preserving on the whole $\operatorname{int}(E_r)$. The map $\overline{\pi_S^{-1}}$ is orientation preserving: This follows from the orientation of S^{ind_q} by Remark 4.4.3. Furthermore, the stretching $v \mapsto \frac{v}{1-\|v\|}$ is orientation preserving where D_1 is oriented as a subspace of $\mathbb{R}^{\operatorname{ind}_q}$(this can be shown by looking at the sign of the determinant of its Jacobian). Hence the whole composition g_q^{-1} is orientation preserving and thereby also g_q is. □

lem:morseFilt r-orientation ClassOfADisk

Lemma 6.2.5. *Let $n \geq 1$ and let $(\Sigma, \partial\Sigma)$ be a compact pair ($\partial\Sigma$ denotes a closed subspace of Σ, and not some kind of boundary. In applications of the Lemma, it will be a boundary) such that $(\Sigma \setminus \partial\Sigma)$ is an oriented C^1-manifold of dimension n. Let $g : S^n \to \Sigma/\partial\Sigma$ be a homeomorphism with $g(S) = [\partial\Sigma]$ (where S denotes the south pole), and let $U \subseteq S^n \setminus S$ be open such that $g|_U$ is an orientation preserving C^1-diffeomorphism onto an open subset of $\Sigma \setminus \partial\Sigma$. Define*

$$o_\Sigma := (g^{-1})^*(\beta_n) \in \tilde{K}^{-n}(\Sigma/\partial\Sigma) = K^{-n}(\Sigma, \partial\Sigma)\,,$$

with β_n being the canonical generator defined in Definition 3.7.16. Let V be

an oriented n-dimensional vector space with K-theoretic orientation β_V and $h : \Sigma/\partial\Sigma \to \overline{V}$ be a pointed map such that for some $z \in V$ we have that $h^{-1}(z) = \{y\}$ with $y \in g(U)$, h is C^1 near y and $\mathrm{d}h|_y$ is invertible. Then

$$h^*(\beta_V) = \operatorname{sign}\det\left(\mathrm{d}h|_y\right) \cdot o_\Sigma$$

Proof. Let $\kappa_V : V \to \mathbb{R}^n$ be the coordinate map of a positively oriented basis, meaning that $\left(\overline{\pi_S^{-1}} \circ \overline{\kappa_V}\right)^* (\beta_n) = \beta_V$ (compare Corollary 4.2.5). Then $\xi := \overline{\pi_S^{-1}} \circ \overline{\kappa_V} \circ h \circ g : S^n \to S^n$ satisfies:

(i) $\xi^{-1}(\pi_S^{-1} \circ \kappa_V(z)) = \{g^{-1}(y)\}$,

(ii) ξ is C^1 near $g^{-1}(y) \in U$,

(iii) by the chain rule, Lemma 4.1.18 and since π_S^{-1}, κ_V and g are orientation preserving we have:

$$\operatorname{sign}\det(\mathrm{d}\xi|_{g^{-1}(y)}) = \operatorname{sign}\det(\mathrm{d}h|_y)\,.$$

Hence by the Single Regular Preimage Lemma 4.4.5 applied to ξ^* we can calculate

$$g^*h^*(\beta_V) = \xi^*(\beta_n) = \operatorname{sign}\det\left(\mathrm{d}h|_y\right) \cdot \beta_n\,.$$

Finally we can apply $(g^{-1})^*$ from the left to finish the proof. □

rem:morseFiltr-canonicalGeneratorsIndependOfChartAndEpsilon

Remark 6.2.6 (Canonical Generators)**.** Let ψ and $\tilde{\psi}$ be two refined oriented morse charts and $\varepsilon, \tilde{\varepsilon} > 0$ such that the boxes

$$(N_q, L_q) = (\psi^{-1}(D^s_\varepsilon \times D^u_\varepsilon)\,,\ \psi^{-1}(D^s_\varepsilon \times \partial D^u_\varepsilon)) \quad \text{and}$$
$$(\tilde{N_q}, \tilde{L_q}) = (\tilde{\psi^{-1}}(D^s_{\tilde{\varepsilon}} \times D^u_{\tilde{\varepsilon}})\,,\ \tilde{\psi}^{-1}(D^s_{\tilde{\varepsilon}} \times \partial D^u_{\tilde{\varepsilon}}))$$

are regular index pairs. Let furthermore

$$\kappa_q := \mathrm{d}\psi^{-1}\big|_0 \circ R_0 \circ s \circ \mathrm{pr}_u \circ \psi\ :\ (\psi^{-1}(D^s_\varepsilon \times D^u_\varepsilon), \psi^{-1}(D^s_\varepsilon \times \partial D^u_\varepsilon)) \to (\overline{T^u_q M}, \{\infty\}) \quad \text{and}$$
$$\tilde{\kappa}_q := \mathrm{d}\tilde{\psi}^{-1}\big|_0 \circ R_0 \circ \tilde{s} \circ \mathrm{pr}_u \circ \tilde{\psi}\ :\ (\psi^{-1}(D^s_\varepsilon \times D^u_\varepsilon), \psi^{-1}(D^s_\varepsilon \times \partial D^u_\varepsilon)) \to (\overline{T^u_q M}, \{\infty\})\,,$$

such that $\alpha_q^{(N_q,L_q)} := \kappa_q^*(\beta_{T^u_q M})$ is a generator of $\tilde{K}^{-\mathrm{ind}(q)}(N_q, L_q)$ and $\tilde{\alpha}_q^{(\tilde{N}_q,\tilde{L}_q)} := \tilde{\kappa}_q^*(\beta_{T^u_q M})$ is a generator of $\tilde{K}^{-\mathrm{ind}(q)}(\tilde{N}_q, \tilde{L}_q)$. Then the induced elements $(\alpha_q^{(N_q,L_q)})_\#$ and $(\tilde{\alpha}_q^{(\tilde{N}_q,\tilde{L}_q)})_\#$ agree.

Proof. We start by assumeing ε small enough such that in the box $\psi(N_q)$, the image of the unstable manifold $\psi(W(q \to) \cap N_q)$ is the graph of a C^1-function $f : D^u_\varepsilon \to D^s_\varepsilon$. (If ε from the Remark was to big we defined a third index pair in this step. We will later show that the inuced generator of this one agrees with all other.) Now we define $\Sigma := f(D^u_\varepsilon)$ to be oriented, via $T_q\Sigma = T^u_q M$. Let $\partial\Sigma := \Sigma \cap L_q$. Now let $j : (\Sigma, \partial\Sigma) \to (N_q, L_q)$ be the inclusion. Then we can apply Lemma 6.2.5 to the two maps:

$$\kappa_q \circ j\,, \quad \tilde{\kappa}_q \circ h \circ j\ :\ \Sigma/\partial\Sigma \to \overline{T^u_q M}\,.$$

- Restricted to Σ the map $pr_u \circ \psi$ is injective with $f(q) = 0$. Furthermore, one calculates that $\mathrm{d}(\kappa_q \circ j)|_q = \frac{1}{\varepsilon}\, \mathbb{1}_{T_q^u M}$.
- Σ lies in $W^u(q)$, which is flow invariant, and on Σ the map h is either the flow φ_{3t} or the collapse. For $\tilde{\varepsilon}$ small, $W^u(q) \cap \tilde{N}_q$ is the graph of the local unstable manifold in the chart $\tilde{\psi}$, so $\mathrm{pr}_u \circ \tilde{\psi}$ vanishes on it only at q. Since $\varphi_{3t}(x) = q$ if and only if $x = q$, we get $(\tilde{\kappa}_q \circ h \circ j)^{-1}(0) = \{q\}$. Near q we have $h = \varphi_{3t}$, hence $\mathrm{d}(\tilde{\kappa}_q \circ h \circ j)|_q = \frac{1}{\tilde{\varepsilon}}\, \mathrm{d}(\varphi_{3t})_q|_{T_q^u M}$. This has positive determinant, as $\tau \mapsto \mathrm{d}(\varphi_{3\tau t})_q|_{T_q^u M}$, $\tau \in [0,1]$, is a path in $\mathrm{GL}(T_q^u M)$ starting at $\mathbb{1}$.

Thereby the Lemma 6.2.5 gives us

$$j^* \circ \kappa_q^*(\beta_{T_q^u M}) = (\kappa_q \circ j)^*(\beta_{T_q^u M}) = o_\sigma = (\tilde{\kappa}_q \circ h \circ j)(\beta_{T_q^u M}) = j^* \circ h^* \circ \tilde{\kappa}_q^(\beta_{T_q^u M})$$

Now j^* is an isomorphism, as it sends the generator $\beta_{T_q^u M}$ to a generator o_σ, letting us apply its inverse to conclude that $\alpha_q^{(N_q, L_q)} = \kappa_q^*(\beta_{T_q^u M}) = h^* \circ \tilde{\kappa}_q^(\beta_{T_q^u M}) = h^*(\alpha_q^{(\tilde{N}_q, \tilde{L}_q)})$. Now two induced elements in a canonical isomorphism class agree, if for any group, the corresponding element agrees. Hence, $(\alpha_q^{(N_q, L_q)})_\# = (\tilde{\alpha}_q^{(\tilde{N}_q, \tilde{L}_q)})_\#$. □

rem:morseFilt r-orientDQwit hGQ

Remark 6.2.7. We can apply the Lemma above to the pair $(D_q, S(q \to))$ together with $g = g_q$, and $U = U_q$ from Lemma 6.2.4 (c).

lem:filt-orie ntedTubularEm bedding

Lemma 6.2.8 (Oriented Tubular Embedding)**.** *Let $B_p := W(\to p) \cap M^c$ and let $J : D_\varepsilon(\mathcal{N}) \to N_p$ be the tubular embedding from Lemma 6.2.3, where $\mathcal{N} \to B_p$ denotes the normal bundle. Now we can orient the normal bundle via the bundle isomorphism $l : \mathcal{N} \to Q^p$ where Q^p denotes the quotient bundle from Definition 6.2.2(the map is given by including the normal bundle into the tangent bundle and projecting then onto the quotient bundle). Then J can be chosen to be fibrewise orientation preserving, meaning that with the linear identification $R_0 : \mathcal{N}_x \to T_0 \mathcal{N}_x$ and the projection bundle map $pr : TM|_{W(\to p) \cap M^c} \to Q^p$ we have that for all x the map*

$$pr \circ \mathrm{d}J_x|_0 \circ R_0 : \mathcal{N}_x \to Q_x^p$$

is orientation preserving.

Proof. This follows directly from $pr \circ \mathrm{d}J_x|_0 \circ R_0 : \mathcal{N}_x \to Q_x^p$ being a bundle isomorphism. Hence it either is orientation preserving in all fibers, or reversing in all fibers (due to the connected base space). If the latter happens, we redefine J by composing it with a map ρ that reflects, say the first coordinate: Here we choose a global orthonormal trivialisation of $\mathcal{N}$ (which exists since the base space B_p is contractible) such that ρ maps the domain of the tubular embedding isomorphically onto itself. □

rem:morseFilt -OrientVjWith Functions

Remark 6.2.9. Let V_j be a fibre of the tubular neighbourhood from Lemma 6.2.3 that intersects $S(q \to)$ in x_j. Now we orient V_j such that the quotient map $T_{x_j} V_j \to$

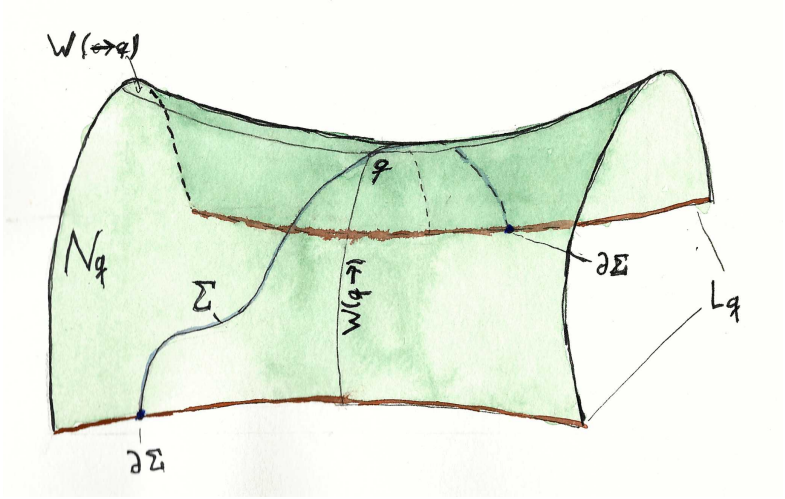


Figure 7: The definition of the transverse Σ from Lemma 6.2.10.

$(T_{x_j}M)/(T_{x_j}W(\to p) = Q^p_{x_j}$ is orientation preserving. Then V_j is the image of a fiber of the tubular embedding $J : \mathcal{N}^\varepsilon \to M$, where $\mathcal{N}^\varepsilon$ denotes the tube arround the 0 section of radius ε in the normal bundle $\mathcal{N}$. Let r_j be the radial map $\mathcal{N} \to \mathcal{N}^\varepsilon_{x_j}$ (the inverse of $v \mapsto (\frac{1}{\varepsilon - \|v\|})$), and let $\mathfrak{p}_j$ be an oriented basis of the normal bundle $\mathcal{N}$. Finally, let J be an oriented Tubular embedding. Then we define

$$g_j := \overline{J_{x_0}} \circ \overline{r_j} \circ \overline{\kappa_{\mathfrak{p}_j}} \circ \overline{\pi_S} : S^n \to V_j/\partial V_j \,.$$

Arround the north pole, the map g_j is a diffeomorphism orientation preserving, as $\overline{\pi_S}$ is, and so is $J_{x_0} \circ r_j \circ \kappa_{\mathfrak{p}_j}$ (which is true by Lemma 6.2.8). Hence, their compactifications are orientation preserving. Now for any oriented vector space V we define $h_j : \overline{\kappa_V^{-1}} \circ \overline{\pi_S} \circ g_j^{-1}$. in this setup, we can apply the Lemma 6.2.5 above with $h^{-1}(0) = \{x_j\}$ to conclude that $h_j^*(\beta_V) = o_{V_j} = (g_j^{-1})^*(\beta_n)$.(It is easy to see that h_j is locally a diffeomorphism and h as invertable differential in the origin.)

lem:morseFilt-canonicalGeneratorsOnTransversalDisks

Lemma 6.2.10. *Let $q \in \mathrm{Crit}(f)$ with $n := \mathrm{ind}(q) \geq 1$, let (N, L) be an index pair of $\{q\}$ and let $\Sigma \subseteq N$ with $\partial\Sigma \subseteq L$ be as in Lemma 6.2.5. Assume*

(i) $\Sigma \cap W(\to p) = \{x\}$ with $x \in g(U)$ and $T_x\Sigma \oplus T_xW(\to p) = T_xM$, ($\Sigma$ is transversal)

(ii) for every $s \geq 0$ there is a neighbourhood U_s of x in Σ with $\varphi_{[0,s]}(U_s) \subseteq N \setminus L$.

Orient $\Sigma \setminus \partial\Sigma$ such that the isomorphism $T_x\Sigma \to Q^p_x$ is orientation preserving.

Then for the inclusion $i : (\Sigma, \partial\Sigma) \to (N, L)$

$$i^*\left(\alpha_q^{(N,L)}\right) = o_\Sigma\,.$$

Remark 6.2.11. The two cases where we apply said lemma are:

1. $\Sigma = D_q,\ x = q,\ (N, L) = (C, B)$ (from Lemma 6.2.4) where $Q_q^q = T_qM/T_q^sM \cong T_q^uM$, so o_{D_q} is taken with respect to the orientation of T_q^uM.
2. $\Sigma = V_j\ x = x_j$ and $(N, L) = (B, A)$.

Proof of Lemma 6.2.10. We prove the Lemma in three steps. First we restriced the statement to the easier case, where $\Sigma \subseteq W(q \to)$. Then we define a new chart such that the stable manifold is not just a graph of a function, but a linear subspace. This makes sure that the projection onto the unstable coordinates collapses the stable part to a point which finally in step 3 generalizes the statement for arbitrary Σ:

Step 1. Let $\chi = (\chi_s, \chi_u)$ be a chart around q such that

1. $W(q \to)$ is the graph of function over the unstable coordinates,
2. the box (N_χ, L_χ) defined as

   $$(\chi^{-1}(D_\varepsilon^s \times D_\varepsilon^u), \chi^{-1}(D_\varepsilon^s \times \partial D_\varepsilon^u))$$

   is a regular index pair of $\{q\}$.
3. ε is small enough such that $W(q \to) \cap N_\chi$ is the graph of the function from above.
4. $\mathrm{d}\chi_q = \mathrm{d}\psi_q$ where ψ is a refined Morse chart.

The existence of such a chart follows from the local unstable manifold theorem 5.0.20.Let κ_χ be the map defined as

$$\kappa_\chi := \mathrm{d}\chi^{-1}\big|_0 \circ R_0 \circ s \circ \mathrm{pr}_u \circ \chi$$

similar to the construction of κ_q in Remark **??**. Let Σ be a disk contained in $W(q \to)$ and hence $\Sigma \cap W(\to q) = q$. Near q and for large $t > 0$ the homotopy equivalence $c : (N, L) \to (N_\chi, L_\chi)$ is given by $y \mapsto [\varphi_{3t}(y)]$(Compare Theorem 6.1.4 and notice that q is in the interior of both $(N \setminus L)$ and $(N_\chi \setminus L_\chi)$). Define $i : (\Sigma, \partial\Sigma) \to (N, L)$ to be the inclusion. Then consider the function

$$h := \underbrace{\mathrm{d}\chi^{-1}\big|_0 \circ R_0 \circ s \circ \mathrm{pr}_u \circ \chi}_{\kappa_\chi} \circ c \circ i : (\Sigma/\partial\Sigma) \to (\overline{T_q^uM})$$

If $h(y) = 0$, then y cannot be $[\partial\Sigma]$ as that point end at ionfinity. By definition of κ_χ we know that $\chi \circ c \circ i(y) \in D_\varepsilon^s \times \{0\}$. And since $y \in \Sigma \subseteq W(q \to)$, and c restricts to a map from $W(q \to)$ to $W(q \to)$ (restrictied to the right domain, and image), we know that

$\chi \circ c \circ i(y) \in \chi(W(q\to)) \cap D^s_\varepsilon \times \{0\}$. Finally, we assumed that $W(q\to)$ was the graph of the unstable coordinates in χ and hence, there is only one $x \in D^s_\varepsilon \times \{0\} \cap \chi(W(q\to))$, which is $(0,0)$. Hence $\varphi_{3t}(y) = q$ which is a fixpoint of the flow and thereby, $y = q$ and 0 has a unique preimage under h. Furthermore, $\mathrm{d}h|_q$ is an endomorphism on $T^u_q M$. We can calculate its determinant by calculating the jacobean determinants of $c \circ i$ and κ_χ in q (resp. in the origin) in a refined Morse chart seperately:

1. The function $\mathrm{d}(c\circ i)|_q$ (restricted in its image to $T^u_q M$) has positive determinant which follows from the flow in a refined morse chart in a critical point restricted to the unstable space has a differential that is diagonal with positive entries. This was calculated in Remark 5.0.17.

2. The function κ_χ can be differentiated in q:

$$\mathrm{d}\kappa_\chi = \mathrm{d}\Big(\mathrm{d}\chi^{-1}\big|_0 \circ R_0 \circ s \circ \mathrm{pr}_u \circ \chi \Big)$$

where $\iota := \mathrm{d}\chi^{-1}\big|_0 \circ R_0 : \{0\} \times \mathbb{R}^n \to T^u_q M$ is a linear isomorphism (since $\mathrm{d}\chi|_q$ maps $T^u_q M$ onto the unstable coordinate plane) and s is the radial stretching with $\mathrm{d}s|_0 = \lambda\,\mathrm{id}$, where $\lambda > 0$. On $T^u_q M$ the map $\mathrm{pr}_u \circ \mathrm{d}\chi|_q$ is the inverse of ι, so, identifying $T_0 T^u_q M$ with $T^u_q M$,

$$\mathrm{d}h|_q = \lambda\ \mathrm{d}\varphi_{3t}|_q \Big|_{T^u_q M} \in \mathrm{End}(T^u_q M)\,.$$

Its determinant is again positive, as the flow restricted to the unstable tangent space of a critical point is diagonal with positive entries.

Hence, $\mathrm{d}h|_q$ has positive determinant and Lemma 6.2.5 gives us the equality

$$i^* \circ c^* \circ \kappa^*_\chi(\beta_{T^u_q M}) = h^*(\beta_{T^u_q M}) = o_\Sigma\,.$$

This concludes the proof for the special case, where Σ is contained in $(q\to)$.

So far we have seen that for any chart χ that satisfies the properties from Step 1 there is the equality above. If we choose χ to be a refined morse chart, then $\kappa^*_\chi(\beta_{T^u_q M}) = \alpha_q^{(N_\chi, L_c hi)}$ is a canonical generator and a member of $(\alpha_q)_\#$.

Step 2 We know by the local unstable manifold theorem, that for our refined morse chart ψ there exists a smooth g_s such that $\psi\big(W(\to q)\big)$ is the graph of g_s with $g_s(0) = 0$ and $\mathrm{d}g_s|_0 = 0$. The chart $\tilde\psi := (\psi_s, \psi_u - g_s \circ \psi_s)$ satisfies $\mathrm{d}\tilde\psi\big|_q = \mathrm{d}\psi|_q$, $W(\to q) = \tilde\psi^{-1}(\mathbb{R}^{m-n} \times \{0\})$ (locally), $W(q\to)$ is still the graph of a function over the unstable coordinates and by Remark 6.1.21 ,for small ε, its box $(N_{\tilde\psi}, L_{\tilde\psi})$ is an index pair. We claim now, that $\kappa^*_{\tilde\psi}(\beta_{T^u_q M})$ is a member of the canonical generator $(\alpha_q)_\#$. Let $E := W(q\to) \cap N_\psi$, be the unstable disk in the box of ψ. In the chart ψ, E is the graph of a map

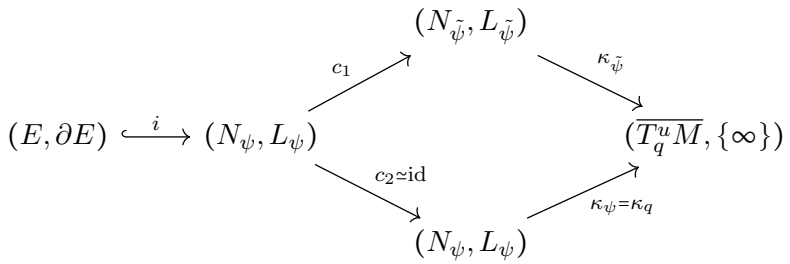
$$e : D^u_\varepsilon \to D^s_\varepsilon$$

Now we define the deformation homotopy for $t \in [0,1]$

Bild einfügen

$$H_t(l,x) := \big((1-t)l + te(x), x\big)$$

that shows that $(E,\partial E)$ is a strong deformation retract of (N_ψ, L_ψ) onto and hence induces an isomorphism $K^{-n}(N_\psi,L_\psi) \to K^{-n}(E,\partial E)$. Now we apply Step 1 to the two charts $\chi = \psi$ and $\chi = \tilde\psi$ with $\Sigma = E$, and $(N,L) = (N_\psi, L_\psi)$. This tells us that pulling back the canonical generator along both paths in the diagram



induce the same o_E. Now $c_2 \simeq \mathrm{id}$ and the inclusion i induces an isomorphism by the deformation above. Hence,

$$c_1^* \circ \kappa_{\tilde\psi}^*(\beta_{T^u_q M}) = \kappa_q^*(\beta_{T^u_q M}) = \alpha_q^{(N_\psi, L_\psi)}.$$

But the canonical generators are compatible with the maps c and c_1^* is an isomorphism, we know that $\alpha_q^{(N_{\tilde\psi}, L_{\tilde\psi})}$ is the unique element with $c_1^*(\alpha_q^{(N_{\tilde\psi}, L_{\tilde\psi})}) = \alpha_q^{(N_\psi, L_\psi)}$. Thereby

$$\kappa_{\tilde\psi}^*(\beta_{T^u_q M}) = \alpha_q^{(N_{\tilde\psi}, L_{\tilde\psi})}.$$

Step 3 Now we assume Σ to just satisfy the Lemmas conditions, use again $\tilde\psi$ and define the map on the quotients

$$h := \underbrace{\mathrm{d}\tilde\psi^{-1}\big|_0 \circ R_0 \circ s \circ \mathrm{pr}_u \circ \tilde\psi}_{\kappa_{\tilde\psi}} \circ c \circ i$$

with $c : (N,L) \to (N_{\tilde\psi}, L_{\tilde\psi})$ being the canonical homotopy equivalence as in Step 1. If $h(y) = 0$, then similar to before, we argue, that $h(y) = 0$ means that $\tilde\psi \circ c \circ i(y) \in D^s_\varepsilon \times \{0\}$ which in turn means that $c(y) = \varphi_{3t}(y) \in \tilde\psi^{-1}(D^s_\varepsilon \times \{0\})$. By definition of $\tilde\psi$ we made sure that $W(\to q) \supseteq \tilde\psi^{-1}(D^s_\varepsilon \times \{0\})$. Thereby, our $y \in \Sigma \cap W(\to q) = \{x\}$ by requirements on Σ. By (ii) and since $\varphi_{[t,3t]}(x)$ are close to q for large t our map h equals $h = \kappa_{\tilde\psi} \circ \varphi_{3t}$ near x. Due to the projection pr_u, $\kappa_{\tilde\psi}$ is constant on $W(\to q)$ (for a small enough box), and the flow φ_{3t} preserves $W(\to q)$. Thereby, the differential factors through the kernel of $\mathrm{d}(\mathrm{pr}_u \circ \tilde\psi)_{\varphi_{3t}(x)} = T_{\varphi_{3t}(x)} W(\to q)$. And since Σ intersects $W(\to q)$ transversally we factor through the composition

$$\begin{array}{ccc} T_x\Sigma & \xrightarrow{\mathrm{d}h|_x} & T^u_q M \\ \cong \big\downarrow & & \big\uparrow [\mathrm{d}\kappa_{\tilde\psi}] \\ Q^q_x & \xrightarrow{[\mathrm{d}\varphi_{3t}]} & Q^q_{\varphi_{3t}(x)} \end{array}$$

Now the first isomorphism is orientation preserving, by the choice of orientation on Σ.

Since φ_s preserves $W(\to q)$, its differential induces a bundle morphism $[\mathrm{d}\varphi_s] : Q^q \to Q^q$ over $\varphi_s|_{W(\to q)}$ which is an isomorphism on every fibre. As $W(\to q)$ is connected, by Proposition 4.1.12 it suffices to check that $[\mathrm{d}\varphi_s]$ is orientation preserving at a single point. We take q: since $\varphi_s(q) = q$, the map $[\mathrm{d}\varphi_s]_q$ is an endomorphism of Q^q_q, and with respect to the basis $([\partial_i(q)])_i$ given by the unstable coordinate vectors of the refined Morse chart it is diagonal with entries $e^{2\mu_i s} > 0$ by Remark 5.0.17. Hence it has positive determinant, and $[\mathrm{d}\varphi_{3t}]$ is orientation preserving (note $3t > 0$).

Similarly, $\kappa_{\tilde{\psi}}$ is constant 0 on the local stable disk $W(\to q) \cap N_{\tilde{\psi}}$, and its differential induces a bundle morphism $[\mathrm{d}\kappa_{\tilde{\psi}}]$ from $Q^q|_{W(\to q)\cap N_{\tilde{\psi}}}$ to the oriented vector space $T^u_q M$ (a bundle over the point 0). It is an isomorphism on every fibre, since $\mathrm{pr}_u \circ \mathrm{d}\tilde{\psi}\big|_y$ is surjective with kernel $T_y W(\to q)$ and $\tilde{\psi}_u(y) = 0$. The local stable disk is connected, and at $y = q$, under the identification $Q^q_q = T_q M / T^s_q M \cong T^u_q M$ by which Q^q is oriented (Proposition 5.1.5), $[\mathrm{d}\kappa_{\tilde{\psi}}]_q$ is the endomorphism $\lambda\,\mathrm{id}$ of $T^u_q M$ with $\lambda > 0$. Hence $[\mathrm{d}\kappa_{\tilde{\psi}}]$ is orientation preserving, and so is the composition $\mathrm{d}h|_x$.

Lemma 6.2.5 gives us

$$o_\Sigma = h^*(\beta_{T^u_q M}) = i^* \circ c^* \circ \kappa^*_{\tilde{\psi}}(\beta_{T^u_q M})$$

Then, Step 2 gives us that $\kappa^*_{\tilde{\psi}}(\beta_{T^u_q M}) = \alpha_q^{(N_{\tilde{\psi}}, L_{\tilde{\psi}})} \in (\alpha_q)_\#$. Thereby $c^* \circ \kappa^*_{\tilde{\psi}}(\beta_{T^u_q M}) = \alpha_q^{(N,L)}$ by definition. This finally concludes the proof. □

Definition 6.2.12. Let $\varphi_{l,r} : S^l \wedge S^r \to S^{l+r}$ be the map from Definition 3.3.27. Then the map

$$\sigma_n : \tilde{K}^{-n}(Y) \to \tilde{K}^{-n+1}(\Sigma Y)$$

given by the identifications

$$\sigma_n : \tilde{K}^{-n}(Y) = \tilde{K}(S^n \wedge Y) \xrightarrow{(\varphi_{n-1,1} \wedge \mathrm{id}_Y)^*} \tilde{K}(S^{n-1} \wedge S^1 \wedge Y) = \tilde{K}^{-n+1}(\Sigma Y)$$

is natural, meaning for any $f : Y \to Y'$ the diagram

$$\begin{array}{ccc} \tilde{K}^{-n}(Y') & \xrightarrow{\sigma_n} & \tilde{K}^{-n+1}(\Sigma Y') \\ \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle \Sigma(f)^*} \\ \tilde{K}^{-n}(Y) & \xrightarrow{\sigma_n} & \tilde{K}^{-n+1}(\Sigma Y) \end{array}$$

comutes.

Proof. This follows directly from the definitions. Remember that $\Sigma(f) : S^1 \wedge Y \to S^1 \wedge Y'$ is given by $\mathrm{id}_{S^1} \wedge f$, and

$$\Sigma(f)^* : \tilde{K}^{-n+1}(\Sigma Y') = \tilde{K}(S^{-n+1} \wedge \Sigma Y') \xrightarrow{(\mathrm{id}_{S^{n-1}} \wedge \mathrm{id}_{S^1} \wedge f)^*} \tilde{K}(S^{-n+1} \wedge \Sigma Y) = \tilde{K}^{-n+1}(\Sigma Y)$$

Now there is the obvious equality

$$(\varphi_{n-1,1}) \circ (\mathrm{id}_{S^{n-1}} \wedge S^1) = (\mathrm{id}_{S^n}) \circ \varphi_{n-1,1}$$

That follows from the definition of $\varphi_{l,r} = \pi_S^{l+r} \circ \Xi^{-1} \circ (\pi_S^l \wedge \pi_S^r)$ together with the naturality of Ξ. But then we can just calculate

$$\Sigma(f)^* \circ \sigma_n = \big((\varphi_{n-1,1} \wedge \mathrm{id}_Y) \circ (\mathrm{id}_{S^{n-1}} \wedge \mathrm{id}_{S^1} \wedge f)\big)^* = \big((\mathrm{id}_{S^n} \wedge f) \circ \varphi_{n-1,1} \wedge \mathrm{id}_Y\big)^* = \sigma_n \circ f^*$$

□

lem:morseFilt-connectingHomomorphismAlongACollar

Lemma 6.2.13 (Connecting Homomorphism along a Collar)**.** *Let (X, Y) be a compact pair (such that the inclusion $Y \to X$ is a cofibration) and X well pointed by $y_0 \in Y$. Let $s : X \to [0,1]$ be continuous with $s^{-1}(0) = Y$ and let $r : s^{-1}([0,1)) \to Y$ be a continuous retraction. Define, with $S^1 = I/\partial I$,*

$$\overline{k} : X/Y \to \Sigma Y = S^1 \wedge Y\,, \quad \overline{k}([z]) := \begin{cases} [1 - s(z), r(z)] & \text{if } s(z) < 1\,, \\ * & \text{if } s(z) = 1\,. \end{cases}$$

Then for all $n \geq 1$ the connecting homomorphism $\delta : \tilde{K}^{-n}(Y) \to K^{-n+1}(X, Y)$ of Definition 3.5.9 satisfies

$$\delta = (-1)^n \cdot \overline{k}^* \circ \sigma_n\,,$$

where $\sigma_n := (\varphi_{n-1,1} \wedge \mathrm{id}_Y)^* : \tilde{K}^{-n}(Y) = \tilde{K}(S^n \wedge Y) \to \tilde{K}(S^{n-1} \wedge S^1 \wedge Y) \cong \tilde{K}^{-(n-1)}(\Sigma Y)$ is the identification via the map $\varphi_{l,r} : S^l \wedge S^r \to S^{l+r}$ from Definition 3.3.27 together with the associativity of the wedge product and $\overline{k}^*$ denotes $(\mathrm{id}_{S^{n-1}} \wedge \overline{k})^*$.

Proof. Let $n = 1$. With the notation of Definition 3.5.9 let

(i) $\overline{m} : X \cup CY \to (X \cup CY)/X \cong CY/Y$

(ii) $l : CY/Y \to \Sigma Y$, given by $[y, t] \mapsto [t, y]$ as in Proposition 3.5.7.

(iii) $\delta = (\pi^*)^{-1} \circ \overline{m}^* \circ l^* \circ \sigma_1$. This is the same map as in Definition 3.5.9. We split θ into two parts and moved one part to our definition of $\overline{m}$ (The part that used to be the map $T : (X \cup CY)/(X) \cong CY/Y$) and the second part is our map l.

Now for $\mu \in [0,1]$ we define the map $H_\mu : X \cup CY \to \Sigma Y$ by

$$H_\mu([y,t]) := [(1-\mu)t, y] \text{ on } CY\,, \quad H_\mu(z) := [(1-\mu) + \mu s(z), r(z)] \text{ on } X$$

(and $*$ if $s(z) = 1$). The map H_μ is well-defined as both formulas agree on Y, continuous (for all, and in μ) as $s(z) \to 1, H_\mu(z) \to [\partial I]$ and pointed. Furthermore, $H_0 = l \circ \overline{m}$:

- On X: $l \circ \overline{m}(x) = *$ and $H_0(x) = [1, r(x)] = *$

- On CY: $l \circ \overline{m}([y,t]) = [t,y]$ and $H_0([y,t]) = [t,y]$,

and $H_1 = (R \wedge 1) \circ \overline{k} \circ \pi$ with the Reflection $R(t) = (1-t)$ and the quotient map $\pi : X \cup CY \to X/Y$:

- On CY: $(R \wedge 1) \circ \overline{k} \circ \pi([y,t]) = (R \wedge 1) \circ \overline{k}([\underbrace{Y}_{=s^{-1}(0)}]) = (R \wedge 1)([1, \underbrace{y'}_{y' \in Y}]) = [0,y'] = [0,y] = H_1([y,t])$
- On X: $(R \wedge 1) \circ \overline{k} \circ \pi(x) = (R \wedge 1)[1-s(x), r(x)] = [s(x), r(x)] = H_1(x)$.

Hence, by the homotopy invariance and Corollary 3.4.16:

$$\overline{m}^* \circ l^* = \pi^* \circ \overline{k}^* \circ (R \wedge 1)^* = -\pi^* \circ \overline{k}^* \quad \Rightarrow \quad \delta = -\overline{k}^* \circ \sigma_1 .$$

For $n \geq 2$, δ is by definition the connecting homomorphism of $(S^{n-1} \wedge X, S^{n-1} \wedge Y)$. Define the map $H'_\mu : (S^{n-1} \wedge X) \cup C(S^{n-1} \wedge Y) \to S^1 \wedge S^{n-1} \wedge Y$:

$$H'_\mu([[v,y],t]) := [(1-\mu)t, v, y], \quad H'_\mu([v,x]) := \begin{cases} [(1-\mu) + \mu s(x), v, r(x)] & \text{if s(x)¡1 ,} \\ * & \text{if s(x)=1 .} \end{cases}$$

Similar to H_μ, the map H' is a pointed homotopy from $l \circ \overline{m}$ to $(R \wedge 1) \circ \overline{k_n} \circ \pi$ where

$$\overline{k_n} : S^{n-1} \wedge (X/Y) \to S^1 \wedge S^{n-1} \wedge Y, [v,x] \mapsto [1-s(x), v, r(x)],$$

so $\delta = -\overline{k_n}^* \circ (\varphi_{1,n-1} \wedge \mathrm{id}_Y)^*$. Now $\overline{k_n} = (P \wedge \mathrm{id}_Y) \circ (\mathrm{id}_{S^{n-1}} \wedge \overline{k})$ where $P : S^{n-1} \wedge S^1 \to S^1 \wedge S^{n-1}$ is given by swapping the factors. Now consider the following commuting diagram

$$\begin{array}{ccc} S^{n-1} \wedge S^1 \wedge Y & \xrightarrow{P \wedge \mathrm{id}_Y} & S^1 \wedge S^{n-1} \wedge Y \\ \downarrow{\scriptstyle \varphi_{n-1,1} \wedge \mathrm{id}_Y} & & \downarrow{\scriptstyle \varphi_{1,n-1} \wedge \mathrm{id}_Y} \\ S^n \wedge Y & \xrightarrow{B} & S^n \wedge Y \\ \downarrow{\scriptstyle \overline{\pi_S} \wedge \mathrm{id}_Y} & & \downarrow{\scriptstyle \overline{\pi_S} \wedge \mathrm{id}_Y} \\ \overline{\mathbb{R}^n} \wedge Y & \xrightarrow{\overline{A} \wedge \mathrm{id}_Y} & \overline{\mathbb{R}^n} \wedge Y \end{array}$$

The middle arrow $B := (\varphi_{1,n-1} \circ P \circ (\varphi_{n-1,1})^{-1}) \wedge \mathrm{id}_Y$ is given by going around the top and commutes by definition. The bottom square commutes by definition of A together with the naturality of the map Ξ that builds up the $\varphi_{l,r} = (\overline{\pi_S^{l+r}})^{-1} \circ \Xi^{-1} \circ (\overline{\pi_S^l} \wedge \overline{\pi_S^r}) = (\overline{\pi_S^{l+r}})^{-1} \circ \overline{(\pi_S^l \times \pi_S^r)} \circ \Xi^{-1}$. (The notation and definition is from Definition 3.3.27) (We omittet the identification of $\mathbb{R}^l \times \mathbb{R}^r = \mathbb{R}^{l+r}$). Now $B = B' \wedge \mathrm{id}_Y$ with

$$B' = (\overline{\pi_S^n})^{-1} \circ \underbrace{\overline{(\pi_S^{n-1} \times \pi_S^1)} \circ \Xi^{-1} \circ P \circ \Xi \circ \overline{(\pi_S^1 \times \pi_S^{n-1})}^{-1}}_{A:\ \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}} \circ (\overline{\pi_S^n})$$

Finally the linear map A is given by moving the last coordinate to the front and hence has determinant $(-1)^{n-1}$. Hence, by Lemma 4.2.3 we conclude that $(\overline{A} \wedge \mathrm{id}_Y)^* = (-1)^{n-1} \cdot \mathrm{id}$. Hence $B^* = (\overline{\pi_S} \wedge \mathrm{id}_Y)^* \circ (-1)^{n-1} \cdot \mathrm{id} \circ (\overline{\pi_S} \wedge \mathrm{id}_Y)^{*-1} = (-1)^{n-1} \cdot \mathrm{id}$, as it is the conjugation of $(\overline{A} \wedge \mathrm{id}_Y)^*$ via an isomorphism (Alternativly one could say that $\pm\mathrm{id}$ commutes with grop homomorphisms). As the top square commutes we have

$$(P \wedge \mathrm{id}_Y)^* \circ (\varphi_{1,n-1} \wedge \mathrm{id}_Y)^* = (-1)^{n-1} (\varphi_{n-1,1} \wedge \mathrm{id}_Y)^* .$$

Putting everything together we conclude that

$$\begin{aligned}
\delta &= -\overline{k_n}^* \circ (\varphi_{1,n-1} \wedge \mathrm{id}_Y)^* \\
&= -(\mathrm{id}_{S^{n-1}} \wedge \overline{k})^* \circ \underbrace{(P \wedge \mathrm{id}_Y)^* \circ (\varphi_{1,n-1} \wedge \mathrm{id}_Y)^*}_{=(-1)^{n-1}(\varphi_{n-1,1} \wedge \mathrm{id}_Y)^*} \\
&= (-1)^n (\mathrm{id}_{S^{n-1}} \wedge \overline{k})^* \circ \underbrace{(\varphi_{n-1,1} \wedge \mathrm{id}_Y)^*}_{\sigma_n}
\end{aligned}$$

□

lem:morseFiltr-bottMapOnOrientedSuspension

Lemma 6.2.14. *Let V be an oriented real vector space of dimension $n \geq 1$, let $g : (0,1) \to \mathbb{R}$ be an orientation preserving diffeomorphism where $(0,1)$ is oriented as a subspace of $\mathbb{R}$ (e.g. $g(t) := \tan(\pi(t - \frac{1}{2}))$) and*

$$G_V : \Sigma\overline{V} = S^1 \wedge \overline{V} \to \overline{\mathbb{R} \oplus V}, \quad [t, v] \mapsto (g(t), v) .$$

Then G_V is a homeomorphism by Lemma 3.3.24. We can orient $\mathbb{R} \oplus V$ by putting $\mathbb{R}$ in front. Let $b = \beta \boxtimes (-)$ be the Bott map. Then there is an $e_n \in \{\pm 1\}$, depending only on n, with

$$b(\sigma_n(\beta_V)) = e_n \cdot G_V^*(\beta_{\mathbb{R} \oplus V}) \quad \in \tilde{K}^{-(n+1)}(\Sigma\overline{V}) ,$$

where $\sigma_n := (\varphi_{n-1,1} \wedge \mathrm{id}_{\overline{V}})^* : \tilde{K}^{-n}(\overline{V}) = \tilde{K}(S^n \wedge \overline{V}) \to \tilde{K}(S^{n-1} \wedge S^1 \wedge \overline{V}) \cong \tilde{K}^{-(n-1)}(\Sigma\overline{V})$ similar to Lemma 6.2.13.

Proof. Both sides of the equation are generators of $\tilde{K}^{-(n+1)}(\Sigma\overline{V}) \cong \mathbb{Z}$, as isomorphisms map generators to generators and the isomorphism to $\mathbb{Z}$ follows from $\Sigma\overline{V} \simeq S^{n+1}$. Hence both sides agree up to a sign $e(V)$:

$$b(\sigma_n(\beta_V)) = e(V) G_V^*(\beta_{\mathbb{R} \oplus V}) \quad \in \tilde{K}^{-(n+1)}(\Sigma\overline{V})$$

In the notation of Lemma 3.3.24, the map G_V is given by $\Xi^{-1} \circ (\overline{g} \wedge \mathrm{id}_{\overline{V}})$, where $\overline{g}$ is the extension $I/\partial I \to \overline{\mathbb{R}}$ of g with $\overline{g}(\infty) = \infty$. Let A be an isomorphism $V \to W$. Then

using the naturality of Ξ (we omit its indices) we can conclude that:

$$\begin{aligned} G_W \circ (\mathrm{id} \wedge \overline{A}) &= \Xi^{-1} \circ (\overline{g} \wedge \mathrm{id}_{\overline{W}}) \circ (\mathrm{id} \wedge \overline{A}) \\ &= \overline{(g \times \mathrm{id}_W)} \circ \Xi^{-1} \circ (\mathrm{id} \wedge \overline{A}) \\ &= \overline{(g \times \mathrm{id}_W)} \circ \overline{(\mathrm{id} \times A)} \circ \Xi^{-1} \\ &= \overline{(\mathrm{id} \oplus A)} \circ \overline{(g \times \mathrm{id}_V)} \circ \Xi^{-1} \\ &= \overline{(\mathrm{id} \oplus A)} \circ \underbrace{\Xi^{-1} \circ (\overline{g} \wedge \overline{\mathrm{id}_V})}_{G_V} \end{aligned}$$

Now by Corollary 4.2.6 we know that $\overline{A}^* \beta_W = \mathrm{sign}(\det(A))\beta_V$ and $\overline{\mathrm{id} \oplus A}^* (\beta_{\mathbb{R} \oplus W}) = \mathrm{sign}\det(A)\beta_{\mathbb{R} \oplus V}$. Now b and σ_n are natural, hence commute with topological induced maps:

$$\begin{aligned} (\mathrm{id} \wedge \overline{A})^* \circ b \circ \sigma_n(\beta_W) &= b \circ \sigma_n \circ (\overline{A}^* \beta_W) = \mathrm{sign}\det(A) \cdot b \circ \sigma_n(\beta_V) \\ (\mathrm{id} \wedge \overline{A})^* \circ G_W^*(\beta_{\mathbb{R} \oplus W}) &= (G_W \circ (\mathrm{id} \wedge \overline{A}))^* \beta_{\mathbb{R} \oplus W} = G_V^* \circ \overline{(\mathrm{id} \oplus A)}^* \beta_{\mathbb{R} \oplus W} \\ &= \mathrm{sign}(\det(A)) \cdot G_V^* \beta_{\mathbb{R} \oplus V} \end{aligned}$$

We now claim that for fixed n the signs $e(V)$ and $e(W)$ defined above agree (and are thereby independent of the chosen vector space). For this we consider the diagram below, that commutes by the considerations above (naturality of b and σ_n and Ξ):

$$\begin{array}{ccccc} \tilde{K}^{-n}(\overline{W}) & \xrightarrow{b \circ \sigma_n} & \tilde{K}^{-(n+1)}(\Sigma \overline{W}) & \xleftarrow{G_W^*} & \tilde{K}^{-(n+1)}(\overline{\mathbb{R} \oplus W}) \\ \downarrow{\scriptstyle \overline{A}^*} & & \downarrow{\scriptstyle (\mathrm{id} \wedge \overline{A})^*} & & \downarrow{\scriptstyle \overline{\mathrm{id} \oplus A}^*} \\ \tilde{K}^{-n}(\overline{V}) & \xrightarrow[b \circ \sigma_n]{} & \tilde{K}^{-(n+1)}(\Sigma \overline{V}) & \xleftarrow[G_V^*]{} & \tilde{K}^{-(n+1)}(\overline{\mathbb{R} \oplus V}) \end{array}$$

This lets us compute:

$$\begin{aligned} (\mathrm{id} \wedge \overline{A})^* \circ (b \circ \sigma_n)(\beta_W) &= \mathrm{sign}\det(A) \cdot \underbrace{b \circ \sigma_n(\beta_V)}_{e(V) G_V^*(\beta_{\mathbb{R} \oplus V})} \\ &= e(V) \cdot \underbrace{\mathrm{sign}\det(A) \cdot G_V^*(\beta_{\mathbb{R} \oplus V})}_{(\mathrm{id} \wedge \overline{A})^* \circ G_W^*(\beta_{\mathbb{R} \oplus W})} \\ &= e(V) \cdot (\mathrm{id} \wedge \overline{A})^* \circ G_W^*(\beta_{\mathbb{R} \oplus W}) \end{aligned}$$

Now applying $((\mathrm{id} \wedge \overline{A})^*)^{-1}$ from the left (which exists as it is induced by a homeomorphism) gives us that the sign $e(V)$ satisfies the definition of $e(W)$ making them equal. Thereby $e(V)$ is independent of V and only depends on n (and potentially g). Finally, it is invariant under the choice of orientation preserving diffeomorphism g, as any two such maps g_0 and g_1 are homotopic via the homotopy $(1-s)g_0 + sg_1$ that extends to a homotopy of the compactifications. Hence $(G_V)^*$ and thus $e(V)$ do not depend on g. □

Remark 6.2.15. Let e_n be the sign from Lemma 6.2.14. Then $e_n = (-1)^n$.

Proof. By Lemma 4.2.7 we know that $\beta_{\mathbb{R}\oplus V} = \Xi^*(\beta_{\mathbb{R}} \boxtimes \beta_V)$. Now with $G_V = \Xi^{-1}(\overline{g} \wedge \mathrm{id})$ and by naturality of $\boxtimes$ we have that

$$G_V^*(\beta_{\mathbb{R}\oplus V}) = (\overline{g} \wedge \mathrm{id})^*(\beta_{\mathbb{R}} \boxtimes \beta_V) = (\overline{g}^* \beta_{\mathbb{R}}) \boxtimes \beta_V = \beta_1 \boxtimes \beta_V$$

On the other side we have $b \circ \sigma_n(\beta_V) = \beta \boxtimes (\sigma_n(\beta_V))$. Now for any $x \in \tilde{K}^{-n}(Y)$ we have

$$\beta \boxtimes \sigma_n(x) = (-1)^n \beta_1 \boxtimes_{1,n} x \in \tilde{K}^{-(n+1)}(\Sigma Y)\,.$$

As both sides are pullbacks of the same product (in the 0th degree) $\beta \boxtimes_{0,0} x \in \tilde{K}(S^1 \wedge S^1 \wedge S^n \wedge Y)$ via two different maps

$$S^{n+1} \wedge S^1 \wedge Y \to S^1 \wedge S^1 \wedge S^n \wedge Y.$$

By definition, σ_n takes the last coordinate of the S^n part and maps it to the S^1 part. If we give all the coordinates names the left side acts as

$$\begin{aligned} S^{n+1} \wedge S^1 \wedge Y &\to S^1 \wedge S^1 \wedge S^n \wedge Y \\ (d_1, \dots d_{n+1}, s_1, y) &\mapsto (\underbrace{d_1}_{S^1}, \underbrace{d_2}_{S^1}, \underbrace{d_3, \dots, d_{n+1}, s_1}_{S^n}, \underbrace{y}_{Y}) \end{aligned}$$

where d_1 and d_2 correspond to β.

The right side is defined in Definition 3.7.8 as the reduced external product pulled back along a switch of coordinates that swiches the sphere S^n coming from $x \in \tilde{K}(S^n \wedge Y)$ with the sphere S^1 corresponding to β_1, or in our notation:

$$\begin{aligned} S^{n+1} \wedge S^1 \wedge Y &\to S^1 \wedge S^1 \wedge S^n \wedge Y \\ (d_1, \dots d_{n+1}, s_1, y) &\mapsto (\underbrace{d_1}_{S^1}, \underbrace{s_1}_{S^1}, \underbrace{d_2, d_{n+1}}_{S^n}, \underbrace{y}_{Y}) \end{aligned}$$

Hence, we can translate $\beta_1 \boxtimes \sigma_n(x)$ into $\beta \boxtimes_{1,n} x$ by pulling back along $P \wedge \mathrm{id}_Y$, whilst P as a linear map between vector spaces has determinant $(-1)^n$ (we need to move s_1 past n basis vectors corresponding to $d_2, \dots d_{n+1}$), letting us apply Lemma 4.2.3 to conclude its pullback is the identity times $(-1)^n$. With $x = \beta_V$ we have that

$$b \circ \sigma_n(\beta_V) = (-1)^n \beta_1 \boxtimes \beta_V = (-1)^n G_V^* \beta_{\mathbb{R}\oplus V}$$

and hence $e_n = (-1)^n$. □

lem:morseFiltr-contributionOfASigleFlowLine

Lemma 6.2.16. *Let $n := \mathrm{ind}_p \geq 1$, $V_j \subseteq S(q \to)$ a fiber of the tubular neighbourhood N_p with center $x_j \in W(q \to p)$ and denote the flow line through x_j by $\gamma_j \in \mathcal{M}(q \to p)$. Orient V_j such that the quotient map $T_{x_j} V_j \to Q^p_{x_j}$ is orientation preserving.(it is easy to see that it is an isomorphism). Let $l_j : S(q \to) \to V_j/\partial V_j$ be the collap and δ be the connecting homomorphism of the pair $(D_q, S(q \to))$. Then*

$$b \circ \delta \circ l^*(o_{V_j}) = e(\gamma) o_{D_q}\,,$$

where $o_{D_q} = (g_q^{-1})^*(\beta_{n+1})$ with $g_q : S^{n+1} \to D_q/\partial S(q\to)$ as in Lemma 6.2.4 (c), and $o_V = (g_j^{-1})^*(\beta_n)$ as in Remark 6.2.9. Said different, the diagram commutes:

$$\begin{array}{ccccc} \tilde{K}^n(V_j/\partial V_j) & \xrightarrow{l_j^*} & \tilde{K}^{-n}(S(q\to)) & \xrightarrow{\delta} & \tilde{K}^{-n+1}(D_q, S(q\to)) \\ & \searrow^{o_{V_j}\mapsto e(\gamma_j)(o_{D_q})} & & & \downarrow b \\ & & & & \tilde{K}^{-(n+1)}(D_q, S(q\to)) \end{array}$$

Proof. By Lemma 6.2.4 we know that the map g_q restricts to an orientation preserving diffeomorphism over the set $\{x \in D_q \mid \tau(x) > T_q\}$ for some $T_q \geq 0$.Let $T_0 > 2T_q$ and define the maps

$$s : D_q \to [1,\infty), \quad x \mapsto \min\{\frac{\tau(x)}{T_0}, 1\}$$

and

$$r : D_q \to S(q\to), \quad x \mapsto \rho(x).$$

Now we consider the vectorspace $Q^p_{x_j}$ and its corresponding generator $o_{V_j} = h_j^*(\beta_{Q^p_{x_j}})$ where h_j is the mapping $(V_j/\partial V_j) \to \overline{Q^p_{x_j}}$ Now we want to calculate the connecting homomorphism along the collar $S(q)$ in D_q letting us apply Lemma 6.2.13, to the pair with the map s and r. Furthermore, the naturality of σ_n and b let us deduce that

$$\begin{aligned} b\circ\delta\circ l_j^*\circ h_j^*(\beta_{Q^p_{x_j}}) &= (-1)^n\cdot b\circ\overline{k}^*\circ\sigma_n\circ l_j^*\circ h_j^*(\beta_{Q^p_{x_j}}) \\ &= (-1)^n\cdot(\Sigma(h_j\circ l_j)\circ\overline{k})^* b\circ\sigma_n(\beta_{Q^p_{x_j}}) \\ &= (\underbrace{G_V\circ\Sigma(h_j\circ l_j)\circ\overline{k}}_{:=F})^*(\beta_{\mathbb{R}\oplus Q^p_{x_j}}) \end{aligned} \tag{49}$$

{eq:morseFiltr-boundaryMorphismInASingleFlowLine}

Now we can calculate $F : D_q/S(q\to) \to \overline{R\oplus Q^p_{x_j}}$ explicitly:

$$F(z) \mapsto \begin{cases} [g(1-s(z)), h_j\circ l_j\circ r(z)] & \text{if } s(z) \text{ and } z\notin S(q\to) < 1 \Leftrightarrow 0 < \tau(x) < T_0 \\ [S(q\to)] & \text{else}, \end{cases}$$

where g denotes a orientation preserving diffeomorphism $(0,1)\to\mathbb{R}$. Gemoetrically, the map $g(1-s(z))$ increases, as we move towards $S(q\to)$. This is the place, where we make sure to point the first basis vector downward, and builds the geometric bridge to the morse co-boundary map. But now $F(z) = 0$, if and only if $s(z) = 1$ meaning that $2\tau(z) = T_0$ and $\rho(z) = x_j$ meaning that $F^{-1}(0) = \{\varphi_{-\frac{1}{2}T_0}(x_j)\} = \{\hat{x_j}\}$. Let $(w_1,\dots,w_n)$ be a positive basis of $T_{x_j}V_j = T_{x_j}S(q\to)$. By Lemma 6.2.4 (a) and (b) we have that $\mathrm{d}(\tau)(-\mathrm{grad}(f)) = -1$ and $s(\hat{x_j}) = \frac{1}{2}$. Hence,

$$\mathrm{d}_{\hat{x_j}}(g(1-s(z)))(-\mathrm{grad}_{\hat{x_j}}(f)) = g'(\frac{1}{2})\cdot\frac{1}{T_0}.$$

And since $\mathrm{d}\rho(-\mathrm{grad}(f)) = 0$ we have

$$\mathrm{d}F|_{\hat{x_j}}\left(-\mathrm{grad}_{\hat{x_j}}(f)\right) = \left(g'(\frac{1}{2})\cdot\frac{1}{T_0}, 0\right).$$

Similar we calculate

$$\mathrm{d}F|_{\hat{x}_j}\left(\mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_i\right) = \left(0, \mathrm{d}h_j|_{x_j} w_i\right).$$

Since h_j is locally a diffeomorphism, and g was choosen to be one, we get basis

$$\begin{aligned}&\left(\mathrm{d}F|_{\hat{x_j}}\left(g(1-s(z))\right)(-\mathrm{grad}_{\hat{x_j}}(f)),\ \mathrm{d}F|_{\hat{x_j}}\left(\mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_1\right),\ \ldots,\ \mathrm{d}F|_{\hat{x_j}}\left(\mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_n\right)\right)\\ &= \mathrm{d}F|_{\hat{x_j}}\left(-\mathrm{grad}_{\hat{x_j}}, \mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_1, \ldots, \mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_n\right)\end{aligned}$$

of $\mathbb{R}\oplus Q^p_{x_j}$. This basis is positive, as long as $\mathrm{d}h_j|_{x_j} : T_{x_j}(V_j) \to Q^p_{x_j}$ is orientation preserving which we already figured ou to be true in Lemma 6.2.9. Hence, we introduced a positive basis via $\mathrm{d}F|_{\hat{x_j}}$. Hence, the map $\mathrm{d}f|_{\hat{x}_j}$ is orientation preserving, if and only if the basis

$$\begin{aligned}&\left(-\mathrm{grad}_{\hat{x}_j}, \mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_1, \ldots, \mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_n\right)\\ &\left(-\mathrm{grad}_{\varphi_{-(T_0/2)}(x_j)}, \mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_1, \ldots, \mathrm{d}_{\varphi_{-(T_0/2)}}\big|_{x_j} w_n\right)\end{aligned}$$

is positive in $T_{\hat{x_j}}W(q\to)$. This basis is part of an oriented local frame over $\{\varphi_{-s}(x_j) \mid s\in[0,\infty)\}\subset D_q$. In $s=0$ this frame gives us the basis

$$(-\mathrm{grad}_{x_j}(f), w_1, \ldots, w_n)$$

Which is an oriented basis of $TW(q\to)$ that carries the same orientation, as the fiber over x_j in the oriented bundle that inherits its orientation via the exact sequence

$$0 \longrightarrow T\gamma_j \xrightarrow{\ i\ } TW(q\to)|_{\gamma_j} \xrightarrow{\ p\ } TM/TW(\to p)\big|_{\gamma_j} \longrightarrow 0$$

Hence, local frame is oriented for all s, iff the frame is oriented for $s=1$, which happens, if the induced orientation of the exact sequence agrees with the one in $W(q\to)$. Hence, the orientability of $\mathrm{d}f|_{\hat{x}_j}$ is equivalent to $e(\gamma_j)=1$.

We can now continue our calculation in equation (49) and apply Lemma 6.2.5 to conclude that

$$b\circ\delta\circ l_j^*\circ\underbrace{h_j^*(\beta_{Q^p_{x_j}})}_{(o_{V_j})} = e(\gamma_j)(\beta_{0_{D_q}}).$$

□

local frames are oriented with respect to the subspace orientation, iff in any fiber, the base is oriented

Proof of Theorem 6.2.1. For all $l\in\mathbb{Z}$ we get the commuting square

$$\begin{array}{ccc} K_{\#}^{-(k-1)-l}(\mathcal{IP}_{k-1}) & \xrightarrow{\Delta^k} & K_{\#}^{-(k)-l}(\mathcal{IP}_k) \\ \downarrow b & & \downarrow b \\ K_{\#}^{-(k-1)-l-2}(\mathcal{IP}_{k-1}) & \xrightarrow{\Delta^k} & K_{\#}^{-(k)-l-2}(\mathcal{IP}_k) \end{array}$$

Hence, we only need to proof the original statement for $l = 0, 1$. For $l = 1$ the bottom row of the diagram 36 is trivial, as the connected simple systems $\mathcal{IP}_k$ consists of spaces that are homotopically equivalent to k-spheres and we know by 3.7.15 that $K^{-k-1}(S^k) \cong \tilde{K}(S^{2k+1}) = 0$. The same argument works for the image of Δ^k for $l = 1$. So for the rest of the proof we assume $l = 0$. Now since N_p is a tubular neighbourhood of $W(\to p) \cap M^c$ we know that $N_p \cap S(q \to)$ consists of finitly many components V_j and each component has one $\{x_j\} = W(q \to p) \cap V_j$. In fact, the proof of Lemma 6.2.3 shows that the V_j are fibers of the normal neighbourhood of $W(\to p)$ inside $W(q \to)$ and hence homeomorphic to $\mathbb{R}$. Now we take the tiple (C, B, A) from Definition 6.2.2. We have a map of triples: referenz!

$$\left(W(q \to) \cap N_q \ , \ S(q \to) \ , \ S(q \to) \setminus \bigcup_j \operatorname{int}(V_j) \right) \to (C, B, A)$$

giving us a commuting diagramm (the square bracets for the input of the K funktors $K[-]$ are choosen for readability and have no deeper meaning)

$$\begin{array}{ccc} K^{-\lambda_p}[B, A] & \xrightarrow{\delta^1_{triple}} & K^{-\lambda_q}[C, B] \\ \downarrow i^*_{B,A} & & \downarrow i^*_{C,B} \\ K^{-\lambda_p}\left[S(q \to), S(q \to) \setminus \bigcup_j \operatorname{int}(V_j)\right] & \xrightarrow{\delta^2_{triple}} & K^{-\lambda_q}[W(q \to) \cap N_q, S(q \to)] \\ i_j^* \downarrow \uparrow c_j^* & \nearrow_{f^j} & \\ K^{-\lambda_p}[V_j, \partial V_j] & & \end{array}$$

diag:morseFiltr-boundaryMapsummonds

Diagram 50

Now $i^*_{C,B}$ is an isomorphism, because the collapsing of the tubular neighbourhood N_q induces a homotopy equivalence

$$\frac{N_q \cup N_p}{N_p \cap L_q} \cong \frac{N_q}{L_q} \to \frac{W(q \to) \cap M_c}{L_q \cap W(q \to)} \cong \frac{W(q \to) \cap N_q}{S(q \to)}$$

that is inverse to the inclusion (once the pairs are viewed as pointed quotients).

Due to the quotient

$$S(q \to) \Big/ S(q \to) \setminus \bigcup_j \operatorname{int}(V_j) \cong \bigvee_j \left(V_j / \partial V_j\right)$$

being a pointed sum, the map δ^2_{triple} is a sum of the functions $f^j \circ i_j^*$ where the $f^j : K^{-\lambda_p}[V_j, \partial V_j] \to K^{-\lambda_q}[W(q\to) \cap N_q, S(q\to)]$ are defined to be $\delta^2_{triple} \circ c_j^*$.

Now we will inspect said composition more closely, for which we need to dive into the definition of the triple connecting homomorphism, which was given by the composition of the connecting homomophism of the pair after the pullback alng the inclusion:

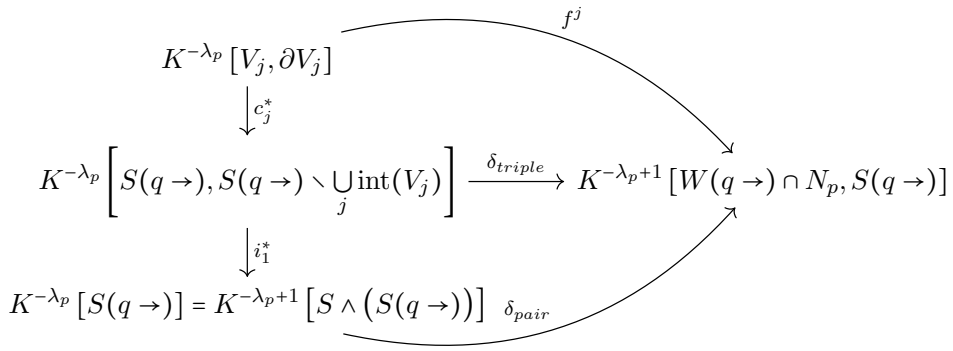


Diagram 51

Now notice that the composition $i_1^* \circ c_j^*$ is induced by the collapsing map

$$l_j : (S(q\to)) \to (V_j / \partial V_j)\,.$$

Hence, $f_j = \delta \circ l_j^*$. Now by diagramm 50 together with Theorem we know that

label additivität von K theorie

$$b \circ \delta^1_{triple} = (i^*_{C,B})^{-1} \sum_j f^j \circ i_j^* \circ i^*_{B,A} = (i^*_{C,B})^{-1} \sum_j \delta \circ l_j^* \left(i_j^* \circ i^*_{B,A}\right).$$

Let $\alpha_p^{(B,A)} \in \alpha_\#(q)$ be a representant of a generator. Then Δ^{k-1} is induced by the sum of maps where each summand is a map given by $b \circ \delta_{triple}$. Hence, we want to calculate:

$$b \circ \delta_{triple}(\alpha_p^{(B,A)}) = (i^*_{C,B})^{-1} \sum_j b \circ \delta \circ l_j^* \left(i_j^* \circ i^*_{B,A} \alpha_p^{(B,A)}\right).$$

Here we used, that b commutes with pullbacks. Now we have the two datums that are adapted to Lemma 6.2.10:

1. $(V_j, \partial V_j)$ where condition one is assured by the Smale condition on f, together with the splitting $T_{x_j} W(\to p) = -\mathrm{grad}_{x_j} + T_{x_j} V_j$ (which is deduced by a dimension count). The second condition follows from continuity of the flow: $f(\varphi_{T+s}(x_j)) > a > a - \varepsilon$ for all $s \geq 0$. Hence by cojntinuity, for every s there is a small neighbourhood U_s such that $\varphi_{[}0, s] U_s \subseteq N_p \setminus L_p$.

2. $(D_q, S(q\to))$, where $D_q \subseteq C$. The first propertie is almost trivial, as $T_q D_q = T_q^u M$ which is a direct sum $T_q M = T_q^u M \oplus T_q^s M$ where the second summand is the tangend space of the stable manifold of q in q. The second property is again true by continuity of the flow, for which q is a fixed point.

The Lemma 6.2.10 gives us then:

1. $\left(i_j^* \circ i_{B,A}^* \alpha_p^{(B,A)}\right) = o_{V_j}$ and
2. $i_{C,B}^* \alpha_q^{C,B} = o_{D_Q}$.

Now we can apply Lemma 6.2.16 to calculate

$$b \circ \delta_{triple}(\alpha_p^{(B,A)}) = (i_{C,B}^*)^{-1} \sum_j b \circ \delta \circ l_j^*\left(o_{V_j}\right) = (i_{C,B}^*)^{-1} \sum_j e(\gamma_j) o_{D_q} = \sum_j e(\gamma_j) \alpha_q^{C,B}.$$

Now we need to sort out the algebraic definition of Δ and then we are done!

□

References

atiyah1989k [Ati89] M. F. Atiyah. *K-theory*. Second. Advanced Book Classics. Notes by D. W. Anderson. Redwood City, CA: Addison-Wesley Publishing Company Advanced Book Program, 1989, pp. xx+216. ISBN: 0-201-09394-4.

banyaga2004lectures [BH04] A. Banyaga and D. Hurtubise. *Lectures on Morse Homology*. Texts in the Mathematical Sciences. Springer Netherlands, 2004. ISBN: 9781402026959. URL: https://books.google.de/books?id=AX-_sbMjOK4C.

Brodzki1996 [Bro96] Jacek Brodzki. *An Introduction to K-theory and Cyclic Cohomology*. 1996. arXiv: funct-an/9606001. URL: https://arxiv.org/abs/funct-an/9606001.

Courtney2004 [Cou04] Dennis Courtney. *A Brief Glance at K-theory*. 2004. URL: https://math.berkeley.edu/~hutching/teach/215b-2004/courtney.pdf.

CliffordPreston1961 [CP61] A. H. Clifford and G. B. Preston. *The Algebraic Theory of Semigroups*. Vol. I. Mathematical Surveys 7. Providence, R.I.: American Mathematical Society, 1961.

Green1965 [Gre65] J. A. Green. *Sets and Groups*. London: Routledge & Kegan Paul, 1965.

hatcher2002algebraic [Hat02] Allen Hatcher. *Algebraic Topology*. Cambridge: Cambridge University Press, 2002. ISBN: 978-0-521-79540-1.

Landweber2005 [Lan05] Gregory D. Landweber. *K-Theory and Elliptic Operators*. 2005. arXiv: math/0504555. URL: https://arxiv.org/abs/math/0504555.

Perko2001 [Per01] Lawrence Perko. *Differential Equations and Dynamical Systems*. 3rd ed. Vol. 7. Texts in Applied Mathematics. Softcover / Paperback edition. New York / Berlin: Springer, 2001, p. 557. ISBN: 978-1461265269.

Salamon1985 [Sal85] Dietmar Salamon. "Connected simple systems and the Conley index of isolated invariant sets". In: *Transactions of the American Mathematical Society* 291.1 (1985), pp. 1–41. DOI: 10.1090/S0002-9947-1985-0797044-3.

Salamon1990 [Sal90] Dietmar A. Salamon. "Morse theory, the Conley index and Floer homology". In: *Bull. London Math. Soc.* 22.2 (1990). Bulletin of the London Mathematical Society, 22(2):113–140 (1990), pp. 113–140. DOI: 10.1112/blms/22.2.113.

Smale1961OnGradient [Sma61] Stephen Smale. "On Gradient Dynamical Systems". In: *Annals of Mathematics* 74.1 (1961), pp. 199–206. DOI: 10.2307/1970311.

Steenrod1967 [Ste67] N. E. Steenrod. "A convenient category of topological spaces". In: *Michigan Mathematical Journal* 14.2 (1967), pp. 133–152. DOI: 10.1307/mmj/1028999711.

strickland_cgwh [Str] Neil P. Strickland. *The Category of CGWH Spaces*. Unpublished note, https://ncatlab.org/nlab/files/StricklandCGWHSpaces.pdf. abgerufen am 25. Juni 2026.

tomdieck2008algebraic

[tom08] Tammo tom Dieck. *Algebraic Topology*. EMS Textbooks in Mathematics. European Mathematical Society (EMS), Zürich, 2008. ISBN: 978-3-03719-048-7. DOI: 10.4171/048.